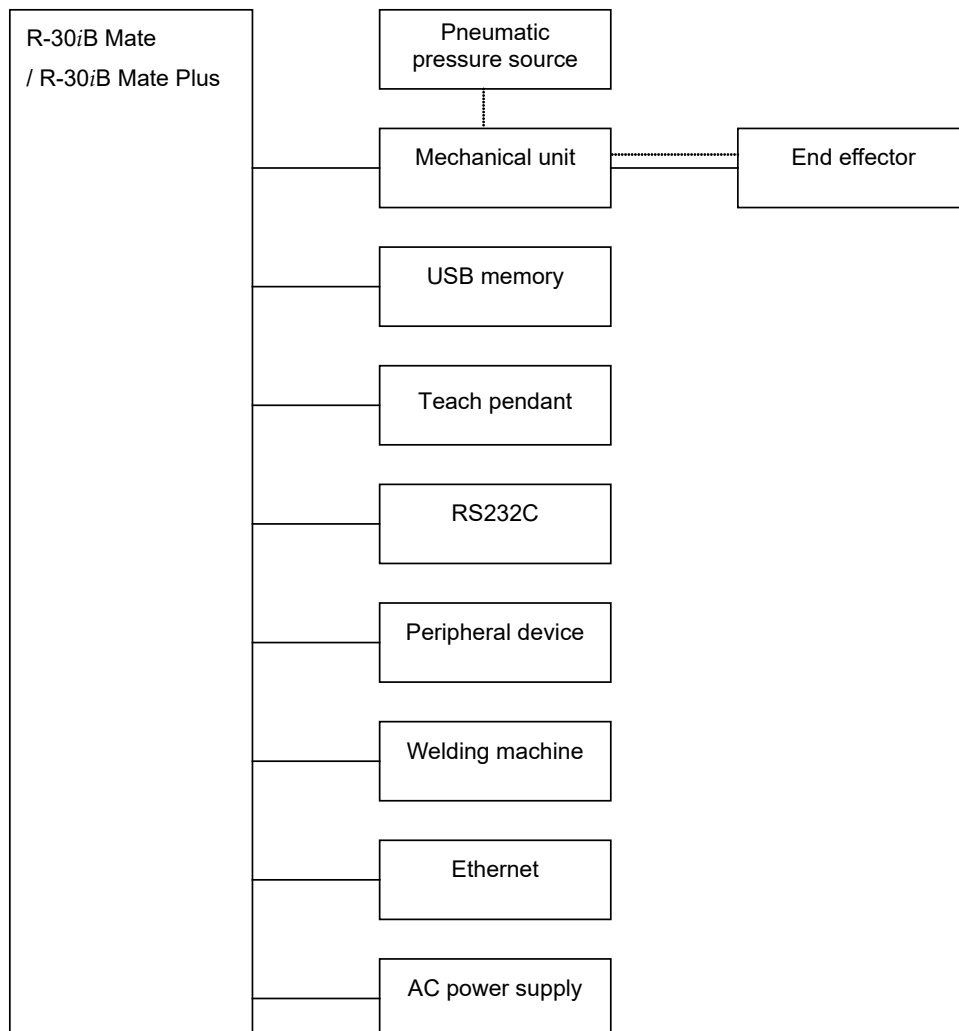


1 OVERVIEW

This section describes the electrical interface connections in the R-30*i*B Mate/ R-30*i*B Mate Plus. It also includes information about installation of the R-30*i*B Mate/ R-30*i*B Mate Plus.

2 BLOCK DIAGRAM

Fig.2 is a block diagram of electrical interface connections with the R-30iB Mate/ R-30iB Mate Plus.

**NOTE**

—— : Indicates electrical connection.

----- : Indicates mechanical connection.

Fig.2 Block diagram of electrical interface connection

3 ELECTRICAL CONNECTIONS

3.1 CONNECTION DIAGRAM BETWEEN MECHANICAL UNITS

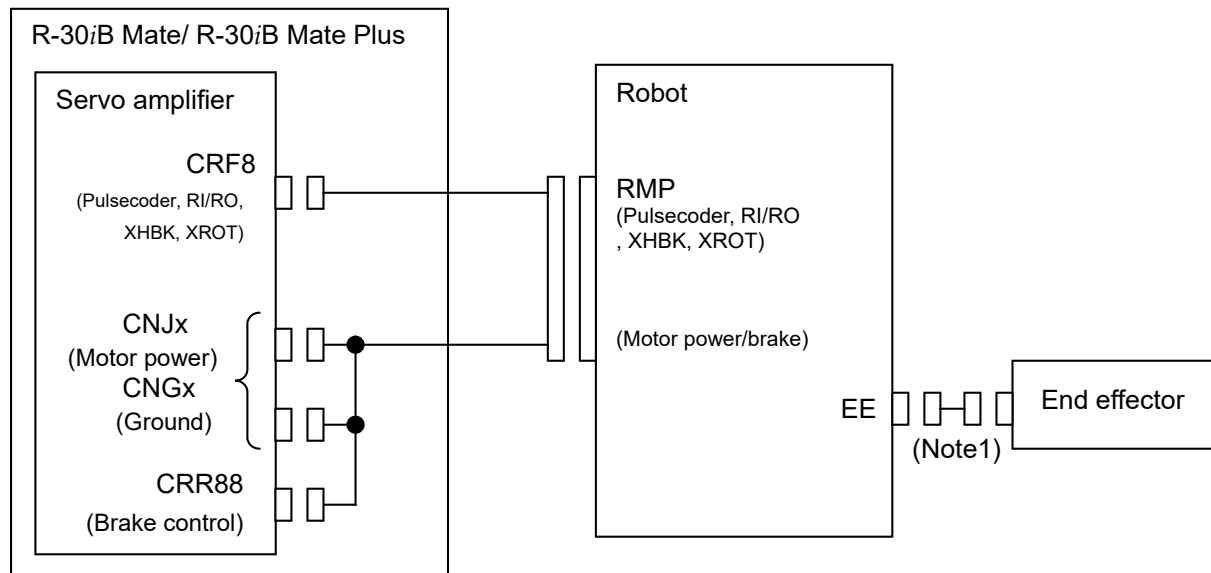
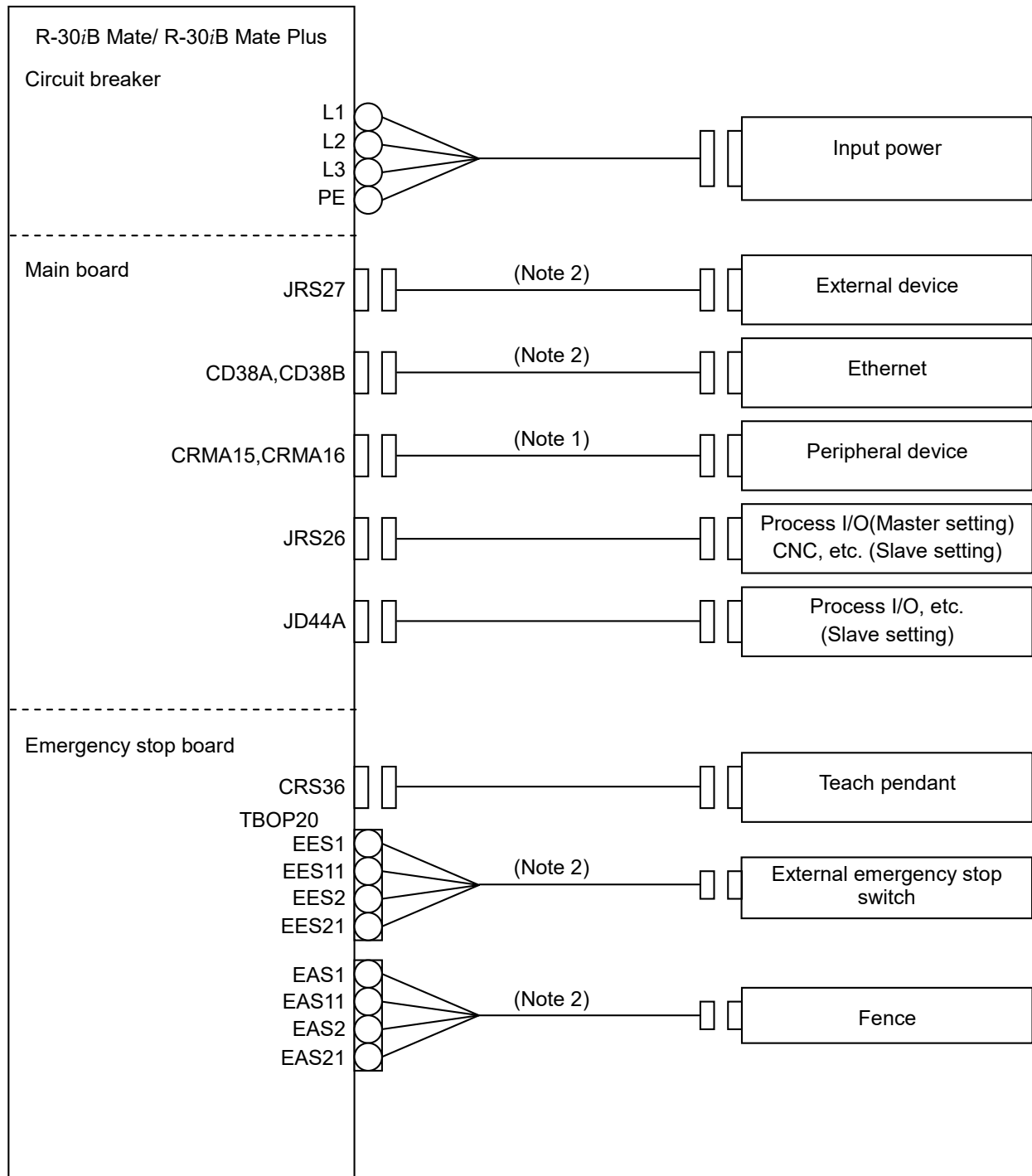


Fig.3.1 (a) Mechanical connection diagram

NOTE

1 This cable is not included. It must be supplied by the customer.

**NOTE**

- 1 For detail of the peripheral device connection, see the section 4.3 of Peripheral device interface.
- 2 This cable is not included. It must be supplied by the customer.

Fig.3.1 (b) Unit-to-unit connection diagram

3.2 CONNECTION TO FANUC I/O Link and FANUC I/O Link *i*

3.2.1 Connection of I/O Link and I/O Link *i* by Using JRS26 Connector

The connection of I/O link and I/O Link *i* by using JRS26 connector is shown below.

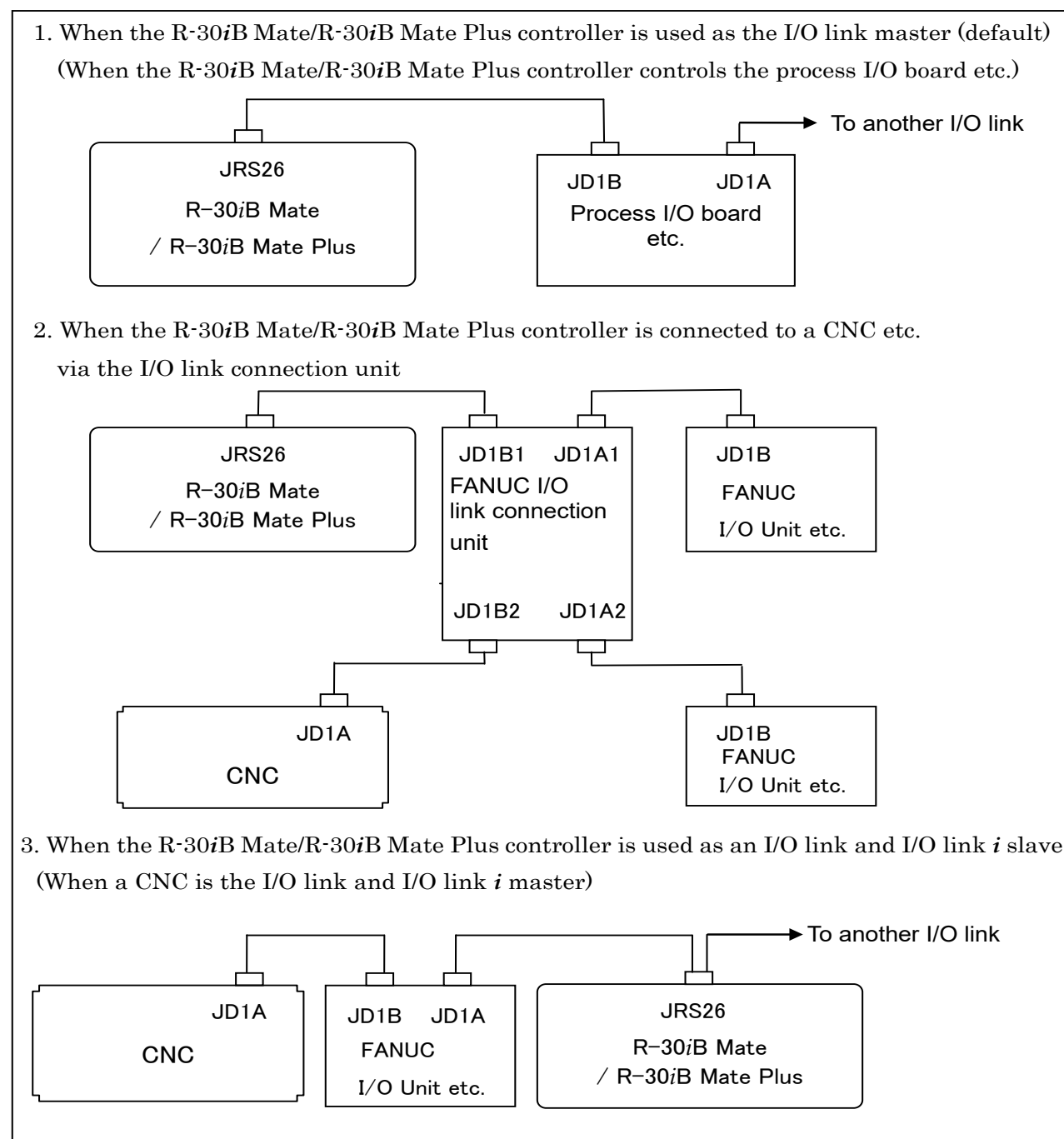


Fig.3.2.1 Connection of I/O link and I/O Link *i* by using JRS26 connector

3.2.1.1 Connection of the I/O Link cable by using JRS26 connector

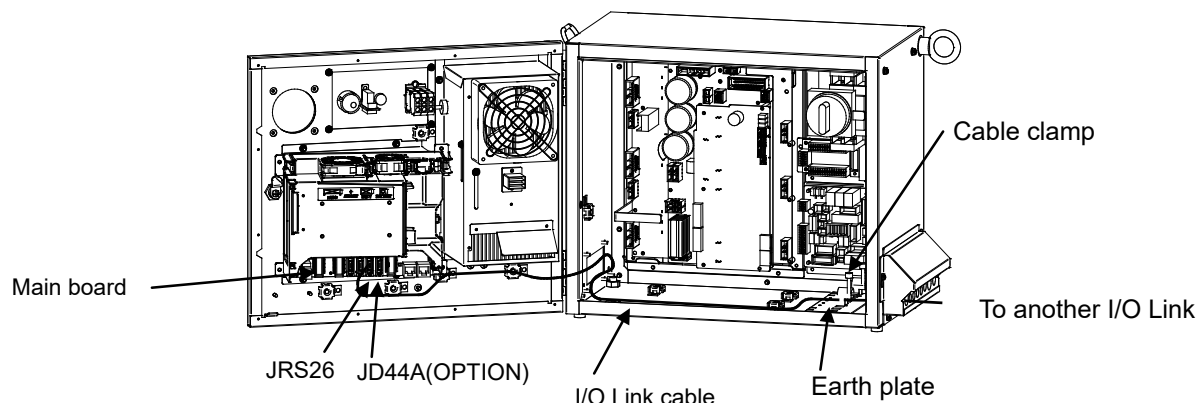


Fig.3.2.1.1 Path of the I/O link cable by using JRS26 connector

1. Connect the cable according to the system. Be sure to perform shielding. Shield the cable collectively and ground the shield on the CNC side.
2. Before connection turn off the power.

NOTE

For connection with the CNC with I/O Link and I/O Link *i*, turn on or off the power of the CNC and the robot controller at the following timing.

- a) Slave units and the master must be powered on or off at the same time.
- b) If the CNC or robot controller is powered off after startup of the system, I/O Link and I/O Link *i* error occur. To successfully make connection with I/O Link and I/O Link *i* again, power off all of the units and then power them on at the timing indicated in a).

When used as master interface JRS26 interface

11	0V	01	RXSLC1	Master interface
12	0V	02	XRSLC1	
13	0V	03	TXSLC1	
14	0V	04	XTXSLC1	
15	0V	05	RXSLC2	
16	0V	06	XRSLC2	
17		07	TXSLC2	
18	(+5V)	08	XTXSLC2	
19	(24V)	09	(+5V)	
20	(+5V)	10	(24V)	

Note: +5V is connected when the optical I/O link adapter is used.

When used as slave interface JRS26 Interface

Refer to item 3 of Fig. 3.2.1

11	0V	01	RXSLC1	From Master controller
12	0V	02	XRSLC1	
13	0V	03	TXSLC1	
14	0V	04	XTXSLC1	
15	0V	05	RXSLC2	To the next I/O link device
16	0V	06	XRSLC2	
17		07	TXSLC2	
18	(+5V)	08	XTXSLC2	
19	(24V)	09	(+5V)	
20	(+5V)	10	(24V)	

Note: +5V is connected when the optical I/O link adapter is used.

3. When the R-30iB Mate/ R-30iB Mate Plus controller is connected to CNC or preceding I/O link *i* slave unit, use a twisted-pair cable in which wires RXSLC1 (Pin No.1 of JRS26) and XRSLC1 (Pin No.2 of JRS26) are paired and wires TXSLC1 (Pin No.3 of JRS26) and XTXSLC1 (Pin No.4 of JRS26) are paired.

4. When the R-30iB Mate/ R-30iB Mate Plus controller is connected to next I/O link *i* slave unit, use a twisted-pair cable in which wires RXSLC2 (Pin No.5 of JRS26) and XRXSLC2 (Pin No.6 of JRS26) are paired and wires TXSLC2 (Pin No.7 of JRS26) and XTXSLC2 (Pin No.8 of JRS26) are paired.

3.2.1.2 Cable connection diagram of the I/O Link cable by using JRS26 connector

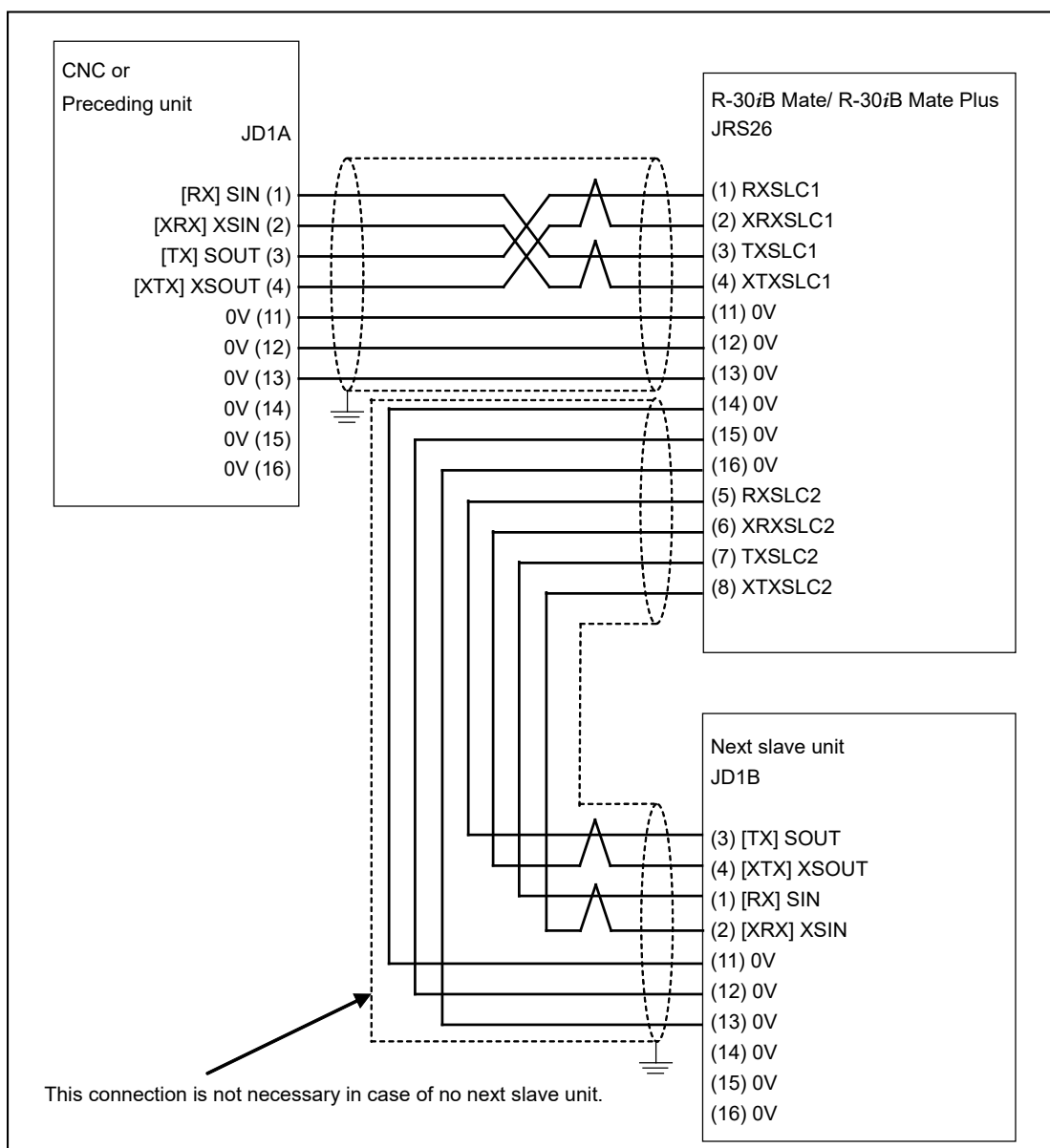


Fig.3.2.1.2 Cable connection diagram of the I/O Link cable by using JRS26 connector

3.2.2 Connection of JD44A Connector(Optional)

JD44A is used to connect the Additional safety I/O board (Mini slot)
The connection of JD44A connector is shown below.

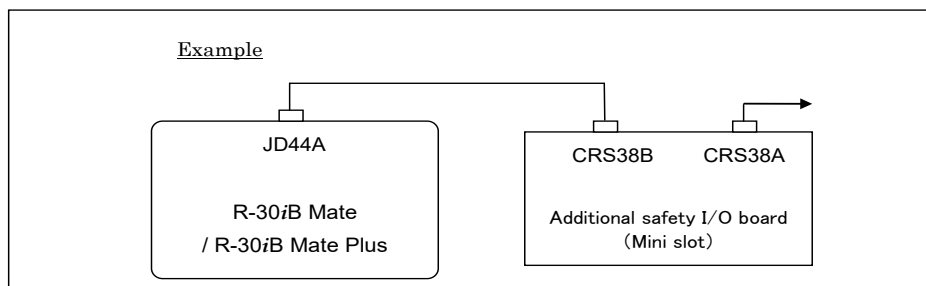


Fig.3.2.2 Connection of JD44A connector

3.2.2.1 Connection of the I/O Link cable by using JD44A connector

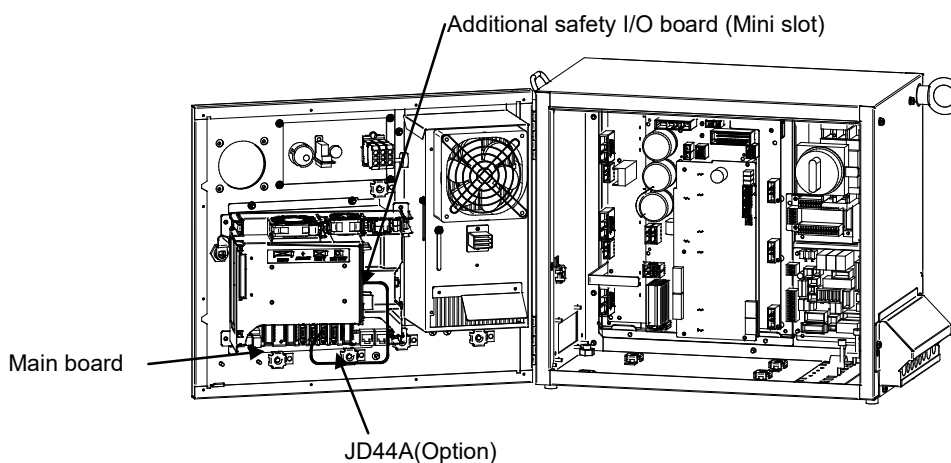


Fig.3.2.2.1 Path of the I/O link cable by using JD44A connector

JD44A Interface

11	0V	01	(Reserve)
12	0V	02	(Reserve)
13	0V	03	(Reserve)
14	0V	04	(Reserve)
15	0V	05	RXSLCS
16	0V	06	XRSLCS
17		07	TXSLCS
18	(+5V)	08	XTXSLCS
19	(24V)	09	(+5V)
20	(+5V)	10	(24V)

Note: +5V is connected when the optical I/O link adapter is used.

When the R-30iB Mate/ R-30iB Mate Plus controller is connected to Additional I/O board, use a twisted-pair cable in which wires RXSLCS (Pin No.5 of JD44A) and XRSLCS (Pin No.6 of JD44A) are paired and wires TXSLCS (Pin No.7 of JD44A) and XTXSLCS (Pin No.8 of JD44A) are paired.

3.2.2.2 Cable connection diagram of the I/O Link cable by using JD44A connector

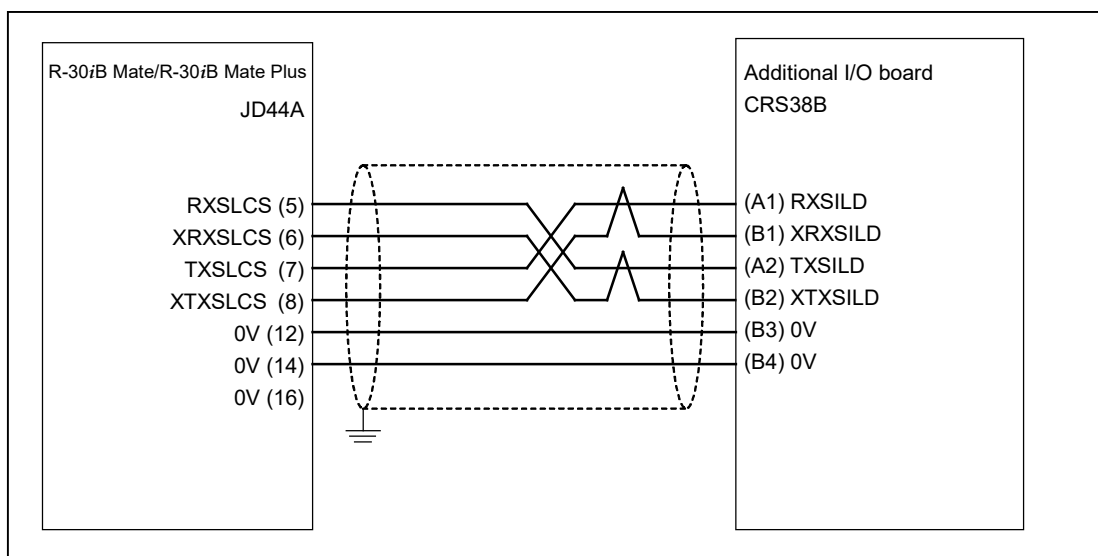


Fig.3.2.2.2 Cable connection diagram of the I/O Link cable by using JD44A connector

3.3 EXTERNAL CABLE WIRING DIAGRAM

3.3.1 Robot Connection Cables



WARNING

Before operating the robot, uncoil the interconnection cables from their shipping position to prevent excessive heat, which may damage the cables.
(Coiled part should be shorter than 10 meter.)

There are two types of the robot connection cable;

Non-flex type: usage is restricted to fixed laying

Flex type: possible to use in the cable track

Specification of cable

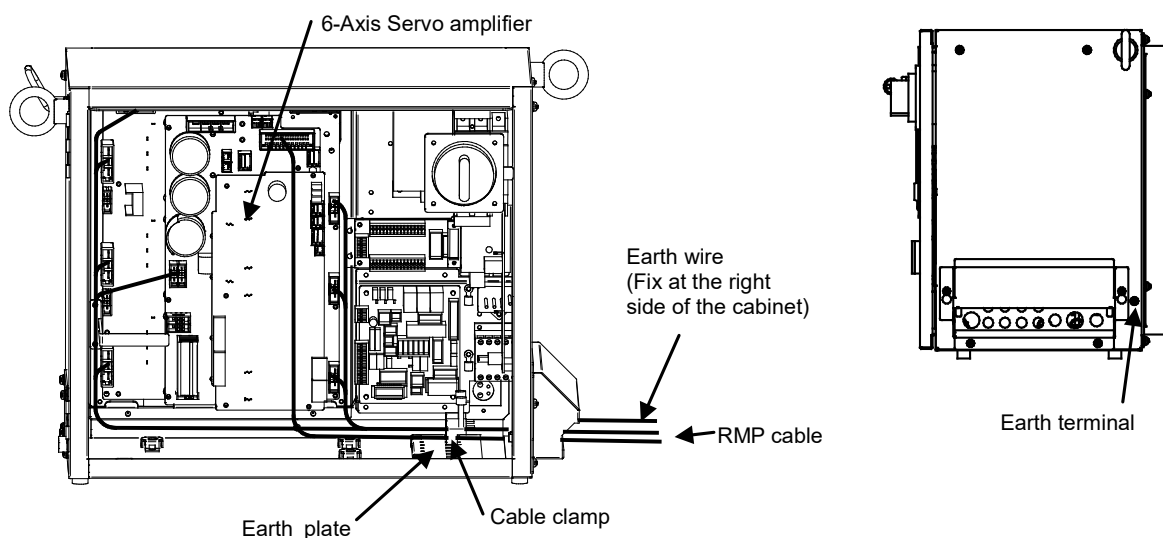
		Robot	Non-flex type			Flex type		
			Diameter (mm)	Weight (kg/m)	Minimum bending radius (mm)	Diameter (mm)	Weight (kg/m)	Minimum bending radius (mm)
RP1		Group 1 Group 11	16.0	0.45	200	20.5	0.71	200
RM1		Group 1 Group 11	26.1	1.22	200	25.4	1.2	200
RMP	RP	Group 8	16.0	0.45	200	20.5	0.71	200
	RM		20.0	0.7	200	18.4	0.7	200
RMP	RP	Group 12	14.2	0.31	200	20.5	0.71	200
	RM		15.5	0.3	200	15.5	0.49	200
EARTH		All models	4.7	0.065	200	4.7	0.065	200

Group 1	M-710iC, R-1000iA, R-2000iC
Group 8	ARC Mate 100iC, ARC Mate 100iD, ARC Mate 120iC, M-10iA, M-10iD, M-20iA, M-20iB, ARC Mate 0iB, R-0iB
Group 11	M-2iA, M-3iA
Group 12	LR Mate 200 iC, LR Mate 200iD, M-1iA, ARC Mate 50iD, CR-4iA, CR-7iA

Using condition of flex type cable

- (1) When routing cables in movable places, use a cable bearer.
- (2) The bending radius (R) of the cable track is more than 200mm.
- (3) The cable should be fixed to the cable track by using the clamp. (e.g. foam rubber)
- (4) The size of the hole to support a cable in the cable track should be more than 110% of the cable size and should have the gap more than 3mm.
- (5) When cables are laid in the cable track, pay attention for the cable not to be twisted.

Cable Route



CAUTION

Signal cable should be clamped to Earth plate by cable clamp.
In case of CE controller, Power/Brake cable should be clamped to Earth plate by cable clamp.

Robot Model: LR Mate 200iC, LR Mate 200iD, M-1iA, ARC Mate 50iD, ARC Mate 100iC, ARC Mate 100D, ARC Mate 120iC, M-10iA, M-10iD, M-20iA, M-20iB, ARC Mate 0iB, R-0iB, CR-4iA, CR-7iA

Detail of cable connection to servo amplifier

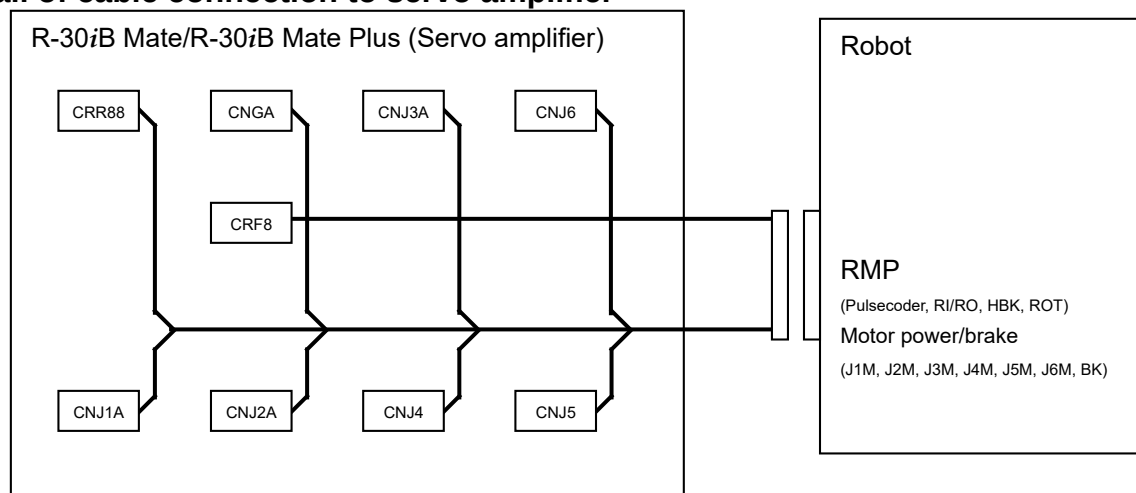


Fig.3.3.1 (a) Robot connection cable

Robot Model: M-2iA, M-3iA, R-1000iA, M-710iC, R-2000iC

Detail of cable connection to servo amplifier

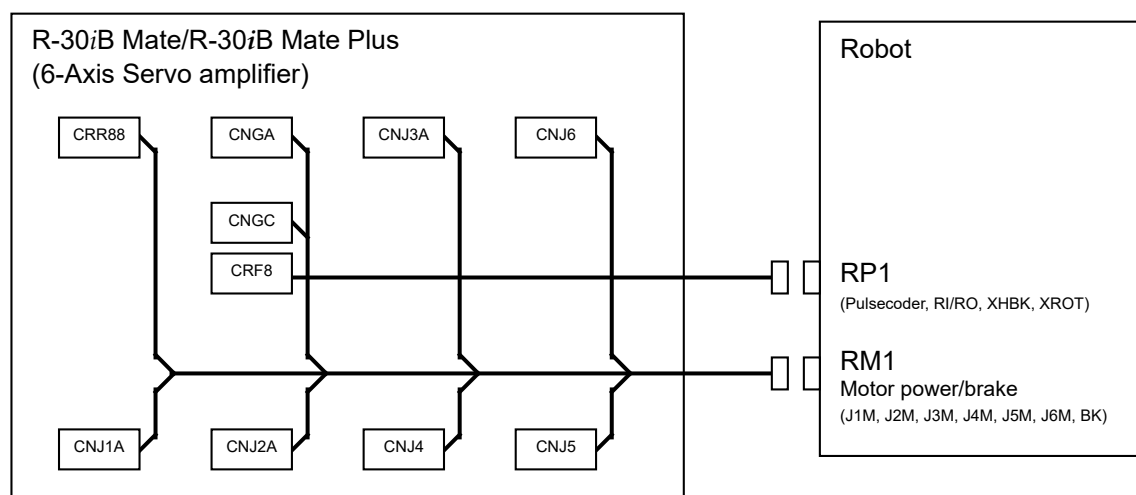


Fig.3.3.1 (b) Robot connection cable

3.3.2 Teach Pendant Cable

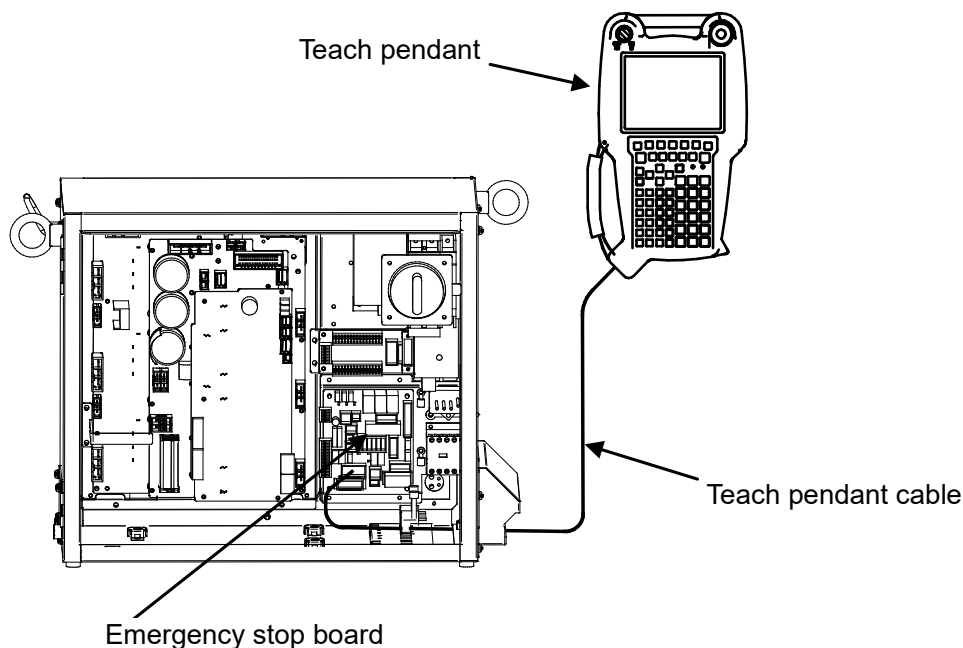


Fig.3.3.2 Teach pendant cable

3.3.3 Connecting the Input Power

3.3.3.1 Connecting the input power cable

- (1) Fig.3.2.3.1 shows the method of connecting the input power supply cable.
- (2) Use the input power cable according to the following Table 3.2.3.1. However, the input power cable according to the breaker or the fuse of the input power supply (power distribution panel) connected to the robot controller must be used.
- (3) Provide a class-D or better ground.
There shall be no switches or disconnects in the grounding conductor.
The resistance to the ground must not exceed 100Ω.
Use a thick wire to withstand the maximum current used.

In case of NRTL controller

Provide a grounding conductor of equivalent gauge as the supply conductors.

Grounding/Bonding to comply with NEC Article 250 or CEC Section 10 as appropriate.

Table 3.3.3.1 Conductor size and terminal size of AC power supply

Input Voltage	Input power source capacity (Refer to CONNECTIONS 5.3)	Conductor size of AC Power supply cable	Terminal size of AC power supply cable	Conductor size of earth cable	Terminal size of earth cable
200V	<12kVA	AWG14 to AWG10 (Note 1)	M5	WARNING 2	M5
200V	12kVA ≤	AWG10 ≤ (Note 1)	M5	WARNING 2	M5

⚠ WARNING

- 1 The input power cable according to the breaker or the fuse of the input power supply (power distribution panel) connected to the robot controller must be used.
- 2 Use conductor of earth cable size is as well as the AC power supply cable size.
- 3 Disconnection of protective earth ground may impair the protection provided by the system.

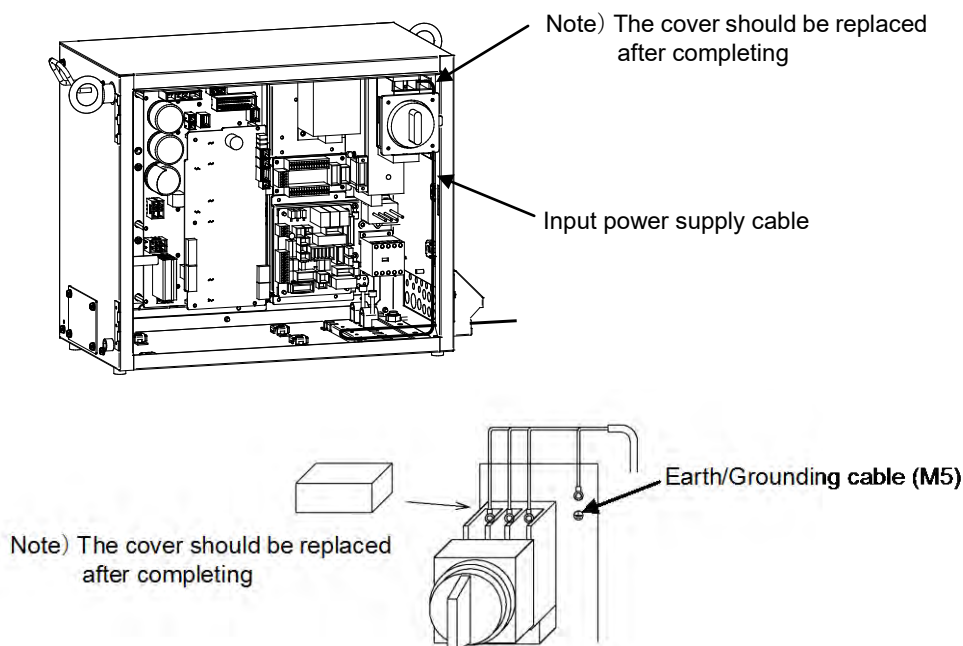


Fig.3.3.3.1 Connecting the input power cable

⚠ WARNING

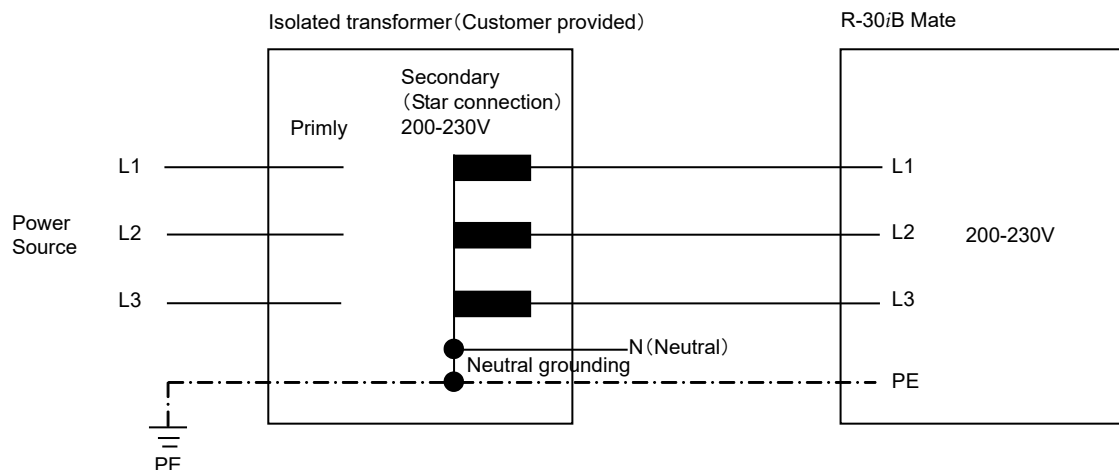
The cover for primly terminal of main breaker should be replaced after completing.

3.3.3.2 Isolated transformer

- (1) According to the voltage of the input power source, an isolated transformer should be required. Refer to table 3.3.3.2. If required, set up isolated transformer between input power source and controller.
- (2) In case of CE controller, the output of transformer should be the star connection with neutral grounding.

Table 3.3.3.2 Isolated transformer

Input power source	BASIC	NRTL	CE
200-230VAC single phase	No	No	No
200-230VAC three phase	No	No	Yes
others	Yes	Yes	Yes

**⚠ WARNING**

In case of CE controller, the output of transformer should be the star connection with neutral grounding.

NOTE

Isolated transformer is provided by customer.
Choose the appropriate transformer according the required power source capacity of the robot.

Example of Isolated transformer

Manufacture	RIST Transformatorenbau GmbH
Specification	25065LK
Capacity	3kVA
Output	AC200V

3.3.3.3 Leakage breaker

- (1) The motor is driven by the PWM inverter system using a power transistor bridge. A high-frequency leakage current flows through the stray capacitance between the ground and the motor coils, power cable, and amplifier. This might cause the leakage-current circuit breaker or leakage-protection relay installed in the path of the power supply to cut out.
Use the following leakage current circuit breaker for inverters to prevent incorrect operation.
- (2) Leakage breaker using robot controller has sensitive electric current of 30mA.

Table 3.3.3.3 Example of leakage current circuit breaker for inverters

Manufacture	Type
Fuji Electric Co., Ltd.	EG A series or later SG A series or later
Hitachi, Ltd.	ES100C type or later ES225C type or later
Matsushita Electric Works, Ltd.	Leakage current circuit breaker, C type or later Leakage current circuit breaker, KC type or later

3.3.4 Connecting the External Emergency Stop

After connecting the safety signals like external emergency stop signal and/or safety fence signal, verify that,

- All safety signals stop the robot as intended.
- There is no mistake in connection of safety signals.

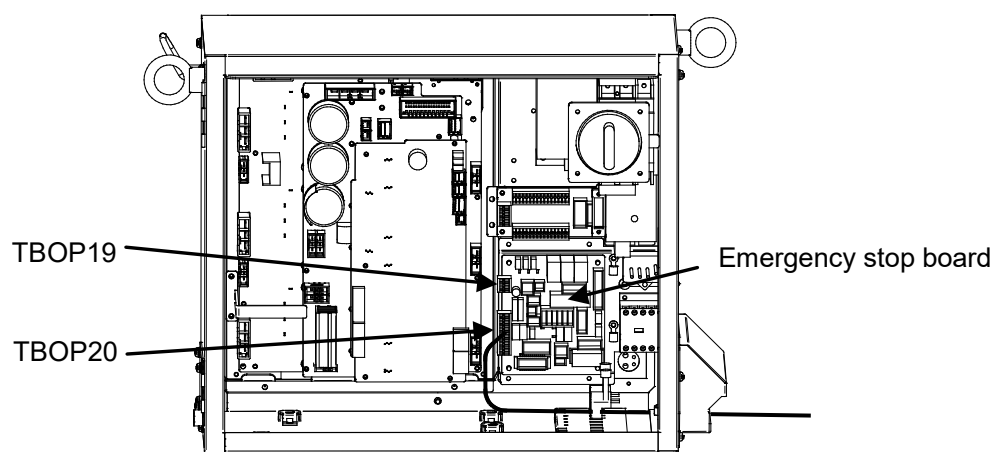


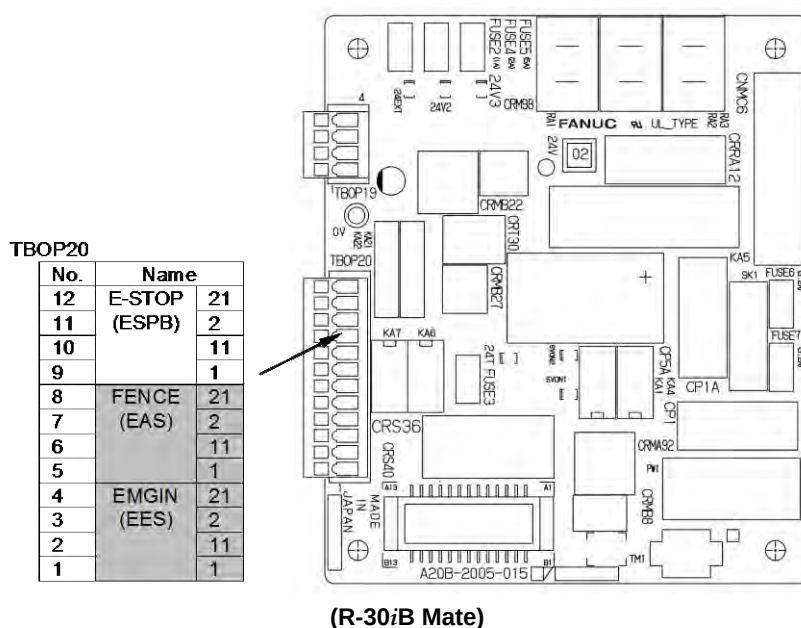
Fig.3.3.4 (a) Connecting the external emergency stop

NOTE

For protection against the noise, the shielded cable is recommended for the connection cable.

Cut part of the jacket of the cable to expose the shield, and fasten this part to the earth plate with the cable clamp.

External emergency stop output



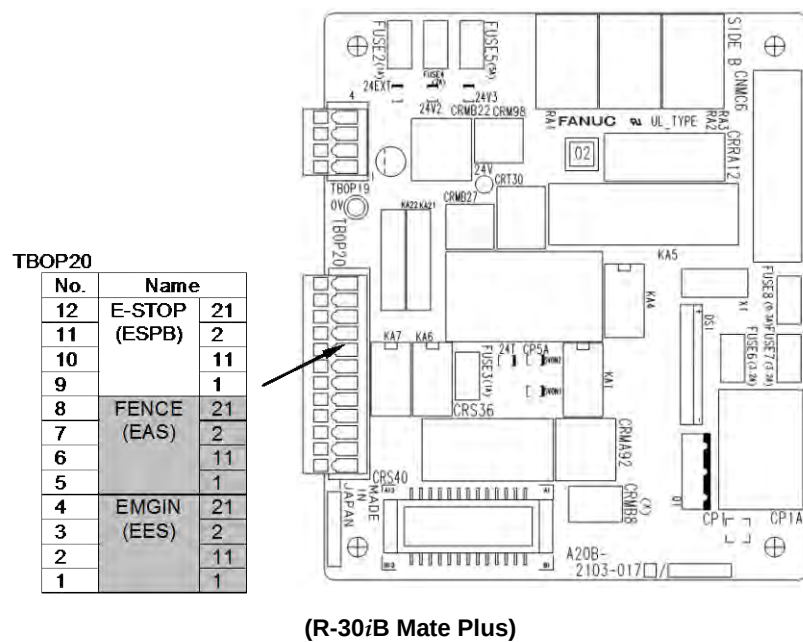


Fig.3.3.4 (b) Emergency stop board

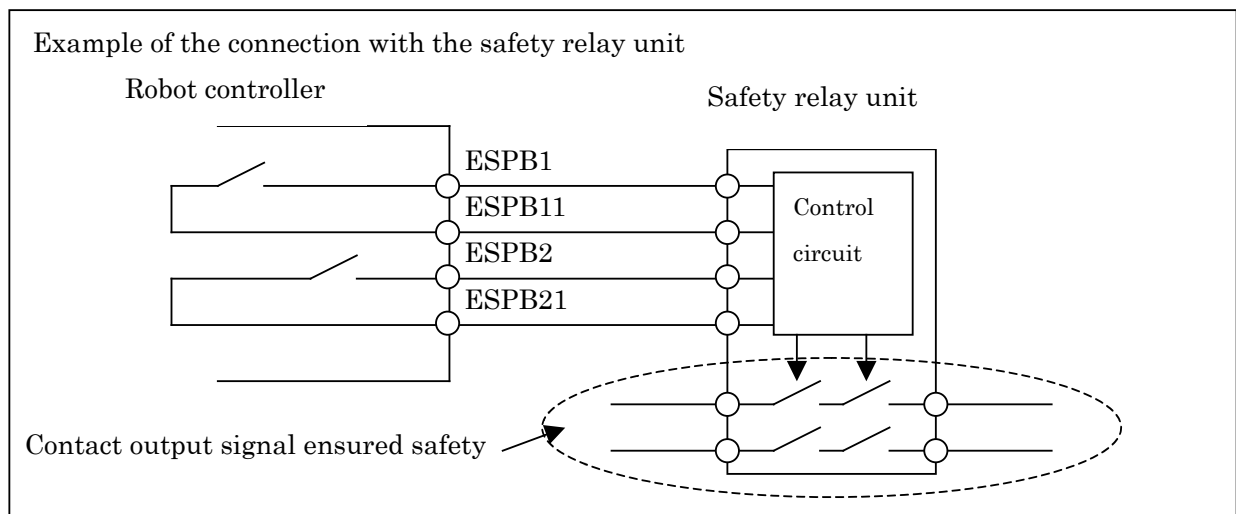
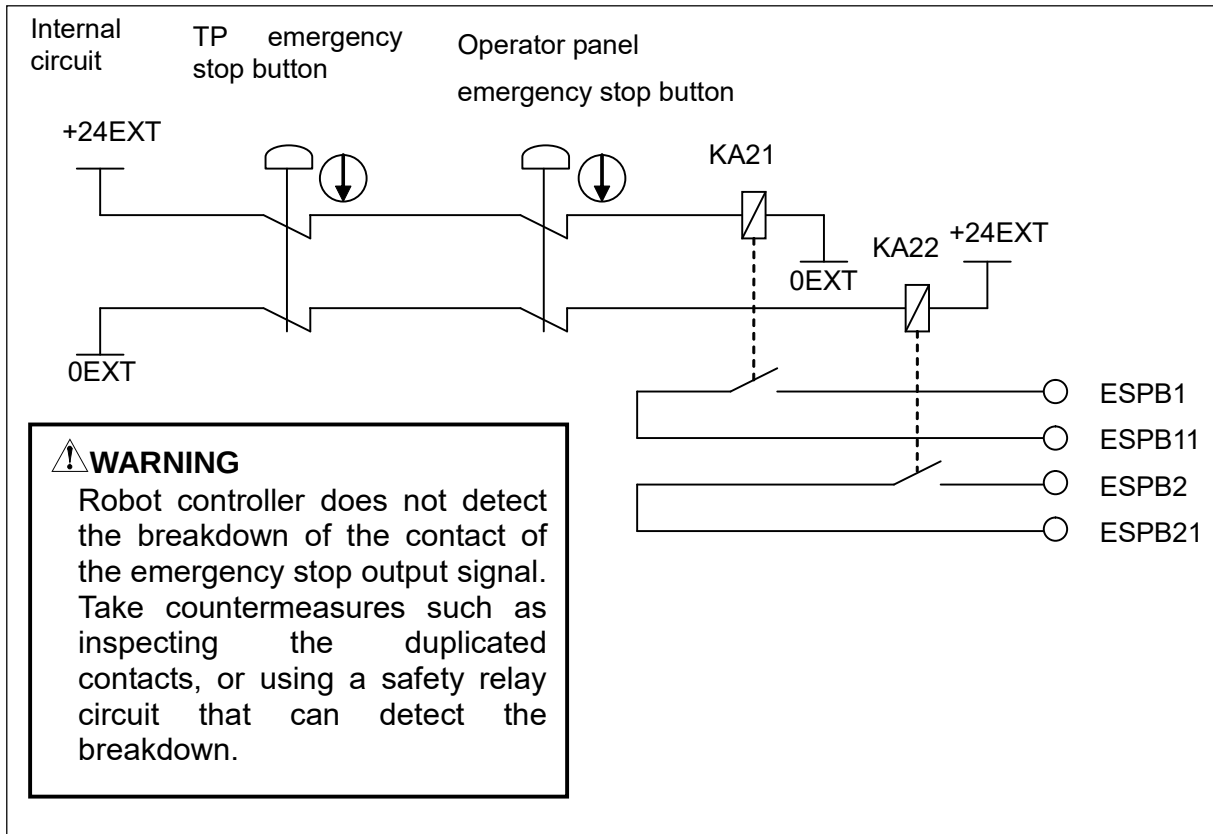
For the circuit, see Fig. B (e) in Appendix B, "TOTAL CONNECTION DIAGRAM".

Signal	Description	Current, voltage	Min. load
ESPB1 — ESPB11 ESPB2 — ESPB21	The contact is open when one of the TP emergency stop button or the Operator panel emergency stop button is pressed. The contact is also open while the controller is powered off regardless of status of emergency stop buttons. By connecting external power supply to the emergency stop circuit, the contact works even while the robot controller is powered off. (See "External power connection" of this section) The contact is closed during normal operation.	Rated contact: 30 VDC, 5 A resistor load	(Reference value) DC5V 10mA

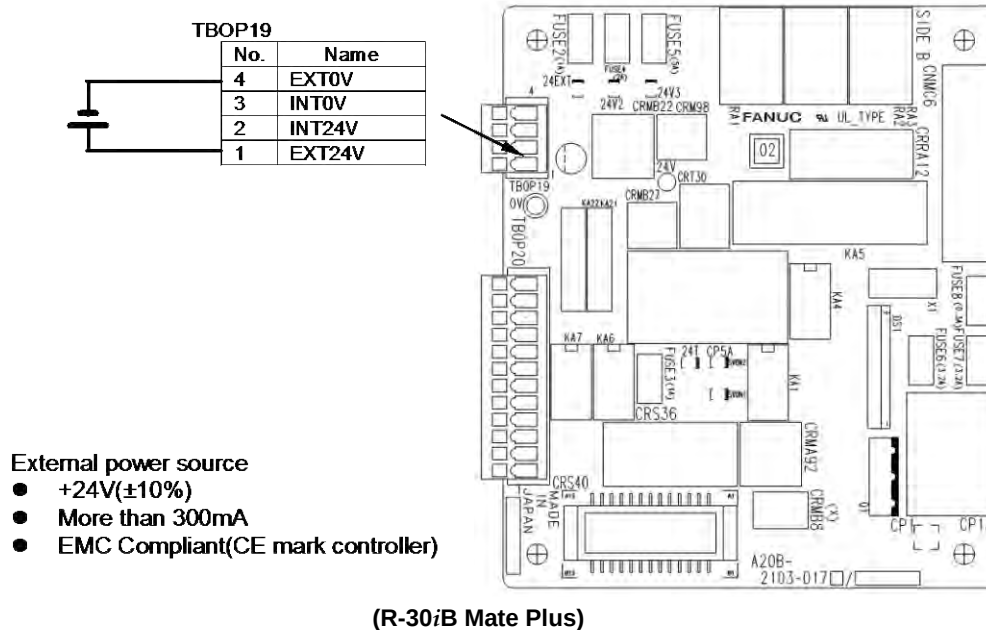
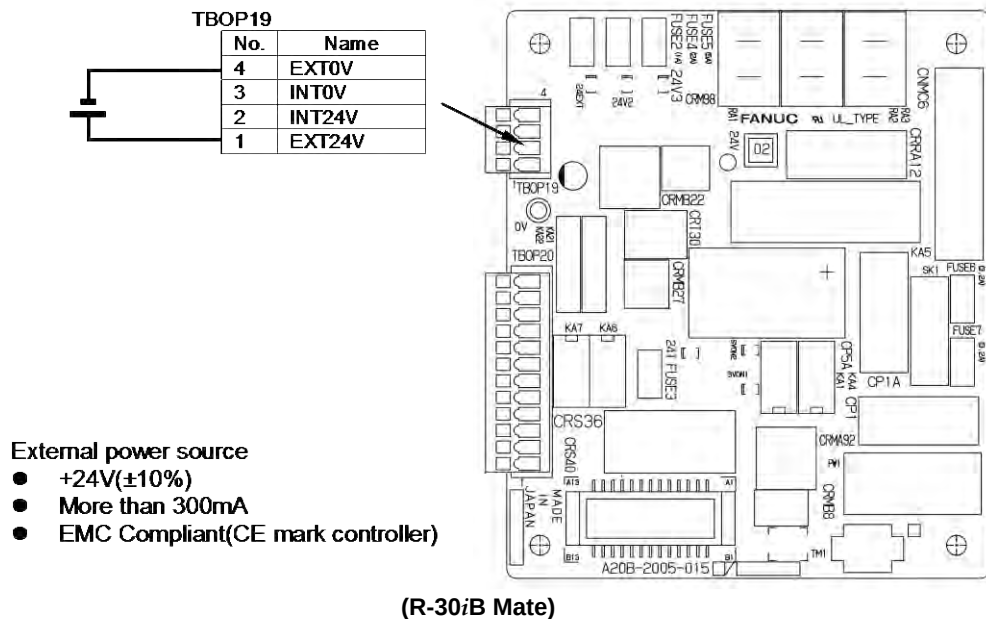
NOTE

For protection against the noise, the shielded cable is recommended for the connection cable.

Cut part of the jacket of the cable to expose the shield, and fasten this part to the earth plate with the cable clamp.

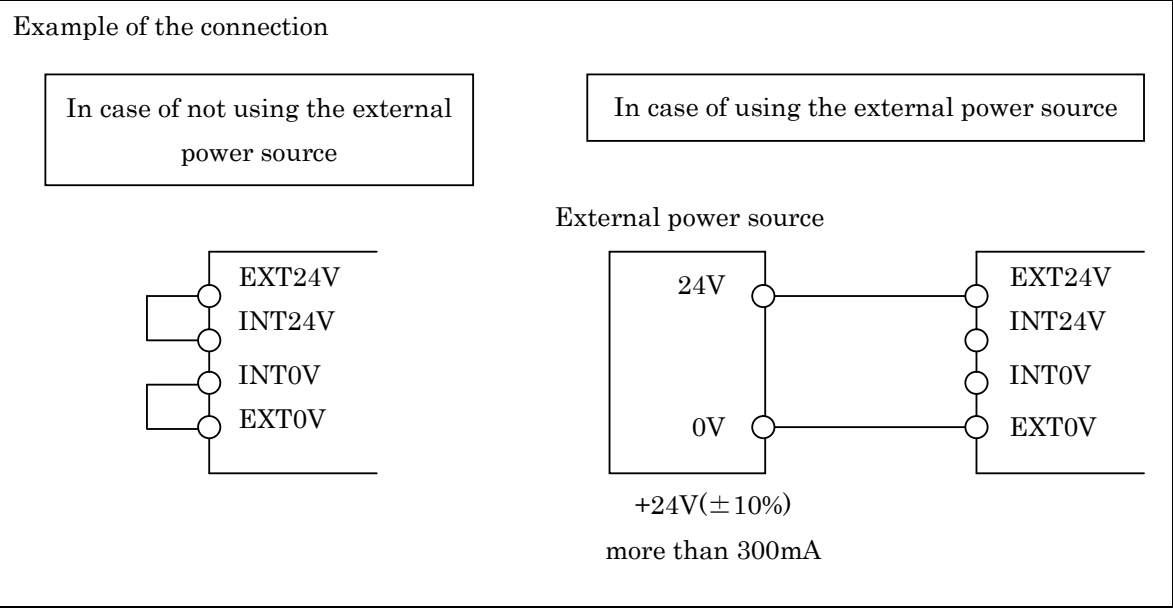


External power connection



Emergency stop board

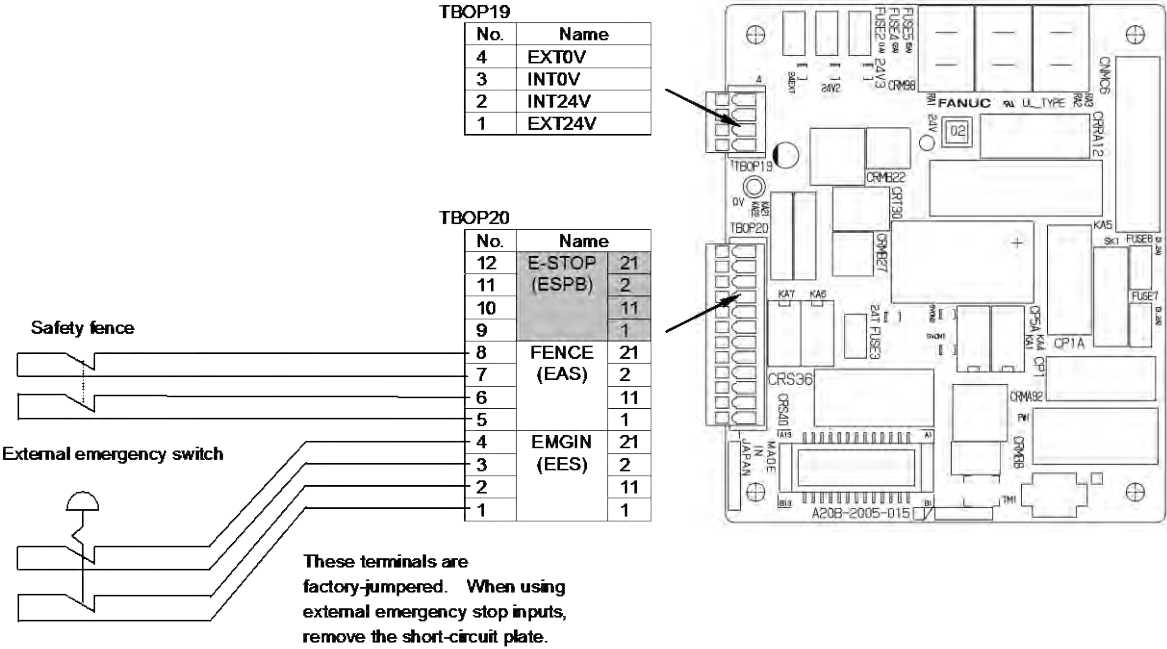
The relays for emergency stop input and output can be separated from controller's power. Please connect external +24V instead of internal +24V, if emergency stop output must not be effected controller's power.



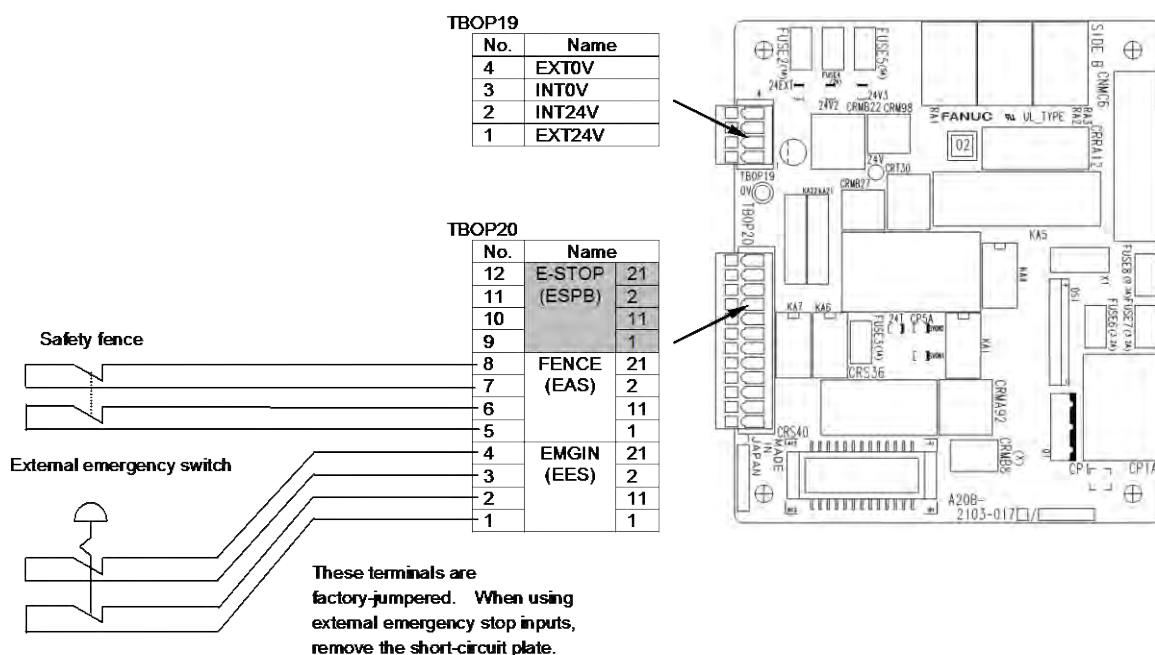
NOTE

For protection against the noise, the shielded cable is recommended for the connection cable.
Cut part of the jacket of the cable to expose the shield, and fasten this part to the earth plate with the cable clamp.

External emergency stop input



(R-30iB Mate)



(R-30iB Mate Plus)

Emergency stop board

Signal	Description	Current, voltage
EES1 EES11 EES2 EES21	Connect the contacts of the external emergency stop switch to these terminals. When the contacts are open, the robot stops according to predetermined stop pattern. (Note 2) When using the contacts of a relay or contactor instead of the switch, connect a spark killer to the coil of the relay or contactor, to suppress noise. When these terminals are not used, jumper them.	Open and close of 24VDC 0.1A (Note 1)
EAS1 EAS11 EAS2 EAS21	These signals are used to stop the robot safely when the safety fence gate is opened during operation in the AUTO mode. When the contacts are open in the AUTO mode, the robot stops according to predetermined stop pattern. (Note 2) In the T1 or T2 mode and the DEADMAN switch is held correct position, the robot can be operated even when the safety fence gate is open. When using the contacts of a relay or contactor instead of the switch, connect a spark killer to the coil of the relay or contactor, to suppress noise. When these terminals are not used, jumper them.	Open and close of 24VDC 0.1A (Note 1)

NOTE

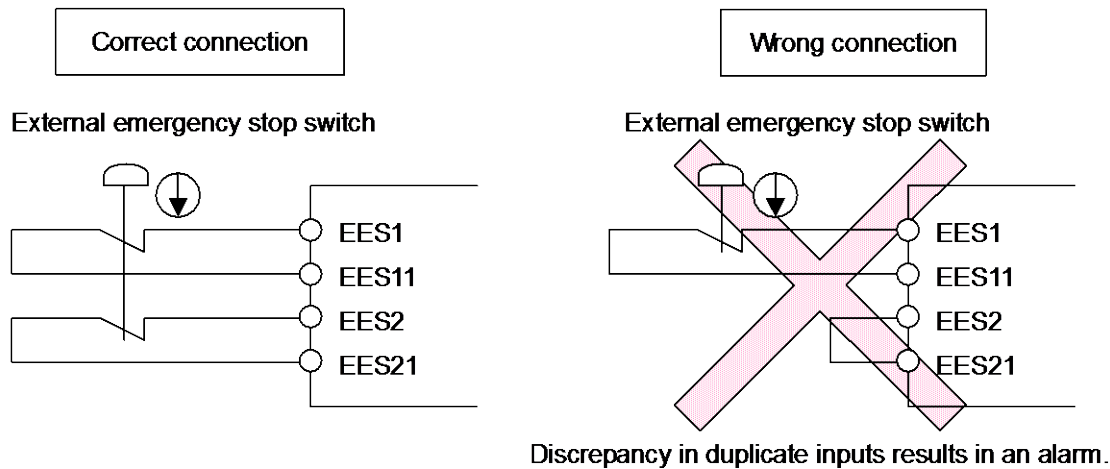
1. Use a contact which minimum load is 5 mA less.
2. See Chapter 7 in SAFETY PRECAUTIONS.

NOTE

For protection against the noise, the shielded cable is recommended for the connection cable.

Cut part of the jacket of the cable to expose the shield, and fasten this part to the earth plate with the cable clamp.

Examples of connection of duplicate safety signals



Input timing of duplicate safety signals

Duplicate inputs are used for signals such as the external emergency stop signal, safety fence signal, and servo off signal so that a response is made even when a single failure occurs. The statuses of these duplicate input signals must always be changed at the same timing according to the timing specifications provided in this section. The robot controller always checks that the statuses of the duplicate inputs are the same, and if the controller finds a discrepancy, it issues an alarm. If the timing specifications are not satisfied, an alarm may be issued because of a signal discrepancy.

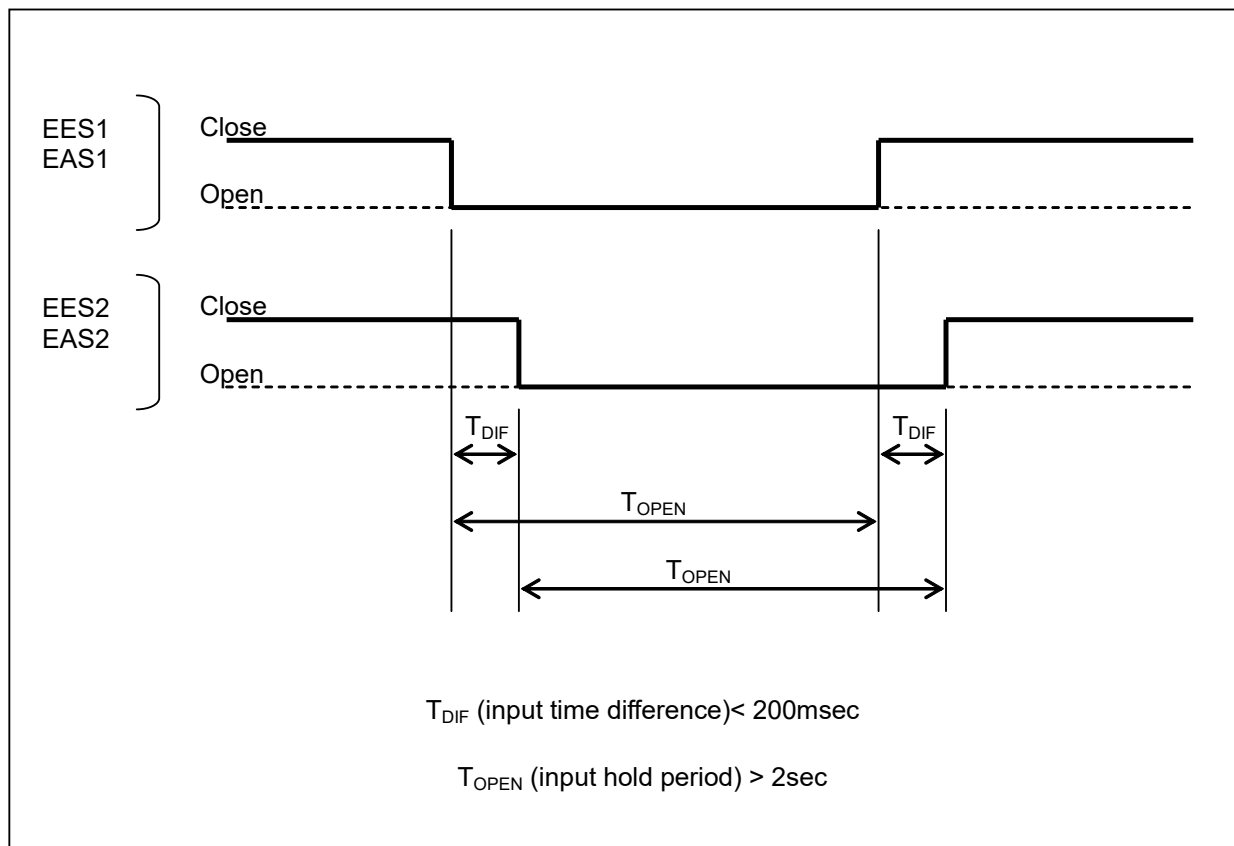


Fig.3.3.4(c) Input timing of duplicate safety signals

Connecting external on/off and external emergency stop signal input/output wires

	FANUC's specification	Manufacturer's specification (WAGO)	Remark
4-pole terminal block (TBOP19)	A63L-0002-0154#104	734-104	
12-pole terminal block (TBOP20)	A63L-0002-0154#112	734-112	
Jumper pin	A63L-0002-0154#402F	734-402F	
Operation lever	A63L-0002-0154#230-M	734-230	2 pieces of 734-230 and operation manual are included in FANUC's specification

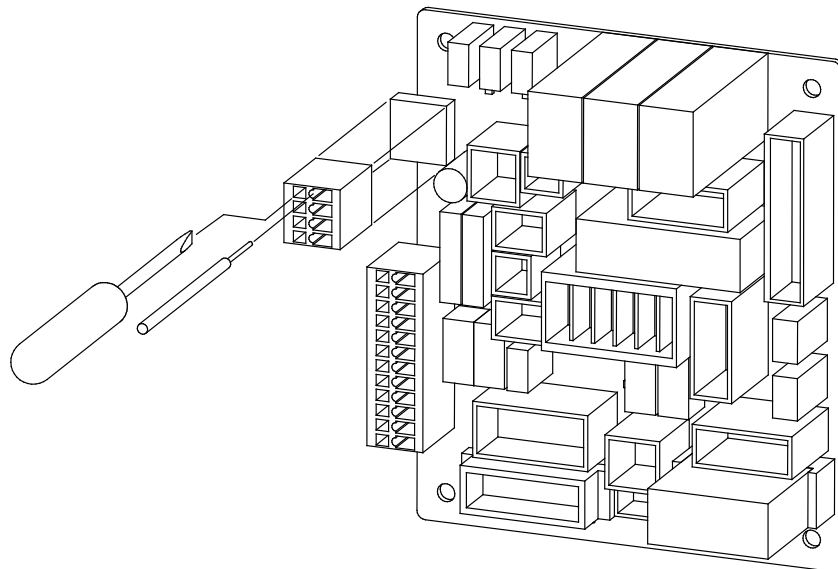
1. Detach the plug connector block from the emergency stop board.
2. Insert the tip of a flat-blade screwdriver into the manipulation slot and push down its handle.
3. Insert the end of the signal wire into the wire slot.
4. Pull out the screwdriver.
5. Attach the plug connector block to the emergency stop board.



CAUTION

Do not insert a wire into the wire hole of a plug connector or pull it out with the plug connector block mounted on the emergency stop board; otherwise, the emergency stop board may be damaged.

FANUC recommends the lever (A05B-2600-K030) for connecting the signal wire to the plug connector block instead of Flat-blade screwdriver.



3.3.5 Connecting the Auxiliary Axis Brake (CRR65 A/B)

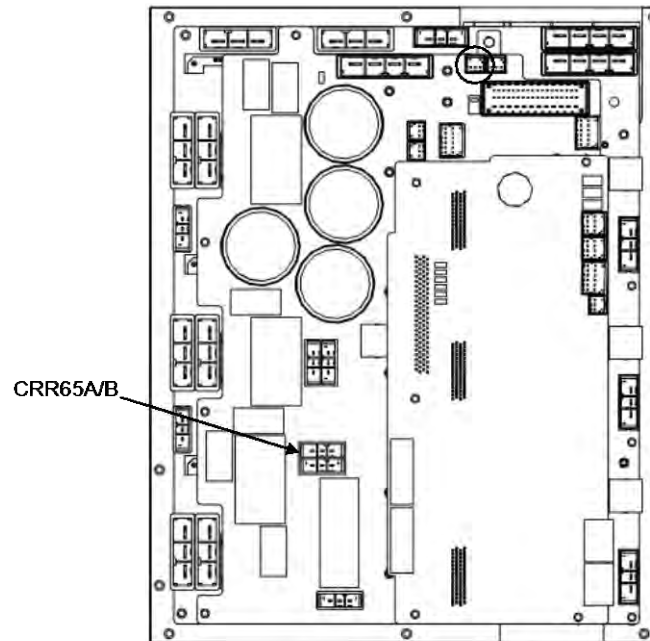


Fig.3.3.5 6-axis servo amplifier

CRR65 A/B

A1	BJA1	B1	BJA2
A2		B2	
A3	COMMON	B3	COMMON

Specification

	TE Connectivity Specification	FANUC Specification
Rece-housing	1-178128-3	A63L-0001-0460#032KSX
Rece-contact (AWG16-20)	175218-2	A63L-0001-0456#ASL

3.3.6

Connecting the Auxiliary Axis Over Travel (CRM68)

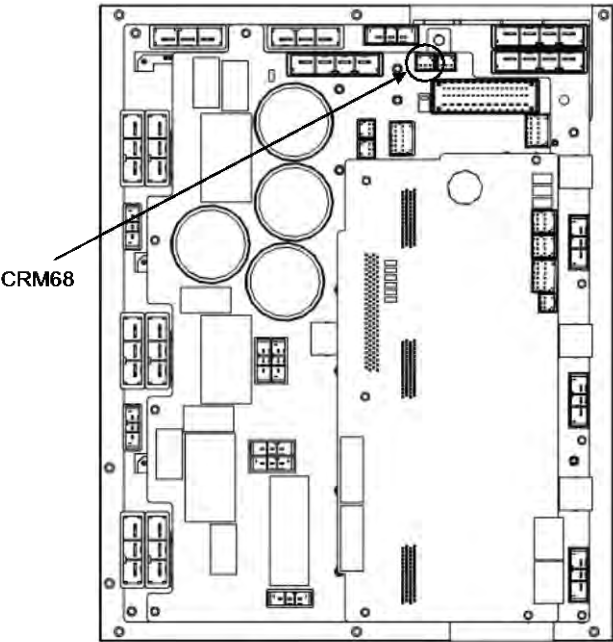


Fig.3.3.6 6-axis servo amplifier

CRM68		
A1		AUXOT1
A2		AUXOT2
A3		

Specification		
	TE Connectivity Specification	FANUC Specification
Rece-housing	1-1318120-3	A63L-0001-0812#R03SX
Rece-contact (AWG18-22)	1318107-1	A63L-0001-0812#CRM

4 PERIPHERAL DEVICE, ARC WELDING, AND EE INTERFACES

R-30iB Mate/R-30iB Mate Plus I/O peripheral device interfaces include printed circuit boards and a unit selected according to the applications. Table 4 lists details of the printed-circuit boards and units. Figure 4 shows the locations of these boards and units.

Table 4 Peripheral device interface types

Table 4-1 Peripheral device interface types							
No.	Name	Drawing number	Peripheral device interface				Remarks
			CRMA15		CRMA16		
			DI	DO	DI	DO	
1a	Main board A (R-30iB Mate)	A05B-2650-H001	20	8 (Source)	8	16 (Source)	Standard
	Main board A (R-30iB Mate Plus)	A05B-2680-H001					
1b	Main board B (R-30iB Mate)	A05B-2650-H002	20	8 (Source)	8	16 (Source)	Vision, Force sensor I/F
	Main board B (R-30iB Mate Plus)	A05B-2680-H002					
1c	Main board C (R-30iB Mate)	A05B-2650-H003	20	8 (Source)	8	16 (Source)	Vision, Force sensor I/F PMC, HDI
	Main board C (R-30iB Mate Plus)	A05B-2680-H003					

No.	Name	Drawing number	Peripheral device interface				Remarks
			CRMA52A		CRMA52B		
			DI	DO	DI	DO	
2	Process I/O board MA	A05B-2650-J060	10	8 (Source)	10	8 (Source)	Option

No.	Name	Drawing number	Peripheral device interface				Remarks
			CRW11				
			WI	WO	D/A	A/D	
3	Process I/O board MB	A05B-2650-J061	5	4(Sink)	2	0	Option

No.	Name	Drawing number	Remarks
4	Connector converter board	A05B-2650-J070	This option board converts peripheral device interfaces CRMA15 and CRMA16 of the main board to the MR connector manufactured by Honda Tsushin Kogyo Co., LTD.
5	Terminal converter board	A05B-2650-J071	This option board converts peripheral device interfaces CRMA15 and CRMA16 of the main board to the terminal blocks manufactured by Phoenix Contact GmbH & Co. KG.

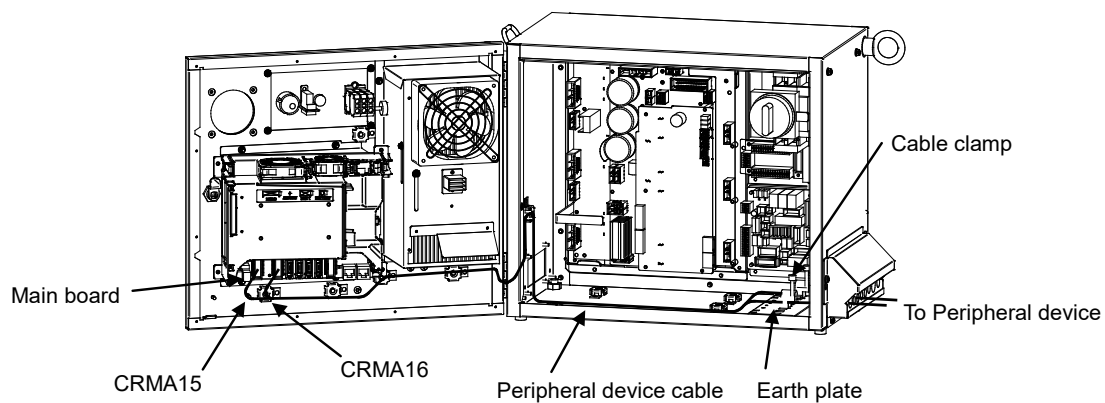
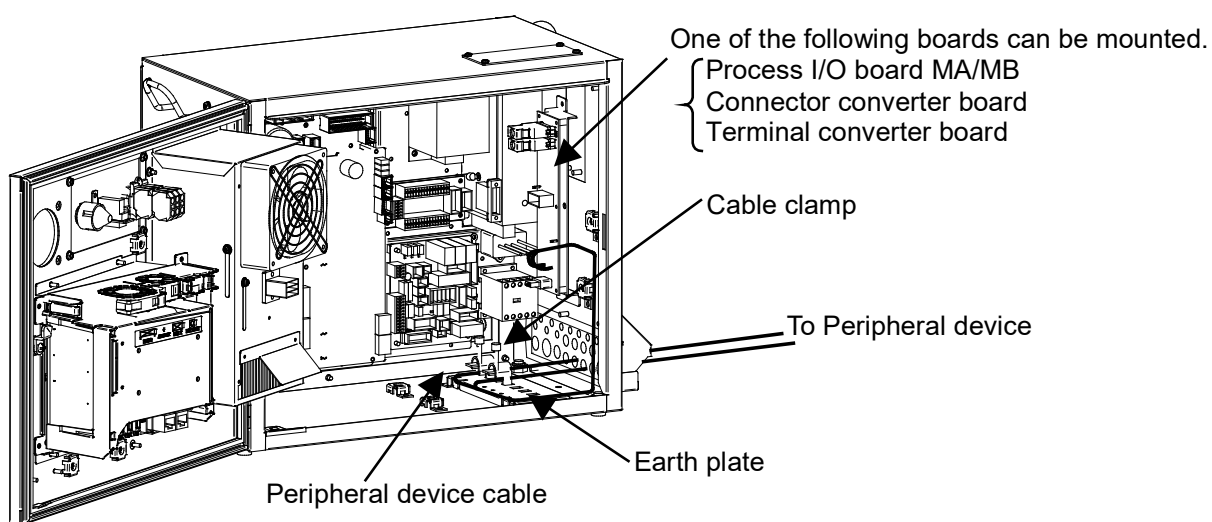
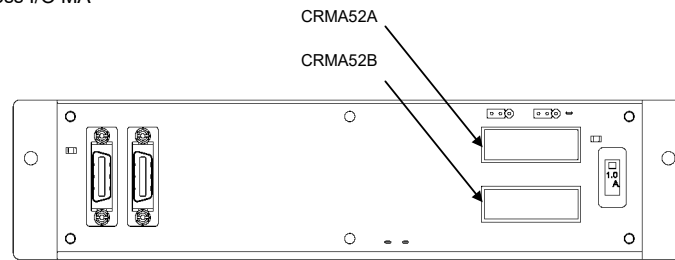


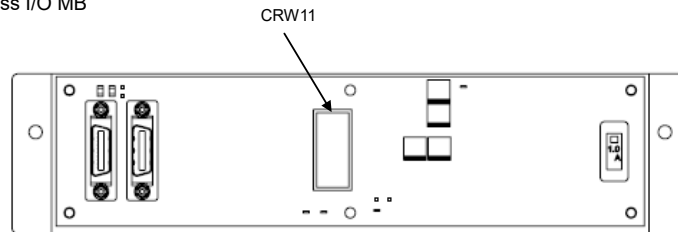
Fig.4(a) Connecting the peripheral device cable (CRMA15, CRMA16)



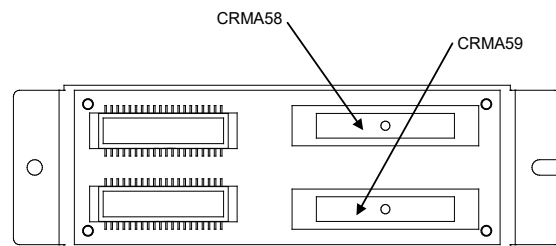
Process I/O MA



Process I/O MB



Connector converter board



Terminal converter board

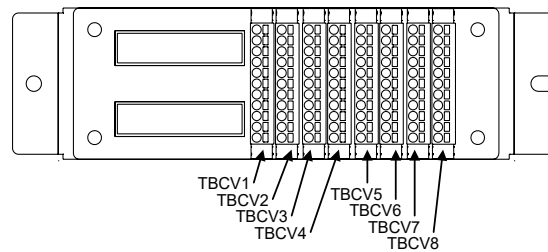


Fig.4(b) Connecting the peripheral device cable
(Process I/O board MA/MB, Connector converter board, Terminal converter board)

4.1 PERIPHERAL DEVICE INTERFACE BLOCK DIAGRAM

Following are a block diagram of the peripheral device interface and the specifications.

4.1.1 In Case of Main Board (CRMA15, CRMA16)

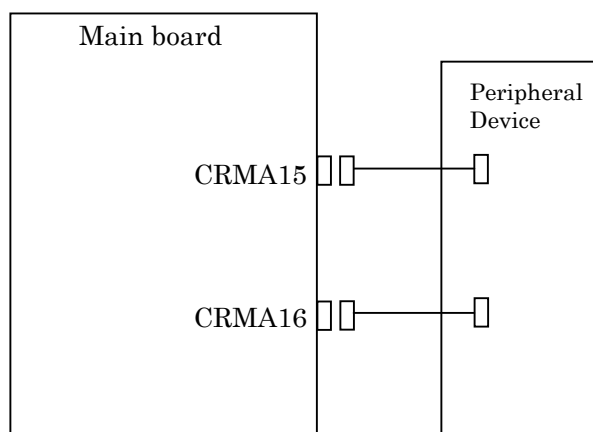
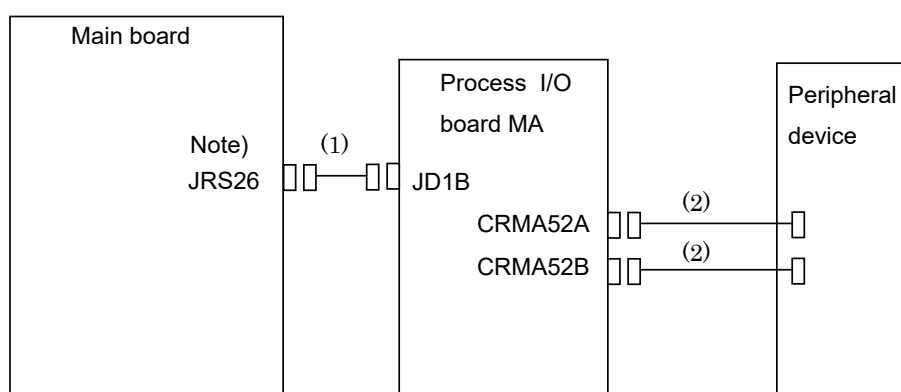


Fig.4.1.1 Block diagram of connecting peripheral device cable

Name	Drawing number	Remarks
Peripheral device connection cable (For main board)	A05B-2650-J100	Length: 10m (CRMA15) Length: 10m (CRMA16)
	A05B-2650-J101	Length: 20m (CRMA15) Length: 20m (CRMA16)

4.1.2 In the Case of the Process I/O Board MA



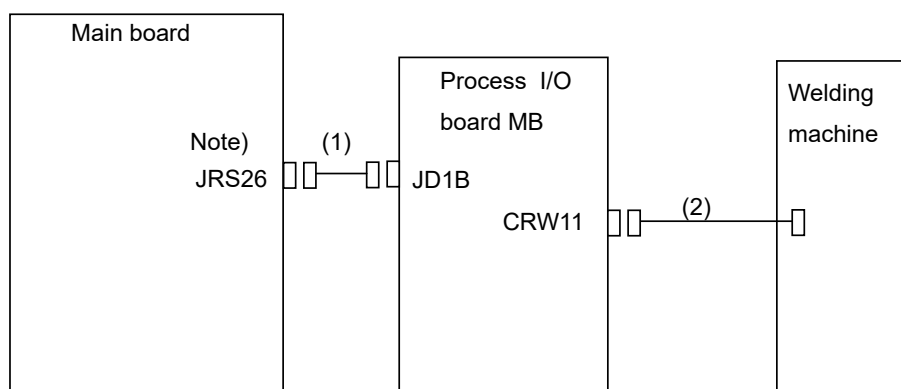
NOTE

The connection depends on whether the R-30iB Mate/R-30iB Mate Plus is the I/O link master or an I/O link slave. For details, see Section 3.2.1

Fig.4.1.2 Block diagram of the process I/O MA

Component	Drawing number	Remark
(1) I/O link cable	-	Included in the process I/O board MA
(2) Peripheral device cable (For process I/O MA)	A05B-2650-J150	Connection length 10m (one): CRMA52
	A05B-2650-J151	Connection length 20m (one): CRMA52

4.1.3 In the Case of the Process I/O Board MB



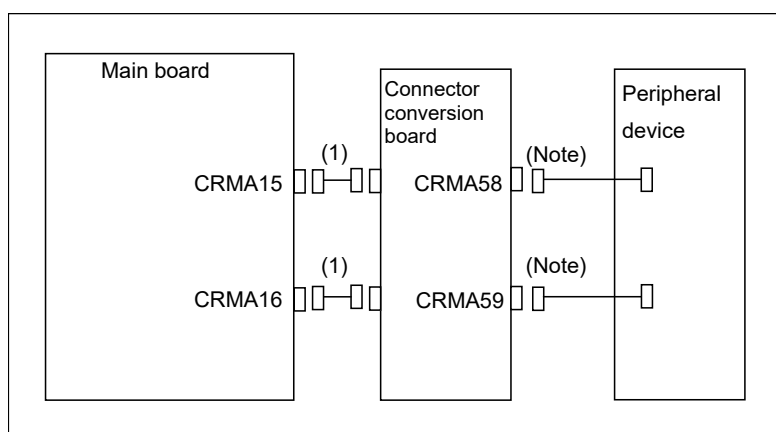
NOTE

The connection depends on whether the R-30iB Mate/R-30iB Mate Plus is the I/O link master or an I/O link slave. For details, see Section 3.2.1

Fig.4.1.3 Block diagram of the process I/O MB

No.	Component	Drawing number	Remark
(1)	I/O link cable	-	Included in the process I/O board MB
(2)	Welding machine connection cable (FANUC interface/elbow type)	A05B-2650-J160	Connection length 3m (one): CRW11
		A05B-2650-J161	Connection length 7m (one): CRW11
		A05B-2650-J162	Connection length 14m (one): CRW11

4.1.4 In the Case of the Connector Conversion Board



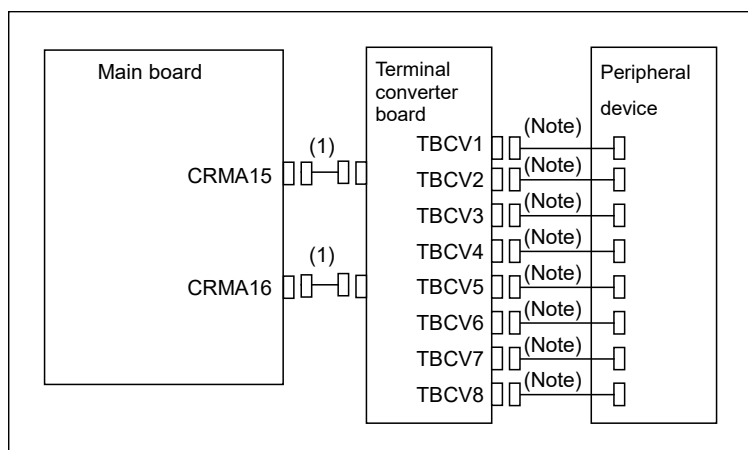
NOTE

This cable is not provided by FANUC. The customer needs to make it. For details on the connection method, see "4.3.2. Connection between the Connector Converter Board and Peripheral Devices".

Fig.4.1.4 Connection diagram of the connector conversion board

No.	Component	Drawing number	Remark
(1)	Connection cable	-	Included in the I/O connector conversion board.

4.1.5 In the Case of the Terminal Converter Board

**NOTE**

This cable is not provided by FANUC. The customer needs to make it. For details on the connection method, see "Connection between the Terminal Converter Board and Peripheral Devices".

Fig.4.1.5 Connection diagram of the terminal converter board

No.	Component	Drawing number	Remark
(1)	Connection cable	-	Included in the terminal converter board.

4.2 I/O SIGNALS OF MAIN BOARD

There are 28 data inputs (DI) and 24 data outputs (DO) on main board.
Table 4.2 shows I/O signals of main board.

Table 4.2 I/O Signals of main board (1/2)

Connector number	Signal name	Description	Remarks
CRMA15-A5	DI101	Peripheral device status	General signal
CRMA15-B5	DI102		
CRMA15-A6	DI103		
CRMA15-B6	DI104		
CRMA15-A7	DI105		
CRMA15-B7	DI106		
CRMA15-A8	DI107		
CRMA15-B8	DI108		
CRMA15-A9	DI109		
CRMA15-B9	DI110		
CRMA15-A10	DI111		
CRMA15-B10	DI112		
CRMA15-A11	DI113		
CRMA15-B11	DI114		
CRMA15-A12	DI115		
CRMA15-B12	DI116		
CRMA15-A13	DI117		
CRMA15-B13	DI118		
CRMA15-A14	DI119		
CRMA15-B14	DI120		
CRMA16-A5	XHOLD	Temporary stop	
CRMA16-B5	RESET	External reset	
CRMA16-A6	START	Start	
CRMA16-B6	ENBL	Operation enabled	
CRMA16-A7	PNS1	Program number	
CRMA16-B7	PNS2		
CRMA16-A8	PNS3		
CRMA16-B8	PNS4		

Table 4.2 I/O Signals of main board (2/2)

Connector number	Signal name	Description	Remarks
CRMA15-A15	DO101	Peripheral device status	General signal
CRMA15-B15	DO102		
CRMA15-A16	DO103		
CRMA15-B16	DO104		
CRMA15-A17	DO105		
CRMA15-B17	DO106		
CRMA15-A18	DO107		
CRMA15-B18	DO108		
CRMA16-A10	DO109		
CRMA16-B10	DO110		
CRMA16-A11	DO111		
CRMA16-B11	DO112		
CRMA16-A12	DO113		
CRMA16-B12	DO114		
CRMA16-A13	DO115		
CRMA16-B13	DO116		
CRMA16-A14	DO117		
CRMA16-B14	DO118		
CRMA16-A15	DO119		
CRMA16-B15	DO120		
CRMA16-A16	CMDENBL	During automatic operation	
CRMA16-B16	FAULT	Alarm	
CRMA16-A17	BATALM	Battery voltage drop	
CRMA16-B17	BUSY	During operation	

4.3 INTERFACE FOR PERIPHERAL DEVICES

4.3.1 Connection between the Main Board (CRMA15, CRMA16) and Peripheral Devices

Peripheral device control interface A1 (source type DO)

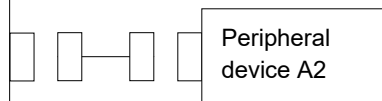
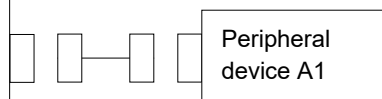
CRMA15

	A	B
01	24F	24F
02	24F	24F
03	SDICOM1	SDICOM2
04	0V	0V
05	DI101	DI102
06	DI103	DI104
07	DI105	DI106
08	DI107	DI108
09	DI109	DI110
10	DI111	DI112
11	DI113	DI114
12	DI115	DI116
13	DI117	DI118
14	DI119	DI120
15	DO101	DO102
16	DO103	DO104
17	DO105	DO106
18	DO107	DO108
19	0V	0V
20	DOSRC1	DOSRC1

Peripheral device control interface A2 (source type DO)

CRMA16

	A	B
01	24F	24F
02	24F	24F
03	SDICOM3	
04	0V	0V
05	XHOLD	RESET
06	START	ENBL
07	PNS1	PNS2
08	PNS3	PNS4
09		
10	DO109	DO110
11	DO111	DO112
12	DO113	DO114
13	DO115	DO116
14	DO117	DO118
15	DO119	DO120
16	CMDENBL	FAULT
17	BATALM	BUSY
18		
19	0V	0V
20	DOSRC2	DOSRC2



SDICOM1 to 3 signal are common selection signal for SDI.

When +24F common is used, connect to 0V.

When 0V common is used, connect to +24F.

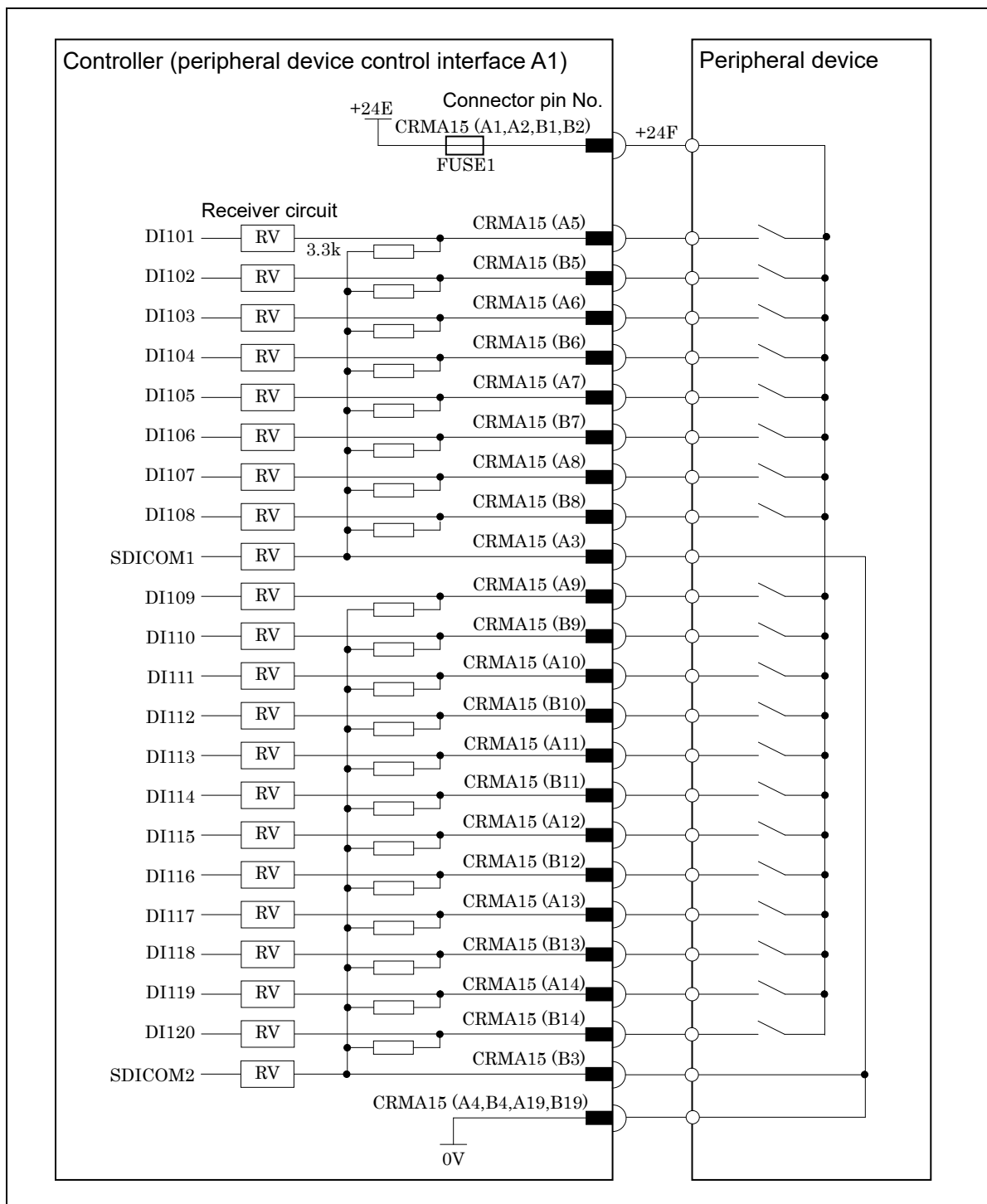
SDICOM1 → Selects a common for DI101 to DI108.

SDICOM2 → Selects a common for DI109 to DI120.

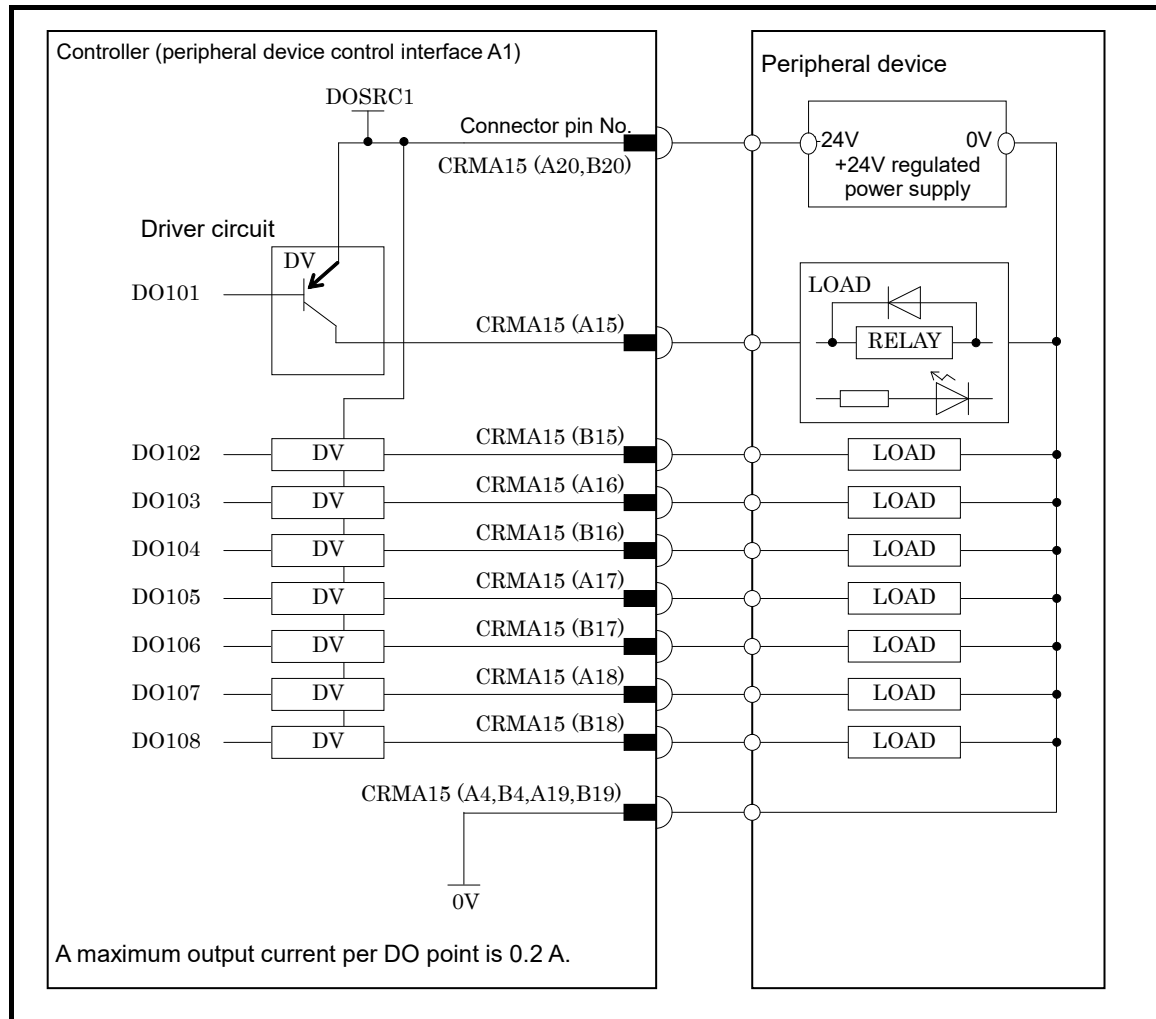
SDICOM3 → Selects a common for XHOLD, RESET, START, ENBL, PNS1 to PNS4.

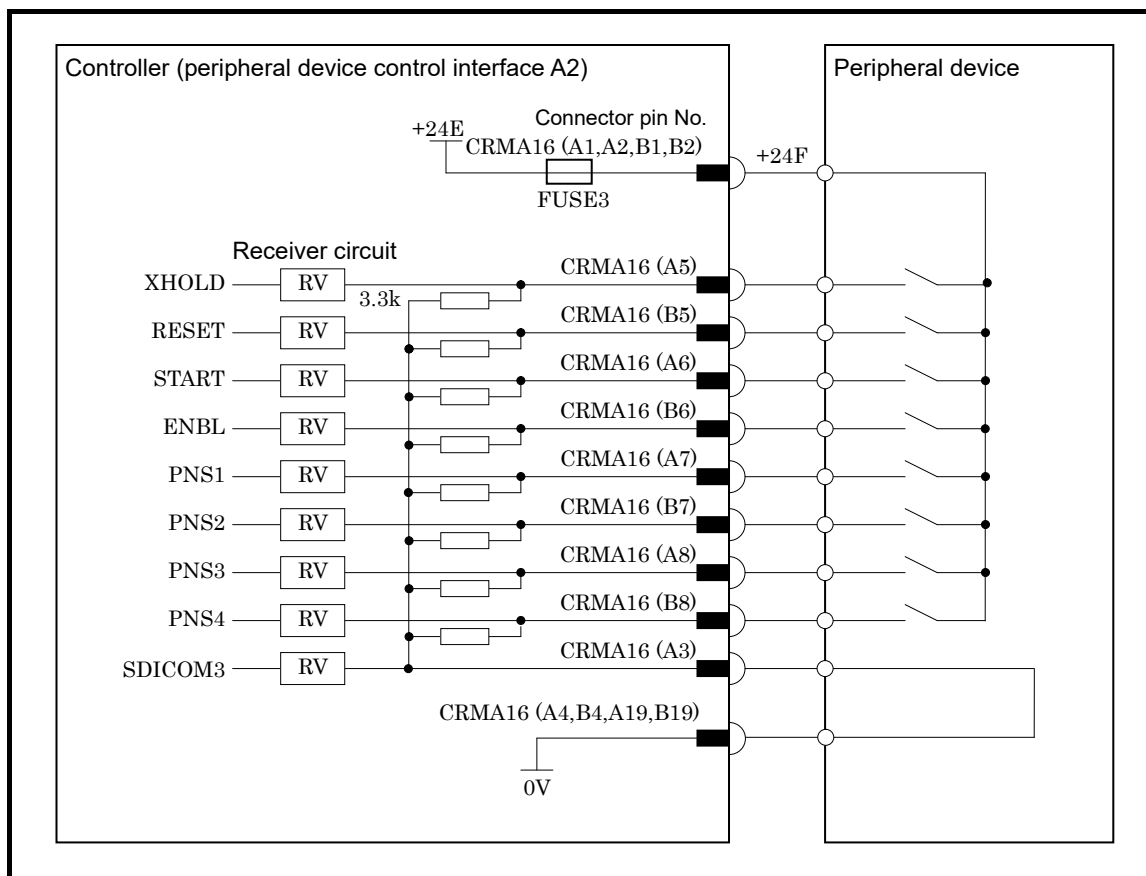
NOTE

- 1 The peripheral device connection cables are optional.
- 2 The DOSRC1 and DOSRC2 pins of the CRMA15 and CRMA16 are pins for supplying power to drivers. (None of these pins can be left open.)

**NOTE**

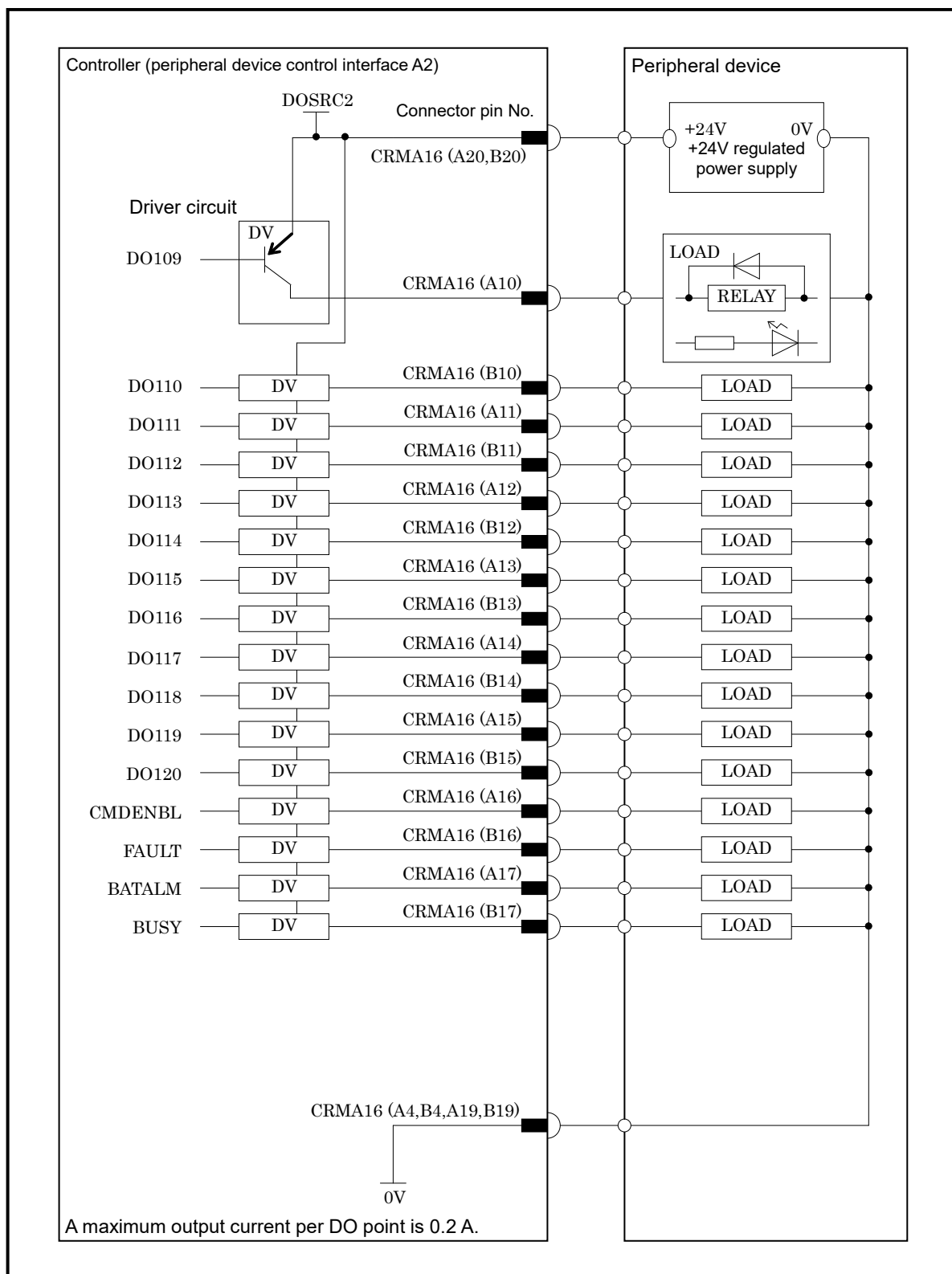
In this diagram, common voltage of input devices is +24V.



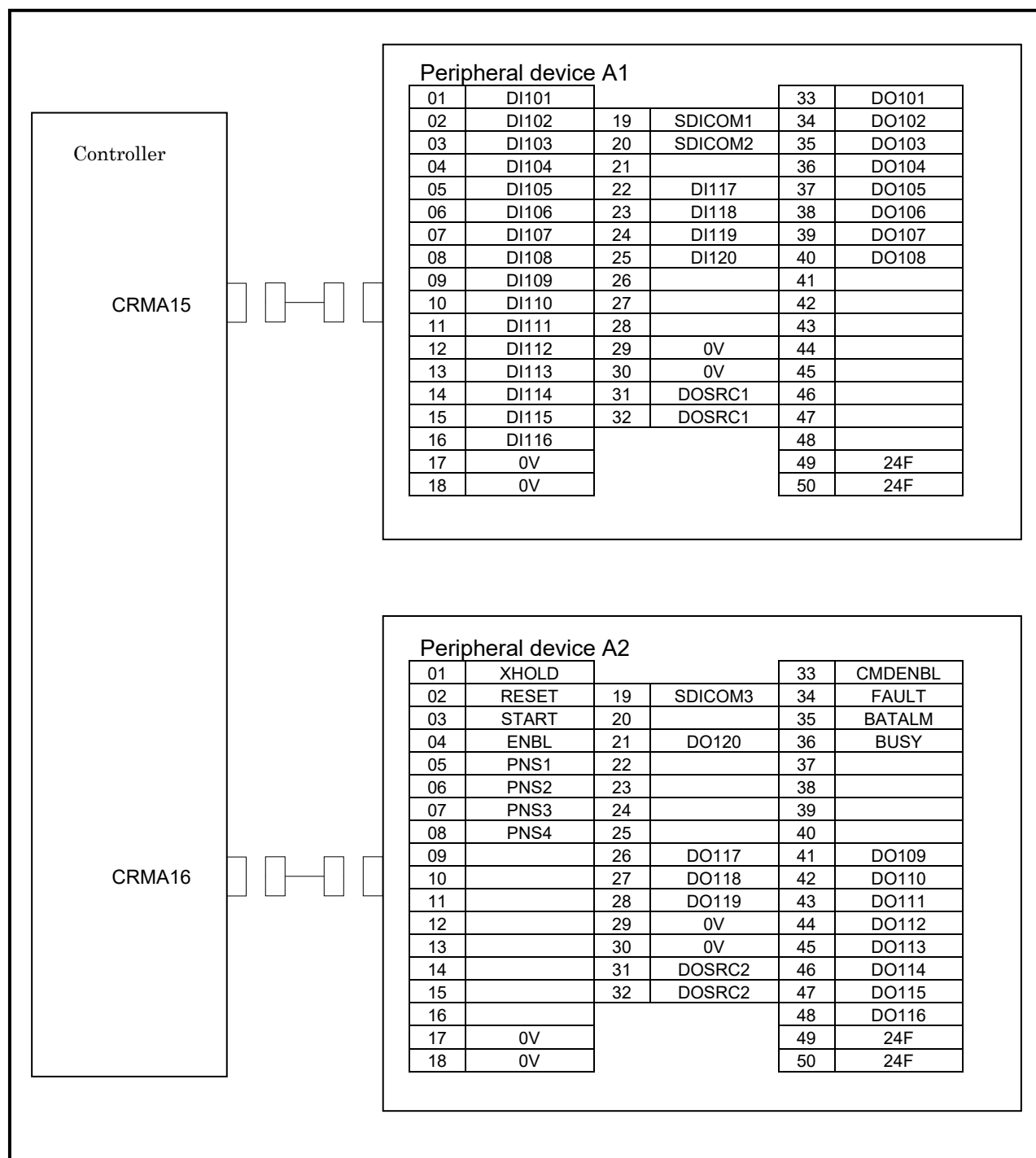


NOTE

In this diagram, common voltage of input devices is +24V.

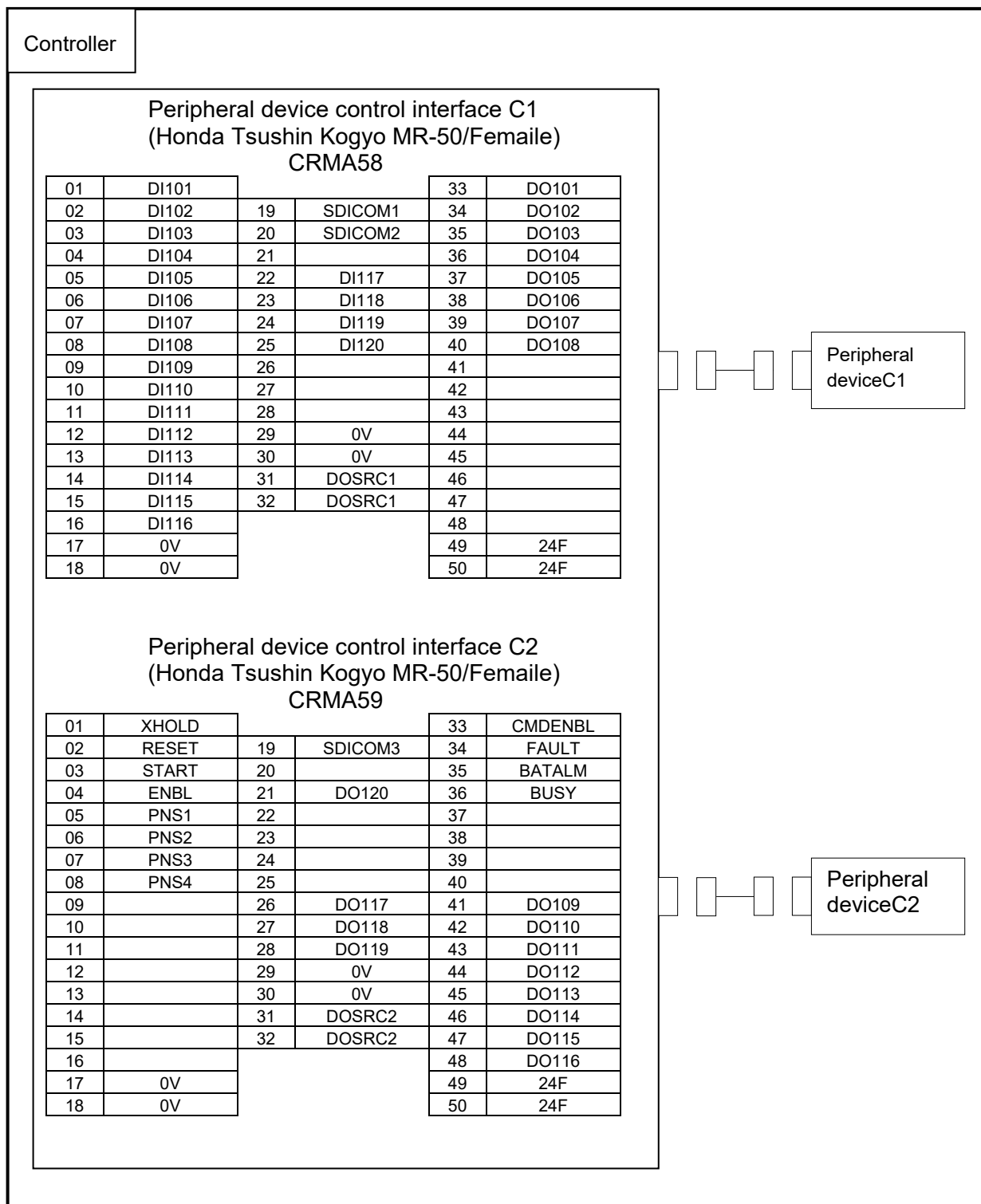


The following shows the connector interface of the optional peripheral device cables on the peripheral device side.



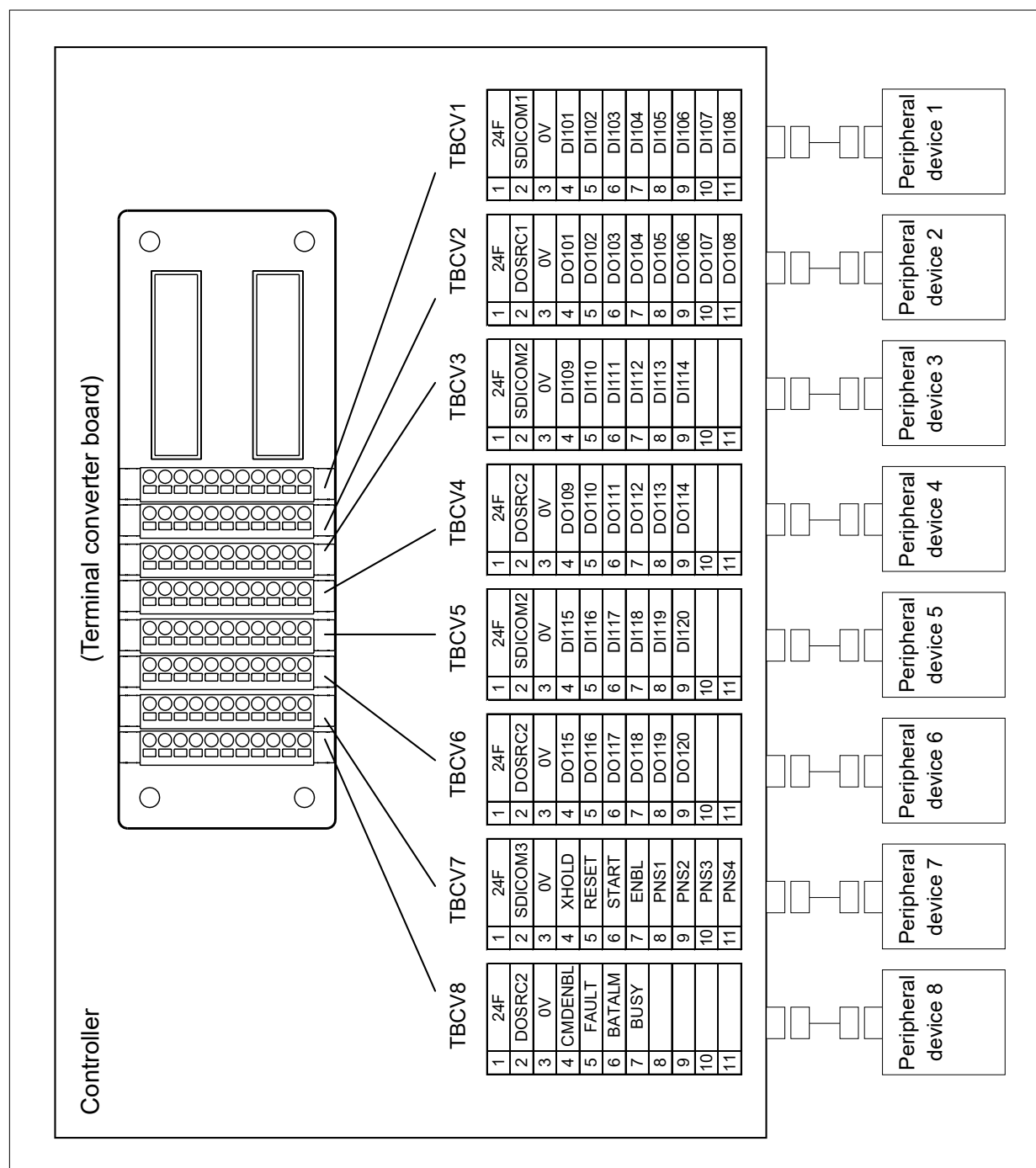
4.3.2 Connection between the Connector Converter Board and Peripheral Devices

The connector interface of the optional connector conversion board is shown below. For electrical connection, see Section 4.3.1.



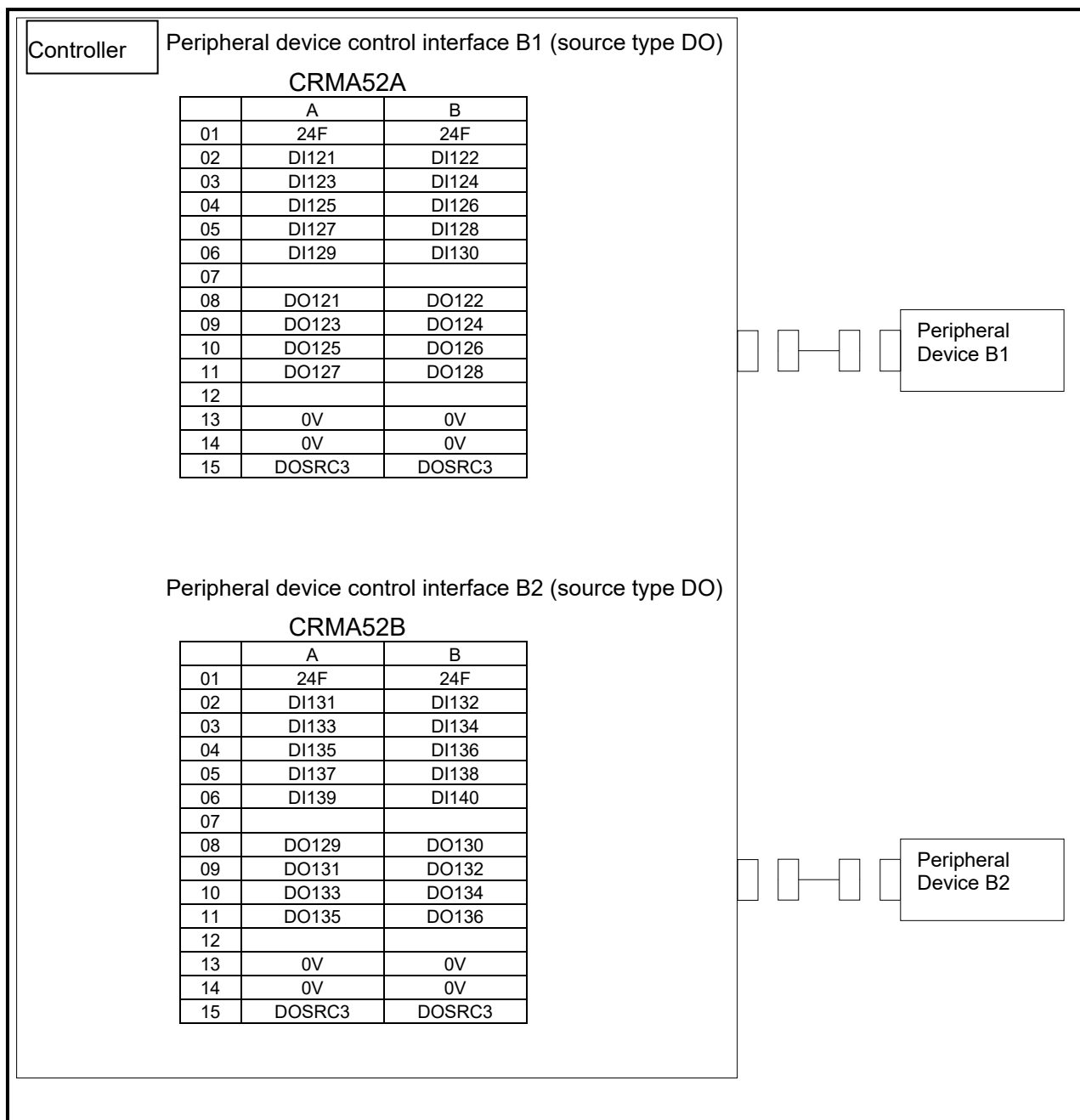
4.3.3 Connection between the Terminal Converter Board and Peripheral Devices

The terminal block interface of the optional terminal converter board is shown below. For electrical connection, see Section 4.3.1.



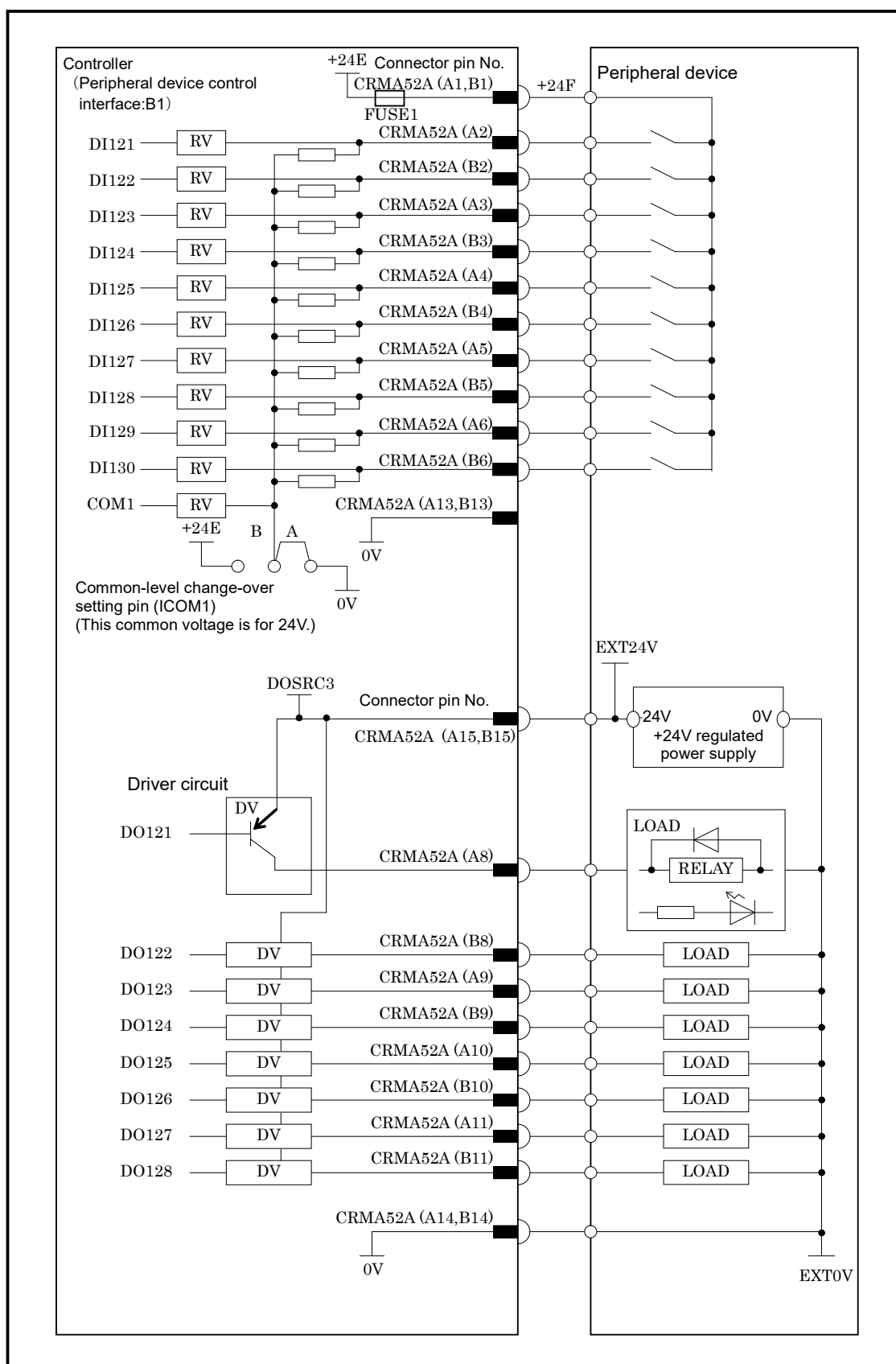
	Manufacturer	Manufacturer's specification	Remark
11-pole terminal block (TBCV1-8)	Phoenix Contact	FMC 1,5/11-ST-3,5-RF AU	Refer to appendix F

4.3.4 Connection between the Process I/O Board MA and Peripheral Devices



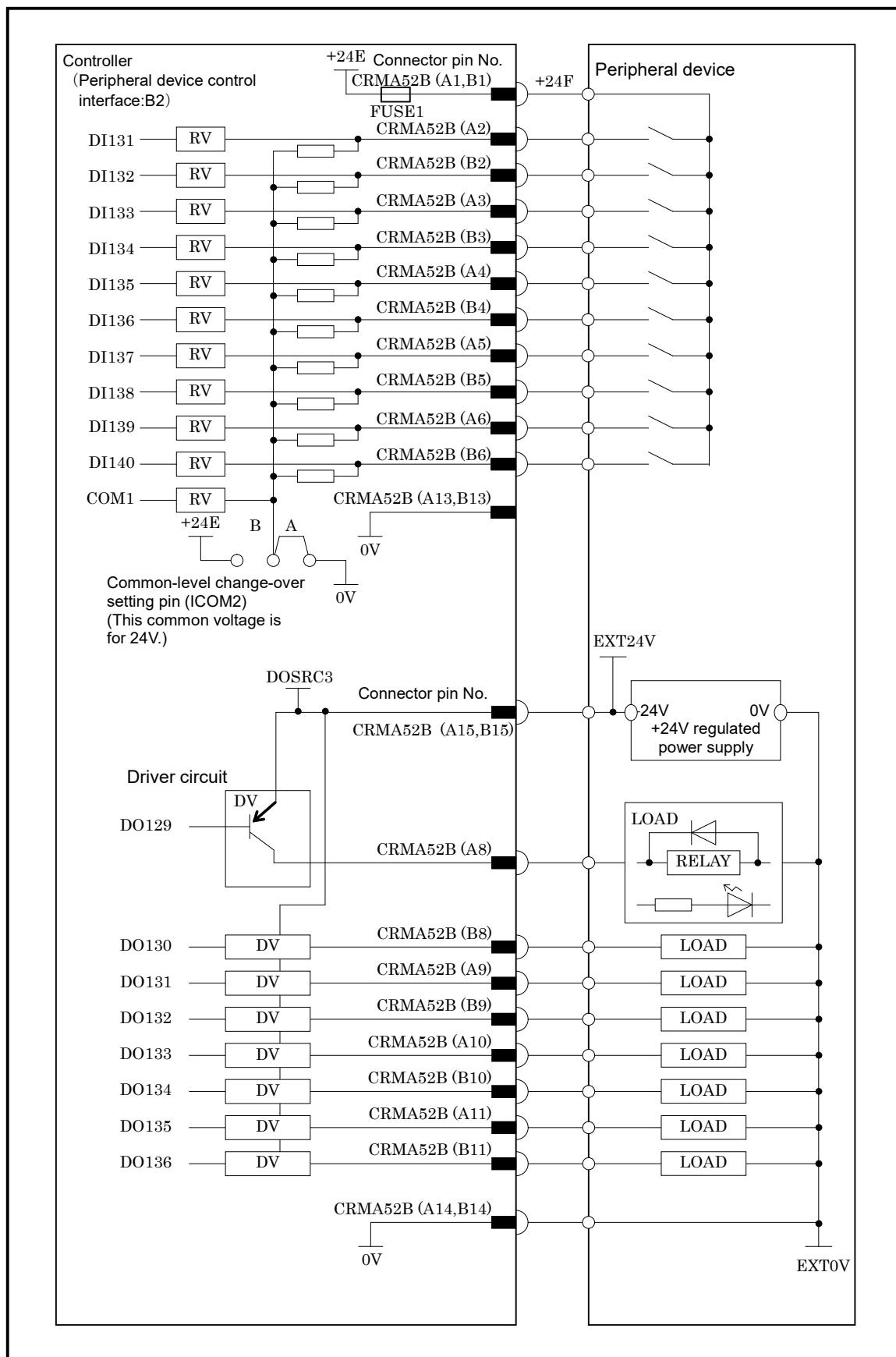
NOTE

- 1 The peripheral device connection cable is optional.
- 2 The DOSRC3 pin of CRMA52A and CRMA52B supply power to the drivers (connect all pins).



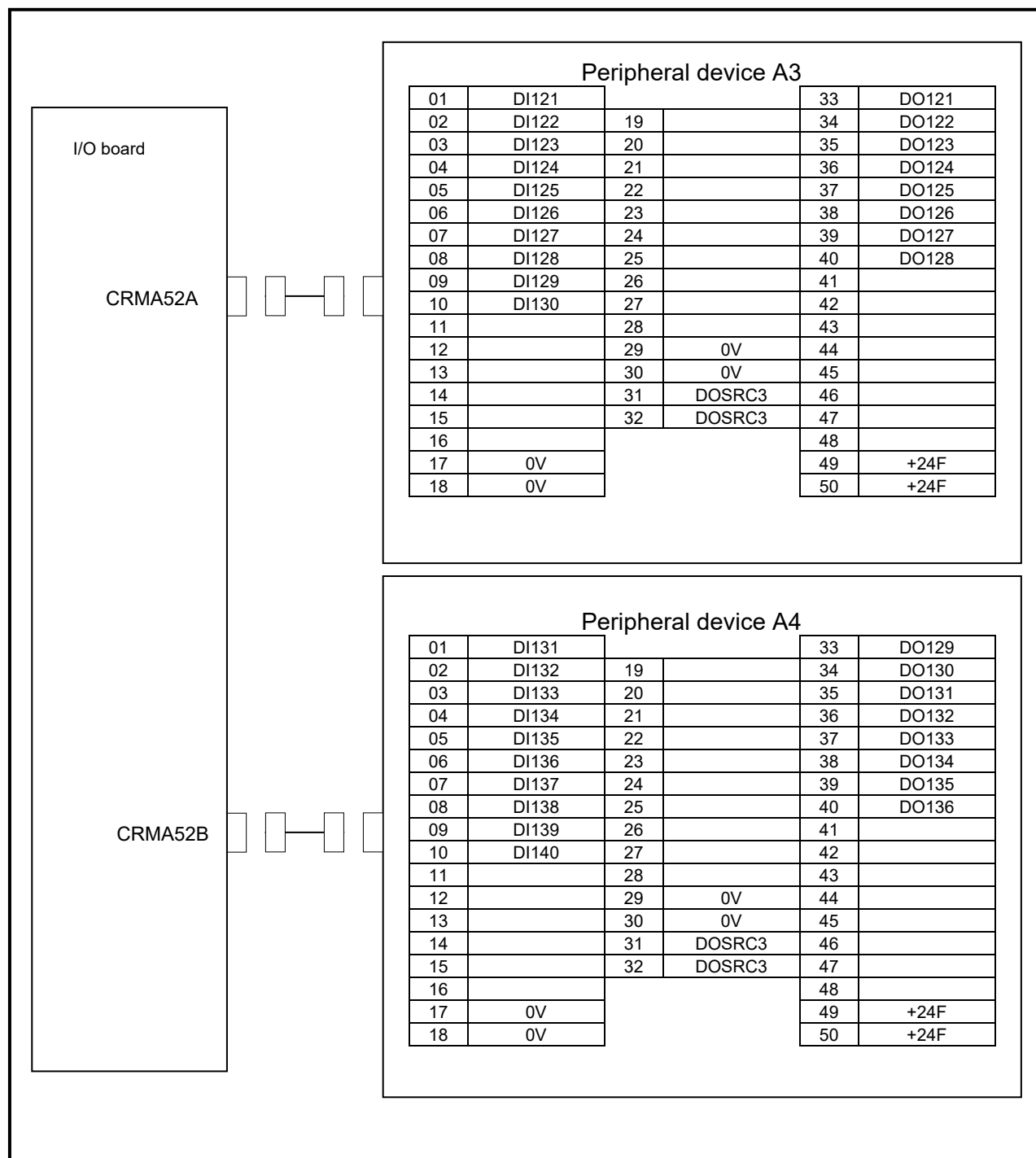
NOTE

In this diagram, common voltage of input device is 24V.

**NOTE**

In this diagram, common voltage of input device is 24V.

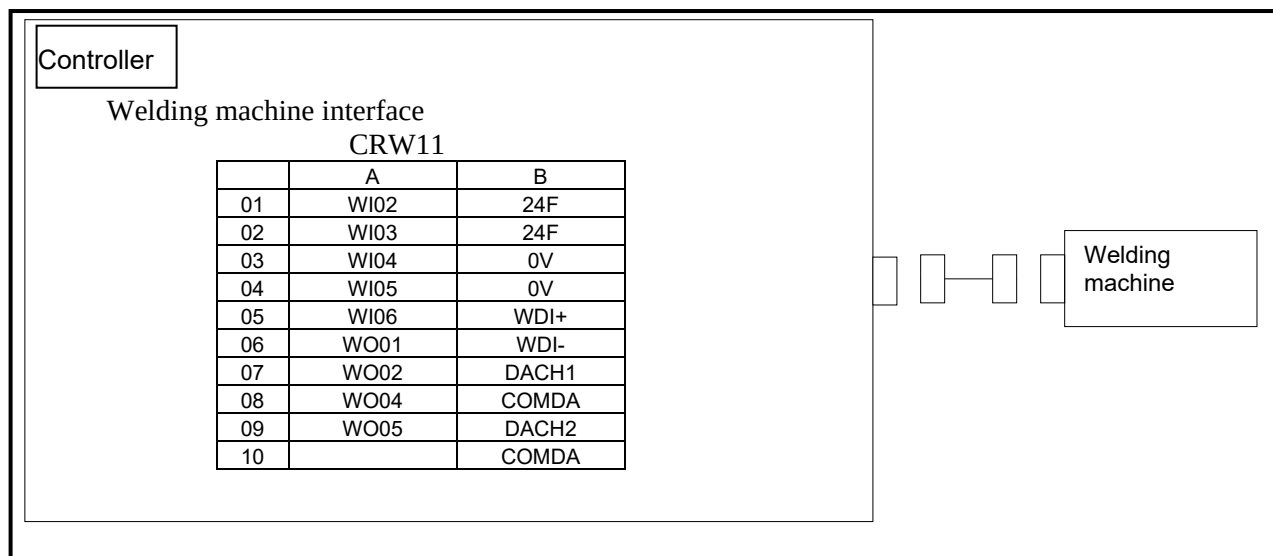
The following shows the connector interface of the optional peripheral device cables on the peripheral device side.

**NOTE**

Refer to the previous page about details of connection.

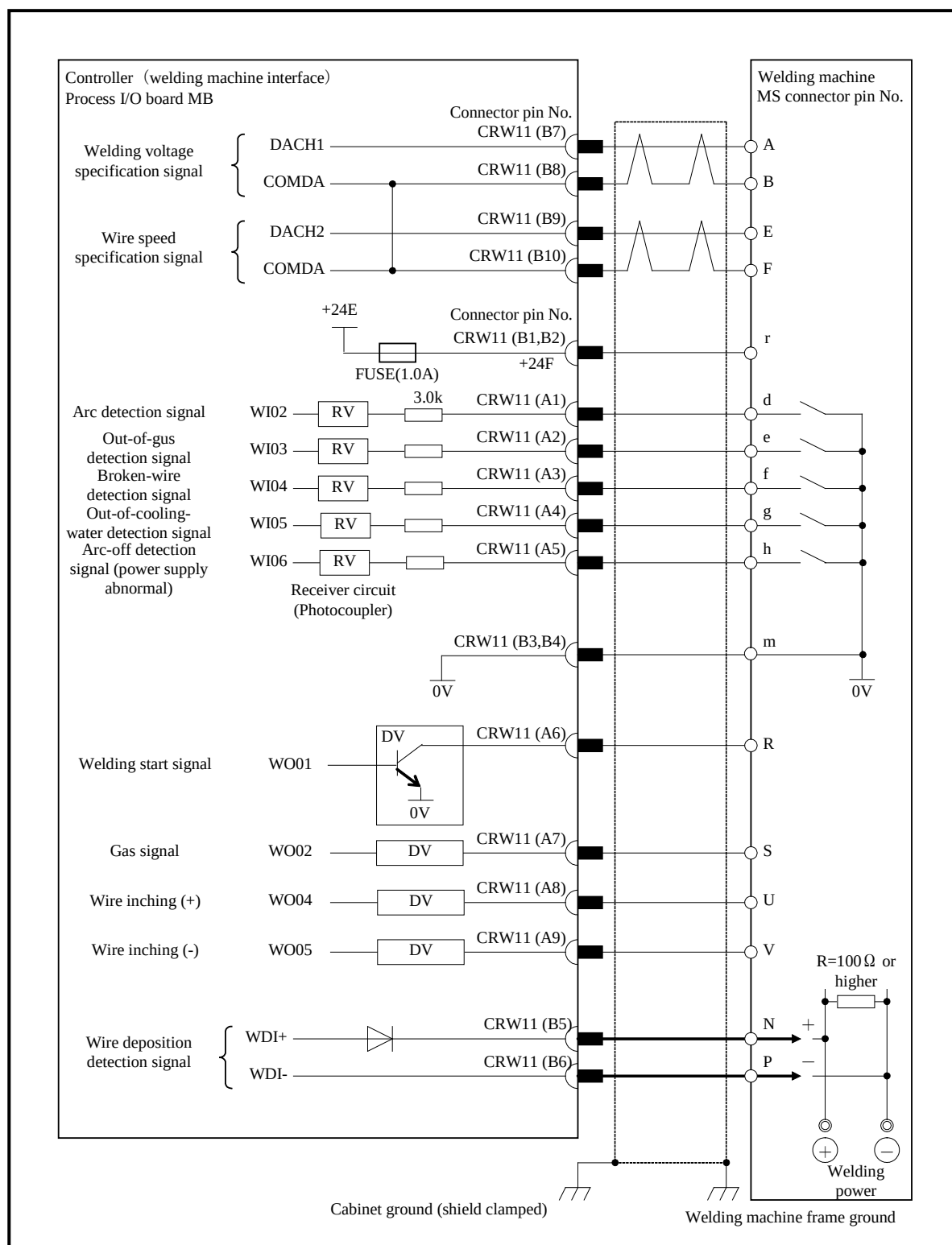
4.4 INTERFACE FOR WELDING MACHINES

4.4.1 Connection between the Process I/O Board MB and Welding Machines



NOTE

- 1 The welding machine connection cable is optional.



**Pin-to-pin connection between CRW11 connector and welding machine connector (FANUC interface)
(analog output, welding wire deposition detection, WI/WO connection)**

4.5 EE INTERFACE

4.5.1 Connection between the Robot and End Effector

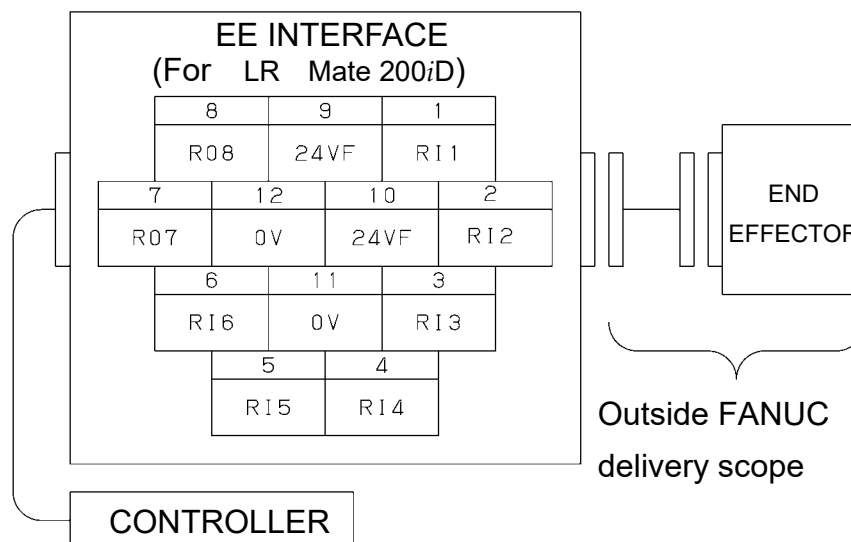


Fig.4.5.1(a) Connection of End effector (In case of LR Mate 200iD)

NOTE

RO1 to RO6 are used as the on/off signals of the solenoid valve option.
For details, refer to the operator's manual of each robot.

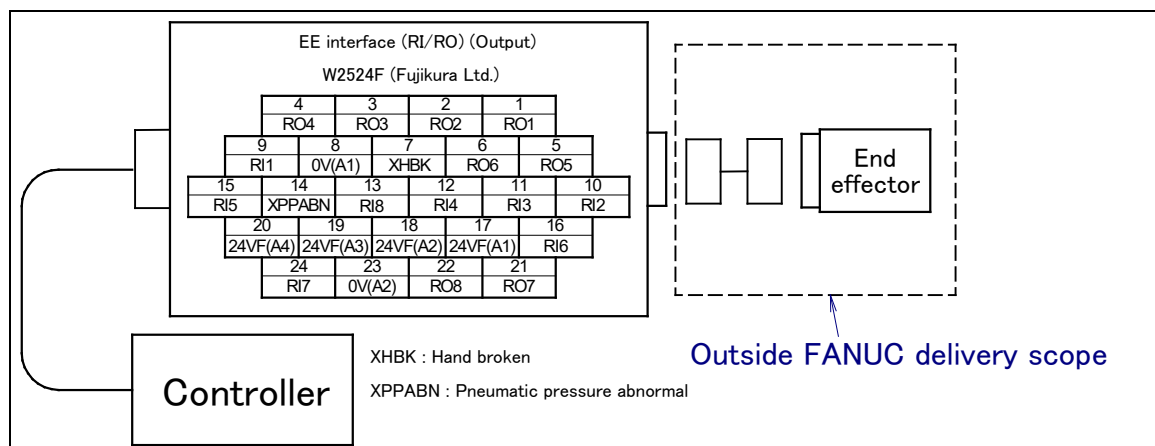
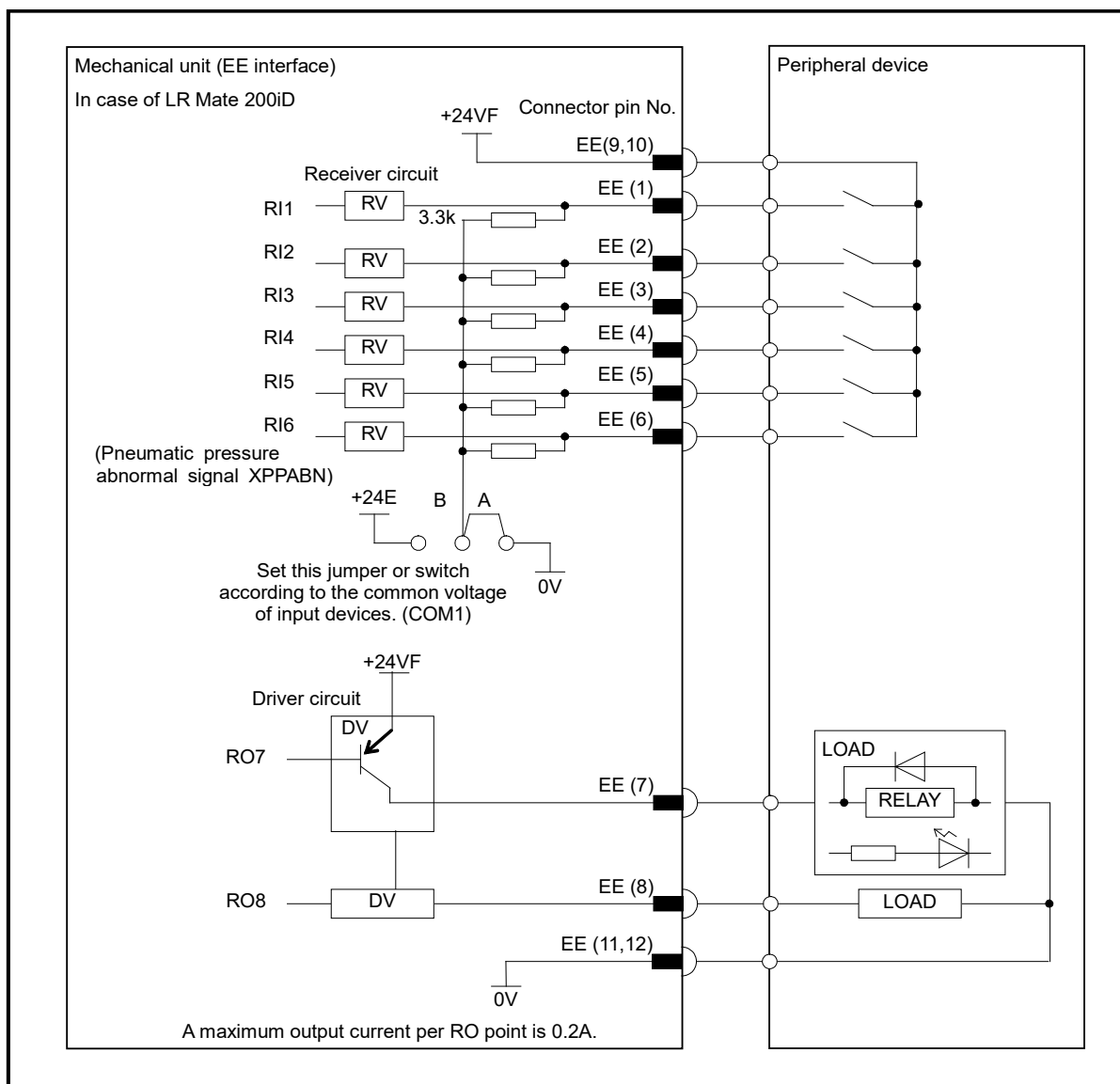


Fig.4.5.1(b) Connection of End effector (In case of R-2000iC)

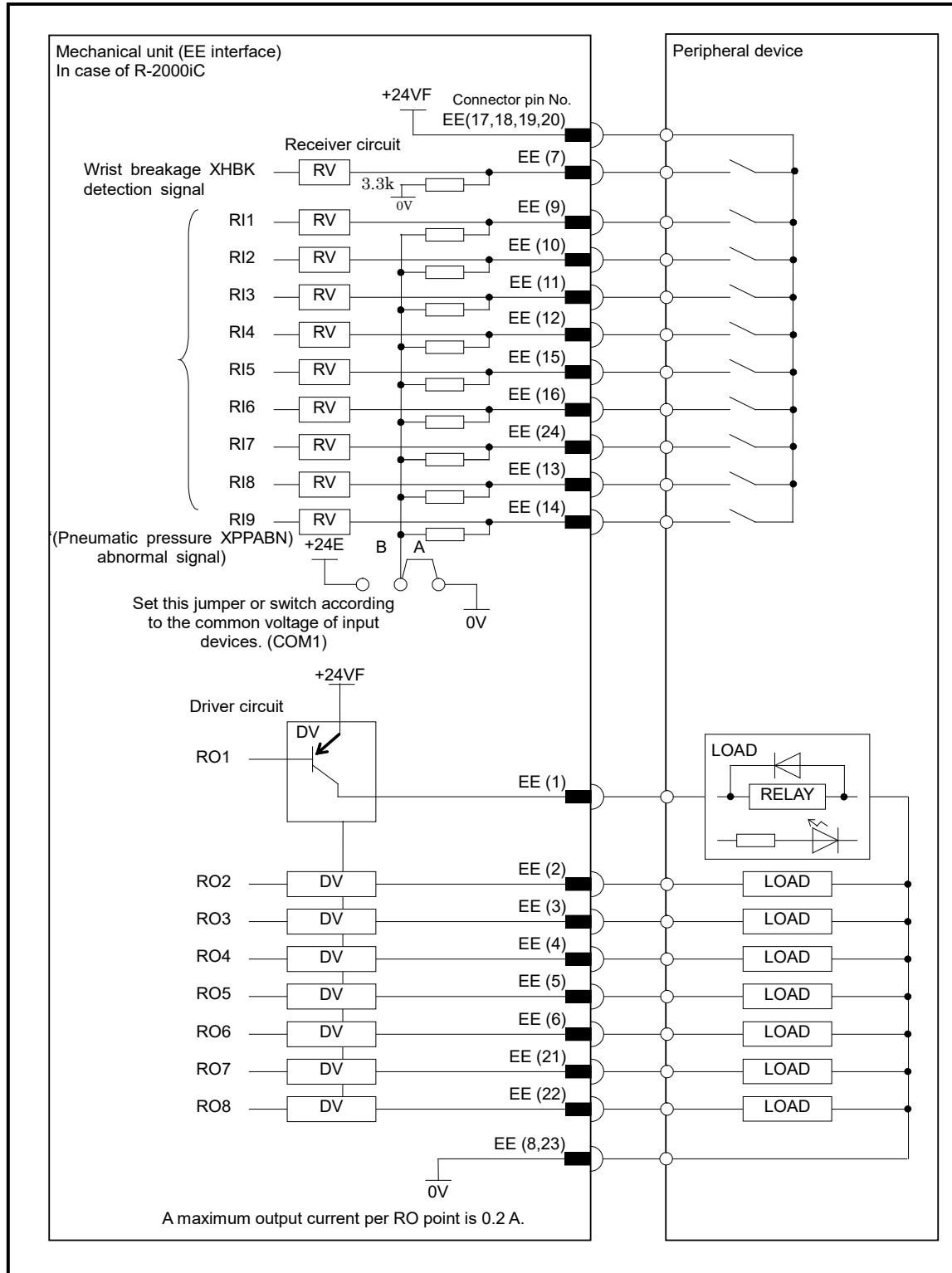
NOTE

EE interface depends on the option of the robot. Refer to the operator's manual of each robot.



NOTE

- 1 In this diagram, common voltage of input devices is +24V.
- 2 The common-level change-over setting pin or switch (COM1) is in the 6-axis servo amplifier.

**NOTE**

- 1 In this diagram, common voltage of input devices is +24V.
- 2 The common-level change-over setting pin or switch (COM1) is in the 6-axis servo amplifier.

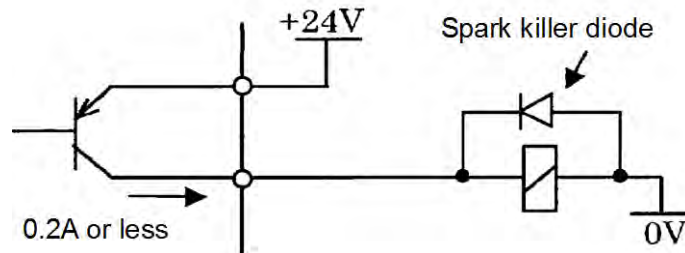
4.6 DIGITAL I/O SIGNAL SPECIFICATIONS

This section describes the specifications of the digital I/O signals interfaced with the peripheral device, end effector, and arc welder.

4.6.1 Peripheral Device Interface A

- (1) Output signals in peripheral device interface A (Source type DO)

- (a) Example of connection



- (b) Electrical specifications

Maximum load current when driver is on:	200mA (including momentary level)
Saturation voltage when driver is on:	1.0V max.
Dielectric strength:	24V $\pm$ 20% (including momentary level)
Leakage current when driver is off:	100 μ A

- (c) The external power supply to output signals must satisfy the following:

Power supply voltage:	+24V $\pm$ 10%
Power supply current:	For each printed circuit board of this type (Total sum of maximum load currents including momentary levels + 100mA or more)

Power-on timing:

At the same time when the controller is turned on or earlier

Power-off timing:

At the same time when the controller unit is turned off or later

- (d) Spark killer diode

Rated peak reverse voltage:	100V or more
Rated effective forward current:	1A or more

- (e) Driver for output signals

In the driver device, the current of each output signal is monitored, and when an overcurrent is detected, the relevant output is turned off. After an output has been turned off by overcurrent, the overcurrent state is released because the output is off, so the output on state is restored. Therefore, in the ground fault or overcurrent state, the output is turned on and off repeatedly. Such a condition is found also when a load with a high surge current is connected.

The driver device also includes an overheat detection circuit, which turns off all outputs of the device when the internal temperature of the device has increased as a result of a continued overcurrent state due to a ground fault of an output and so on. The outputs are held off, but their normal states can be restored by turning the power to the controller on and off after the internal temperature of the device has lowered.

- (f) Note on use

Do not use the +24V power supply of the robot.

When adding a relay, solenoid, or the like directly to the circuit, connect a diode for counter electromotive voltage protection in parallel to the load.

When using a load, such as a lamp, that generates surge current when it is turned on, install a protection resistor.

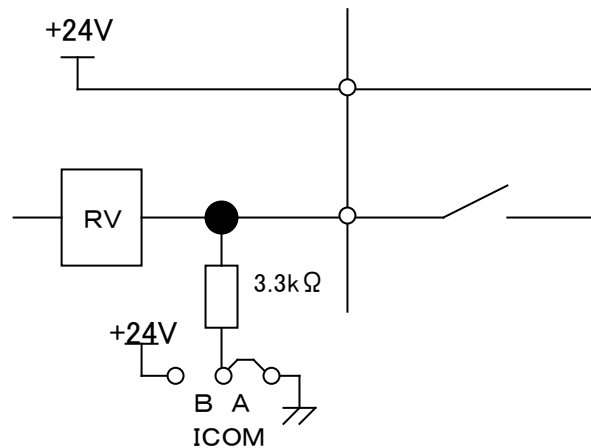
(g) Applicable signals

Output signals of main board I/O board CRMA15 and CRMA16
CMDENBL, FAULT, BATALM, BUSY, DO101 to DO120

Output signals of Process I/O board CRMA52A and CRMA52B
DO01 to DO16

(2) Input signals in peripheral device interface A

(a) Example of connection

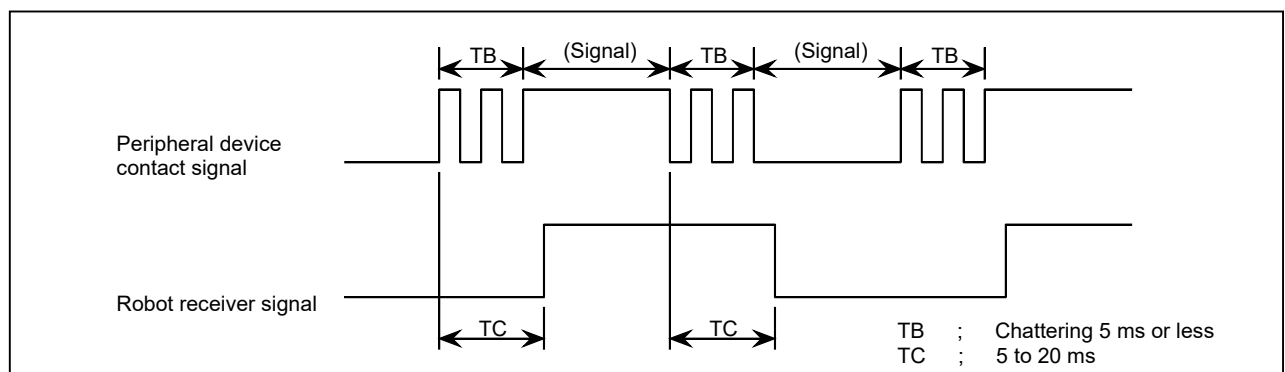


(b) Electrical specifications of the receiver

Type:	Grounded voltage receiver
Rated input voltage:	Contact close : +20V to +28V Contact open : 0V to +4V
Maximum applied input voltage:	+28VDC
Input impedance:	3.3kΩ (approx.)
Response time:	5ms to 20ms

(c) Specifications of the peripheral device contact

Voltage and Current:	DC24V, 0.1A (Use a contact which minimum load is 5mA or less.)
Input signal width:	200ms or more (on/off)
Chattering time:	5ms or less
Closed circuit resistance:	100Ω or less
Opened circuit resistance:	100kΩ or more

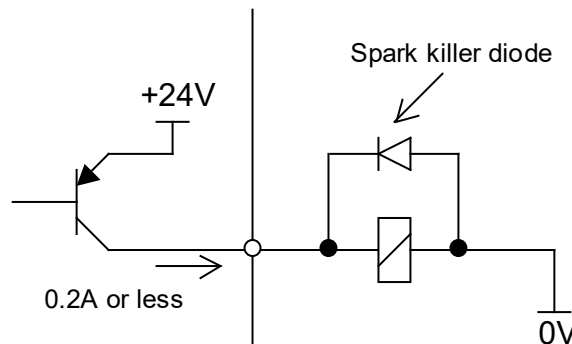


- (d) Note on use
Apply the +24 V power at the robot to the receiver.
However, the above signal specifications must be satisfied at the robot receiver.
- (e) Applicable signals
Input signals of main board CRMA15 and CRMA16
XHOLD, FAULT RESET, START, HOME, ENBL
DI101 to DI120
Input signals of Process I/O board CRMA52A and CRMA52B
DI01 to DI20

4.6.2 EE Interface

(1) Output signals in EE interface

(a) Example of connection



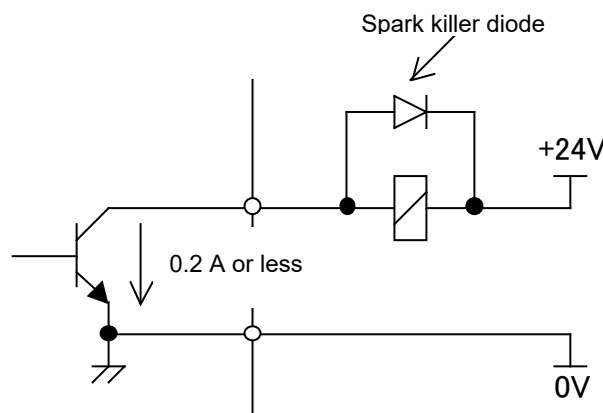
- (b) Electrical specifications

Maximum load current when driver is on:	200mA (including momentary level)
Saturation voltage when driver is on:	1.0V max.
Dielectric strength:	24V $\pm$ 20% (including momentary level)
Leakage current when driver is off:	100 μ A
- (c) Power supply to output signals
The +24V power supply on the robot side can be used if the total current level, including the current of the welding interface, is 0.7A or less.
- (d) Driver for output signals
In the driver device, the current of each output signal is monitored, and when an overcurrent is detected, the relevant output is turned off. After an output has been turned off by overcurrent, the overcurrent state is released because the output is off, so the output on state is restored. Therefore, in the ground fault or overcurrent state, the output is turned on and off repeatedly. Such a condition is found also when a load with a high surge current is connected.
The driver device also includes an overheat detection circuit, which turns off all outputs of the device when the internal temperature of the device has increased as a result of a continued overcurrent state due to a ground fault of an output and so on. The outputs are held off, but their normal states can be restored by turning the power to the controller on and off after the internal temperature of the device has lowered.
- (e) Note on use
When adding a relay, solenoid, or the like directly to the circuit, connect a diode for counter electromotive voltage protection in parallel to the load.
When using a load, such as a lamp, that generates surge current when it is turned on, install a protection resistor.

- (f) Applicable signals
RO1 to RO8
- (2) Input signal in peripheral device interface
The input signals are the same as those of other I/O boards. (Refer to Subsection 4.5.1 in CONNECTIONS.)
 - (a) Applicable signals
RI1 to RI8, XHBK, XPPABN

4.6.3 I/O Signal Specifications for ARC-Welding Interface (A-cabinet/Process I/O Board MB)

- (1) Specification for arc welding machine interface digital output signals
 - (a) Example of connection



- (b) Electrical specifications

Rated voltage:	24VDC
Maximum applicable voltage:	30VDC
Maximum load current:	200mA (including momentary level)
Transistor type:	Open-collector NPN
Saturation voltage when the circuit is on:	Approximately 1.0V
- (c) Spark killer diode

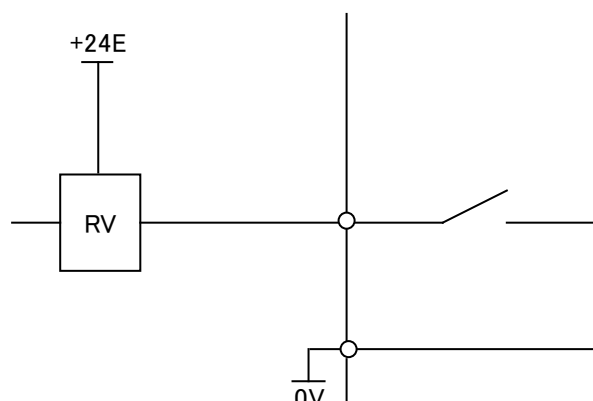
Rated peak-to-peak reverse withstand voltage:	100V or higher
Rated effective forward current:	1A or more
- (d) Caution for use

The arc welding machine interface can use the +24V power supply of the robot unless the sum of its sink current and that of the EE interface exceeds 0.7A. When using a relay or solenoid directly as a load, connect the load and a back electromotive force voltage prevention diode in parallel.

When using a load, such as a lamp, that generates surge current when it is turned on, install a protection resistor.
- (e) Applicable signals
Arc welding machine interface output signals
[WO1, 2,4,5]

(2) Specification for arc welding machine interface digital input signals

(a) Example of connection

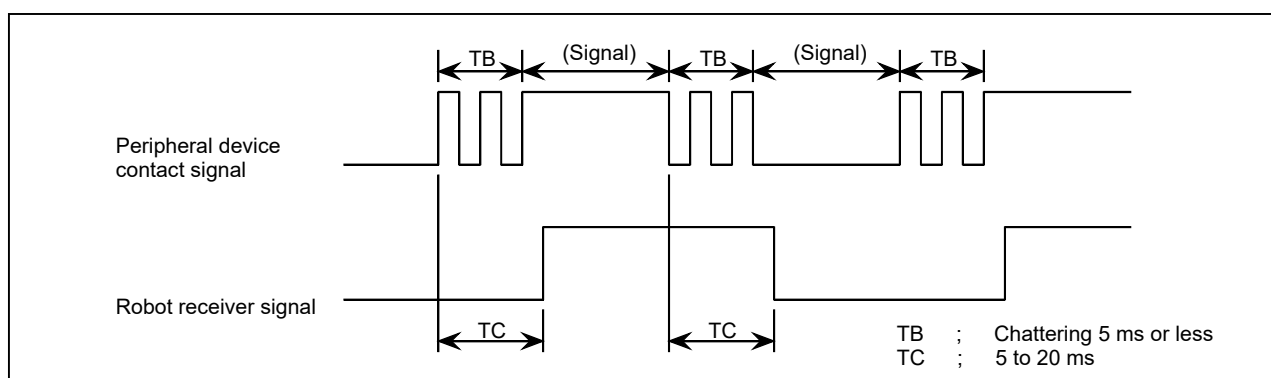


(b) Electrical specifications of the receiver

Type:	Grounded voltage receiver
Rated input voltage:	Contact close +20V to +28V Contact open 0V to +4V
Maximum applied input voltage:	+28VDC
Input impedance:	3.0k Ω (approx.)
Response time:	5ms to 20ms

(c) Specifications of the peripheral device contact

Voltage and Current:	DC24V, 0.1A (Use a contact which minimum load is 5mA less.)
Input signal width:	200ms or more (on/off)
Chattering time:	5ms or less
Closed circuit resistance:	100 Ω or less
Opened circuit resistance:	100k Ω or more



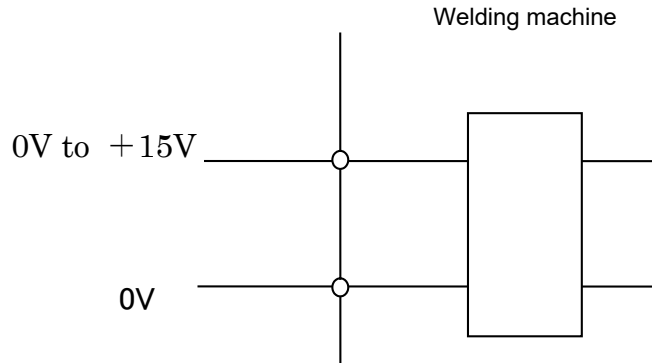
(d) Note on use

Apply the +24 V power at the robot to the receiver.
However, the above signal specifications must be satisfied at the robot receiver.

(e) Applicable signals

Arc welding machine interface input signals
[WI2 to 6]

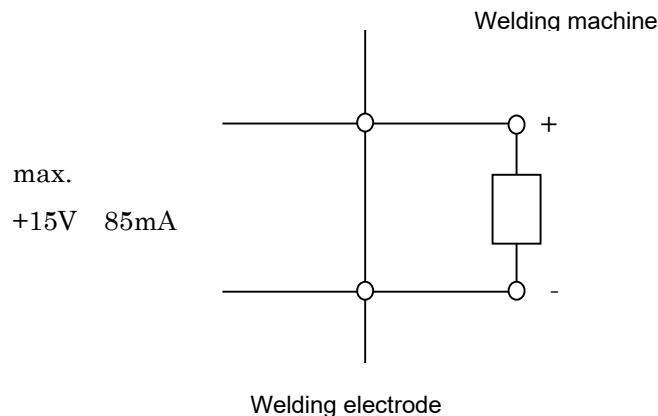
- (3) Specification for arc welding machine interface analog output signals (welding voltage and wire feed speed specification signals)
- (a) Example of connection



- (b) Caution for use
- Input impedance: 3.3k Ω or higher
- Install a high-frequency filter.

(Wire deposit detection: WDI+ and WDI-)

- (a) Example of connection



(Wire deposition detection: WDI+, WDI-)

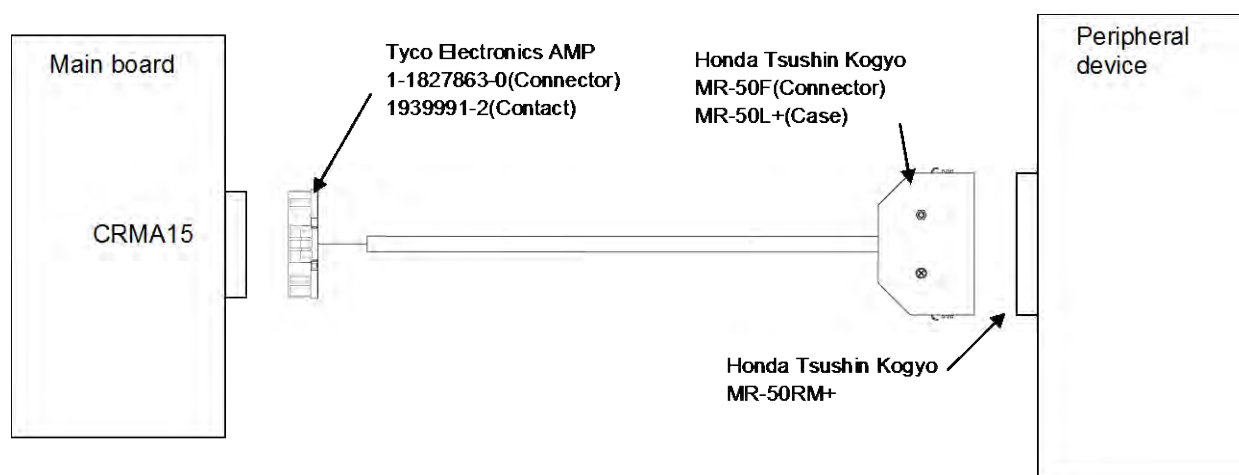
- (b) Caution for use
- The resistance between the + and - terminals of the welding machine must be 100 Ω or higher.
- The TIG welding deposition detection circuit must be isolated from the welding circuit (high frequency).
- This circuit can withstand up to 80 V.

4.7 SPECIFICATIONS OF THE CABLES USED FOR PERIPHERAL DEVICES AND WELDERS

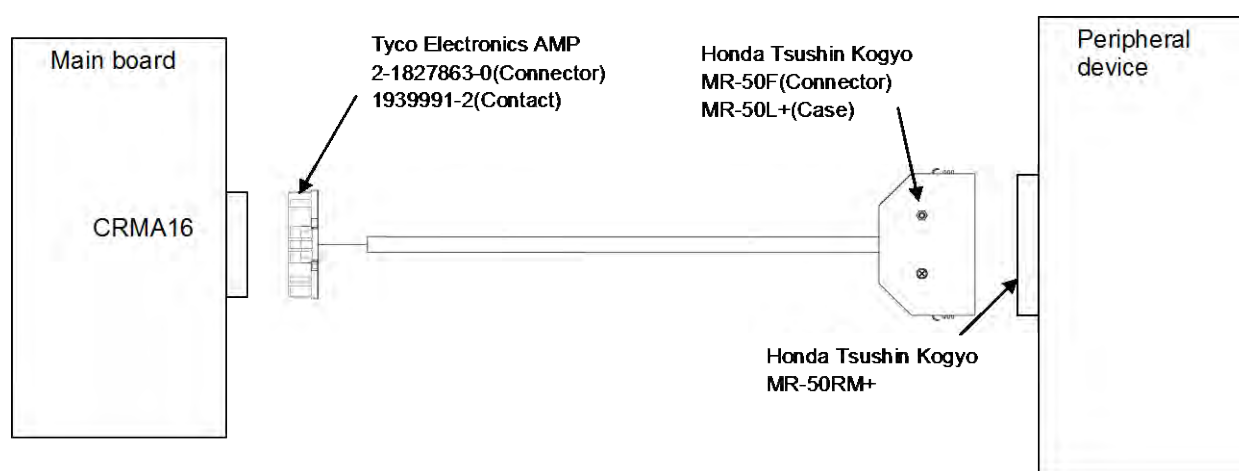
If the customer manufactures cables, make sure they conform to the FANUC standard cables described in this section.

(See the description in "Peripheral Device Interface" in this manual for the specifications of the FANUC standard cables.)

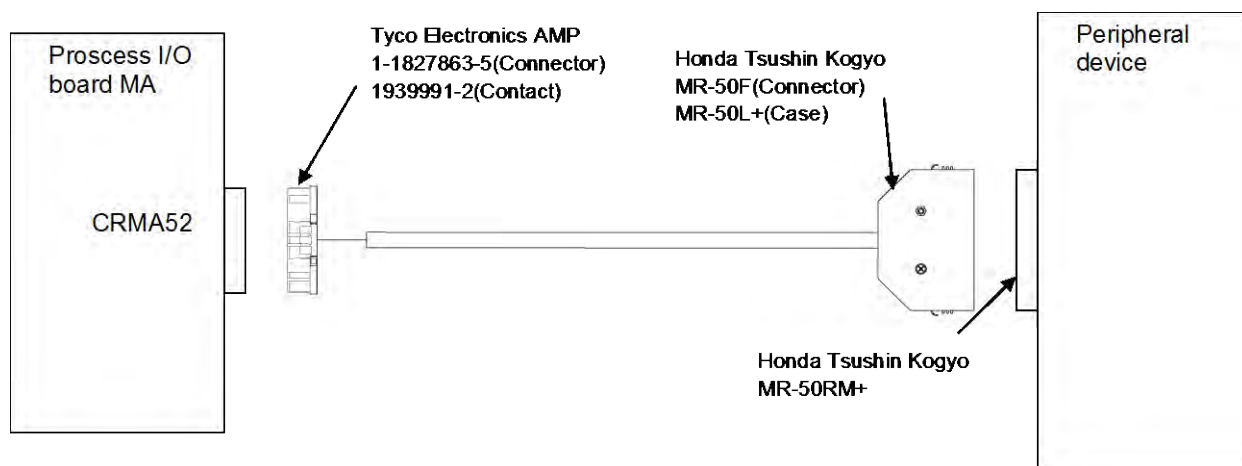
4.7.1 Peripheral Device Interface A1 Cable (CRMA15: Tyco Electronics AMP, 40 pins)



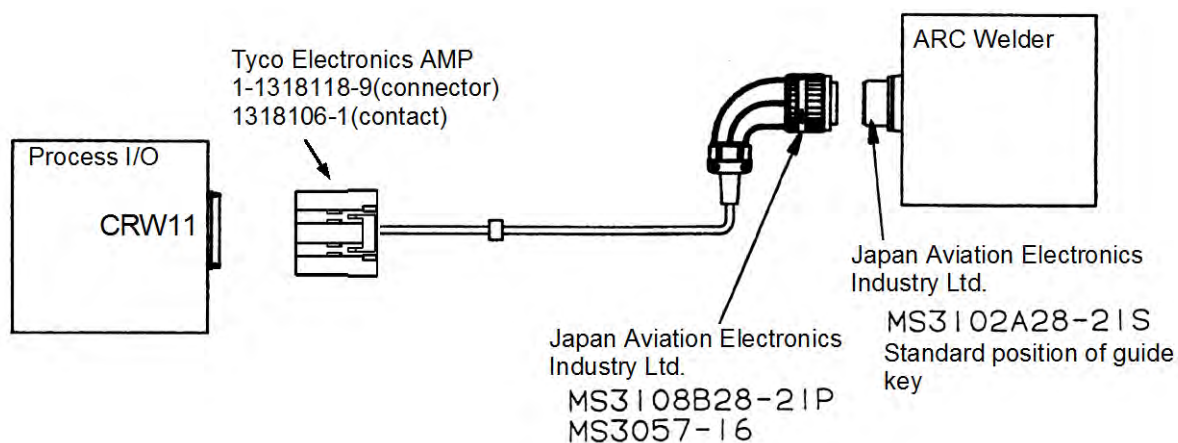
4.7.2 Peripheral Device Interface A2 Cable (CRMA16: Tyco Electronics AMP, 40 pins)



4.7.3 Peripheral Device Interface B1 and B2 Cables (CRMA52; Tyco Electronics AMP, 30 pin)



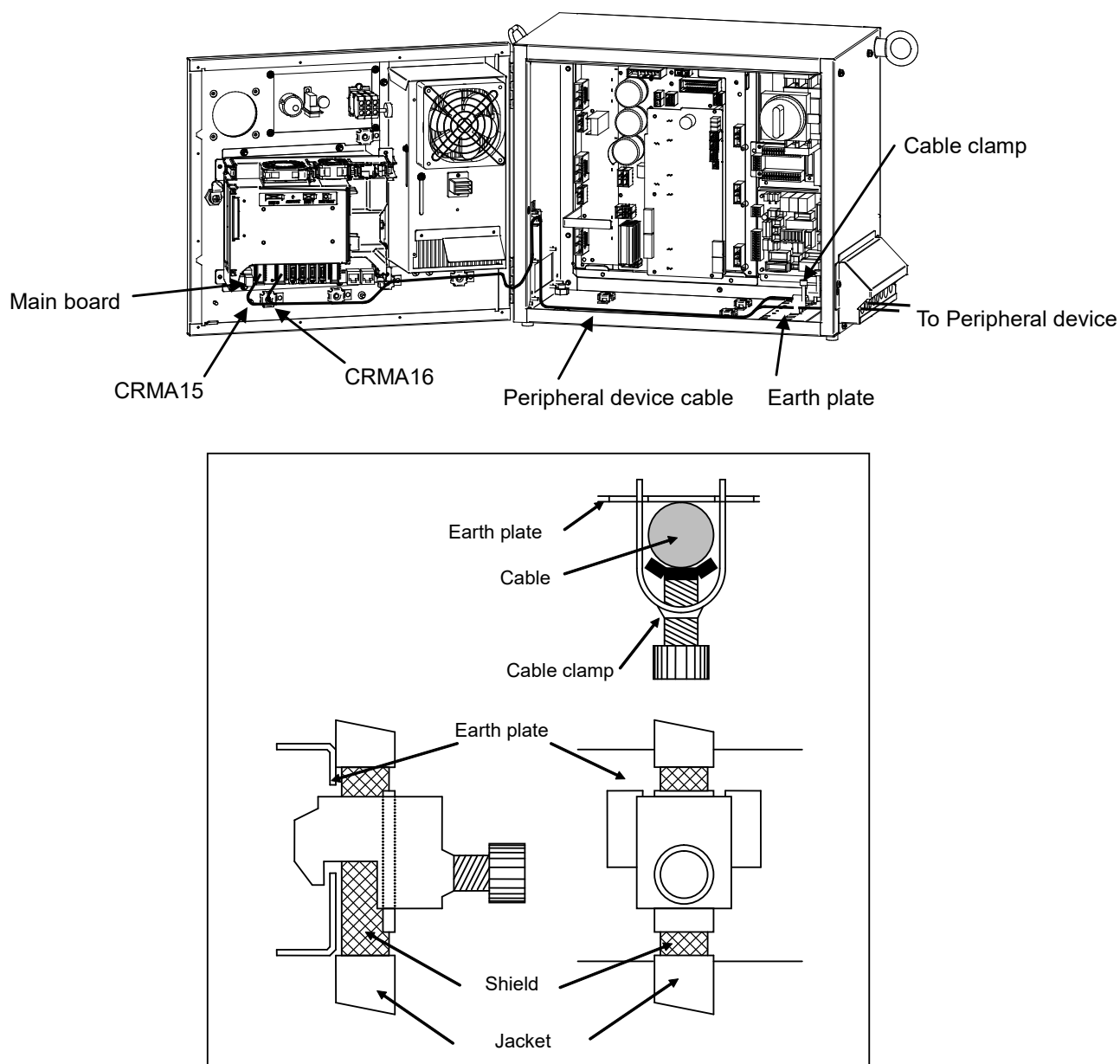
4.7.4 ARC Weld Connection Cables (CRW11; Tyco Electronics AMP, 20 pin)



4.8 CABLE CONNECTION FOR THE PERIPHERAL DEVICES, END EFFECTORS, AND ARC WELDERS

4.8.1 Peripheral Device Connection Cable

Fig.4.8.1 shows the connection of the peripheral device cable in the cabinet.

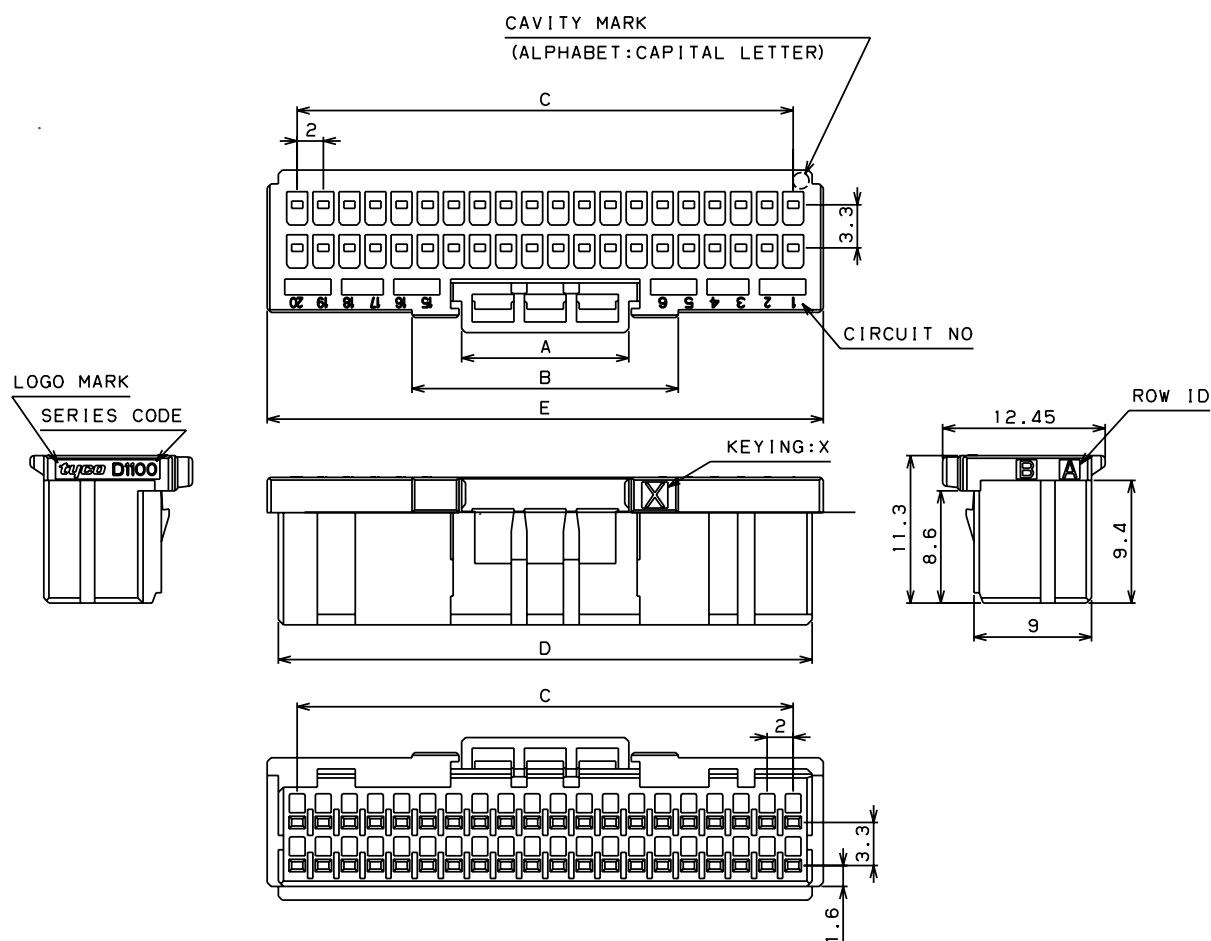
**NOTE**

For protection against the noise, cut part of the jacket of the connection cable to expose the shield, and fasten this part to the earth plate with the cable clamp.

Fig.4.8.1 Peripheral Device Cable Connection

4.8.2 Peripheral Device Cable Connector

- (1) The connector for peripheral device cables (Robot controller side).



Connector specifications	Applicable interface	Dimensions					Remark
		A	B	C	D	E	
1-1827863-0 (Connector)	CRMA15	12.8	20.4	38.0	40.9	42.6	Tyco Electronics AMP 40 pins (X-key)
2-1827863-0 (Connector)	CRMA16						Tyco Electronics AMP 40 pins (Y-key)
1939991-2 (Contact)	CRMA15 CRMA16						Tyco Electronics AMP

Maintenance tool

Hand tool (for crimping contact)

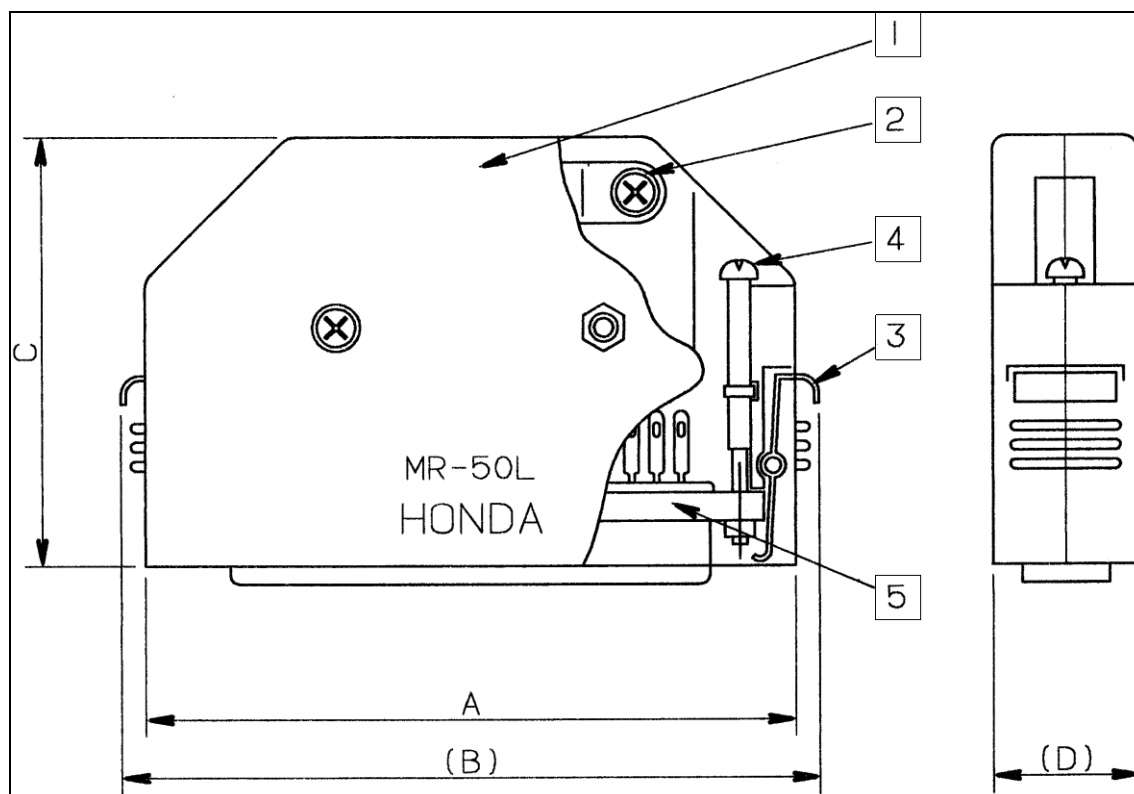
2119141-1:A05B-2550-K064

Extraction tool

1891526-1:A05B-2550-K061

Fig.4.8.2 (a) Peripheral device cable connector

- (2) The connector for peripheral device cables (Peripheral device side).

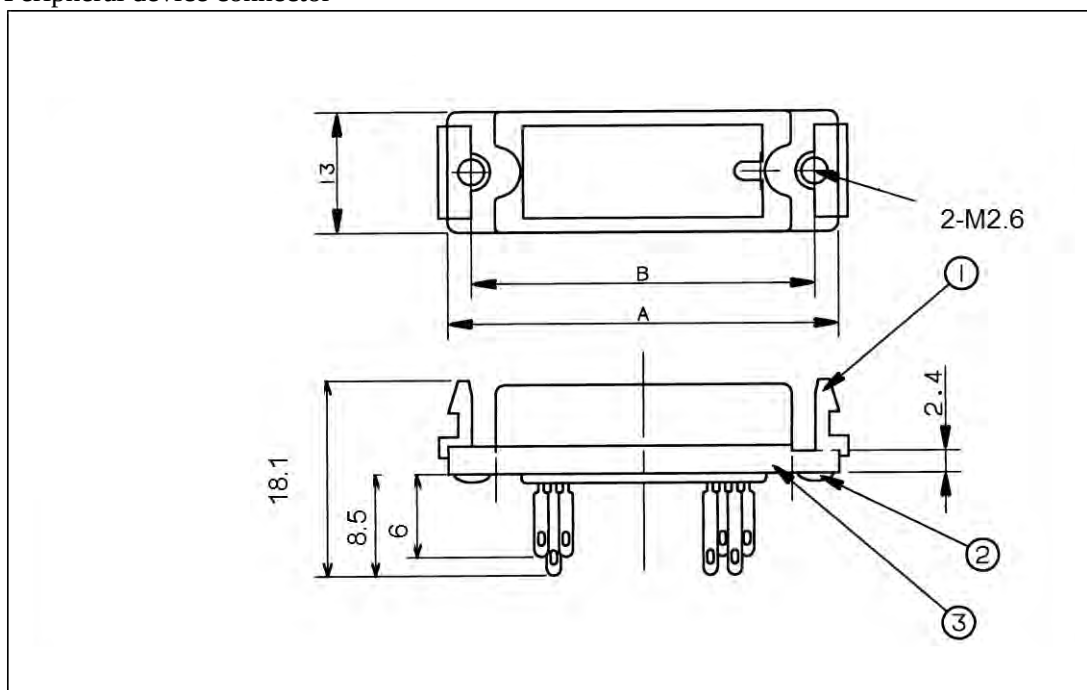


Connector specifications	Applicable interface	Dimensions				Remark
		A	(B)	C	(D)	
MR-50F(Connector) MR-50L+(Case)	Peripheral Device	67.9	73.5	44.8	18	Honda Tsushin Kogyo, 50 pins (F) Solder type

Symbol	Name
1	Connector cover
2	Cable clamp screw
3	Connector clamp spring
4	Connector clamp screw
5	Connector

Fig.4.8.2 (b) Peripheral device cable connector

(3) Peripheral device connector



Connector specifications	Applicable device	Dimensions		Remark
		A	B	
MR-50RM+	Peripheral Device	61.4	56.4	Honda Tsushin Kogyo, 50 pins (F) Solder type

Symbol	Name
(1)	Connector clamp screw
(2)	Screw M2.6 x 8
(3)	Connector

Fig.4.8.2 (c) Peripheral device connector

4.8.3 Recommended Cables

(1) Peripheral device connection cable

Connect a peripheral device using a completely shielded, heavily protected cable conforming to the specifications in Table 4.8.3 (a).

Allow an extra 50 cm for routing the cable in the controller.

The maximum cable length is 30 m.

Table 4.8.3 (a) Recommended Cable (for Peripheral Device Connection)

Number of wires	Wire specifications (FANUC specifications)	Conductor		Sheath thickness (mm)	Effective outside diameter (mm)	Electrical characteristics	
		Diameter (mm)	Configuration			Conductor resistance (Ω /km)	Allowable current (A)
50	A66L-0001-0042	ϕ 1.05	7/0.18 AWG24	1.5	ϕ 12.5	106	1.6A
20	A66L-0001-0041	ϕ 1.05	7/0.18 AWG24	1.5	ϕ 10.5	106	1.6A

(2) End effector connection cable

Connect an end effector using a heavily protected cable with a movable wire conforming to the specifications in Table 4.8.3 (b).

The cable length is determined so that the cable will not interfere with the end effector and the wrist can move through its full stroke.

Table 4.8.3 (b) Recommended Cable (for End Effector Connection)

Number of wires	Wire specifications (FANUC specifications)	Conductor		Sheath thickness (mm)	Effective outside diameter (mm)	Electrical characteristics	
		Diameter (mm)	Configuration			Conductor resistance (Ω/km)	Allowable current (A)
6	A66L-0001-0143	$\phi 1.1$	40/0.08 AWG24	1.0	$\phi 5.3$	91	3.7
20	A66L-0001-0144	$\phi 1.1$	40/0.08 AWG24	1.0	$\phi 8.6$	91	2.3
24	A66L-0001-0459	$\phi 0.58$	40/0.08 AWG24	1.0	$\phi 8.3$	93	2.3

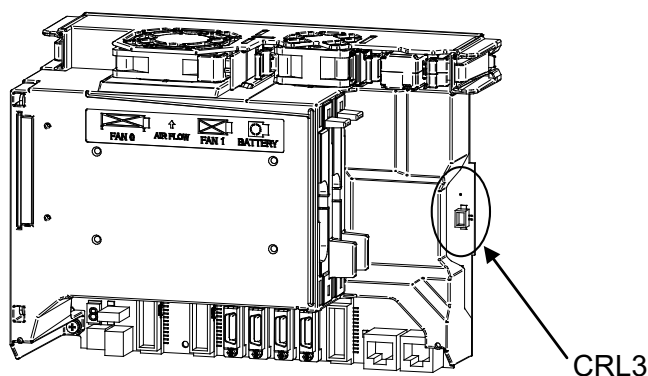
NOTE

For protection against the noise, cut part of the jacket of the connection cable to expose the shield, and fasten this part to the earth plate with the cable clamp.

4.9 CONNECTION OF HDI

4.9.1 Connecting HDI

The HDI signals are used in combination with special application software. The HDI signals cannot be used as general-purpose DIs.



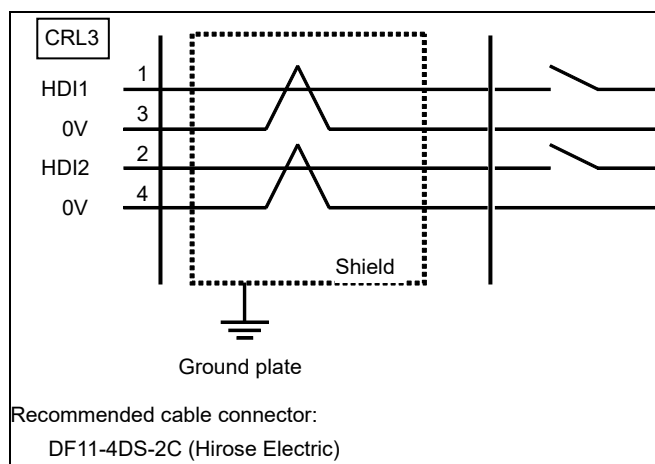
R-30/B Mate/R-30/B Mate Plus

Main board

CRL3

1	HDI1	2	HDI2
3	0V	4	0V

Cable connections

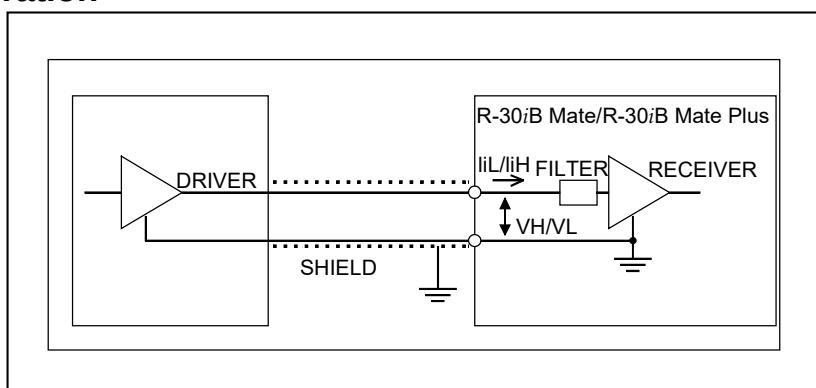


NOTE

For protection against the noise, cut part of the jacket of the connection cable to expose the shield, and fasten this part to the earth plate with the cable clamp.

4.9.2 Input Signal Rules for the High-speed Skip (HDI)

Circuit configuration



Absolute maximum rating

Input voltage range V_{in} : -3.6 to +10 V

Input characteristics

Unit	Symbol	Specification	Unit	Remark
High level input voltage	VH	3.6 to 11.6	V	
Low level input voltage	VL	0 to 1.0	V	
High level input current	liH	2 (max)	mA	$V_{in}=5$ V
		11 (max)	mA	$V_{in} = 10$ V
Low level input current	liL	-8.0 (max)	mA	$V_{in} = 0$ V
Input signal pulse duration		20 (min)	μ s	
Input signal delay or variations		20 (max)	μ s	

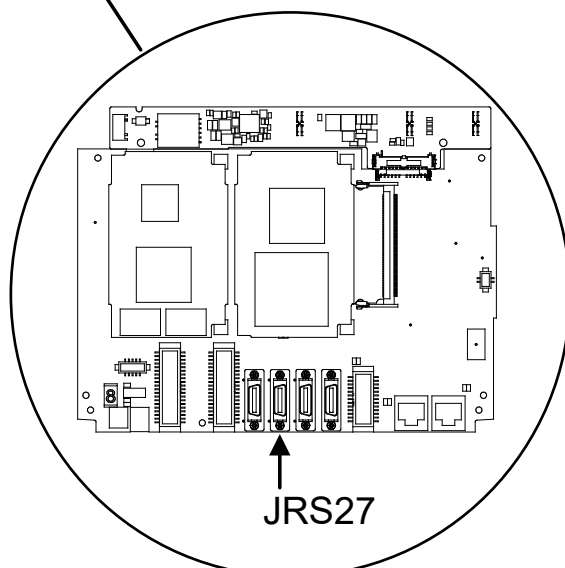
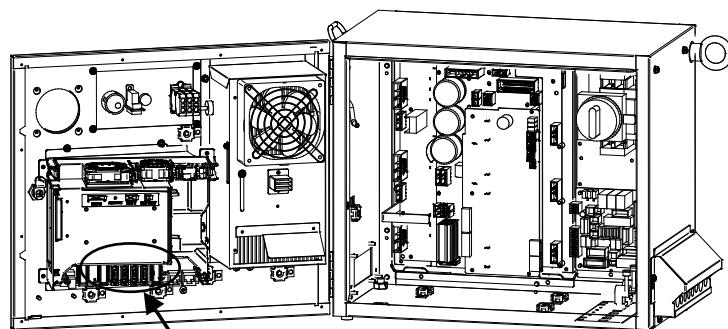
NOTE

- 1 The plus (+) sign of liH/liL represents the direction of flow into the receiver. The minus (-) sign of liH/liL represents the direction of flow out of the receiver.
- 2 The high-speed skip signal is assumed to be 1 when the input voltage is at the low level and 0 when it is at the high level.

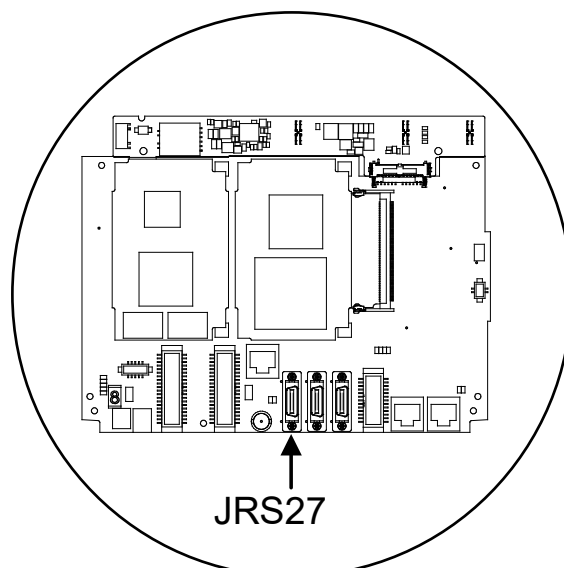
4.10 CONNECTING THE COMMUNICATION UNIT

4.10.1 RS232C Interface

4.10.1.1 Interface



(R-30iB Mate)



(R-30iB Mate Plus)

JRS27					
1	RD (RXDA)	11	SD (TXDA)	Honda Tsushin Kogyo CONNECTOR: PCR-E20FS COVER: PCR-V20LA, or compatible connector	
2	SG (0V)	12	SG (0V)		
3	DR (DSRA)	13	ER (DTRA)		
4	SG (0V)	14	SG (0V)		
5	CS (CTSA)	15	RS (RTSA)		
6	SG (0V)	16	SG (0V)		
7		17			
8		18			
9		19	+24V (24V-3)		
10	+24V (24V-3)	20			

NOTE

- 1 +24 V can be used as the power supply for FANUC RS-232-C equipment.
- 2 Do not connect anything to those pins for which signal names are not indicated.

4.10.1.2 RS232C interface signals

Generally signals as follows are used in RS232C interface.

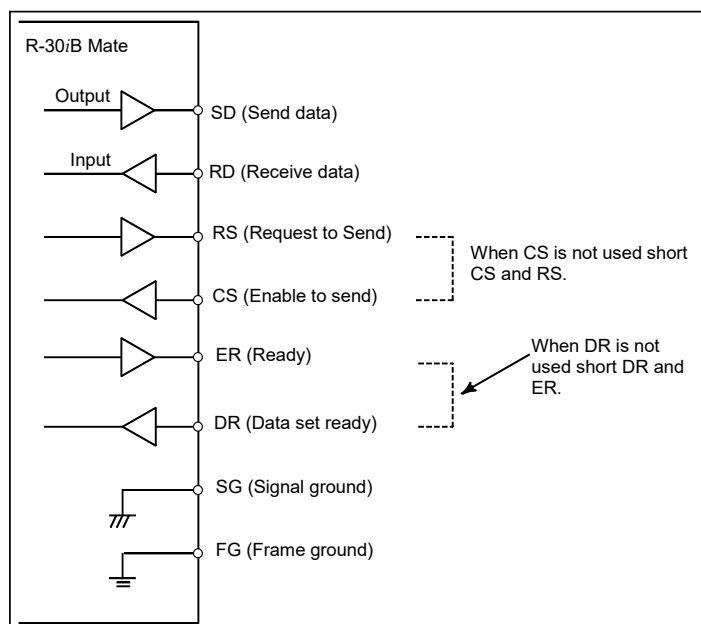
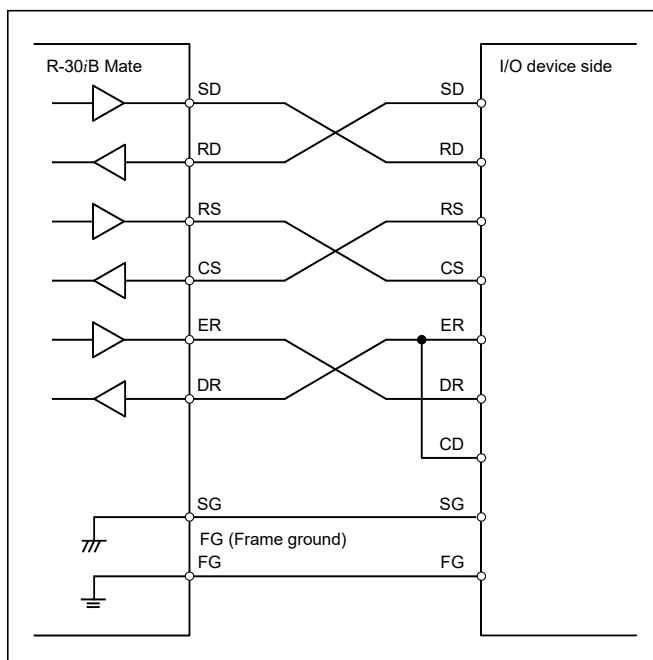


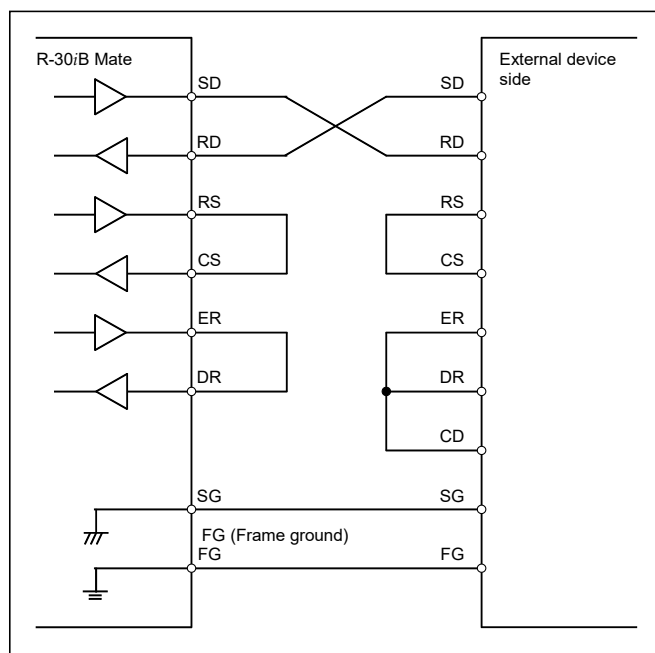
Fig.4.10.1.2 RS232C interface

4.10.1.3 Connection between RS232C interface and I/O device

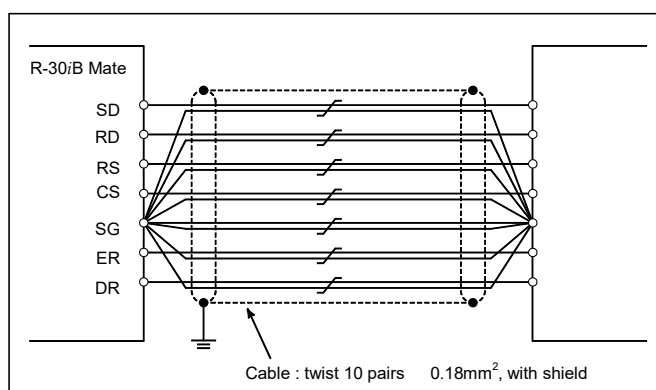
The figure below shows a connection with the handshaking of the ER/DR, RS/CS signals.



- The figure below shows a connection without the handshaking of the RS/CS, ER/DR signals.



Cable connection



Pair each signal with SG.

NOTE

For protection against the noise, cut part of the jacket of the connection cable to expose the shield, and fasten this part to the earth plate with the cable clamp.

4.10.2 Ethernet Interface

This section describes information relating to the physical Ethernet connection.



CAUTION

- 1 Before connecting or disconnecting the Ethernet cable, make sure that the power to the robot controller is turned off.
- 2 Please inquire of each manufacturer (of hub, transceiver, cable etc.) about the construction of network or the condition of using the equipment. When configuring your network, you must take other sources of electrical noise into consideration to prevent your network from being influenced by electrical noise. Make sure that network wiring is sufficiently separated from power lines and other sources of electrical noise such as motors, and ground each of the devices as necessary. In addition, high and insufficient ground impedance may cause interference during communications. After installing the robot, conduct a communications test before you actually start operating the robot. We cannot ensure operation that is influenced by network trouble caused by a device other than the robot controller.

4.10.2.1 Connection to Ethernet

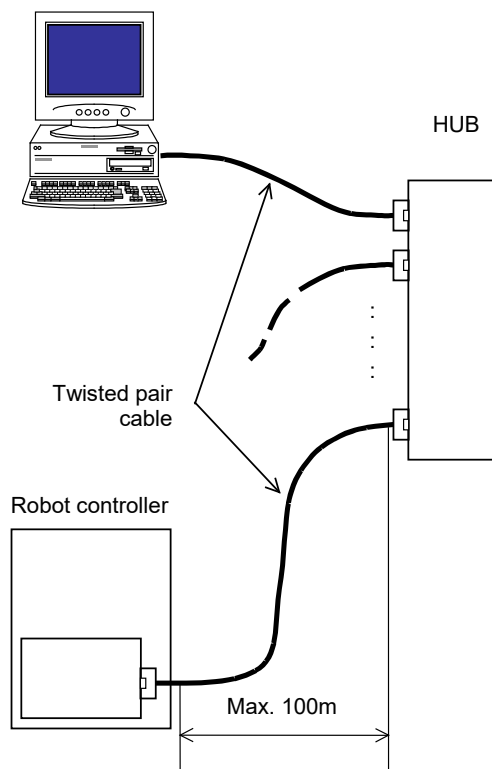
The robot controller is provided with a 100BASE-TX interface and 1000BASE-T (Only R-30iB Mate Plus) interface.

Prepare a hub for connecting the controller to the Ethernet trunk. The following shows an example of a general connection.

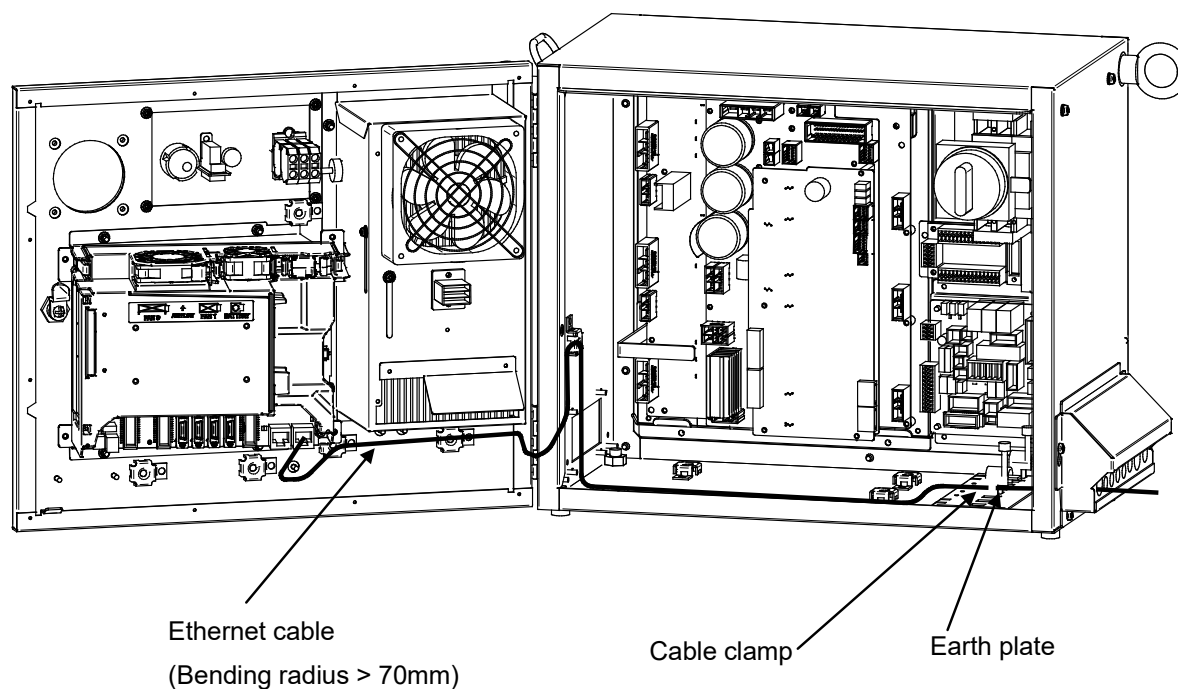
To connect the control unit to the CD38C (for R-30iB Mate Plus) Ethernet interface in a half duplex communication mode, use a hub which satisfies the following conditions:

- Supports 100BASE-TX/1000BASE-T.
- Has an auto-negotiation function.
- Supports store-and-forward switching.
- Supports flow control.

Some devices (hub, transceiver, etc.) that are needed for building a network do not come in a dust-proof construction. Using such devices in an atmosphere where they are subjected to dust or oil mist will interfere with communications or damage the robot controller. Be sure to install such devices in a dust-proof cabinet.



4.10.2.2 Routing of the Ethernet cable



The Ethernet cable must be fastened by a cable clamp to prevent tension being applied to the modular connector (RJ-45) that connects the cable to the controller even if the Ethernet cable is pulled directly. This clamp is also used to ground the cable shield.

4.10.2.3 100BASE-TX connector (CD38A, CD38B) / 1000BASE-T Connector (CD38C) pin assignments

CD38A,CD38B (R-30iB Mate, R-30iB Mate Plus)

Pin No.	Signal name	Description
1	TX+	Send +
2	TX-	Send -
3	RX+	Receive +
4		Not used
5		Not used
6	RX-	Receive -
7		Not used
8		Not used

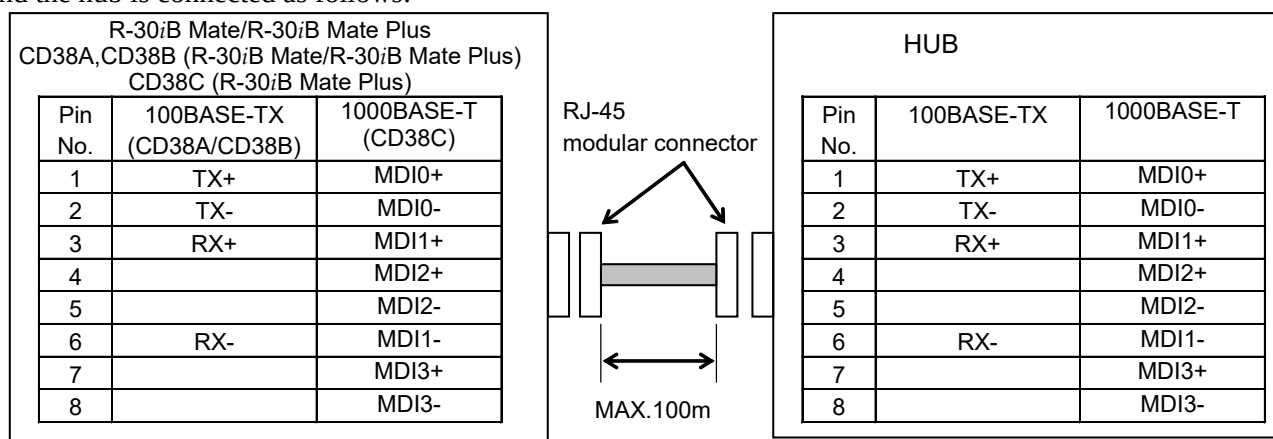
CD38C (1000BASE-T, R-30iB Mate Plus only)

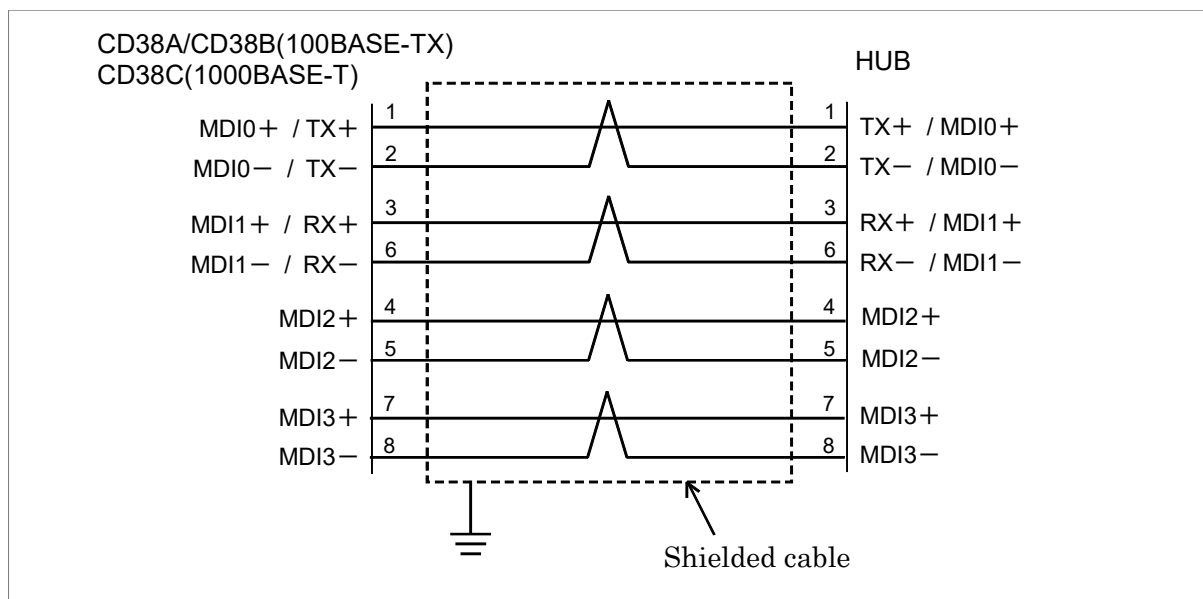
Pin No.	Signal name	Description
1	MDI0+	Bi-directional Data 0+
2	MDI0-	Bi-directional Data 0-
3	MDI1+	Bi-directional Data 1+
4	MDI2+	Bi-directional Data 2+
5	MDI2-	Bi-directional Data 2-
6	MDI1-	Bi-directional Data 1-
7	MDI3+	Bi-directional Data 3+
8	MDI3-	Bi-directional Data 3-

4.10.2.4 Twisted-pair cable specification

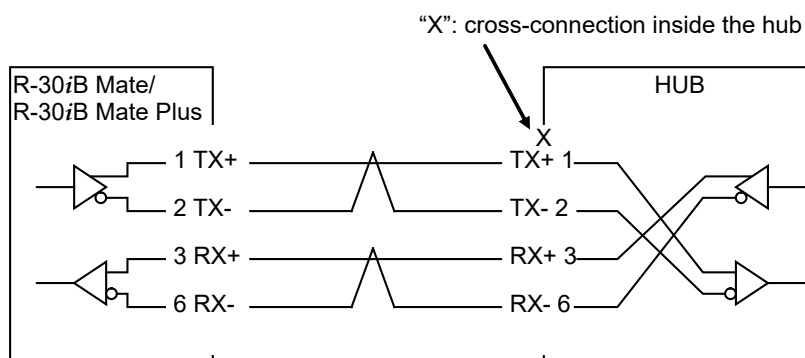
Cable Connection

The cable used for connection between the 100BASE-TX/1000BASE-T interface, CD38, of the controller and the hub is connected as follows:





- Keep the total cable length within 100 m.
Do not extend the cable more than is necessary.
- The figure above shows the cable connection when cables are crossed in the hub.
"X" is usually indicated at the port of the hub to signify that cables are crossed in the hub.



(1) Cable Materials



CAUTION

Unshielded cable (UTP cable) is commercially available as 100BASE-TX/1000BASE-T twisted-pair cable: You should, however, use shielded Category 5(100BASE-TX) / 5e(1000BASE-T) twisted-pair cable (STP cable) to improve the resistance to electrical noise in an FA environment.

Recommended Cables (Non-flexible cable, 100BASE-TX)

Manufacturer	Specification	Remarks
FURUKAWA ELECTRIC CO., LTD.	DTS5087C-4P	Twisted-pair cable
NISSEI ELECTRIC CO., LTD.	F-4PFWMF	Single-conductor cable

Inquiries

Manufacturer		Contact address
FURUKAWA ELECTRIC CO., LTD. Sales Headquarters		2-6-1 Marunouchi, Chiyoda-ku. Tokyo 100-8322 TEL: 03-3286-3126 FAX: 03-3286-3979
NISSEI ELECTRIC CO., LTD. Machida Branch		3F MU Bldg., 1-9-1 Minami-narise, Machida City, Tokyo 194-0045 TEL: 0427-29-2531 FAX: 0427-29-3375
	Overseas Sales Office	IWATANI International Corporation Tokyo Head Office 21-8 Nishi-shinbashi 3-chome, Minato-ku, TOKYO, 105-8458, JAPAN TEL: 03-5405-5810 FAX: 03-5405-5666 Telex: 2524256 IWATYO J
	Remarks	A finished cable with connectors at both ends can be offered.

NOTE

The recommended cables cannot be used for moving parts.

Recommended cable (for movable parts, dedicated to FANUC)

Manufacturer	Specification	Remarks
Oki Electric Cable Co., Ltd.	AWG26 4P TPMC-C5E (S-HFR) K	CAT5e (1Gbps capable, For 1000BASE-T)
Oki Electric Cable Co., Ltd.	AWG26 4P TPMC-C5-F (SB)	CAT3, CAT5 (100Mbps capable, For 100BASE-TX)
Shinko Electric Industrial Co., Ltd.	FNC-118	CAT3, CAT5 (100Mbps capable, For 100BASE-TX)

Specification

1. Manufacture: Oki Electric Cable Co., Ltd.

Manufacture's model number: AWG26 4P TPMC-C5E(S-HFR) K

- Electrical characteristic:
Complying with EIA/TIA 568B.2 Category 5e.
- Structure:
Common-shield braided cable with drain wire. The conductors of the cable are AWG26 annealed-copper strand wire, with a sheath 0.6 mm thick and an outer diameter of 6.8 mm.
- Fire resistance:
UL1581 VW-1
- Oil resistance:
As per Fanuc's internal standard (Equivalent to conventional oil-resistant electrical cable)
- Flexing resistance:
Sliding: 3 millions or more sliding cycles with a bending radius of 50 mm.
bending: 300 thousands or more bending cycles with a bending radius of 20 mm
Twisting: 5 millions or more sliding cycles. (+/- 180degrees)
- UL style No. :
AWM20276 (80°C/30V/VW-1)

2. Manufacture: Oki Electric Cable Co., Ltd.

Manufacture's model number: AWG26 4P TPMC-C5-F (SB)

Manufacture: SHINKO ELECTRIC INDUSTRIES CO., LTD.

Manufacture's model number: FNC-118

- Electric characteristics:
Conforms to EIA/TIA 568A Category 3 and Category 5.
From the viewpoint of attenuation performance, ensure that the length to the hub is 50 m or less.
- Structure:
Group shielded (braided shield). A drain wire is available.
The conductor is an AWG26 annealed copper twisted wire, with a sheath thickness of 0.8 mm and an outer diameter of 6.7 mm $\pm$ 0.3 mm.
- Fire retardancy
UL1581 VW-1
- Oil resistance
Conforms to the FANUC internal standards (equivalent to the conventional oil-resistant electric cables).
- Flexing resistance:
1,000,000 times or more with a bending radius of 50 mm (U-shaped flex test)
- UL style No.
AWM 20276 (80°C/30V/VW-1)

NOTE

Be sure to use the connector TM21CP-88P (03) manufactured by HIROSE ELECTRIC CO., LTD. for this cable.

Inquiries

Manufacturer	Contact address
Oki Electric Cable Co., Ltd.	Nagano Sales Office TEL:0266-27-1597
Shinko Electric Industrial Co., Ltd.	Tokyo Sales Office TEL:03-3492-0073

Cable assembly

Oki Electric Cable Co., Ltd. can also supply the cable assembly mentioned above.

Contact Oki Electric directly to determine the specifications (length, factory test, packing, and so forth) for purchase.

(2) Connector Specification

Use an 8-pin modular connector (RJ-45) with the twisted-pair cable for the Ethernet connection. The following connectors or equivalents must be used.

Non-Flex	Specification	Manufacturer	Remarks
Solid wire	5-569530-3	TE Connectivity	
Solid wire	MS8-RS2T-EMC	SK KOHKI CO., LTD.	Special tools required
Twisted-pair cable	5-569552-3	TE Connectivity	
Twisted-pair cable	TM11AP-88P	HIROSE ELECTRIC CO., LTD.	Special tools required

Flex	Specification	Manufacturer	Remarks
AWG26 4P TPMC-C5-F (SB) AWG26 4P TPMC-C5E (S-HFR) K, or FNC-118	TM21CP-88P (03)	HIROSE ELECTRIC CO., LTD.	Note

NOTE

Information about TM21CP-88P (03):
Connector (standard product of the manufacturer)
Drawing number: A63L-0001-0823#P
Manufacturer: HIROSE ELECTRIC CO., LTD.
Manufacturer type number: TM21CP-88P (03)
Conforms to EIA/TIA 568A Category 3 and Category 5.
For assembly with a cable, contact HIROSE ELECTRIC CO., LTD. directly.
(From HIROSE ELECTRIC CO., LTD., "TM21CP-88P (03) Connection
Procedure Manual (Technical Specification No. ATAD-E2367)" is available as a
technical document.)

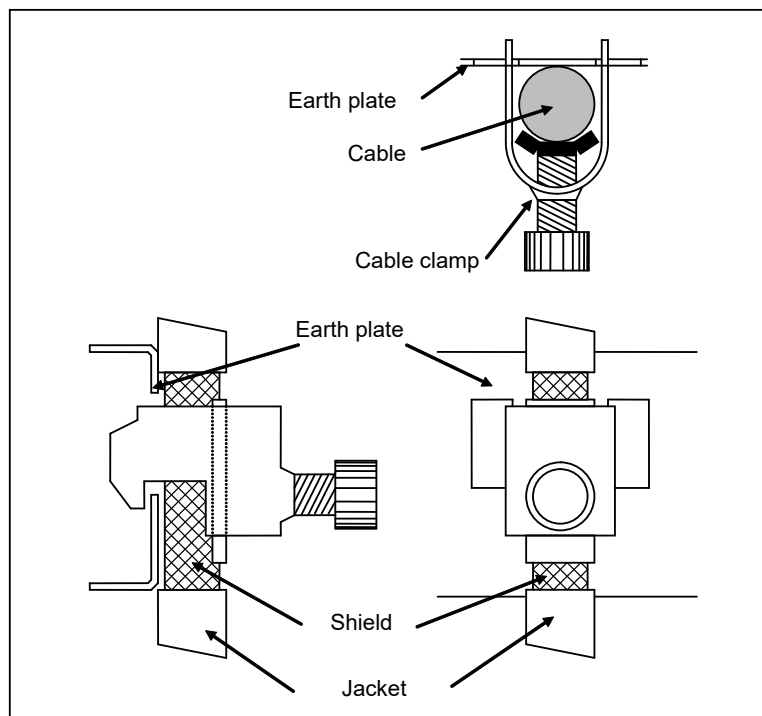
4.10.2.5 Electrical noise countermeasures

Clamping and Shielding Cables

Clamp an Ethernet twisted pair cable according to the method described below, as with cables that need to be shielded. Clamping cables provides support and shielding and is extremely important to the safe operation of the system. Never overlook cable clamping.

Peel off part of the jacket as shown in the figure to expose the outer coating of the shield, and press this outer coating against the earth plate with the cable clamp.

The machine manufacturer must prepare the ground plate and install it as follows:

**NOTE**

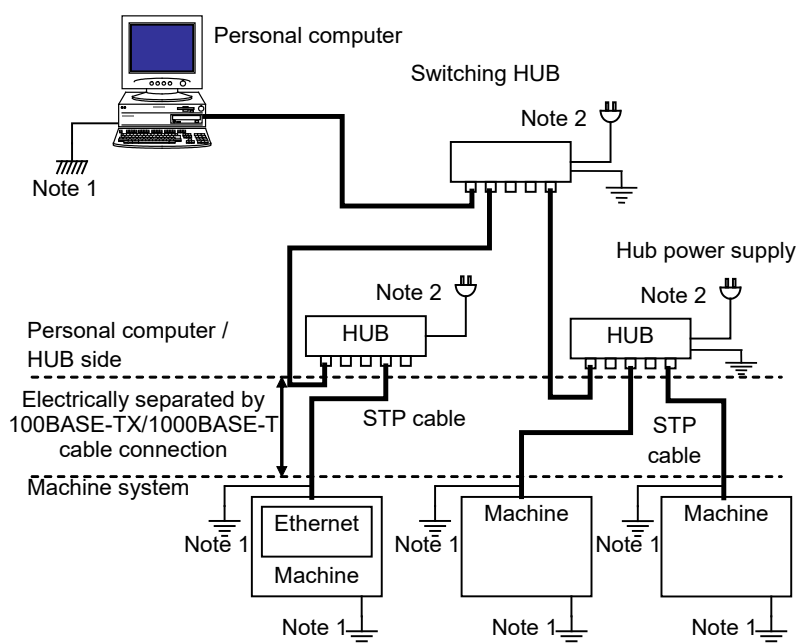
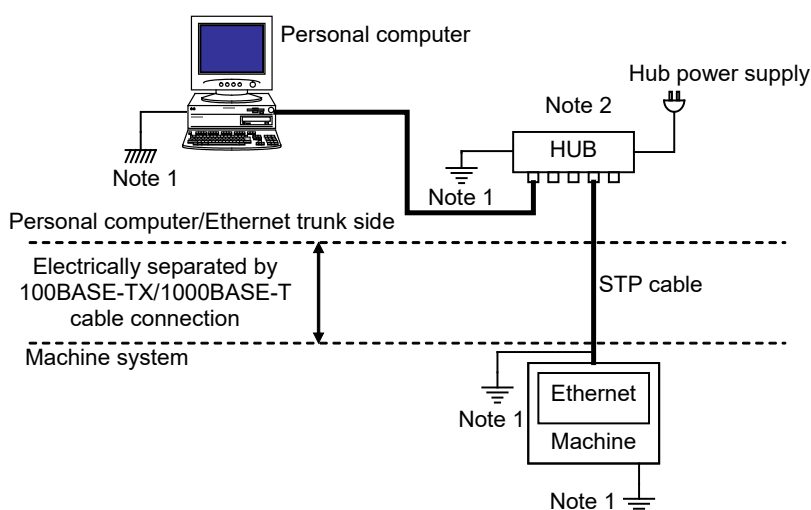
To ensure the safe operation of the system, clamp and shield the cables.

NOTE

- 1 To secure fast response, FL-net communication is not provided with a retransmission process at intervals of several seconds, unlike normal Ethernet communication. It is, therefore, necessary to provide more noise resistance than that provided by general Ethernet wiring work.
- 2 After the laying of cables, conduct satisfactory communication tests not only before system operation but after system operation from the viewpoint of noise prevention measures.

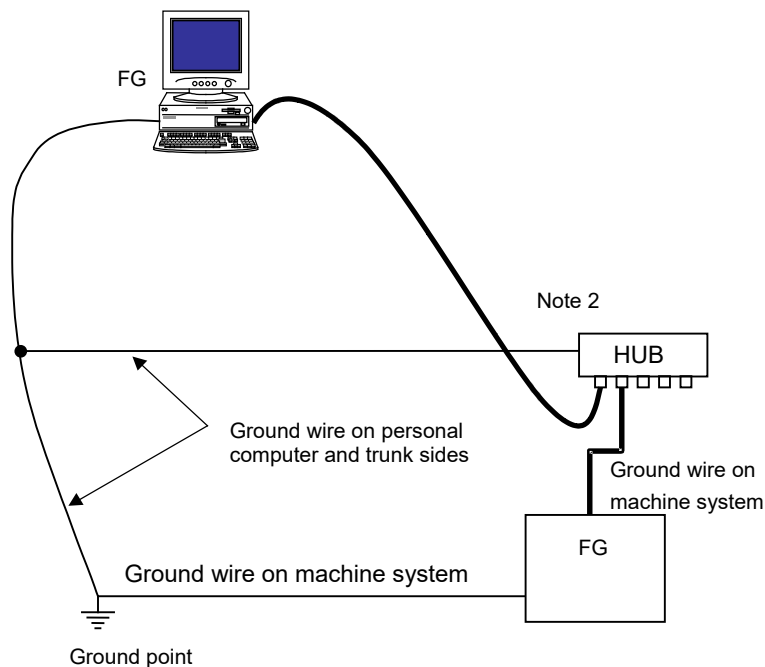
Grounding the Network

Even if the grounding condition on the machine side is satisfied, the communication line can pick up noise from the machine, depending on the machine installation condition and environment, thus resulting in a communication error. To protect against such noise, the machine should be separated and insulated from the Ethernet trunk cable and personal computer. Examples of connection are given below.

**Large-Scale Network****Small-Scale Network**

NOTE

- 1 The ground between PC/HUB side and machine system side must be separated. If it is impossible to separate the ground because there is only one grounding point, connect the ground cable for each system to the grounding point independently. (See figure below.)
The resistance for grounding must be less than 100-ohm (Class D). The thickness of the ground cable is the same as the thickness of AC power cable or more. At least thickness of 5.5mm^2 is necessary.
- 2 Note that the number of allowable hub-to-hub connections depends on the type of hub.
- 3 There is possibility that noise makes the obstacle of communication even if the ground is separated using the 100BASE-TX/1000BASE-T. In the case of using the FAST Ethernet/FAST Data Server under the worst environment, please separate between the PC/Trunk line side and machine system side completely using the 100BASE-FX/1000BASE-SX/LX (Optical fiber media).



Wiring on a single ground point

4.10.2.6 Check items at installation

The following table lists check items at installation.

Check item	Description	Check
Ethernet cable		
Type	Use cables which satisfies all the following conditions:	
	1) With shielding	
	2) Twisted-pair cable	
	3) Category 5	
Length	The cable length shall be within 100 m (50 m (100BASE-TX) or 40m (1000BASE-T) for a movable cable recommended by FANUC).	
Connection	For a twisted-pair cable of 100BASE-TX, the following pins shall be paired:	
	1) Pin No. 1 (TX+) – pin No. 2 (TX-)	
	2) Pin No. 3 (RX+) – pin No. 6 (RX-)	
	For a twisted-pair cable of 1000BASE-T, the following pins shall be paired:	
	1) Pin No. 1 (MDI0+) – pin No. 2 (MDI0+)	
	2) Pin No. 3 (MDI1+) – pin No. 6 (MDI1+)	
	3) Pin No. 4 (MDI2+) – pin No. 5 (MDI2+)	
	4) Pin No. 7 (MDI3+) – pin No. 8 (MDI3+)	
Separation	The Ethernet cables shall be bound separately from the following cables or covered with an electromagnetic shield ^(Note) :	
	1) Group A: AC power lines, power lines for motors, and others	
	2) Group B: Current DC (24 VDC) and others	
Shielding	For a shielded cable, the part of which outer coating is peeled off and exposed shall be fixed to the ground plate with a clamp fixture.	
Clamping	The ground plate shall be located as nearest to the CNC as possible (to make the cable between the ground plate and CNC hard to be affected by noise).	
Connectors	Any cable connector shall not be pulled (to prevent poor contact of the connector).	
Wiring	No cable shall be laid under a heavy object.	
Bending radius	The bending radius shall be at least four times as long as the diameter of the cable.	
For movable part	For a movable part, a cable for a movable part shall be used.	
HUB		
Use conditions	The "cautions on use" of the hub shall be observed (A terminating resistor shall be mounted properly if required).	
Grounding	The hub shall be grounded.	
Cabinet	The hub shall be installed in an enclosed cabinet.	
Vibration	The hub shall be installed so that it is not affected by vibration.	
Bending radius	The bending radius shall be at least four times as long as the diameter of the cable.	

NOTE

Covering a group with an electromagnetic shield means that shielding is provided between groups with grounded steel plates.

5 TRANSPORTATION AND INSTALLATION

This chapter describes the transportation and installation for the controller.

5.1 TRANSPORTATION

The controller is transported by a crane. Attach a sling to eyebolts at the top of the controller.

Crane capacity: Minimum 150kg
Sling capacity: Minimum 150kg

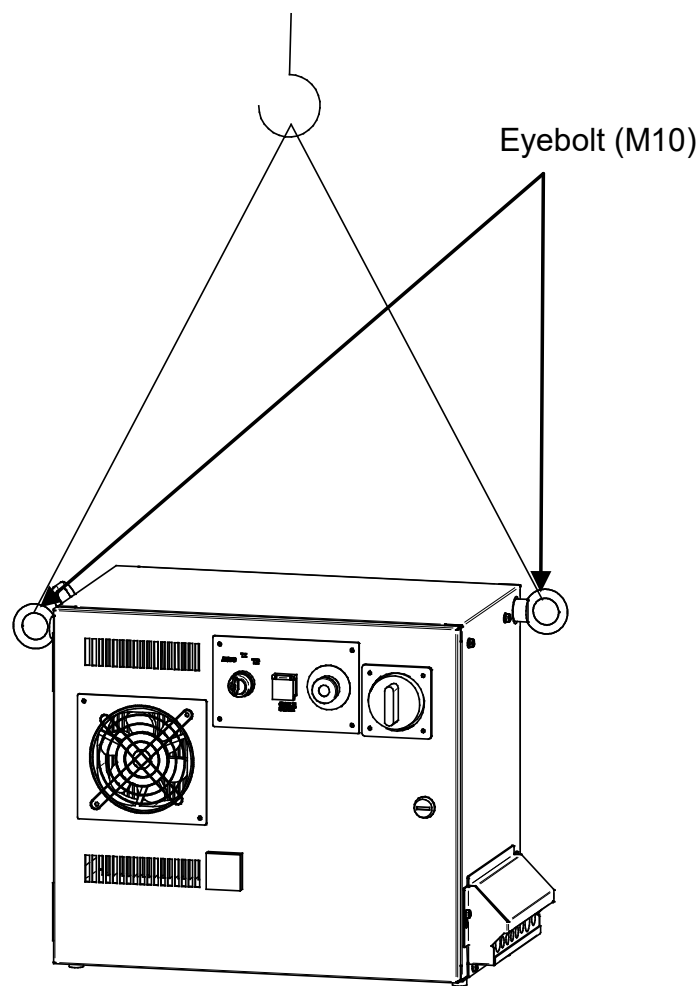


Fig.5.1 Transportation

5.2 INSTALLATION

5.2.1 Installation Method

Following is the installation method for cabinet.

When installing the controller, allow the space for maintenance shown in the following figure.

	Munsell	Color
Body	5GY3.5/0.5	Gray
Door	3.0GY8.2/0.9	White
Op. panel	N1.5	Black

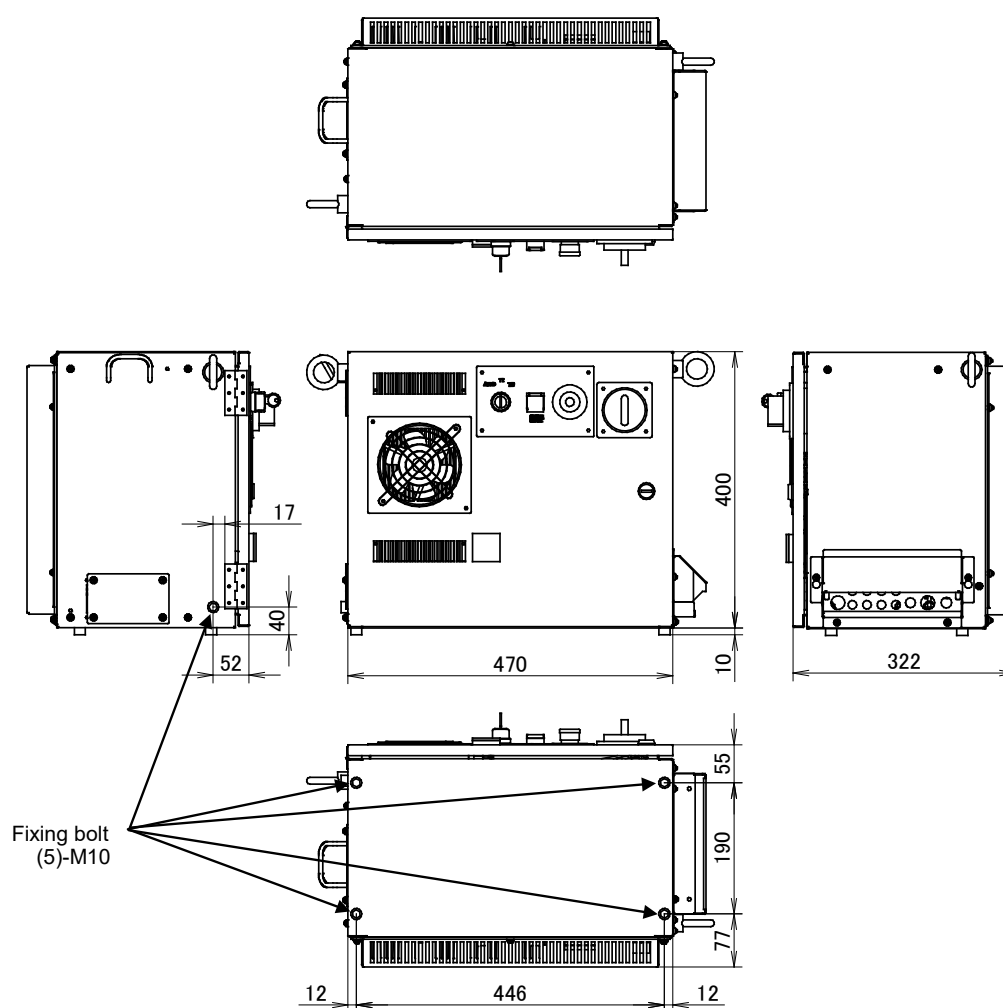


Fig.5.2.1(a) External dimensions (Small size)

	Munsell	Color
Body	5GY3.5/0.5	Gray
Door	3.0GY8.2/0.9	White
Op. panel	N1.5	Black

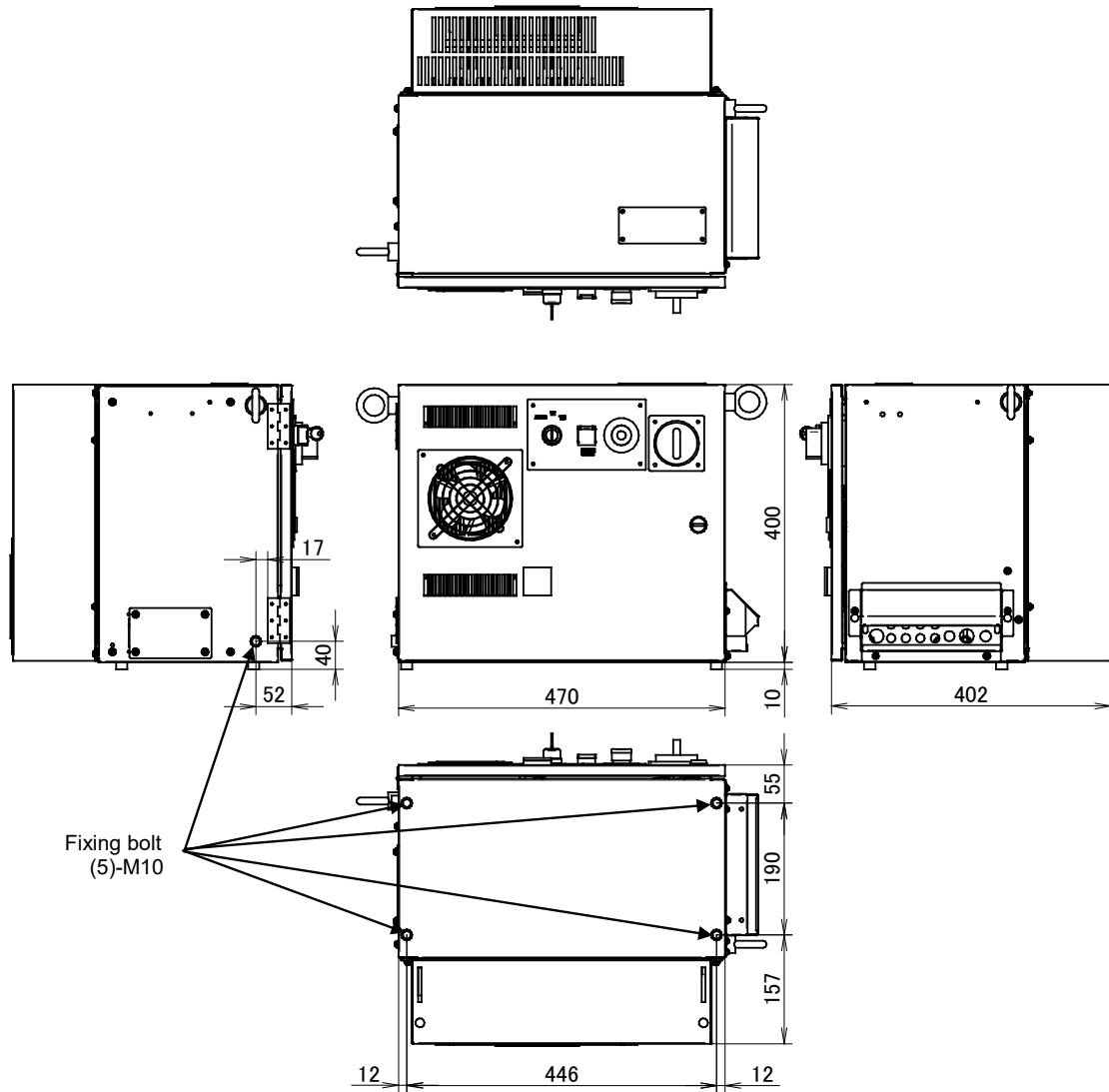


Fig.5.2.1(b) External dimensions (Medium / Large size)

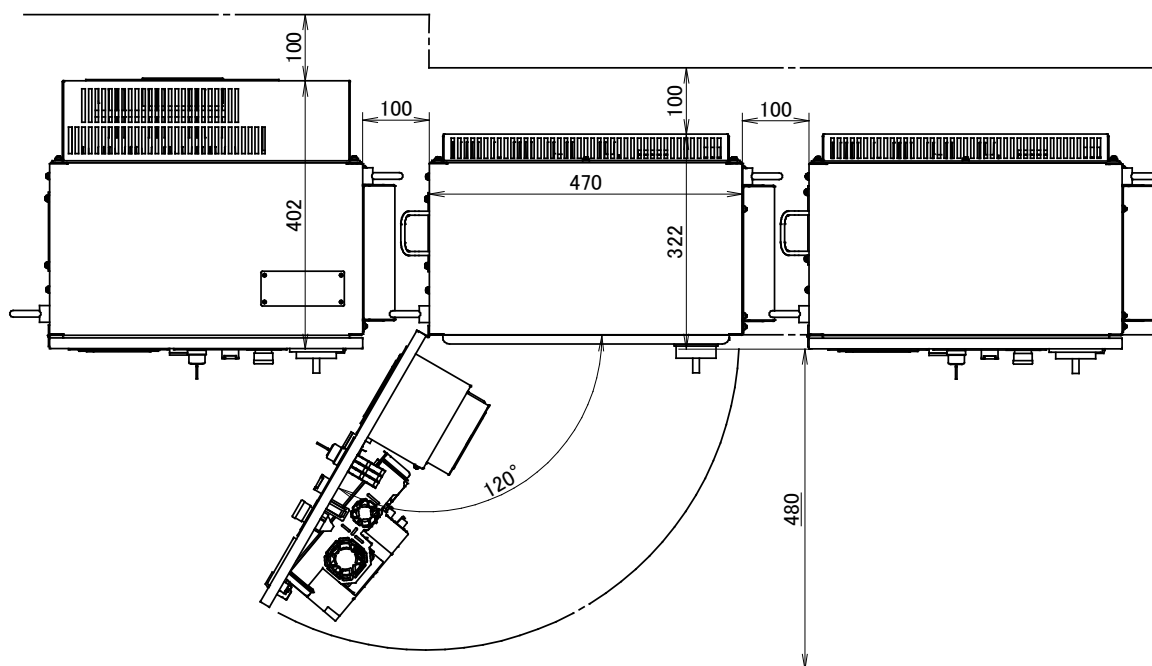


Fig.5.2.1 (c) Installation dimension

NOTE

Keep this area for maintenance and the radiation of heat.

5.2.2 Assemble at Installation

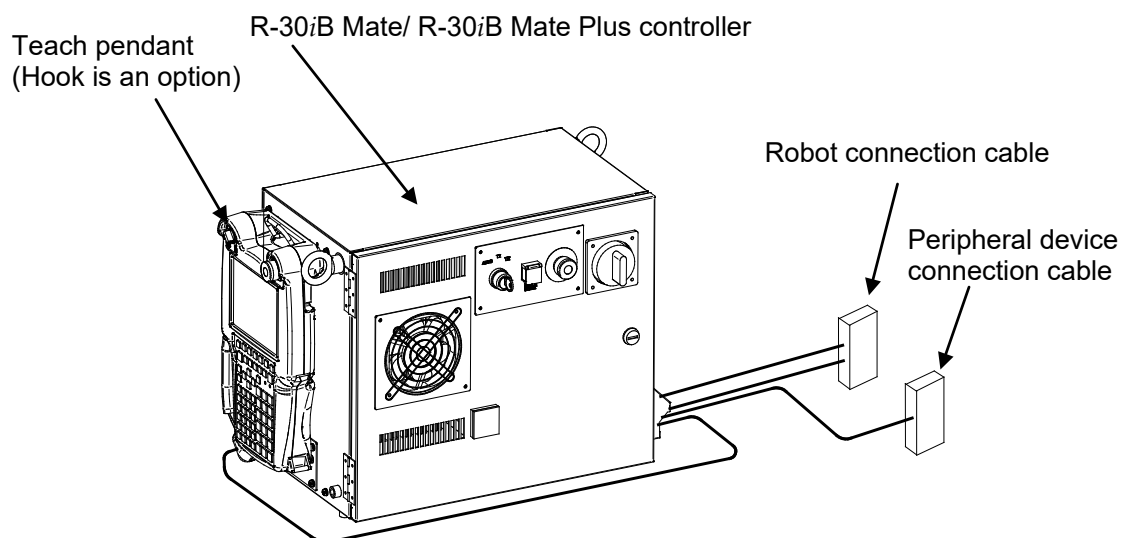


Fig.5.2.2 Assemble at installation

5.3 INSTALLATION OF TEACH PENDANT HOOK (OPTION)

Following is external dimensions for Teach Pendant HOOK (Ordering specification: A05B-2650-K050).

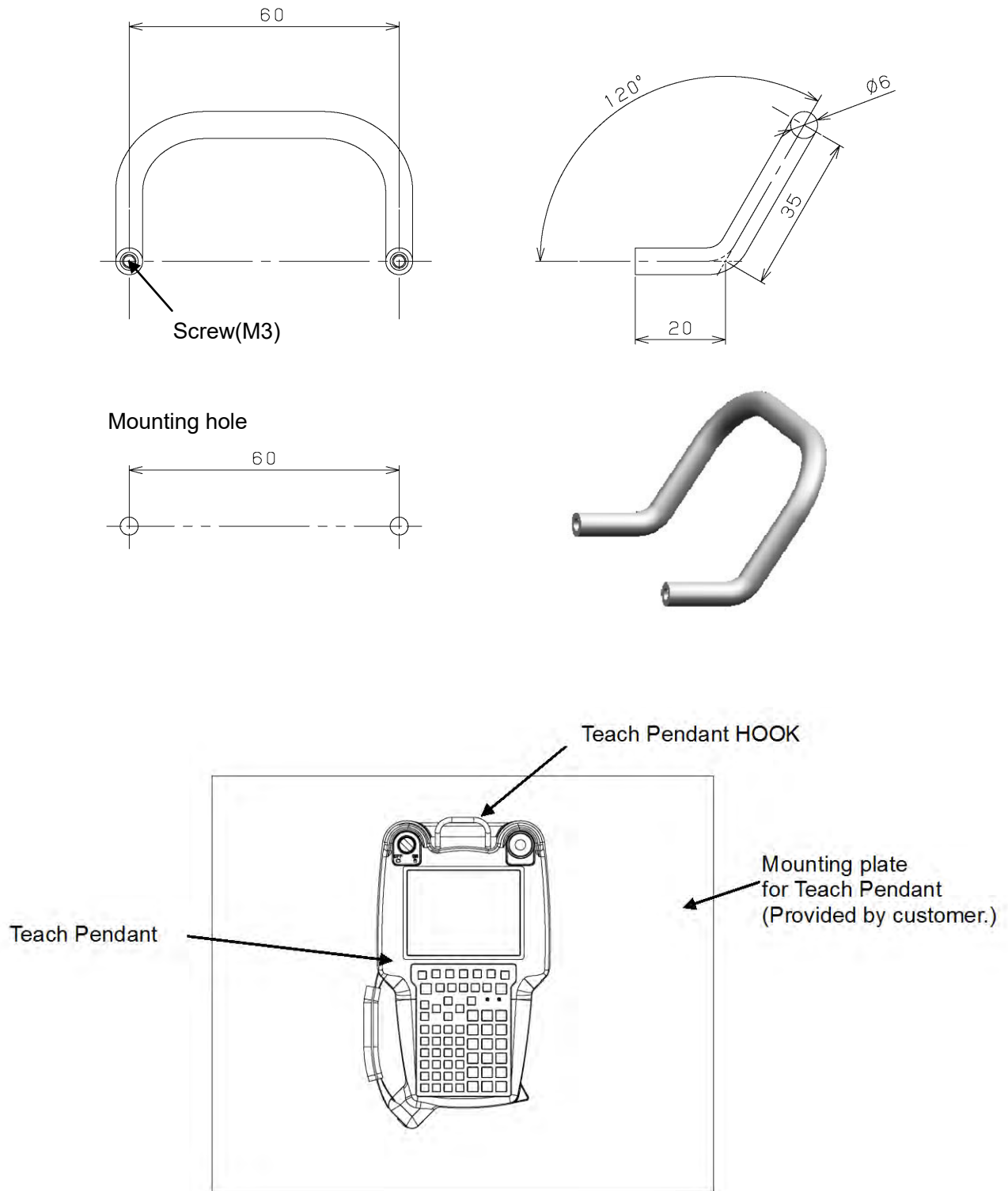


Fig.5.3 External dimensions of Teach Pendant HOOK

5.4 INSTALLATION CONDITION

Item	Model	Specification/condition
Rated Voltage	All models	200-230VAC 50/60Hz Single phase
		200-230VAC 50/60Hz 3 phase
Tolerant fluctuation	All models	Tolerant voltage fluctuation: +10% -15% Tolerant frequency fluctuation: ± 1 Hz
Input power source capacity	R-2000iC, R-1000iA, M-710iC, M-3iA, M-2iA	12kVA
	ARC Mate 120iC, M-20iA, M-20iB	3kVA
	ARC Mate 100iC, ARC Mate 100iD, M-10iA, M-10iD, ARC Mate 0iB, R-0iB	2kVA
	LR Mate 200iC, LR Mate 200iD, ARC Mate 50iD, CR-4iA, CR-7iA	1.2kVA
	M-1iA	1kVA
Average power consumption	R-2000iC, R-1000iA, M-710iC, M-3iA, M-2iA	2.5kW
	ARC Mate 100iC, ARC Mate 120iC, M-10iA, M-20iA, M-20iB, ARC Mate 0iB, R-0iB	1kW
	LR Mate 200iC, LR Mate 200iD, ARC Mate 50iD, CR-4iA, CR-7iA	0.5kW
	M-1iA	0.2kW
Permissible ambient temperature	All models	Operating 0°C to 45°C Storage, Transport -20°C to 60°C Temperature change 0.3°C/minute or less
Permissible ambient humidity	All models	Normal: 75%RH or less, no condensation Short period (less than 1 month): 95%RH or less, no condensation
Atmosphere	All models	An additional protective provision is necessary if the machine is installed in an environment in which there are relatively large amounts of contaminants (dust, dielectric fluid, organic solvent, acid, corrosive gas, salt, etc.).
Installation Category	All models	Installation Category III, Pollution Degree 3, IEC60664-1 and IEC61010-1
Vibration acceleration	All models	4.9m/s ² (0.5G) or less When using the robot in a location subject to serious vibration, consult with your FANUC sales representative.
Altitude	All models	Operating: Up to 1000m Non-operating: Up to 12000m
Ionized and non-ionized radiation	All models	A shielding provision is necessary if the machine is installed in an environment in which it is exposed to radiation (microwave, ultraviolet rays, laser beams, and/or X-rays).
Mass of controller	All models	40kg

Item	Model	Specification/condition
Degree of protection	All models	IP54
	Teach pendant	

NOTE

The power rating indicated above is sufficient as the continuous rating. However, when the robot is rapidly accelerating, the instantaneous requirement may increase to several times the continuous rating.

If the acceleration/deceleration override (ACC) greater than 100% is set in the robot program, the extreme current may flow to the robot controller instantaneously and the input voltage of robot controller will drop.

In this case, if the supply voltage is decreased 10% or more per rated voltage, Power supply alarm, Move error excess alarm, DCLV alarm of servo amplifier may occur.

NOTE**In case of CE controller**

R-30iB Mate controller is a group 1, class A product according to IEC55011.

This means that this product does not generate and/or use intentionally radio-frequency energy, in the form of electromagnetic radiation, inductive and/or capacitive coupling, for the treatment of material or inspection / analysis purpose and that it is suitable for use in all establishments other than domestic and those directly connected to a low voltage power supply network which supplies buildings used for domestic purposes.

There may be potential difficulties in ensuring electromagnetic compatibility in environments other than industrial, due to conducted as well as radiated disturbances.

This product must not be used in residual areas.

This product may cause interference if used in residential areas. Such use must be avoided unless the user takes special measures to reduce electromagnetic emissions to prevent interference to the reception of radio and television broadcasts.

In case that an earth leakage breaker in the power panel tripped (CE/3 phase)

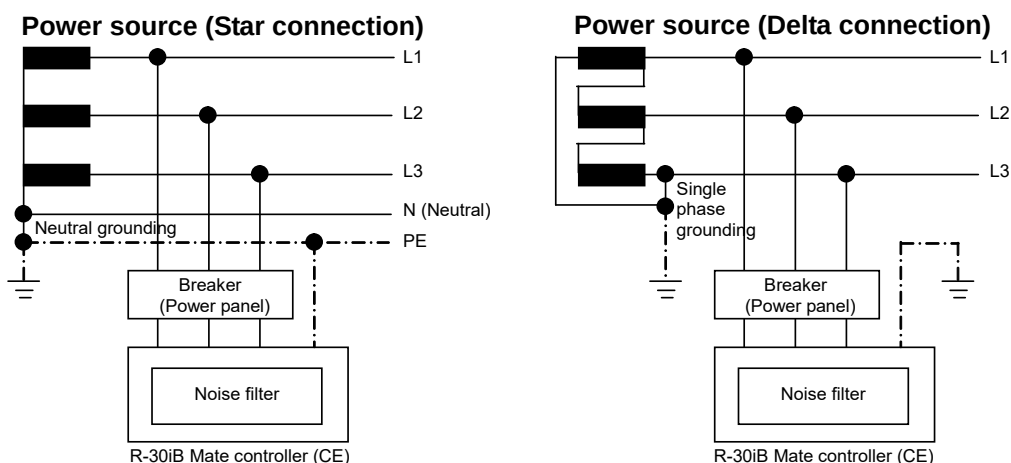
A noise filter is installed in the R-30iB Mate/ R-30iB Mate Plus controller (CE/3 phase). The noise filter is designed to connect to star connection / neutral grounding power system (TN-power system, NOTE.1). In case that the controller is connected to delta connection / single-phase grounding power system (TT-power system NOTE.1), there is a possibility that small leakage current flows via noise filter.

Therefore, if leakage breaker (Rated leakage current: less than 30mA) is installed in a power panel of the building and many controllers are connected to one leakage breaker, the leakage breaker may trip by amount of the leakage current.

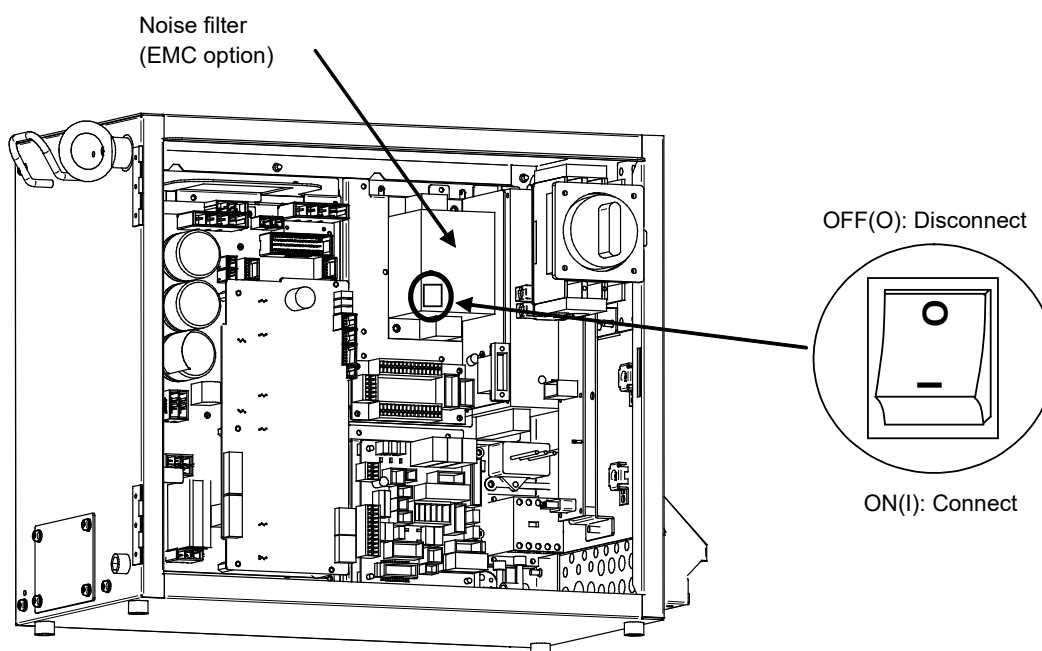
In this case, following countermeasures will be effective.

- (1) Use leakage breaker that has large rated leakage current. Or reduce the number of controllers that are connected to one leakage breaker.
- (2) Install isolated transformer between input power and controller.
- (3) It is possible to reduce the leakage current by turning off a switch at the noise filter. (NOTE.2)
If the controller used in region where CE mark is required, this switch should be ON because European power system is star connection / neutral grounding.

NOTE.1) The TN-power system and TT-power system are based on the AC power distribution system standard IEC60364.



NOTE.2) R-30iB Mate/ R-30iB Mate Plus controller (CE) has a noise filter with switch that connect/disconnect the internal capacitor to the ground. In case that the controller connected to delta connection / single-phase grounding power system (TT-power system), it is possible to reduce the leakage current to less than 10mA by turning off (O-side) the switch at the noise filter.



5.5 ADJUSTMENT AND CHECKS AT INSTALLATION

Adjust the robot according to the following procedure at installation.

No.	Description
1	Visually check the inside and outside of the controller.
2	Check the screw terminals for proper connection.
3	Check that the connectors and printed circuit boards are firmly connected.
4	Connect controller and mechanical unit cables.
5	The breaker off and connect the input power cable.
6	Check the input power voltage.
7	Press the EMERGENCY STOP button on the operator panel and turn on the controller.
8	Check the interface signals between controller and robot mechanical unit.
9	Check the parameters. If necessary, set them.
10	Release the EMERGENCY STOP button on the operator panel.
11	Check the movement along each axis in manual jog mode.
12	Check the signals of EE interface.
13	Check the peripheral device control interface signals.

5.6 RESETTING OVERTRAVEL AND EMERGENCY STOP AT INSTALLATION

An overtravel and emergency stop occur when the robot is operated for the first time after it is installed and the mechanical and controllers are wired. This section describes how to reset the overtravel and emergency stop.

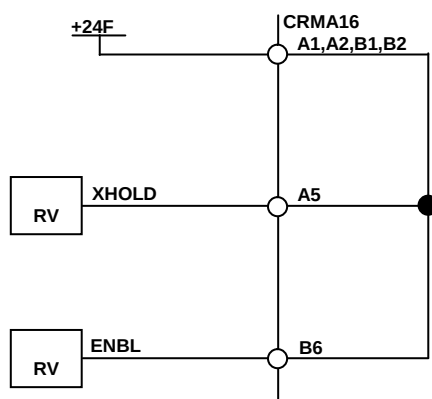
Remove the red plate fastening the swiveling axis beforehand.

The J2 and J3 axes are pressed against the hard stops at shipment. Therefore, an overtravel alarm occurs when the power is turned on after installation.

The robot can also be in an emergency stop state if the peripheral device control interface is not connected.

5.6.1 Peripheral Device Interface Processing

If signals XHOLD and ENBL are not used, connect these signals as follows.



5.6.2 Resetting Overtravel

- 1) Select [OT release] on the overtravel release screen to release each robot axis from the overtravel state.
- 2) Hold down the shift key, and press the alarm release button to reset the alarm condition.
- 3) Still hold down the shift key, and jog to bring all axes into the movable range.

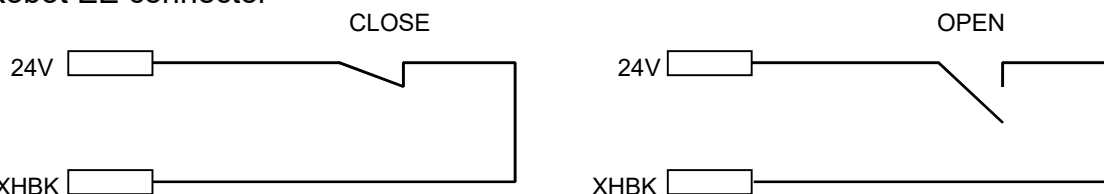
5.6.3 How to Disable/Enable HBK

- 1) Press the [MENU] key on the teach pendant.
- 2) Select [NEXT].
- 3) Select [SYSTEM].
- 4) Press "F1" (TYPE) on the teach pendant.
- 5) Select "Config" to disable/enable HBK.

Status	Hand Broken enable/disable setting	HBK (*1)	HBK detection	Robot operation	Message
1	Enable	CLOSE	Yes	Possible	None
2	Enable	OPEN	Yes	Impossible	SRVO-006
3	Disable	CLOSE	No (*2)	Impossible	SRVO-302
4	Disable	OPEN	No	Possible	At cold start, SRVO-300

NOTE

- 1 Robot EE connector



- 2 The moment the HBK circuit is closed, alarm "Servo 302" occurs. HBK setting needs to be valid manually. When the HBK setting is valid and the HBK circuit is opened, causing alarm "Servo 006".
- 3 If the power is turned off and on again under the condition stated in *2, status 4 is entered, so the alarm condition is removed.

5.6.4 How to Disable/Enable Pneumatic Pressure Alarm (PPABN)

- 1) Press the [MENU] key on the teach pendant.
- 2) Select [NEXT].
- 3) Select [SYSTEM].
- 4) Press "F1" (TYPE) on the teach pendant.
- 5) Select "Config" to disable/enable PPABN.

APPENDIX

A SPECIFICATION LIST

Name	Ordering Specification	FANUC Specification	Note
Main board	A05B-2650-H001	A20B-8200-0790	Standard, Ethernet 1ch
	A05B-2650-H002	A20B-8200-0791	Ethernet 2ch, Vision I/F, Force sensor
	A05B-2650-H003	A20B-8200-0792	Ethernet 2ch, Vision I/F, Force sensor, PMC, HDI
	A05B-2650-H004	A20B-8201-0420	Standard, Ethernet:1ch For I/O Link <i>i</i> slave
	A05B-2650-H005	A20B-8201-0421	Ethernet:2ch, Vision I/F, Force sensor I/F For I/O Link <i>i</i> slave
	A05B-2650-H006	A20B-8201-0422	Ethernet:2ch, Vision I/F, Force sensor I/F, PMC, HDI For I/O Link <i>i</i> slave
	A05B-2680-H001 (R-30iB Mate Plus)	A20B-8201-0750	Standard, Ethernet:2ch For I/O Link <i>i</i> slave
	A05B-2680-H002 (R-30iB Mate Plus)	A20B-8201-0751	Ethernet:3ch, Vision I/F, Force sensor I/F For I/O Link <i>i</i> slave
	A05B-2680-H003 (R-30iB Mate Plus)	A20B-8201-0752	Ethernet:3ch, Vision I/F, Force sensor I/F, PMC, HDI For I/O Link <i>i</i> slave
CPU card	A05B-2600-H020	A20B-3300-0686	Standard / SDRAM 32Mbyte
		A17B-3301-0106	
	A05B-2600-H021	A20B-3300-0687	Standard / SDRAM 64Mbyte
		A17B-3301-0107	
	A05B-2600-H022	A20B-3300-0688	Standard / SDRAM 128Mbyte
		A17B-3301-0108	
	A05B-2600-H023	A20B-3300-0683	High speed / SDRAM 32Mbyte
		A17B-3301-0103	
	A05B-2600-H024	A20B-3300-0684	High speed / SDRAM 64Mbyte
		A17B-3301-0104	
	A05B-2600-H025	A20B-3300-0685	High speed / SDRAM 128Mbyte
		A17B-3301-0105	
	A05B-2600-H026	A17B-3301-0109	Standard / SDRAM 32Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H027	A17B-3301-0110	Standard / SDRAM 64Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H028	A17B-3301-0111	Standard / SDRAM 128Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H029	A17B-3301-0112	High speed / SDRAM 32Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H030	A17B-3301-0113	High speed / SDRAM 64Mbyte For I/O Link <i>i</i> slave
	A05B-2600-H031	A17B-3301-0114	High speed / SDRAM 128Mbyte For I/O Link <i>i</i> slave
	A05B-2670-H020 (R-30iB Mate Plus)	A17B-3301-0250	Standard / DRAM 1GB For I/O Link <i>i</i> slave

Name	Ordering Specification	FANUC Specification	Note
Axis control card	A05B-2600-H040	A20B-3300-0664	6-axis
		A20B-3300-0774	
	A05B-2600-H041	A20B-3300-0663	12-axis
		A20B-3300-0773	
	A05B-2600-H042	A20B-3300-0662	18-axis
		A20B-3300-0772	
	A05B-2600-H043	A20B-3300-0661	24-axis
		A20B-3300-0771	
	A05B-2600-H044	A20B-3300-0660	36-axis
		A20B-3300-0770	
	A05B-2670-H040 (R-30iB Mate Plus)	A20B-3300-0819	6-axis
	A05B-2670-H041 (R-30iB Mate Plus)	A20B-3300-0818	12-axis
	A05B-2670-H042 (R-30iB Mate Plus)	A20B-3300-0817	18-axis
	A05B-2670-H043 (R-30iB Mate Plus)	A20B-3300-0816	24-axis
	A05B-2670-H044 (R-30iB Mate Plus)	A20B-3300-0815	36-axis
FROM/SRAM module	A05B-2600-H060	A20B-3900-0283	FROM 32M/ SRAM 1M
		A20B-3900-0297	
	A05B-2600-H061	A20B-3900-0284	FROM 32M/ SRAM 2M
		A20B-3900-0298	
	A05B-2600-H062	A20B-3900-0285	FROM 32M/ SRAM 3M
		A20B-3900-0299	
	A05B-2600-H063	A20B-3900-0286	FROM 64M/ SRAM 1M
	A05B-2600-H064	A20B-3900-0287	FROM 64M/ SRAM 2M
	A05B-2600-H065	A20B-3900-0288	FROM 64M/ SRAM 3M
	A05B-2600-H066	A20B-3900-0280	FROM 128M/ SRAM 1M
	A05B-2600-H067	A20B-3900-0281	FROM 128M/ SRAM 2M
	A05B-2600-H068	A20B-3900-0282	FROM 128M/ SRAM 3M
	A05B-2600-H069 (R-30iB Mate Plus)	A20B-3900-0293	FROM 256M/ SRAM 1M
	A05B-2600-H070 (R-30iB Mate Plus)	A20B-3900-0295	FROM 256M/ SRAM 2M
	A05B-2600-H071 (R-30iB Mate Plus)	A20B-3900-0296	FROM 256M/ SRAM 3M
Battery	A05B-2650-K030	A98L-0031-0028	For memory backup
Fan board	A05B-2650-H001	A20B-8200-0669	
	A05B-2650-H002		
	A05B-2650-H003		
Emergency stop board		A20B-2005-0150	
		A20B-2103-0170	For R-30iB Mate Plus
4-pole terminal block (TBOP19)		A63L-0002-0154#104	Manufacturer's specification (WAGO):734-104
12-pole terminal block (TBOP20)		A63L-0002-0154#112	Manufacturer's specification (WAGO):734-112
Jumper pin		A63L-0002-0154 #402F	Manufacturer's specification (WAGO):734-402F

Name	Ordering Specification	FANUC Specification	Note
Operation lever	A05B-2600-K030	A63L-0002-0154 #230-M	Manufacturer's specification (WAGO): 734-230 2 pieces of 734-230 and operation manual are included in FANUC's specification
Process I/O board MA	A05B-2650-J060	A20B-2004-0381	DI/DO=20/16(Source type)
Process I/O board MB	A05B-2650-J061	A20B-2101-0731	WI/WO=5/4(Sink type), D/A=2
Peripheral device cable (Process I/O MA)	A05B-2650-J150		Connected length: 10m (one): CRMA52
	A05B-2650-J151		Connected length: 20m (one): CRMA52
Welding machine connection cable (FANUC I/F/elbow type) (Process I/O MB)	A05B-2650-J160		Connected length: 3m (one): CRW11
	A05B-2650-J161		Connected length: 7m (one): CRW11
	A05B-2650-J162		Connected length: 14m (one): CRW11
Connector conversion board	A05B-2650-J070	A20B-2004-0411	CRMA15, CRMA16, Honda MR50 connector
Terminal conversion board	A05B-2650-J071	A20B-1009-0690	CRMA15, CRMA16, PHOENIX CONTACT terminal
6-axis servo amplifier	A05B-2652-H031	A06B-6400-H102	
	A05B-2654-H030	A06B-6400-H002	
	A05B-2657-H030		
	A05B-2652-H030	A06B-6400-H003	
	A05B-2651-H030	A06B-6400-H005	
	A05B-2661-H030		
Discharge resistor	A05B-2650-C100		
	A05B-2650-C101		
CRR65A/B connector Rece-housing		A63L-0001-0460 #032KSX	Auxiliary Axis Brake Manufacturer's specification (TE connectivity): 1-178128-3
CRMA96 connector Rece-contact (AWG16-20)		A63L-0001-0456 #ASL	Auxiliary Axis Brake Manufacturer's specification (TE connectivity): 175218-2
CRM68 connector Rece-housing		A63L-0001-0812 #R03SX	Auxiliary Axis Over Travel Manufacturer's specification (TE connectivity): 1-1318120-3
CRM68 connector Rece-contact (AWG18-22)		A63L-0001-0812 #CRM	Auxiliary Axis Over Travel Manufacturer's specification (TE connectivity): 1318107-1
Peripheral device cable (For Main board)	A05B-2650-J100		Connected length: 10m (one): CRMA15 Connected length: 10m (one): CRMA16
	A05B-2650-J101		Connected length: 20m (one): CRMA15 Connected length: 20m (one): CRMA16
Brake release unit connection cable		A660-2005-T559	M-710iC, R-1000iA, R-2000iC, M-2iA, M-3iA
		A660-2006-T881	M-10iA, M-20iD, M-20iA, M-20iB, ARC Mate 100iC, ARC Mate 100iD, ARC Mate 120iC ARC Mate 0iB, R-0iB, LR Mate 200iD, ARC Mate 50iD, CR-4iA, CR-7iA
		A660-2006-T474	M-1iA, LR Mate 200iC
		A660-2005-T711	Aux. Axis
Sensor I/F unit	A05B-2650-C200		CR-4iA, CR-7iA

B TOTAL CONNECTION DIAGRAM

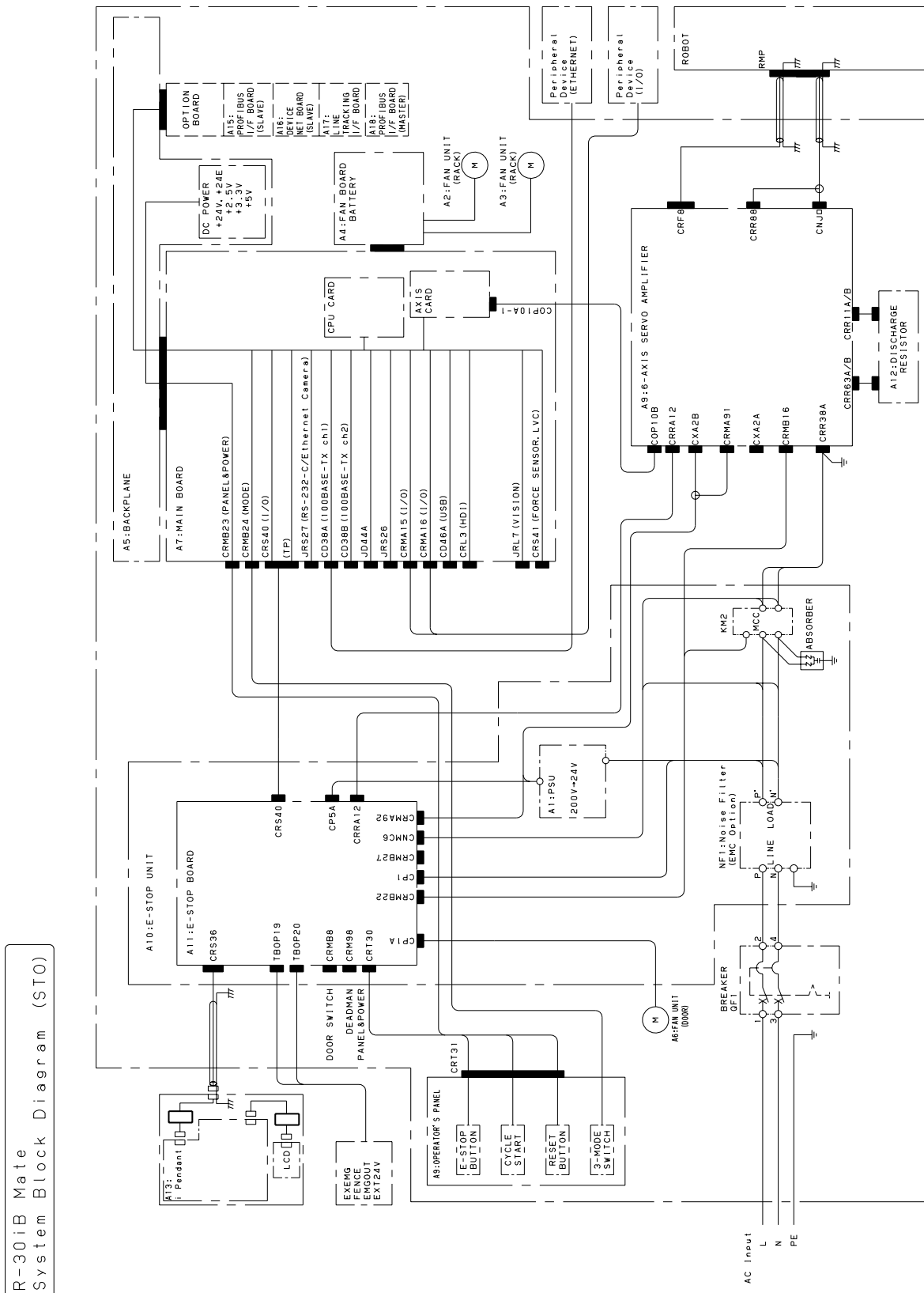
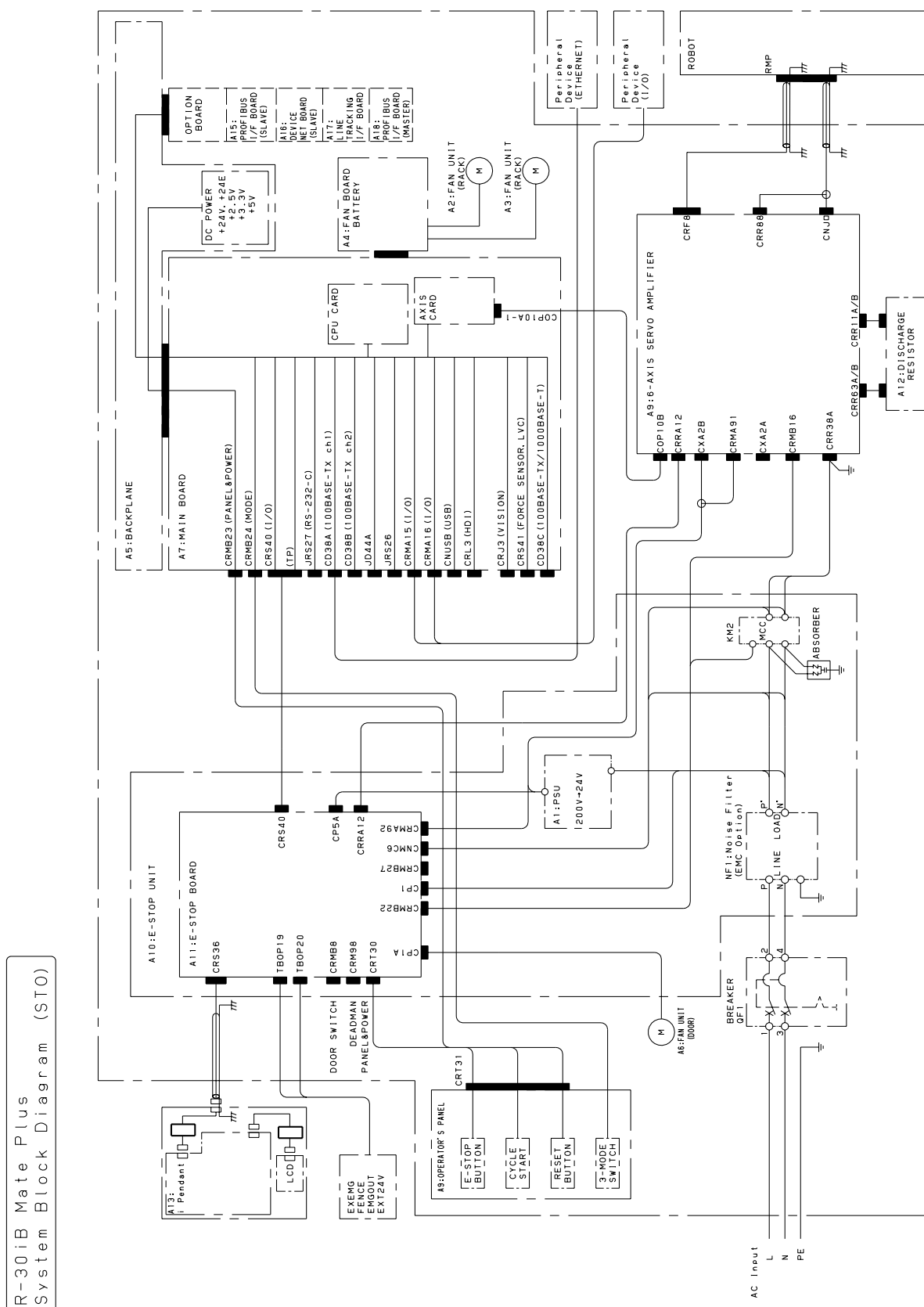


Fig.B (a) System block diagram (R-30iB)

(This diagram shows the single phase input circuit. In case of three phase input, See the Fig.B(c))



3 PHASE POWER SOURCE

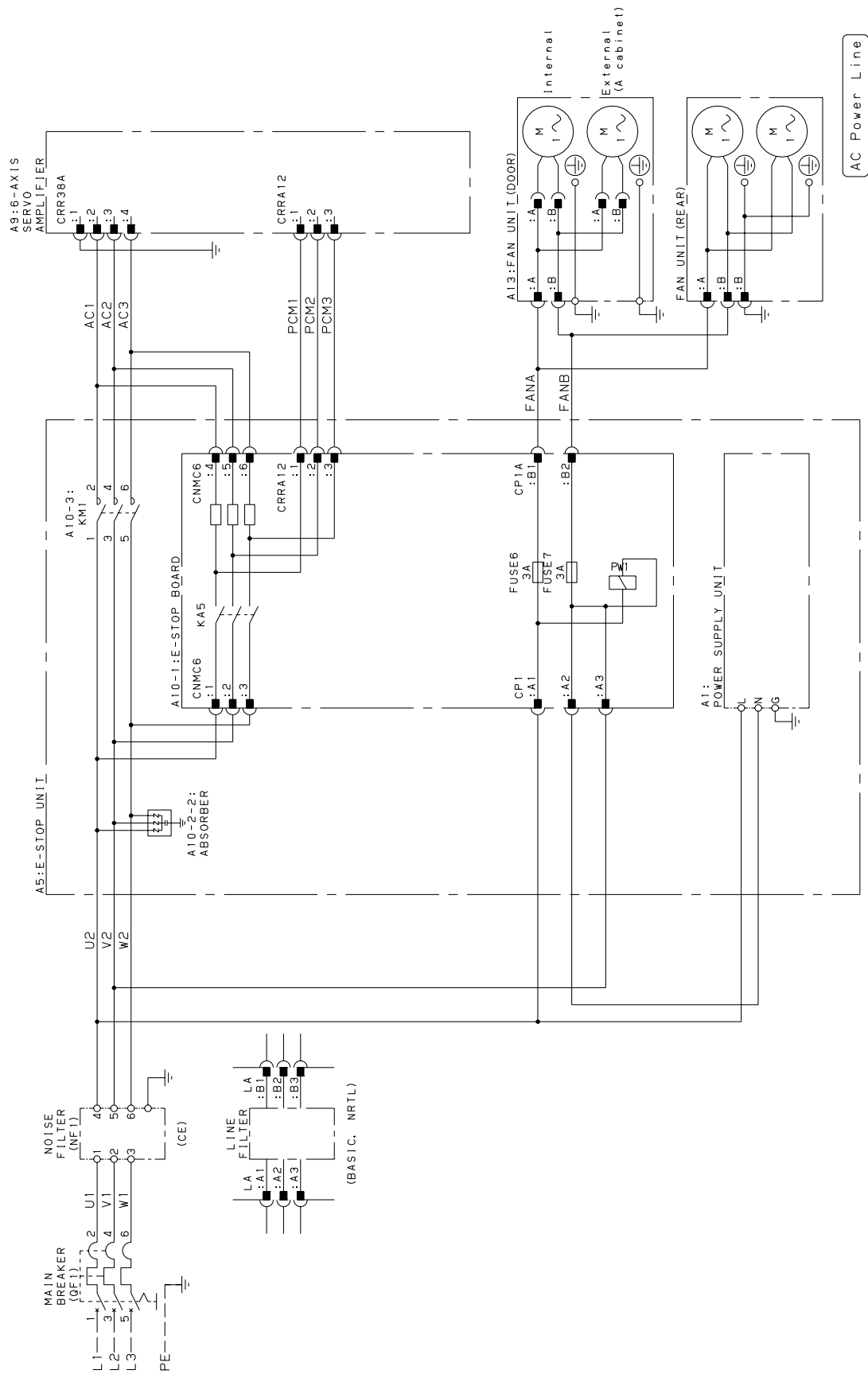


Fig.B(c) AC power line connection diagram (R-30iB Mate, in case of three phase input)

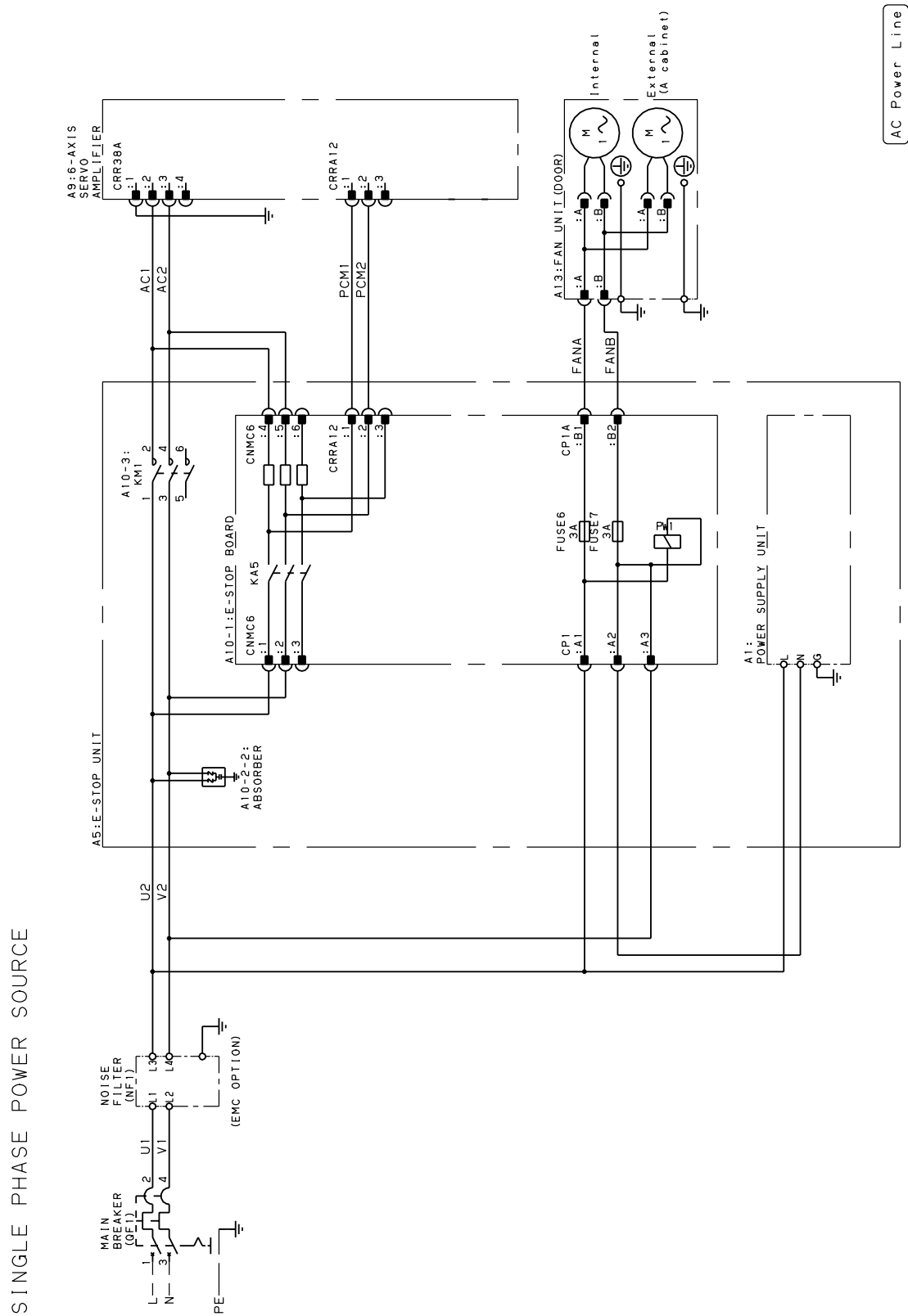


Fig.B(d) AC power line connection diagram (R-30iB Mate, in case of single phase input)

3 PHASE POWER SOURCE

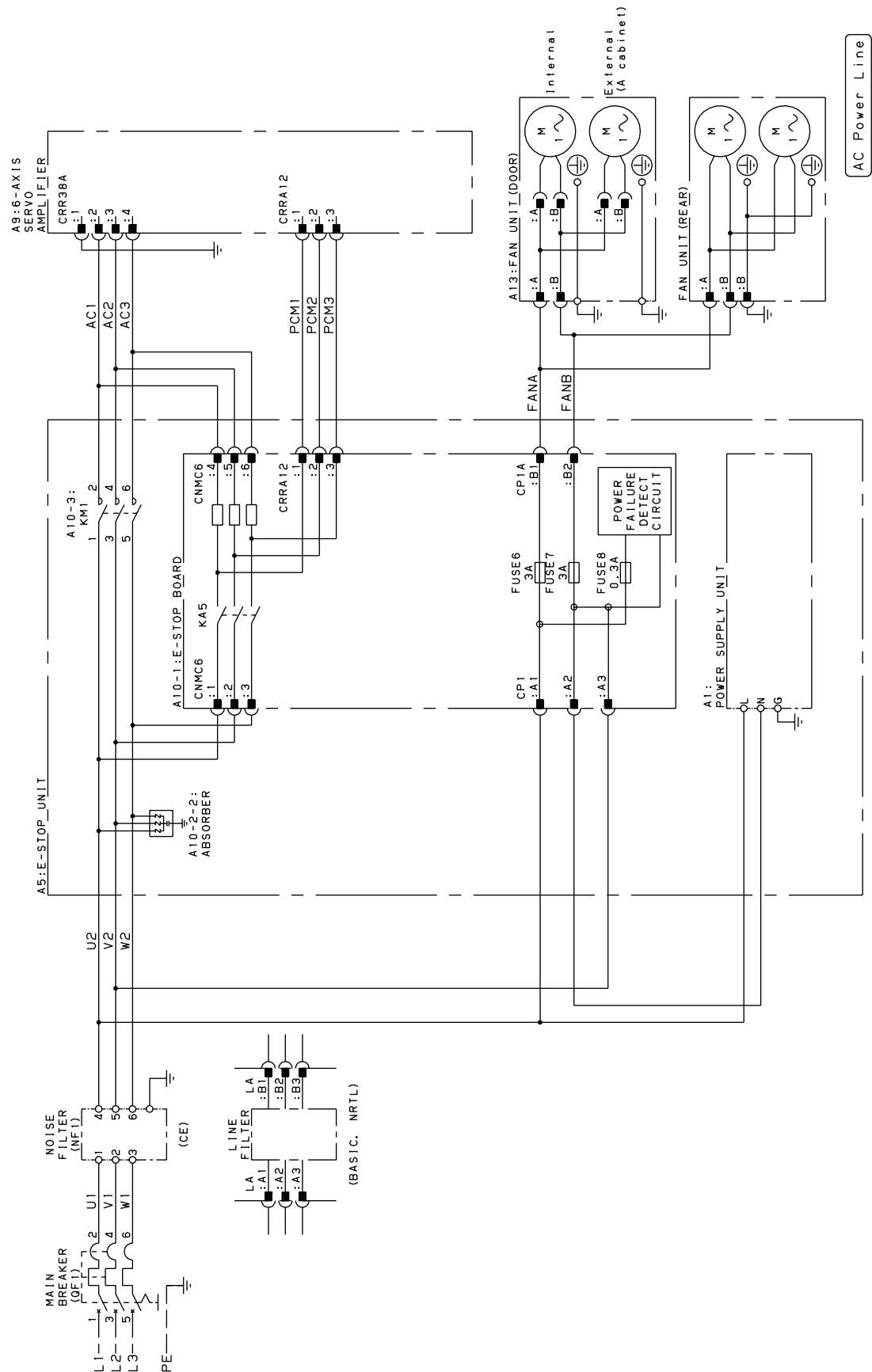


Fig.B(e) AC power line connection diagram (R-30iB Mate Plus, in case of three phase input)

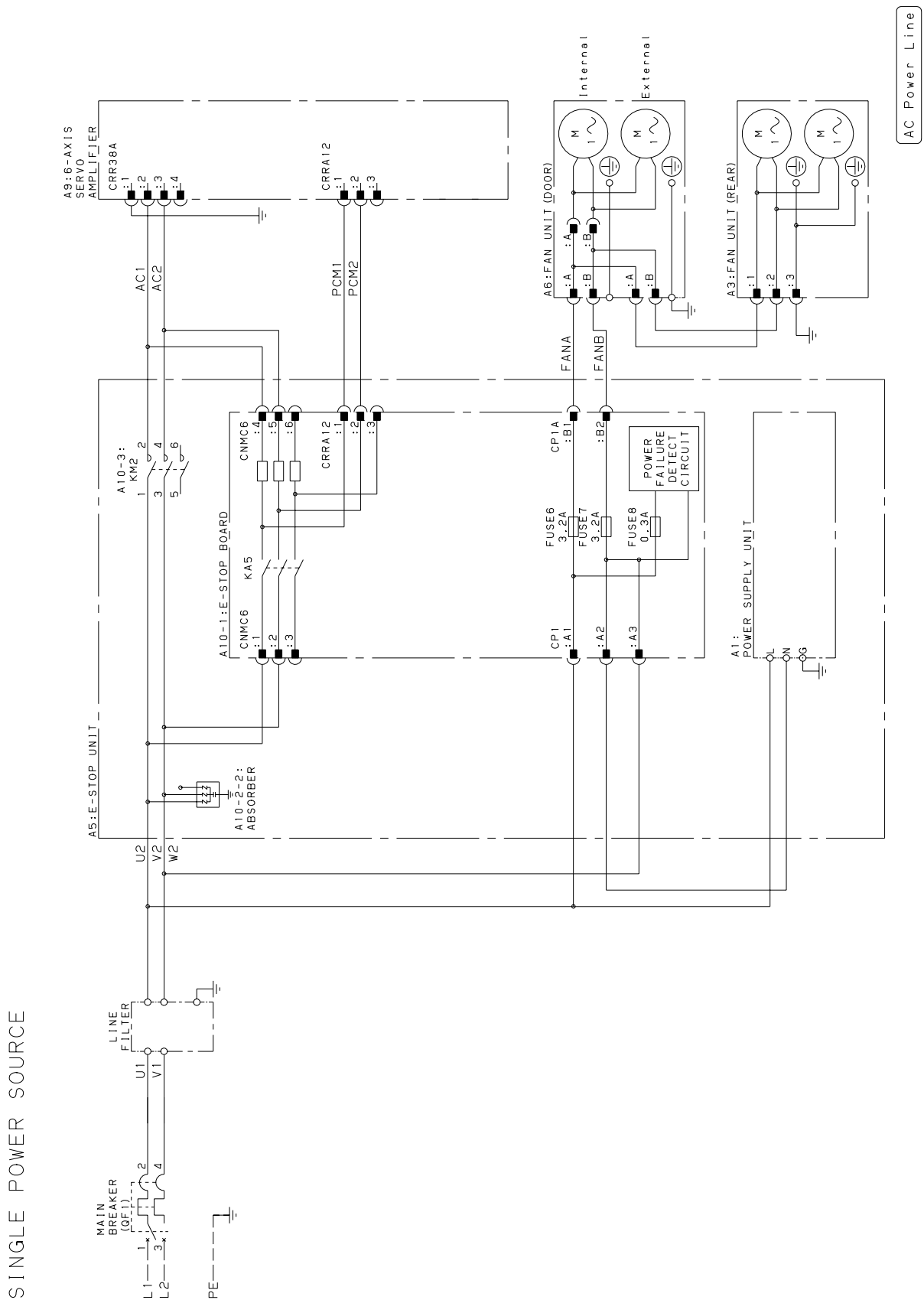


Fig.B(f) AC power line connection diagram (R-30/B Mate Plus, in case of single phase input)

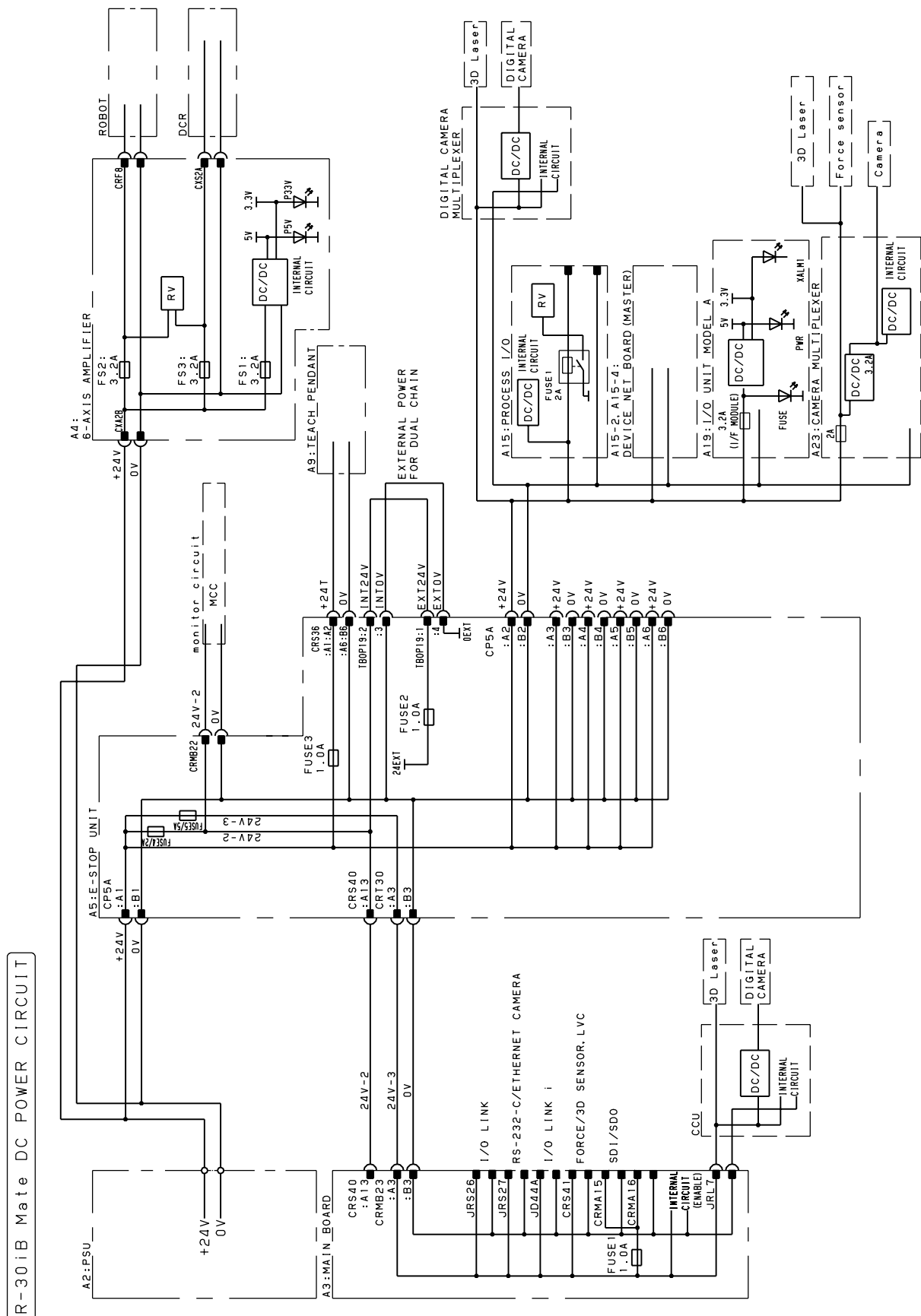


Fig.B(g) DC power line connection diagram (R-30iB Mate)

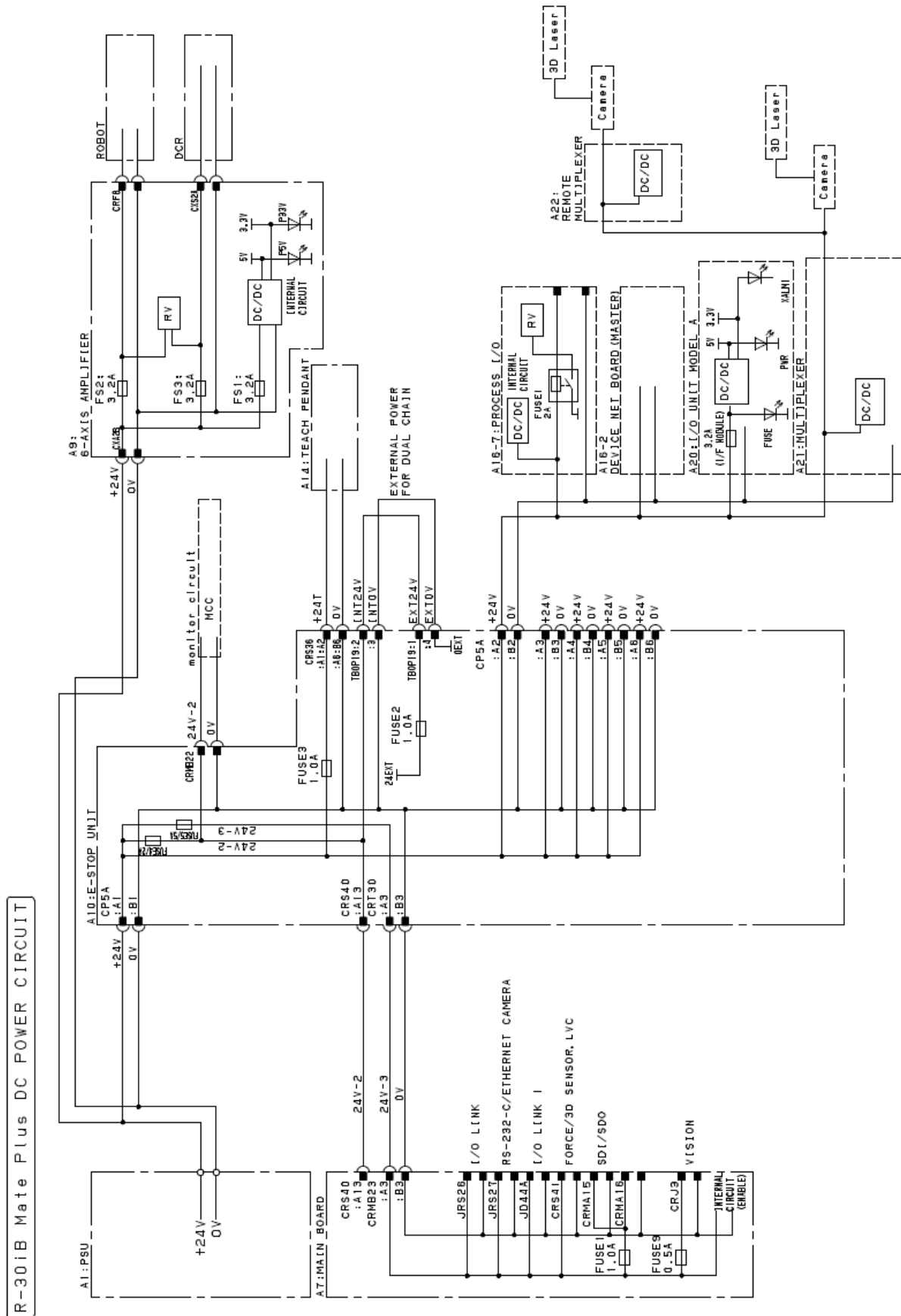


Fig.B(h) DC power line connection diagram (R-30iB Mate Plus)

E-STOP CIRCUIT <R-30iB Mate>

Dual check safety

Single MCC with STO (Single Phase)

DI: Simple DI

PI: Photo coupler DI

Not showing the diodes to protect from reverse electric power.

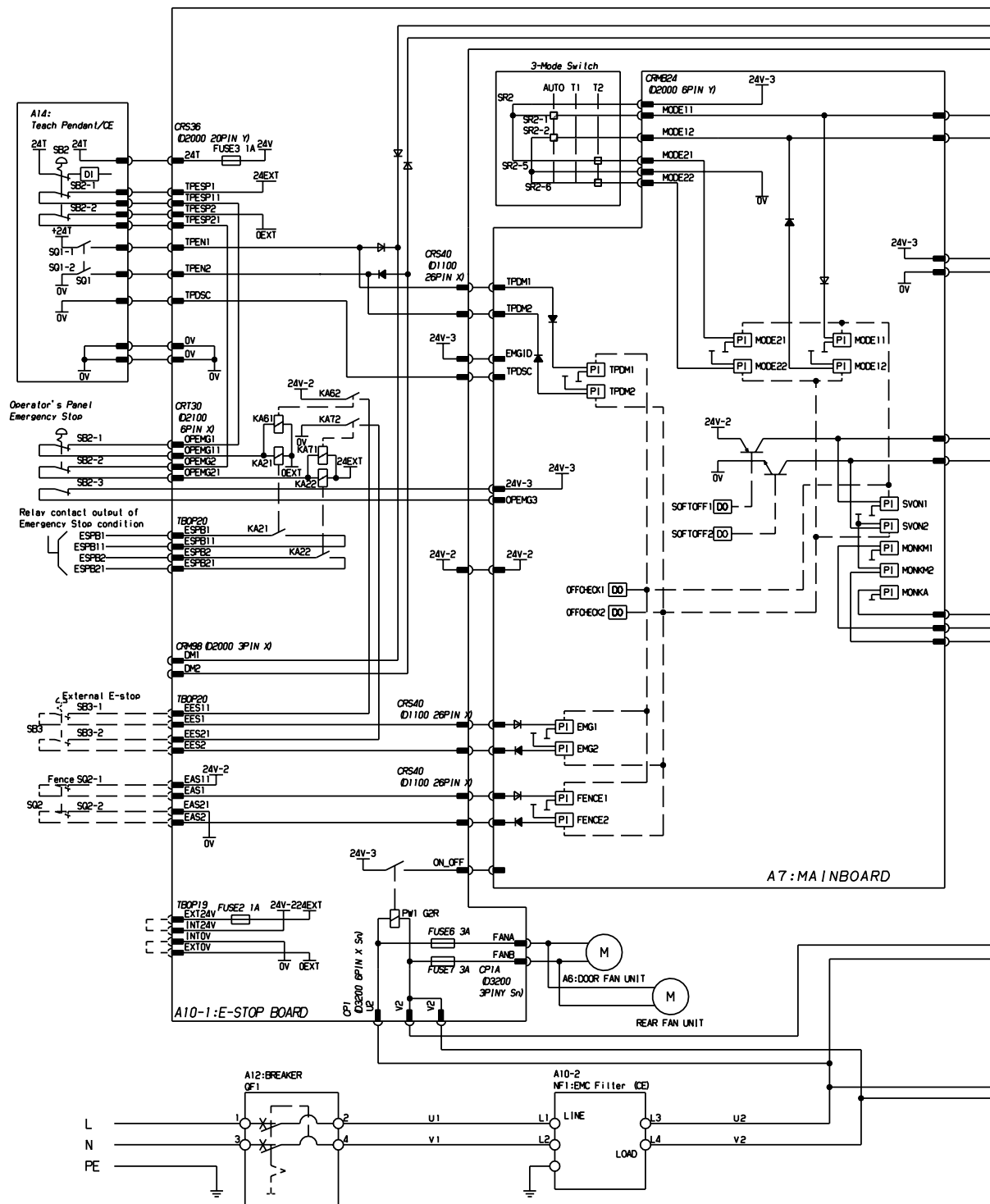


Fig.B(i) R-30iB Mate Emergency stop circuit connection diagram

(This diagram shows the single phase input circuit. In case of three phase input, See the Fig.B(b))



E-STOP CIRCUIT <R-30iB Mate Plus>
Dual check safety
Single MCC with STO (Single Phase)

DI: Simple DI

PI: Photo coupler DI

Not showing the diodes to protect from reverse electric power.

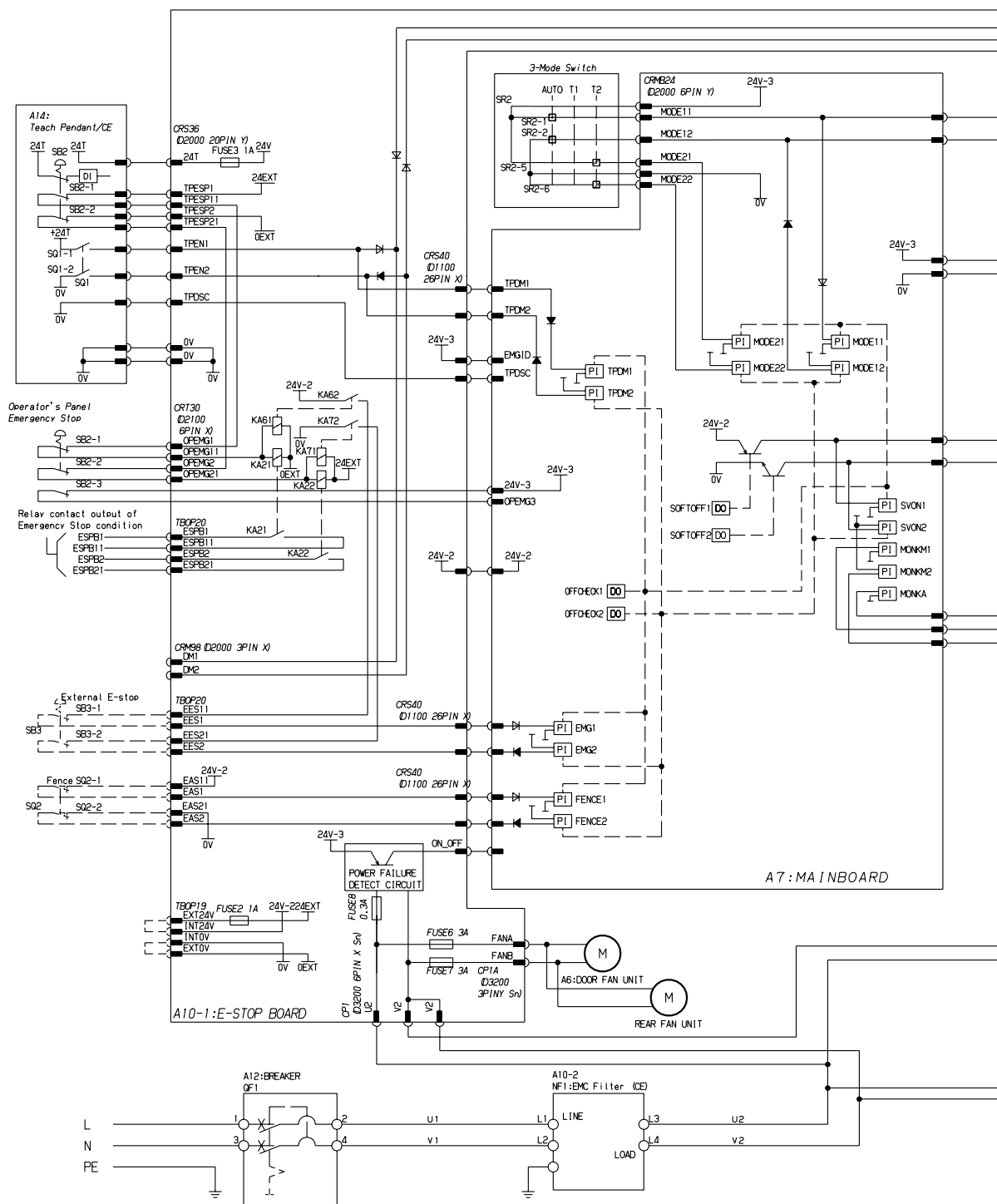


Fig.B(j) R-30iB Mate Plus Emergency stop circuit connection diagram
(This diagram shows the single phase input circuit. In case of three phase input, See the Fig.B(d))



R-30iB Mate E-STOP Board Connector Table

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5	TPDSC	B5	0V
A6	0V	B6	0V
A7	TXTP	B7	TXN_TP
A8	TXTP	B8	TXP_TP
A9	XRTP	B9	RXN_TP
A10	RXTP	B10	RXP_TP

CRS40

D1100D (X)

MAIN BOARD I/F

A1	RXTP	B1	TXTP
A2	XRTP	B2	TXTP
A3	RXP_TP	B3	TXP_TP
A4	RXN_TP	B4	TXN_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	ENGID
A11	EES1	B11	SVON1
A12	EES2	B12	SVON2
A13	24V-2	B13	ON_OFF

CRS40

D1100D (X)

MAIN BOARD I/F

A1	OPMG1	B1	OPENG11
A2	OPMG2	B2	OPENG21
A3	24V-3	B3	0V

CRS40

D1100D (X)

6-AXIS SERVO AMPLIFIER

A1	MON2	B1	BRKON
A2	XSVEMG	B2	BKDLY
A3	XMCCON	B3	MON1
A4	XPCON	B4	KM2ON

CRS36

D2100D (Y)

TEACH PENDANT

A1	+24T	B1	TPESP21
A2	+24T	B2	TPESP2
A3	TPEN2	B3	TPESP11
A4	TPEN1	B4	TPESP1
A5</			

Fig.B(k) Emergency stop board connector table (R-30iB Mate)

R-30iB Mate Plus E-STOP Board
Connector Table

CRS36	A1	+24T	B1	TPESP21
D2100D (Y)	A2	+24T	B2	TPESP2
TEACH PENDANT	A3	TPEN2	B3	TPESP11
	A4	TPEN1	B4	TPESP1
	A5	TPDSC	B5	OV
	A6	OV	B6	OV
	A7	XTXTP	B7	TXN_TP
	A8	TXTP	B8	TXP_TP
	A9	XRTP	B9	RXN_TP
	A10	RXTP	B10	RXP_TP

CP5A	A1	24V	B1	0V
D3500	A2	24V	B2	0V
EXTERNAL DC24V	A3	24V	B3	0V
	A4	24V	B4	0V
	A5	24V	B5	0V
	A6	24V	B6	0V

CRMB8	A1	SVON1	B1	SVON11
DOOR SWITCH	A2	SVON2	B2	SVON21

CRMB22	A1	KA31	B1	KA32
D2100D (Y)	A2	KA41	B2	KA42
MCC I/F	A3		B3	OV
	A4	KM1ON	B4	KM2ON

CRS40	A1	RXTP	B1	TXTP
D1100D (X)	A2	XRTP	B2	XTXTP
MAIN BOARD I/F	A3	RXP_TP	B3	TXP_TP
	A4	RXN_TP	B4	TXN_TP
	A5		B5	OV
	A6	MODE11	B6	MONKM1
	A7	TPDM1	B7	MONKM2
	A8	TPDM2	B8	MONKMA
	A9	EAS1	B9	TPDSC
	A10	EAS2	B10	EMG1D
	A11	EES1	B11	SVON1
	A12	EES2	B12	SVON2
	A13	24V-2	B13	ON_OFF

CRRT30	A1	OPEMG1	B1	OPEMG11
D2100D (X)	A2	OPEMG2	B2	OPEMG21
E-STOP BUTTON	A3	24V-3	B3	OV

CRM92	A1	MON2	B1	BRKON
D2100D (X)	A2	XSVEMG	B2	BKDL
	A3	XMCCON	B3	MON1
	A4	XPCON	B4	KM2ON

CP1A	A1	V21N	B1	FANA
D3200 (XY)	A2	V20UT	B2	FANB
CP1A:FAN UNIT POWER (ROW B, Y-KEY)	A3	U2	B3	

CP1/CP1A	A1	U2	B1	AC1
D3200 (XY)	A2	U2	B2	AC2
CP1 : AC200V INPUT (ROW A, X-KEY)	A3	U2	B3	AC3

CRMB9	A1	MON2	B1	BRKON
D2100D (X)	A2	XSVEMG	B2	BKDL
	A3	XMCCON	B3	MON1
	A4	XPCON	B4	KM2ON

CRMB6	A1	KA31	B1	KA32
D2100D (Y)	A2	KA41	B2	KA42
MUL I1 ARM/AUX. AXIS	A3		B3	OV
	A4	KM1ON	B4	KM2ON

CRMB27	A1	AUXNON2	B1	
D2100D (Y)	A2	24V-2	B2	
MUL I1 ARM/AUX. AXIS	A3	KM2ON_DO	B3	

CRM98	A1	DM1	B1	
D2100D (X)	A2	DM2	B2	
DEADMAN SW. OUTPUT	A3		B3	

CNM6	A1	U2	B1	
D3200 (X)	A2	V2	B2	
DEADMAN SW. OUTPUT	A3	W2	B3	
	A4	AC1	B4	
	A5	AC2	B5	
	A6	AC3	B6	

CRRA12	A1	PCM1	B1	
D3200 (Z)	A2	PCM2	B2	
AC200V MONITOR	A3	PCM3	B3	

TBOP19	A1	EXT24V	B1	
EXTERNAL DC24V	A2	INT24V	B2	
	A3	INT0V	B3	
	A4	EXT0V	B4	

TBOP20	A1	EES1	B1	
EXTERNAL E-STOP/FENCE	A2	EES11	B2	
	A3	EES2	B3	
	A4	EES21	B4	
	A5	EAS1	B5	
	A6	EAS11	B6	
	A7	EAS2	B7	
	A8	EAS21	B8	
	A9	ESPBI	B9	
	A10	ESPBI1	B10	
	A11	ESPBI2	B11	
	A12	ESPBI21	B12	

Fig.B(I) Emergency stop board connector table (R-30iB Mate Plus)

MAIN BOARD

JRS26
PCR20

I/O Link (CH.1)	
11	0V
12	0V
13	0V
14	0V
15	0V
16	0V
17	0V
18	5V
19	24V-3
20	5V

JRS27
PCR20

RS232-C, ETHERNET CAMERA

11	TXDA
12	0V
13	0V
14	0V
15	RTSA
16	0V
17	CAMTX+
18	CAMTX-
19	24V-3
20	0V

JRL 7
PCR20

11	CAMD02
12	0V
13	CAMD03
14	0V
15	CAMD11
16	CAMD12
17	CAMD10
18	CAMD00
19	PI2V
20	CAMD01

JD44A
PCR20

I/O Link i (CH.2)	
11	0V
12	0V
13	0V
14	0V
15	0V
16	0V
17	0V
18	5V
19	24V-3
20	5V

CRS40
D1100D (X)

E-STOP BOARD I/F

A1	B1
A2	B2
A3	RXP_TP
A4	RXN_TP
A5	B5
A6	MODE11
A7	TPDM1
A8	TPDM2
A9	EAS1
A10	EAS2
A11	EES1
A12	EES2
A13	24V-2

CRS41
DF11-10DS-2C (HIROSE)

1	SDATA
3	RV_LVC
5	24V-3
7	DV_LVC
9	5V

CRMB23
D2100D (X)

OP. PANEL SWITCH	
A1	BUSY
A2	START
A3	24V-3

CRMB24
D1200D (Y)

MODE SWITCH	
A1	MODE11
A2	MODE12
A3	24V-3

CRL3

DF11-4DS-2C (HIROSE)

1	XHD10
3	0V

CRMA15
D1/DO

1	24F
2	24F
3	SDICOM1
4	0V
5	D1101
6	D1103
7	D1105
8	D1107
9	D1109
10	D1111
11	D1113
12	D1115
13	D1117
14	D1119
15	D0101
16	D0103
17	D0105
18	D0107
19	0V
20	D0SRC1

CRMA16
D1/DO

1	24F
2	24F
3	SDICOM3
4	0V
5	D1121
6	D1123
7	D1125
8	D1127
9	
10	D0109
11	D0111
12	D0113
13	D0115
14	D0117
15	D0119
16	D0121
17	D0123
18	
19	0V
20	D0SRC2

CD38A
Ethernet 100Base-TX

1	TXA+
2	TXA-
3	RXA+
4	
5	
6	RXA-
7	
8	

CD38B
Ethernet 100Base-TX

1	TXB+
2	TXB-
3	RXB+
4	
5	
6	RXB-
7	
8	

Fig.B(m) Main board connector table (R-30iB Mate)

MAIN BOARD

JRS26
PCR20
I/O Link (CH.1)

11	0V	1	RXSLCB
12	0V	2	XRSLCB
13	0V	3	TXSLCB
14	0V	4	XTXSLCB
15	0V	5	RXSLCC
16	0V	6	XRSLCC
17		7	TXSLCC
18	5V	8	XTXSLCC
19	24V-3	9	5V
20	5V	10	24V-3

JRS27
PCR20
RS232-C

11	TXDA	1	RXDA
12	0V	2	0V
13	DTRA	3	DSRA
14	0V	4	0V
15	RTSA	5	CTSA
16	0V	6	0V
17		7	
18		8	
19	24V-3	9	
20		10	24V-3

CRJ3
SENSOR INTERFACE
(VIDEO INTERFACE)

1	CAMERA
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JD44A
PCR20
I/O Link I (CH.2)

11	0V	1	(Reserve)
12	0V	2	(Reserve)
13	0V	3	(Reserve)
14	0V	4	(Reserve)
15	0V	5	RXSLCS
16	0V	6	XRSLCS
17		7	TXSLCS
18	5V	8	XTXSLCS
19	24V-3	9	5V
20	5V	10	24V-3

CRS40
D1100D (X)
E-STOP BOARD I/F

A1		B1	
A2		B2	
A3	RXP_TP	B3	TXP_TP
A4	RXL_TP	B4	TXL_TP
A5		B5	0V
A6	MODE11	B6	MONKM1
A7	TPDM1	B7	MONKM2
A8	TPDM2	B8	MONKMA
A9	EAS1	B9	TPDSC
A10	EAS2	B10	EMG1D
A11	EES1	B11	SVOM1
A12	EES2	B12	SVOM2
A13	24V-2	B13	ON_OFF

CRS41
DF11-10DS-2C (HIROSE)
FORCE/3D SENSOR, LVC

1	SDATA	2	XSDATA
3	RV_LVC	4	XRV_LVC
5	24V-3	6	0V
7	DV_LVC	8	XDV_LVC
9	5V	10	0V

CRMB23
D2100D (X)
OP. PANEL SWITCH

A1	BUSY	B1	RESET
A2	START	B2	OPMG3
A3	24V-3	B3	INOV

CRMB24
D1100D (Y)
MODE SWITCH

A1	MODE11	B1	MODE21
A2	MODE12	B2	MODE22
A3	24V-3	B3	0V

CRL3
DF11-4DS-2C (HIROSE)
HD1

1	XHD10	2	XHD11
3	0V	4	0V

CRMA15
DI/DO

	A	B
1	24F	24F
2	24F	24F
3	SDICOM1	SDICOM2
4	0V	0V
5	D1101	D1102
6	D1103	D1104
7	D1105	D1106
8	D1107	D1108
9	D1109	D1110
10	D1111	D1112
11	D1113	D1114
12	D1115	D1116
13	D1117	D1118
14	D1119	D1120
15	D0101	D0102
16	D0103	D0104
17	D0105	D0106
18	D0107	D0108
19	0V	0V
20	D0SRC1	D0SRC1

CD38A
Ethernet
100Base-TX

1	TPTXA+
2	TPTXA-
3	TPRXA+
4	NCA1
5	NCA1
6	TPRXA-
7	NCA2
8	NCA2

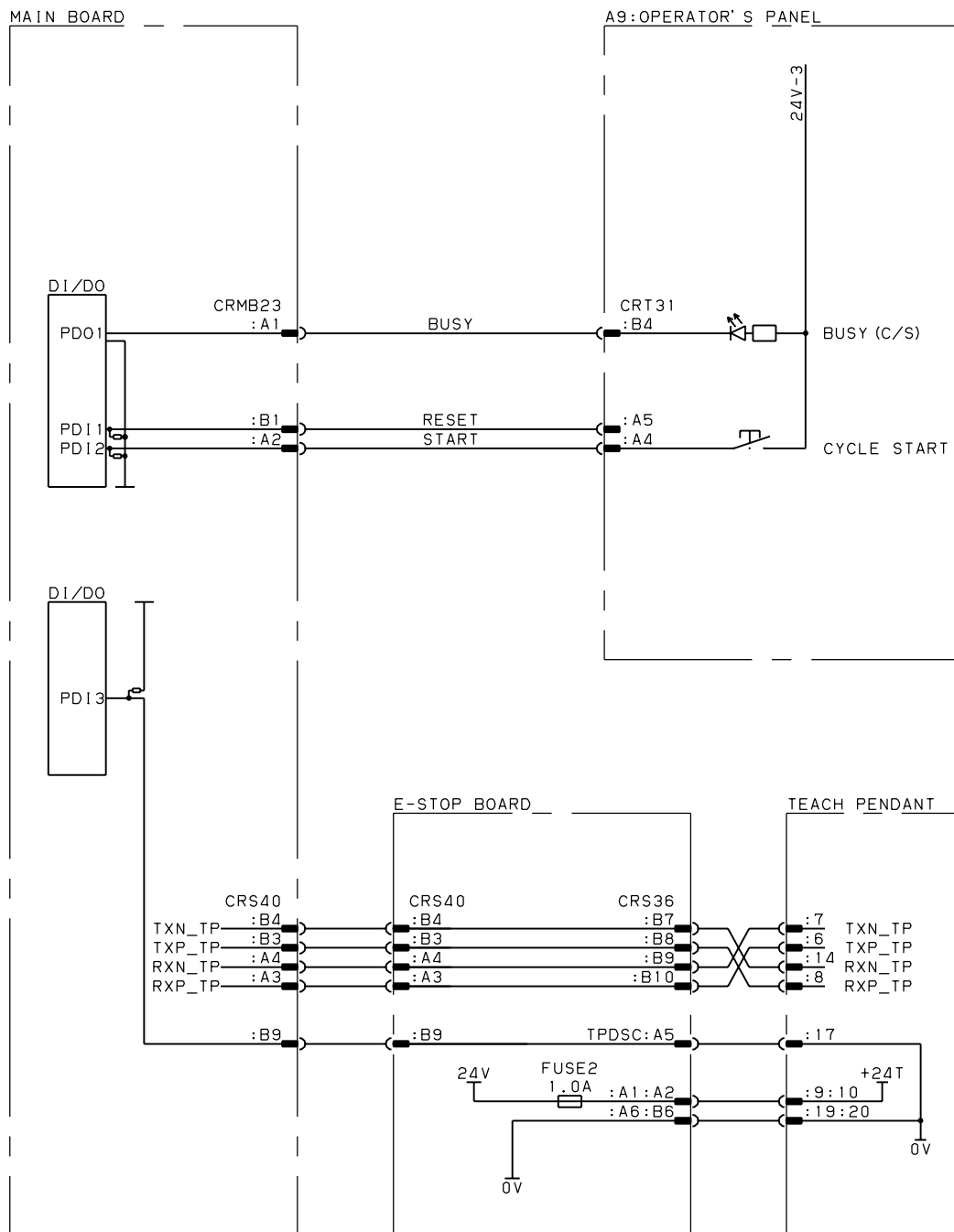
CD38B
Ethernet
100Base-TX

1	TPTXB+
2	TPTXB-
3	TPRXB+
4	NCB1
5	NCB1
6	TPRXB-
7	NCB2
8	NCB2

CD38C
Ethernet
100Base-TX/
1000Base-T

1	TDPA
2	TDMA
3	TDPB
4	TDPC
5	TDMC
6	TDMB
7	TDPD
8	TDMD
9	FG
10	FG

Fig.B(n) Main board connector table (R-30i/B Mate Plus)



A8:
Operator's Panel

Fig.B(o) Operator's panel connection diagram

Line filter
Connector Table

LA
D5200 (XX)

A1	U1	B1	U2
A2	V1	B2	V2
A3	W1	B3	W2

Teach pendant
Connector Table

CONNECTOR ON THE TEACH PENDANT

4	3	2	1
10	+24T	9	+24T
16	TPESP21	15	TPESP2
		20	0V
		19	0V
		18	TPEN2
		17	TPDSC
		16	TPESP1
		15	TPESP11
		14	TPESP11
		13	TPESP11
		12	TPESP1
		11	TPEN1
		10	DRAIN

Operator's panel
Connector Table

CRT31
D2100D (X)
OP. PANEL SWITCH

A1	OPENG1	B1	OPENG11
A2	OPENG2	B2	OPENG21
A3	24V-3	B3	OPENG3
A4	START	B4	BUSY
A5	RESET	B5	
A6		B6	

Fig.B(p) Operator's panel/Teach pendant connector table

CRF8 D-2600				
	1	1	A	B
	1	1	XPRQ1	PRQ1
	2	2	XPRQ2	PRQ2
CRRA11A D3200 (Y)	1		DCRA1	
	2			
	3		DCRA2	
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM91 D2100D (X)	A1		BRKNTM	B2
	A3		XSVEMG	B3
	A4		MON2	B4
	A5		XPCCHON	B5
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (Y)	A1		XBRKRLS2	B1
	A2		XBRKRLS3	B2
	A3		XBRKRLS4	B3
	A4		GUNCHG	B4
CRM97 D2100D (				

Servo Amplifier
Connector Table

Fig.B(q) 6-Axis Servo amplifier connector table

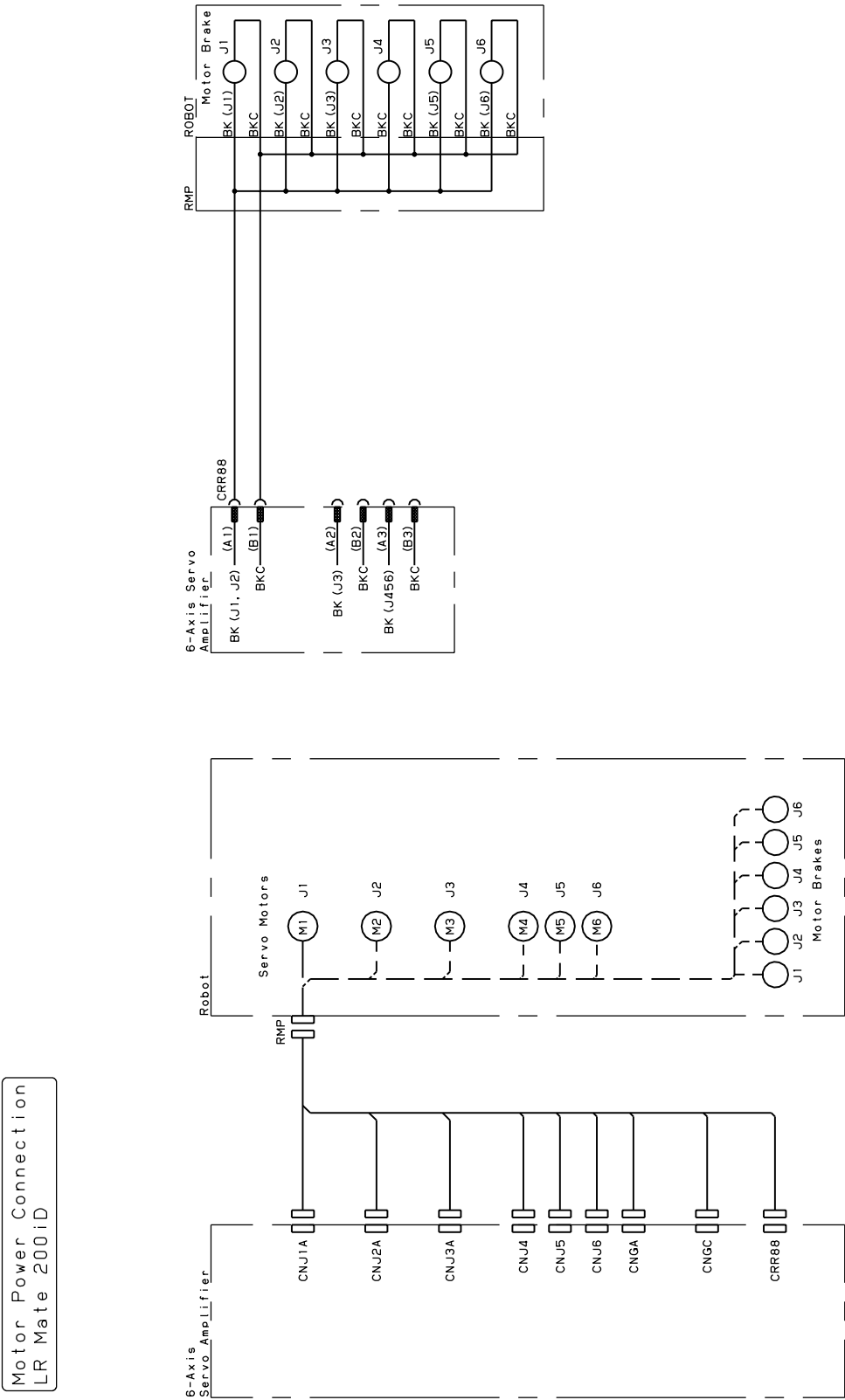


Fig.B (r) Motor power connection
LR Mate 200iC, LR Mate 200iD (6-Axis) ARC Mate 50iD, M-1iA(6-Axis), CR-4iA, CR-7iA

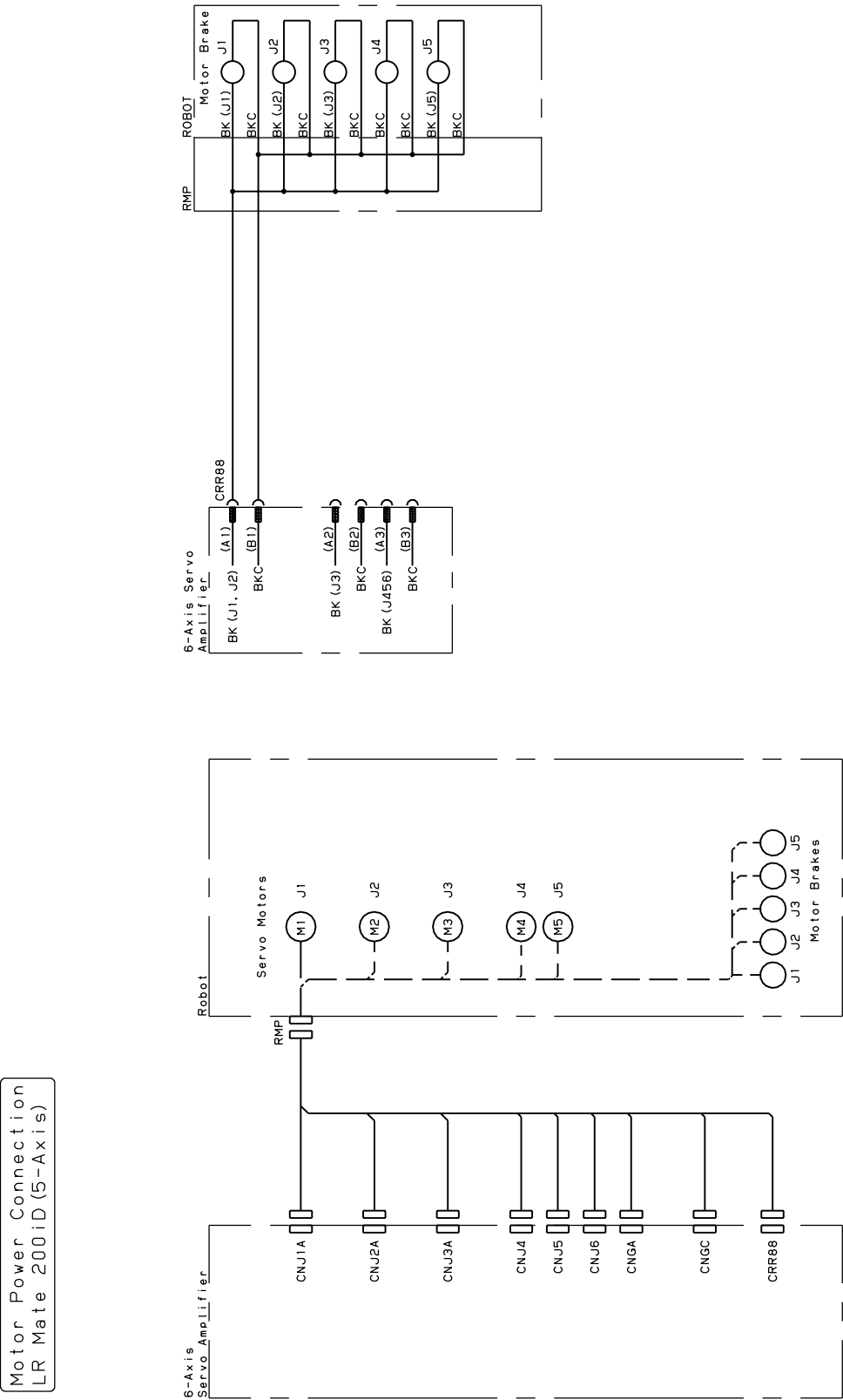


Fig.B(s) Motor power connection
LR Mate 200iD (5-Axis)

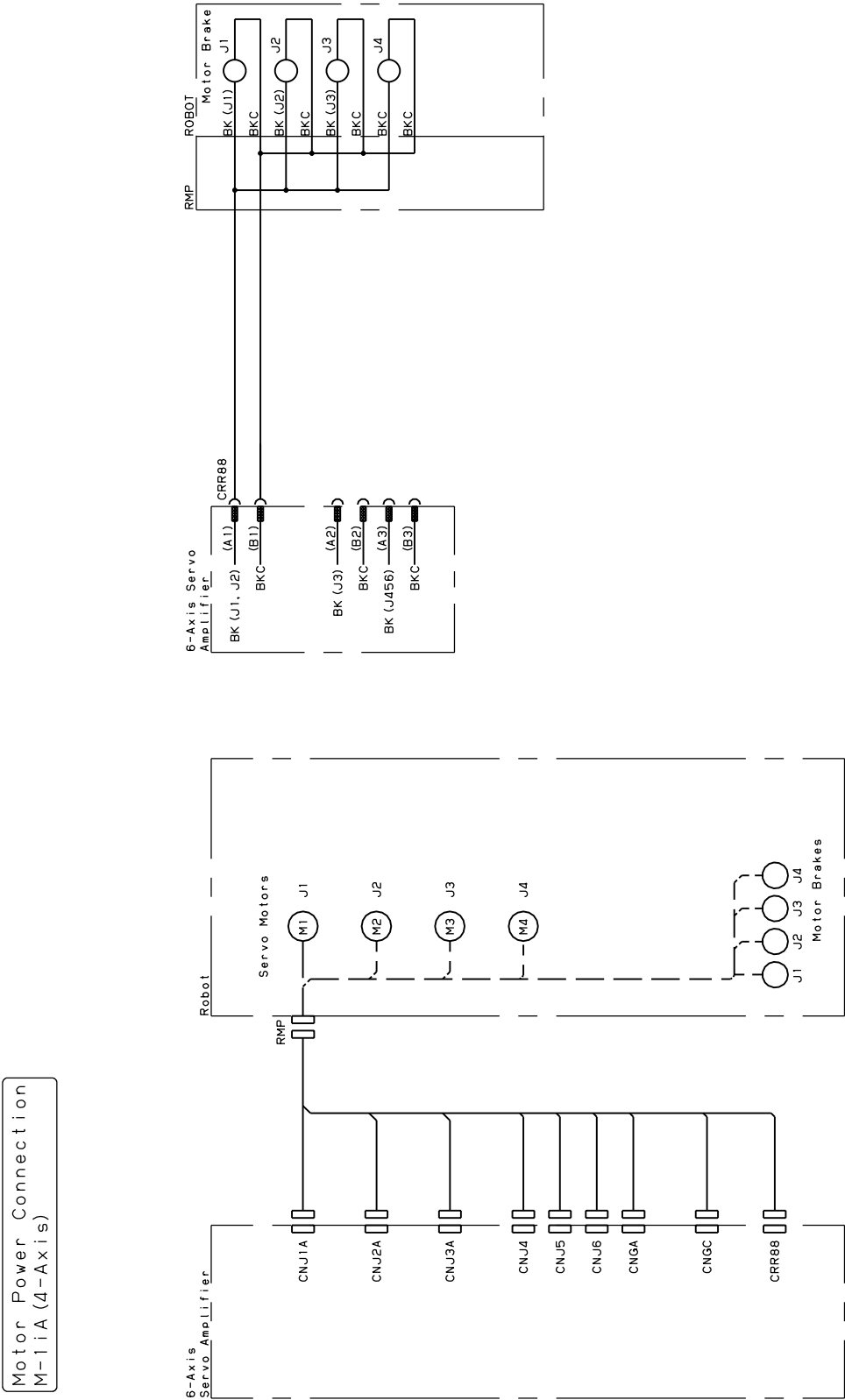


Fig.B(t) Motor power connection
M-1iA (4-Axis)

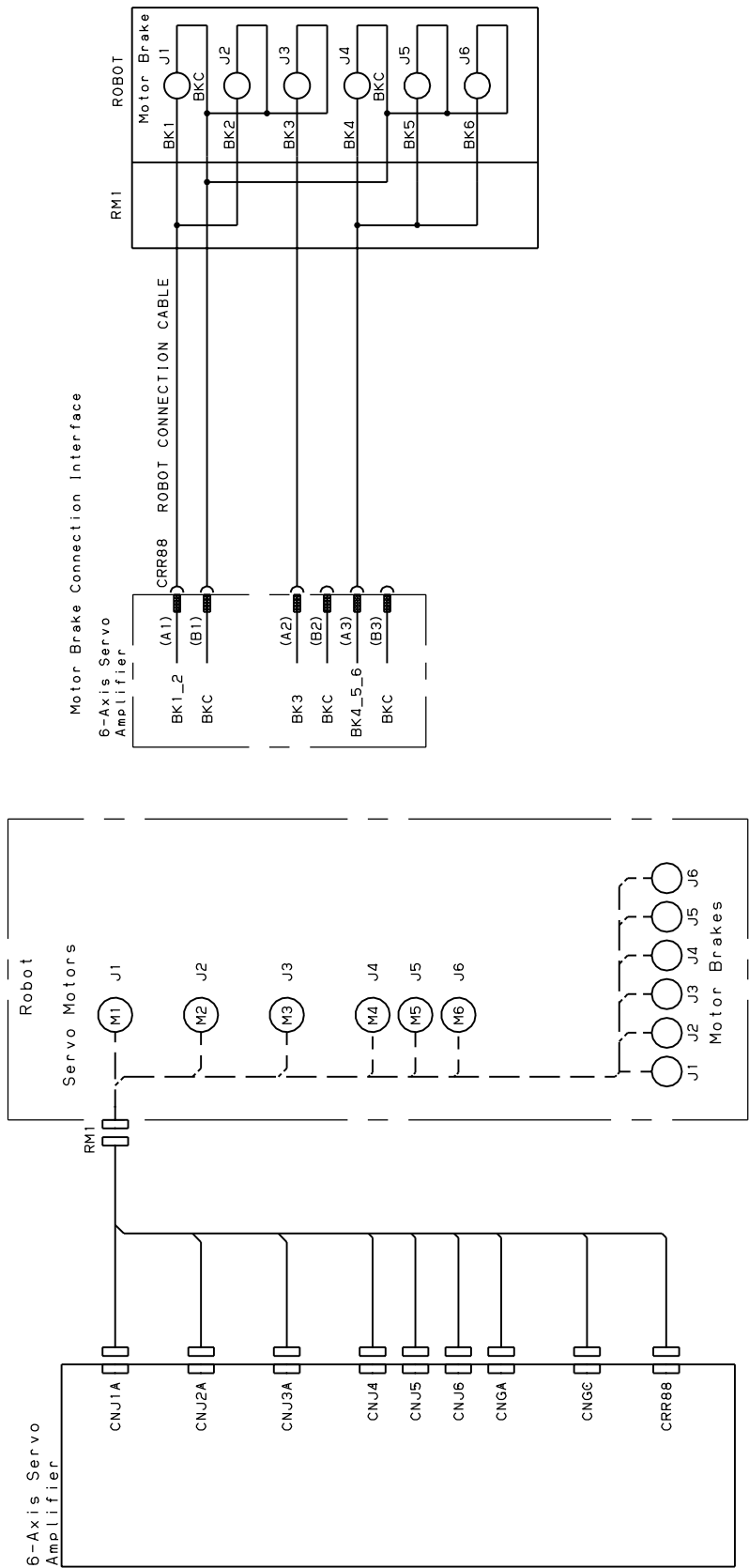


Fig.B(u) Motor power connection
R-2000iC, R-1000iA, M-710iC, M-2iA, M-3iA

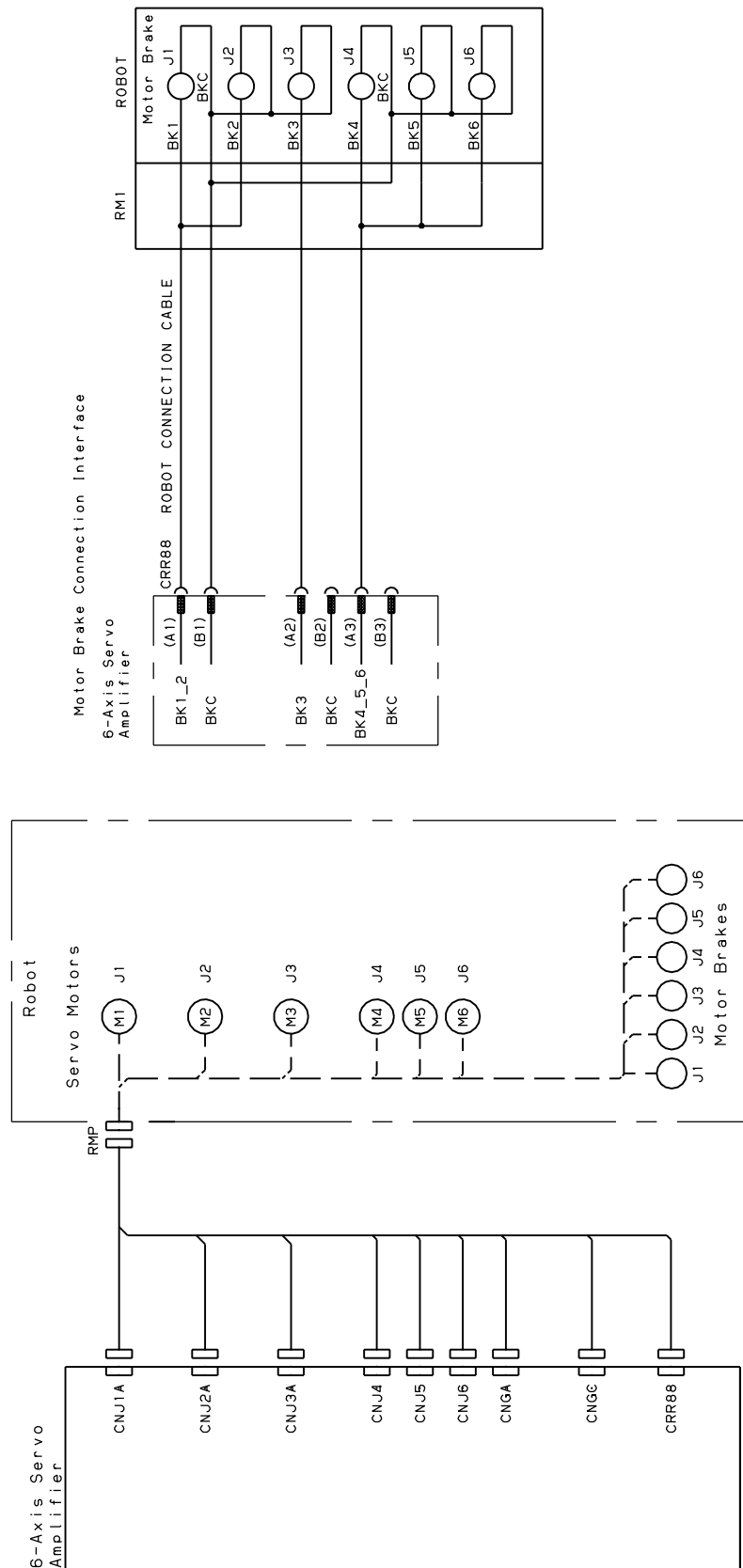
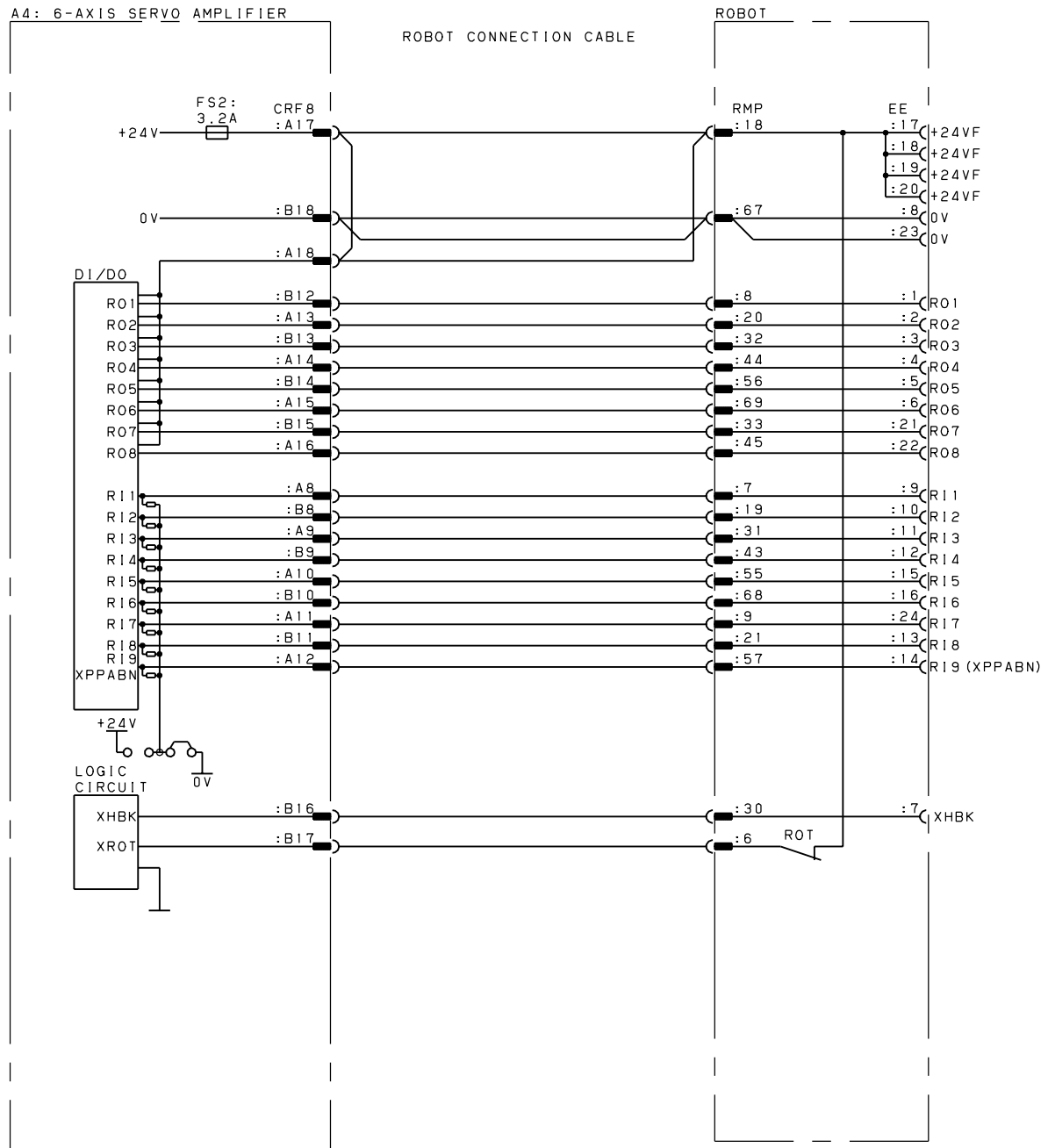


Fig.B(v) Motor power connection

ARC Mate 100iC, ARC Mate 100iD, ARC Mate 120iC, M-10iA, M-10iD, M-20iA, M-20iB, ARC Mate 0iB, R-0iB



RI/RO

Fig.B(w) RI/RO connection diagram
 (LR Mate 200iD, ARC Mate 50iD, ARC Mate 100iC, ARC Mate 100iD, ARC Mate 120iC,
 M-10iA, M-10iD, M-20iA, M-20iB, ARC Mate 0iB, R-0iB, CR-4iA, CR-7iA)
 (There are many type EE connector of mechanical unit. The detail is shown on the mechanical unit manual.)

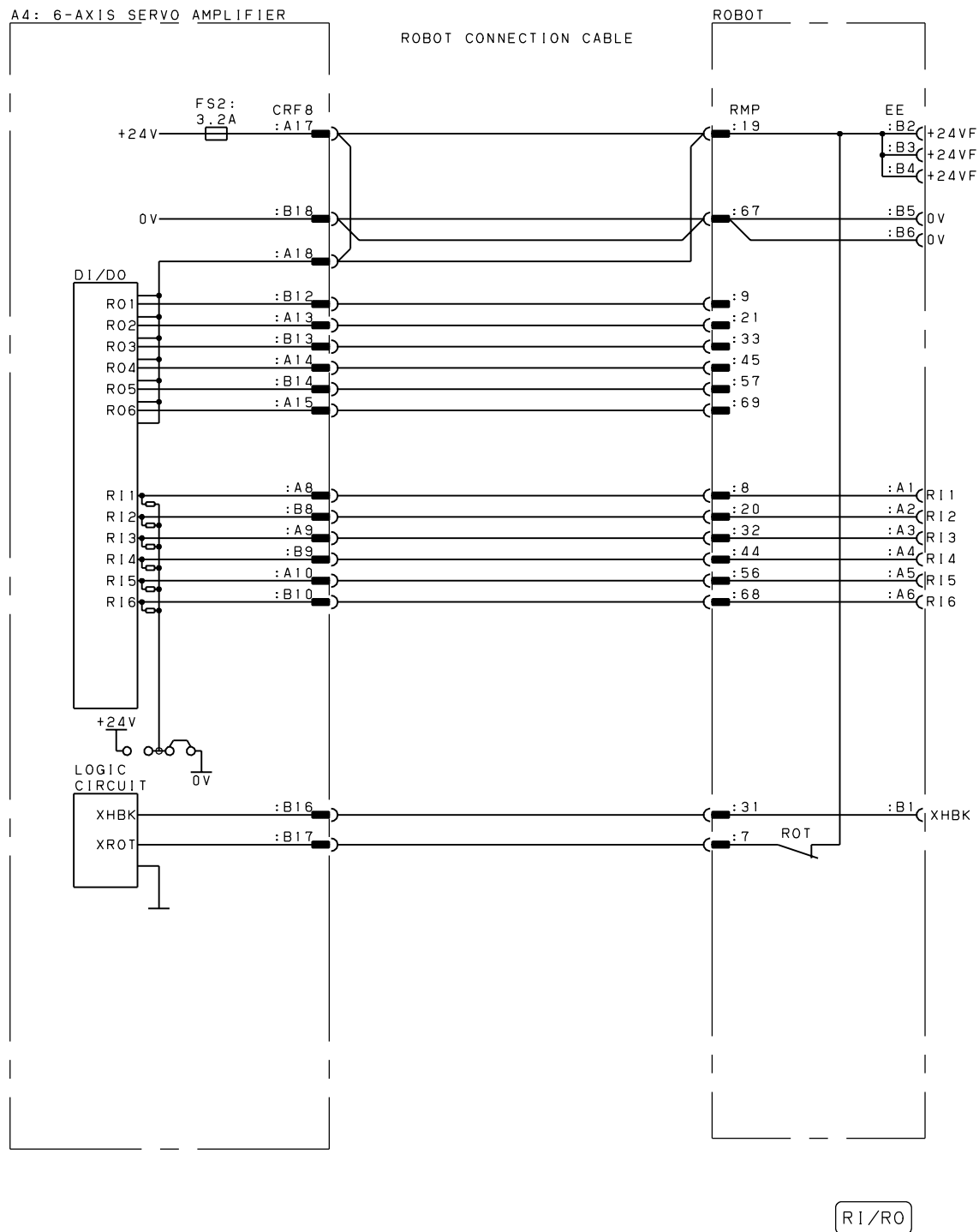
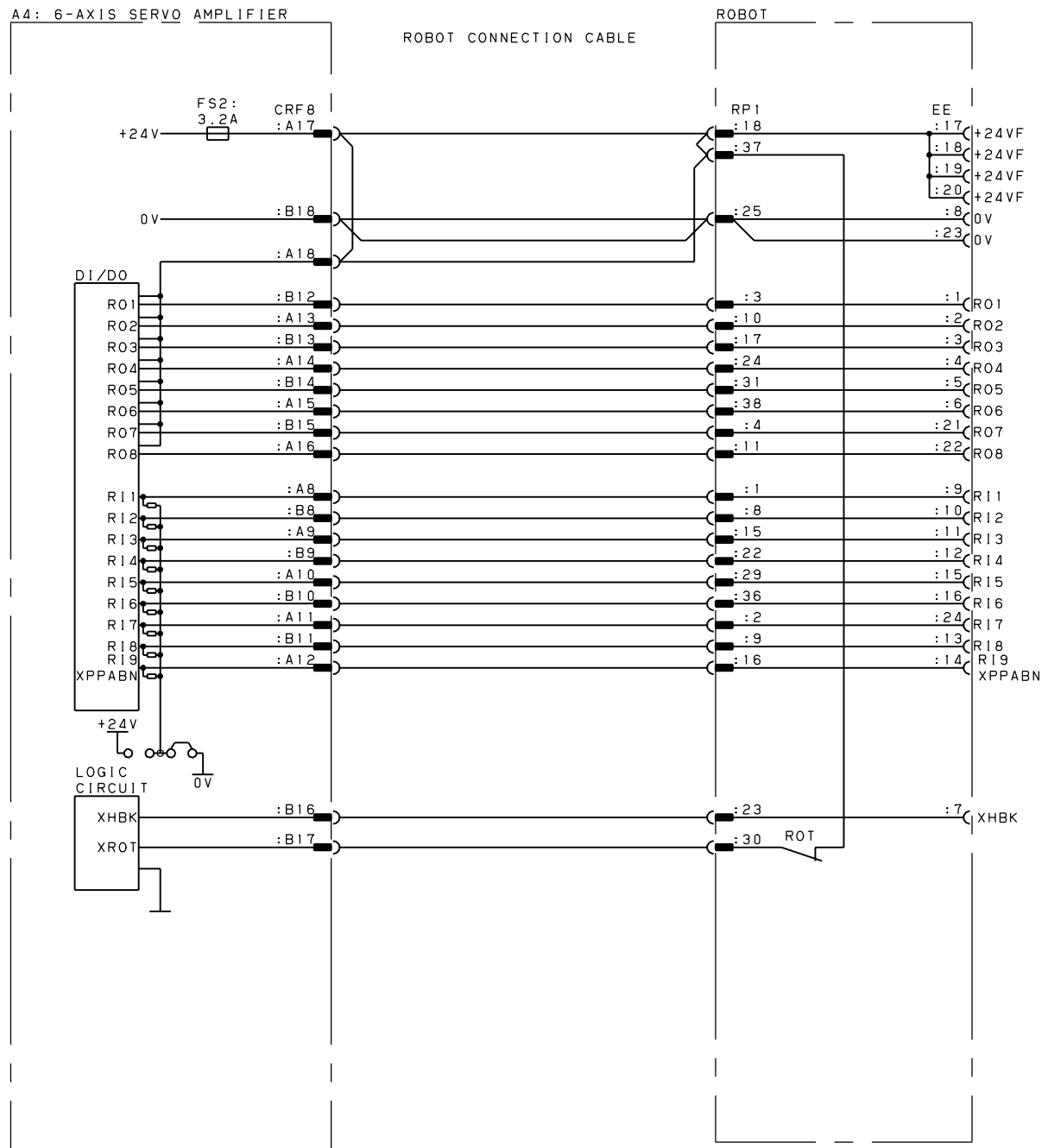


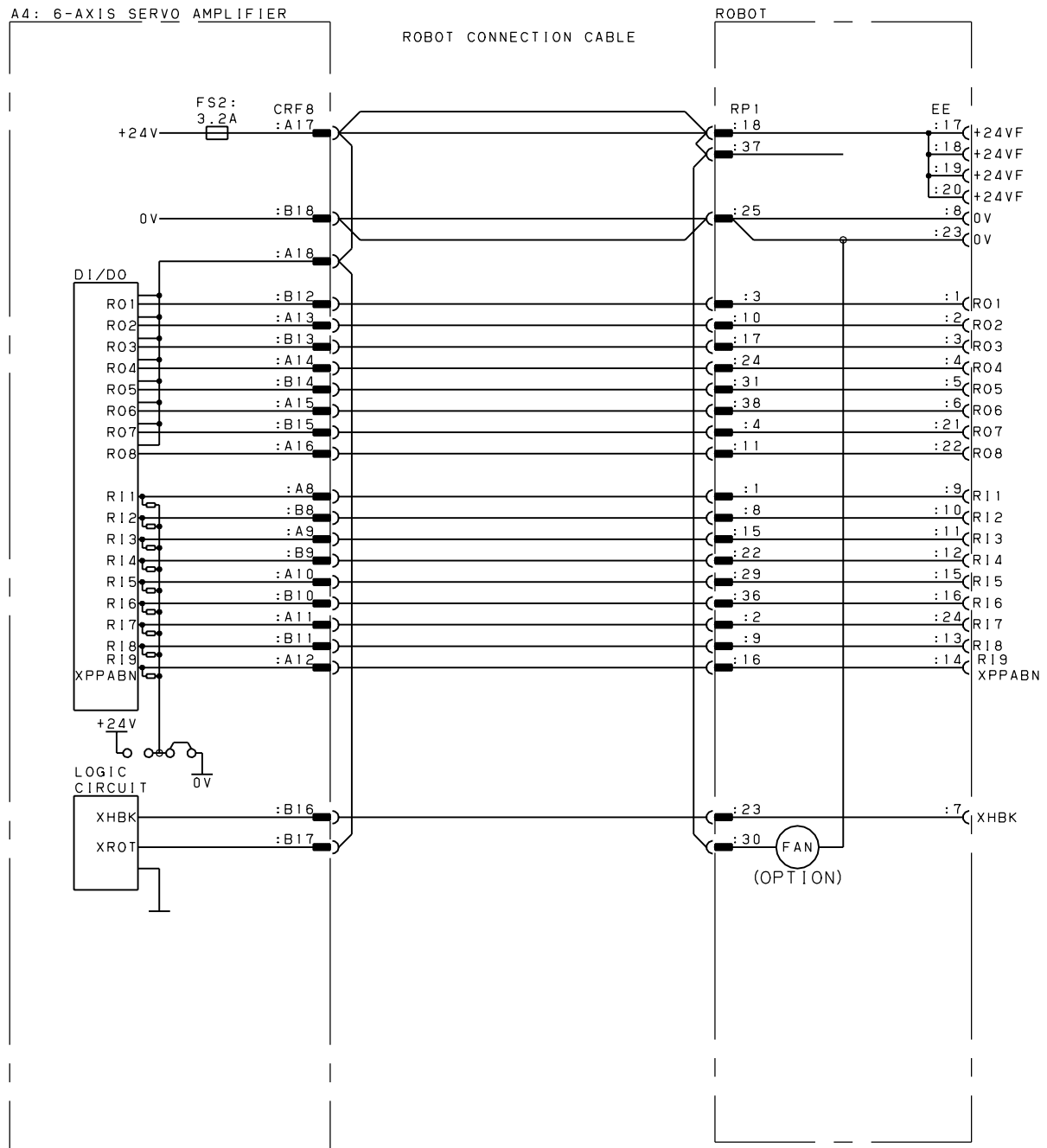
Fig.B(x) RI/RO connection diagram (M-1iA)

(There are many type EE connector of mechanical unit. The detail is shown on the mechanical unit manual.)



RI/RO

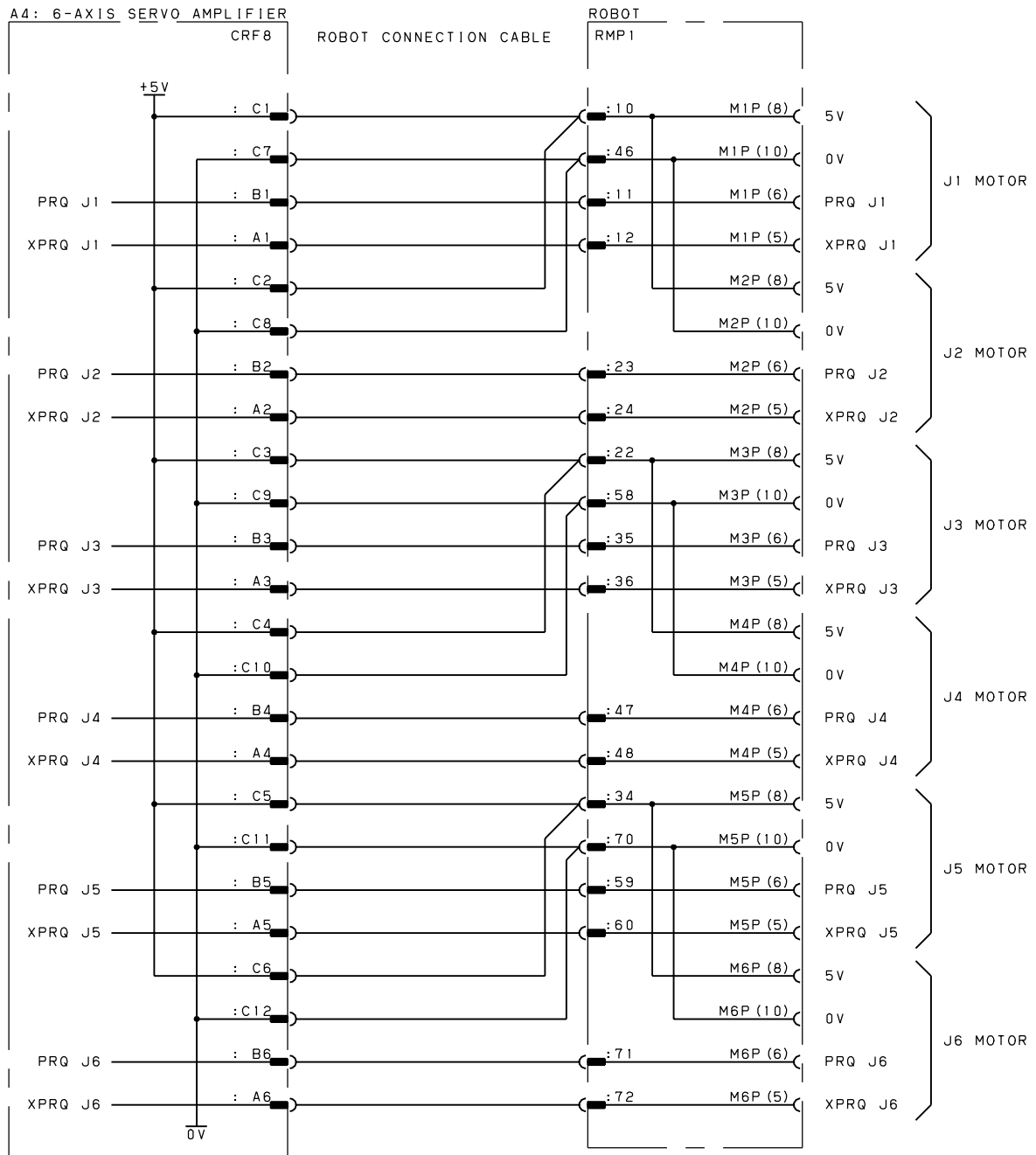
Fig.B(y) RI/RO connection diagram (R-2000iC, R-1000iA, M-710iC)
 (There are many type EE connector of mechanical unit. The detail is shown on the mechanical unit manual.)



RI/RO

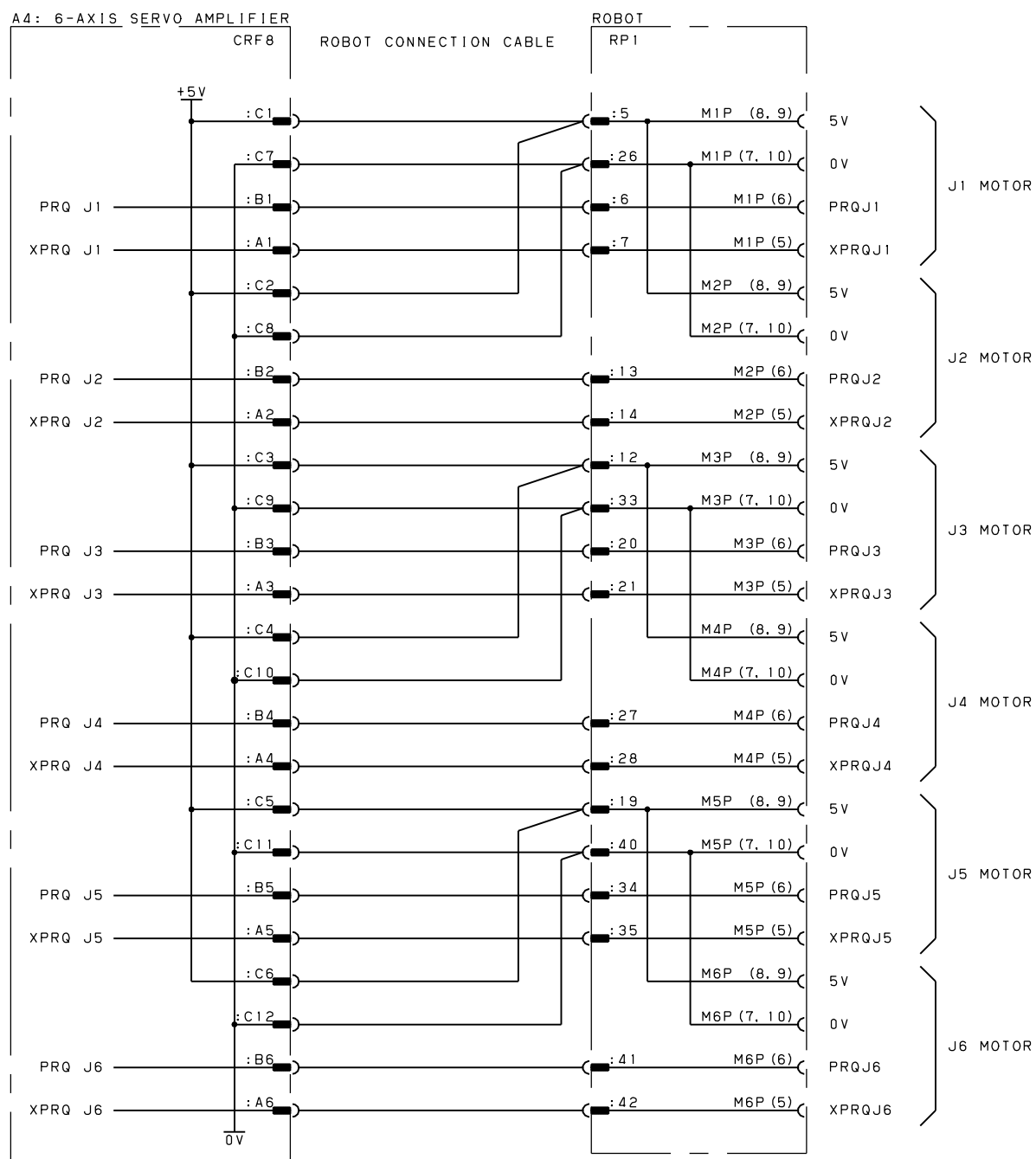
Fig.B(z) RI/RO connection diagram (M-2iA, M-3iA)

(There are many type EE connector of mechanical unit. The detail is shown on the mechanical unit manual.)



SERVO MOTOR F/B

Fig.B(aa) Pulsecoder signal connection diagram
 (LR Mate 200iD, ARC Mate 50iD, ARC Mate 100iC, ARC Mate 100iD, ARC Mate 120iC,
 M-10iA, M-10iD, M-20iA, M-20iB, ARC Mate 0iB, R-0iB, CR-4iA, CR-7iA)



SERVO MOTOR F/B

Fig.B(ab) Pulse encoder signal connection diagram
(R-2000iC, R-1000iA, M-710iC, M-2iA, M-3iA)

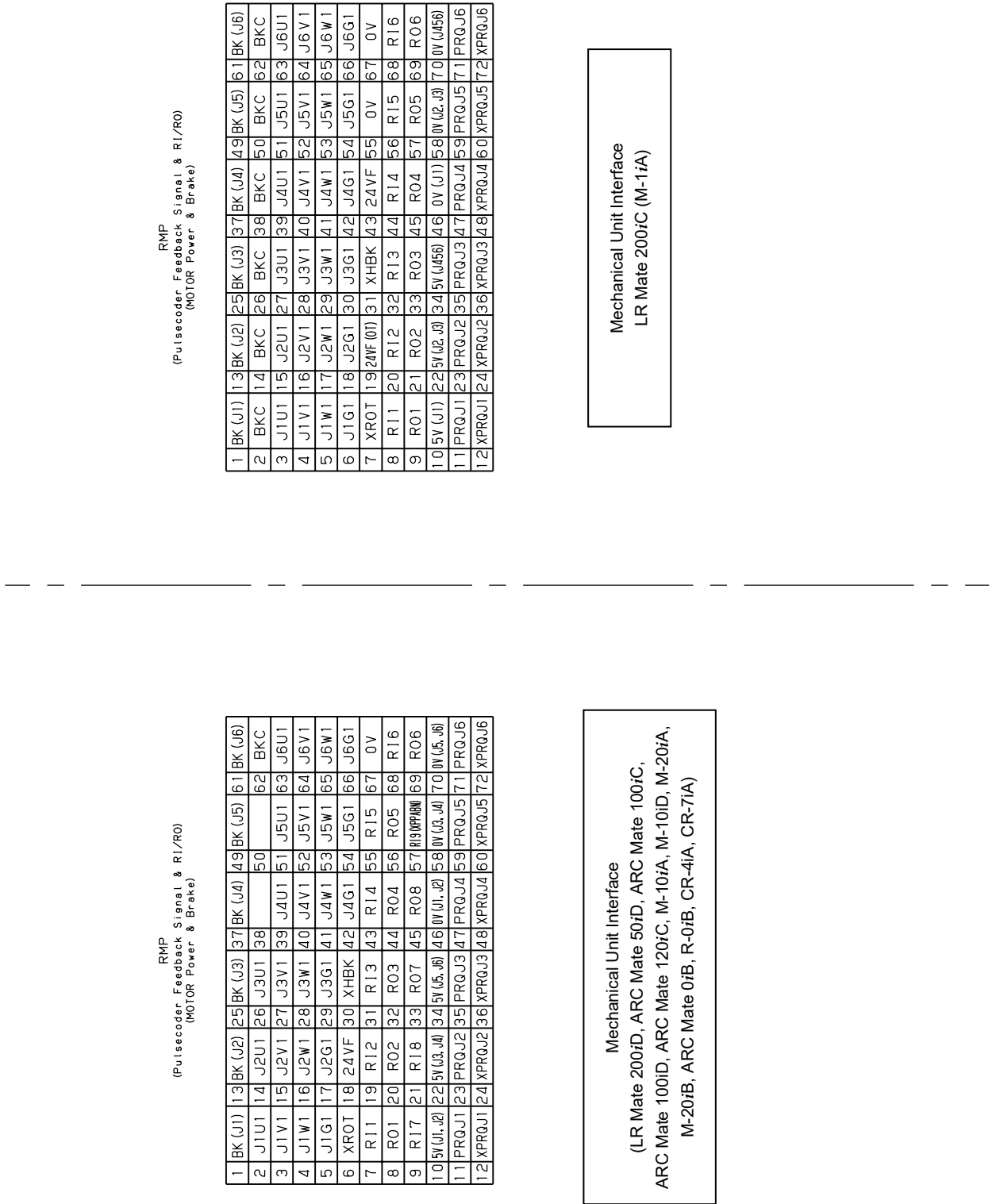


Fig.B(ac) Mechanical unit interface

RP1 (Pulse/Encoder Feedback Signal & RI/RO)															
1	R11	8	R12	15	R13	22	R14	29	R15	36	R16				
2	R17	9	R18	16	R19	23	XHBK	30	24VF (FAN)	37	24VF-3				
3	R01	10	R02	17	R03	24	R04	31	R05	38	R06				
4	R07	11	R08	18	24VF-1 24VF-2	25	0V-1 0V-2	32	S2+	39	S2-				
5	5V-1 5V-2	12	5V-3 5V-4	19	5V-5 5V-6	26	0V-3 0V-4	33	0V-5 0V-6	40	0V-7 0V-8				
6	PRQJ1	13	PRQJ2	20	PRQJ3	27	PRQJ4	34	PRQJ5	41	PRQJ6				
7	XPRQJ1	14	XPRQJ2	21	XPRQJ3	28	XPRQJ4	35	XPRQJ5	42	XPRQJ6				

RM1 (MOTOR Power & Brake)															
1		14	J1V1	24	J1W1	34									
2	J1U1	15	J1V1	25	J1W1	35	J1G1								
3	J1U1	16	J2V1	26	J2W1	36	J1G1								
4	J2U1	17	J2V1	27	J2W1	37	J2G1								
5	J2U1	18	J3V1	28	J3W1	38	J2G1								
6	J3U1	19	J3V1	29	J3W1	39	J3G1								
7	J3U1	20	J4V1	30	J4W1	40	J3G1								
8	J4U1	21	J5V1	31	J5W1	41	J4G1								
9	J5U1	22	J6V1	32	J6W1	42	J5G1								
10	BK1,2 (J1)	23	BK1,2 (J2)	33	BK3	43	J6G1								
11	BK1,2 (J1)	24	BK1,2 (J2)	34	BK3	44	BK4,5,6 (J4)								
12	BK1,2 (J1)	25	BK1,2 (J2)	35	BK3	45	BK4,5,6 (J5)								
13	BK1,2 (J1)	26	BK1,2 (J2)	36	BK3	46	BK4,5,6 (J6)								

Mechanical Unit Interface
M-2 iA, M-3 iA

RP1 (Pulse/Encoder Feedback Signal & RI/RO)															
1	R11	8	R12	15	R13	22	R14	29	R15	36	R16				
2	R17	9	R18	16	R19	23	XHBK	30	XR0T	37	24VF-3				
3	R01	10	R02	17	R03	24	R04	31	R05	38	R06				
4	R07	11	R08	18	24VF-1 24VF-2	25	0V-1 0V-2	32	S2+	39	S2-				
5	5V-1 5V-2	12	5V-3 5V-4	19	5V-5 5V-6	26	0V-3 0V-4	33	0V-5 0V-6	40	0V-7 0V-8				
6	PRQJ1	13	PRQJ2	20	PRQJ3	27	PRQJ4	34	PRQJ5	41	PRQJ6				
7	XPRQJ1	14	XPRQJ2	21	XPRQJ3	28	XPRQJ4	35	XPRQJ5	42	XPRQJ6				

RM1 (MOTOR Power & Brake)															
1		14	J1V1	24	J1W1	34									
2	J1U1	15	J1V1	25	J1W1	35	J1G1								
3	J1U1	16	J2V1	26	J2W1	36	J1G1								
4	J2U1	17	J2V1	27	J2W1	37	J2G1								
5	J2U1	18	J3V1	28	J3W1	38	J2G1								
6	J3U1	19	J3V1	29	J3W1	39	J3G1								
7	J3U1	20	J4V1	30	J4W1	40	J3G1								
8	J4U1	21	J5V1	31	J5W1	41	J4G1								
9	J5U1	22	J6V1	32	J6W1	42	J5G1								
10	J6U1	23	BK2	33	BK3	43	J6G1								
11	BK1	24	BK2	34	BK3	44	BK4								
12	BK1	25	BK2	35	BK3	45	BK5								
13	BK1	26	BK2	36	BK3	46	BK6								

Mechanical Unit Interface
(R-2000iC, R-1000iA, M-710iC)

Fig.B(ad) Mechanical unit interface

C

SPECIFICATIONS OF PERIPHERAL DEVICE INTERFACE

C.1 SIGNAL

The following table lists the I/O signals used for the peripheral device interface in the R-30iB Mate/R-30iB Mate Plus controller.

Input signals (Refer to C.3.1)	
Signal	Description
*IMSTP	Instantaneous stop signal
*HOLD	Hold signal
*SFSPD	Safety speed signal
CSTOPI	Cycle stop signal
FAULT_RESET	Alarm release signal
START	Cycle start signal
HOME	Return to home position
ENBL	Enabling signal
RSR1/PNS1	Robot service request/program number select signal (*1)
RSR2/PNS2	Robot service request/program number select signal (*1)
RSR3/PNS3	Robot service request/program number select signal (*1)
RSR4/PNS4	Robot service request/program number select signal (*1)
RSR5/PNS5	Robot service request/program number select signal (*1)
RSR6/PNS6	Robot service request/program number select signal (*1)
RSR7/PNS7	Robot service request/program number select signal (*1)
RSR8/PNS8	Robot service request/program number select signal (*1)
PNSTROBE	PNS strobe signal
PROD_START	Automatic operation start signal
DI01	General-purpose input signal
DI02	General-purpose input signal
DI03	General-purpose input signal
DI04	General-purpose input signal
DI05	General-purpose input signal
DI06	General-purpose input signal
DI07	General-purpose input signal
DI08	General-purpose input signal
DI09	General-purpose input signal
DI10	General-purpose input signal
DI11	General-purpose input signal
DI12	General-purpose input signal
DI13	General-purpose input signal
DI14	General-purpose input signal
DI15	General-purpose input signal
DI16	General-purpose input signal
DI17	General-purpose input signal
DI18	General-purpose input signal
DI19	General-purpose input signal
DI20	General-purpose input signal
DI21	General-purpose input signal
DI22	General-purpose input signal

NOTE

*1: RSR: Robot Service Request (RSR5 to RSR8 are optional)
 PNS: Program Number Select Input (optional)
 Whether RSR is used or PNS is used can be preset.

Output signals (Refer to C.3.2)

Signal	Description
CMDENBL	Command acceptance enabled signal
SYSRDY	System ready signal
PROGRUN	Program run signal
PAUSED	Program paused signal
HELD	Held signal
FAULT	Alarm signal
ATPERCH	Reference point signal
TPENBL	Teach pendant enabled signal
BATALM	Battery alarm signal
BUSY	Operating signal
ACK1/SNO1	RSR acknowledge/Selected program number signal
ACK2/SNO2	RSR acknowledge/Selected program number signal
ACK3/SNO3	RSR acknowledge/Selected program number signal
ACK4/SNO4	RSR acknowledge/Selected program number signal
ACK5/SNO5	RSR acknowledge/Selected program number signal
ACK6/SNO6	RSR acknowledge/Selected program number signal
ACK7/SNO7	RSR acknowledge/Selected program number signal
ACK8/SNO8	RSR acknowledge/Selected program number signal
SNACK	PNS acknowledge signal
_____	Not used (for future expansion)
DO01	General-purpose output signal
DO02	General-purpose output signal
DO03	General-purpose output signal
DO04	General-purpose output signal
DO05	General-purpose output signal
DO06	General-purpose output signal
DO07	General-purpose output signal
DO08	General-purpose output signal
DO09	General-purpose output signal
DO10	General-purpose output signal
DO11	General-purpose output signal
DO12	General-purpose output signal
DO13	General-purpose output signal
DO14	General-purpose output signal
DO15	General-purpose output signal
DO16	General-purpose output signal
DO17	General-purpose output signal
DO18	General-purpose output signal
DO19	General-purpose output signal
DO20	General-purpose output signal

C.2 SETTING COMMON VOLTAGE

All process I/O boards have a jumper to set the common voltage of input signals to 0 V or 24 V. The system automatically adjusts the polarity by software according to the status of this pin. Therefore, you can operate the system without being concerned about the setting of the common voltage.

To ensure safety, the common reference voltage of the following four signals, is remains at +24V.

*IMSTP
*HOLD
*SFSPD
CSTOPI

C.3 I/O SIGNALS

C.3.1 Input Signals

This section describes the specifications of each input signal.

- (1) Instantaneous stop signal (input) *IMSTP
Effective: At any time
Function: Use the normally closed switch because it is a reverse signal. The system turns off power to the servo unit when the *IMSTP is open (turned off). Do not use *IMSTP as safety relevant signal. For safety purpose, use the external emergency stop signal.
- (2) Alarm release signal (input) FAULT RESET
Effective: In the alarm status
Function: The FAULT RESET signal releases the alarm status. If the servo unit has been turned off, it also turns on the unit. At the same time, the alarm display on the teach pendant (the top line) is cleared.
Description: This signal releases only the alarm status. It does not re-start execution of the program. The robot will keep running if the signal is triggered "ON" during operation.
- (3) Hold signal (input) *HOLD
Effective: At any time
Function: Use the normally-closed switch because it is a reverse signal. The *HOLD signal has the same function as the hold button on the teach pendant. It halts the current program and stops the operation of the robot. While this signal is being input, the held signal (output) HELD is turned on and the robot cannot be operated.
- (4) Start signal (input) START
Effective: When the command acceptance enabled signal (output) CMDENBL is turned on. See the description of CMDENBL in Section C.3.2 (1) for details.
Function: This input signal starts the selected program at the falling edge when the signal is turned off after being turned on. Its function differs according to the setting of parameter \$SHELL_CFG.\$CONT_ONLY.
 - If parameter \$SHELL_CFG.\$CONT_ONLY is set to DISABLED, the START signal starts the program which has been selected from the teach pendant. By default, the program starts from the current cursor position.
 - If parameter \$SHELL_CFG.\$CONT_ONLY is set to ENABLED, the START signal only resumes the execution of the temporarily held program. To execute an inactivated program from the start, input the PROD_START signal.

(5) Cycle stop signal (input) CSTOPI

Effective: At any time

Function:

- If parameter \$SHELL_CFG.\$USE_ABORT is set to DISABLED, the CSTOPI signal releases the program from the wait status caused by an RSR. It does not stop the execution of the current program and allows it to continue processing (by default).
- If parameter \$SHELL_CFG.\$USE_ABORT is set to ENABLED, the CSTOPI signal immediately cancels the execution of the current program. The program returns to the status in which it was before execution, and the information for the subprogram to return to the main program is lost. At the same time, this signal also releases the program from the wait status caused by RSR.

(6) Enabling signal (input) ENBL

Effective: At any time

Function: If the ENBL signal is turned off, the operation of the robot or the activation of a program is inhibited, and the execution of the current program is suspended.

(7) Safety speed signal (input) *SFSPD

Effective: At any time

Function:

- Use the normally-closed switch because it is a reverse signal. Usually this switch should be connected to safety fence. It must be set normally on.
- Since the *SFSPD signal is counted as a remote condition, such input signals as RSR and START to the peripheral device interface cannot take effect unless this signal is turned on.
- If this signal is turned from on to off during robot operation, the execution of the current program is suspended. At the same time, the overriding value is switched to a preset value (parameter \$SCR.\$FENCEOVER.)
- As long as this signal is off, the overriding value cannot be increased beyond the preset value (\$SCR.\$SFJOGOVLIM: For jog, \$SCR.\$SFRUNOVLIM: For test execution.)

(8) Robot service request signal (input) RSR1/RSR2/RSR3/RSR4

Effective: When the command acceptance enabled signal (output) CMDENBL is turned on. See the description of CMDENBL in Section C.3.2 (1) for details.

Function:

- The user can choose between RSR and PNS (optional), although they cannot be used simultaneously.
- Four input signals, RSR1 to RSR4, are used.
- If a signal is input to an RSR input, a specified program is started. The program number can be set by a menu.
- If another program has already started processing, the newly activated program enters the wait status. As soon as the current program terminates, the waiting program starts processing.
- By using an RSR instruction, each RSR in a program can be enabled or disabled.
- A menu is provided to register the program number of a specified program when each RSR is input. (Refer to the application manual for details of the menu).

////////////////////////////////////		
	1/8	
1 Job selection:	RSR	RSR or PNS
2 RSR1 program number:	12	0..9999
3 RSR2 program number:	23	0..9999
4 RSR3 program number:	5	0..9999
5 RSR4 program number:	64	0..9999
6 Base number:	100	0..9999
7 Acknowledge:	Enabled	Enabled or disabled
8 Acknowledge pulse width:	250 msec	0..9999msec
=		
[TYPE]		
+		

- When an RSR is input, the program whose program name consists of the specified program number plus a base value is started. For example, if a signal is input to RSR2 when program number 23 is registered in RSR2, the program to be started is the one with the program name calculated from the expression RSR + (RSR2 program number + base number), i.e., RSR0123.
The base number is stored in parameter \$SHELL_CFG.\$JOB_BASE, and can be changed in a program with a parameter instruction. (For example, \$SHELL_CFG.\$JOB_BASE =100). In this way, the combination of programs which can be started by RSRs can be changed.
- Whether the system should output an acknowledge signal to an RSR can be selected from the menu. If so specified, a pulse is output from the signal corresponding to the RSR, one of signals ACK1 to ACK4, when the input of the RSR is accepted. From the same menu, the width of the pulse can also be specified.
It is possible to accept other RSRs while outputting an acknowledge signal.
- Input of a CSTOPIT signal can clear the program queue waiting for execution after acceptance of RSRs.

(9) PNS/PNSTROBE (input)

Signal name: PNS: Program number select

PNSTROBE: Strobe input for PNS

Effective: When the command acceptance enabled signal (output) CMDENBL is turned on. See the description of CMDENBL in Section C.3.2 (1) for details.

Function:

- The PNS/PNSTROBE signal selects whether the RSR function is used or the PNS function (optional) is used. If the PNS function is enabled, the RSR function cannot be used.
- The eight signals PNS1 to PNS8 are used to specify a program at the instant the strobe signal PNSTROBE rises.
- A menu is provided to specify the information about PNS.

<pre> //////////////////////////////////// 1 Job selection: PNS 1/3 2 Base number: 100 3 Acknowledge pulse width: 250 msec </pre>			<pre> RSR or PNS 0..9999 0..9999msec </pre>
<pre> = [TYPE] = </pre>			

If a number other than zero is entered to PNS input, a program is selected whose program number is the entered value plus the base number. For example, if the PNS value is 23, the program to be started has the program name calculated from the expression

PNS + (entered PNS value + base number), i.e., PNS0123.

If zero is entered to PNS input, it is cleared as if no selection has been made.

- A PNS signal, which can only select a program, cannot execute the selected program. The execution of the selected program can only be started after input of automatic operation start signal PROD_START.
- For safety, the selected program cannot be changed from the teach pendant unless PNSTROBE is turned off.
- If a program is selected by PNS, the program number is output to selected program number signal (output) SNO, and a pulse is output to program selection acknowledge signal SNACK. Using these signals, peripheral devices can confirm the correct program has been selected. For the timing of these signals, see the sections describing SNO and SNACK.
- The following operations are effective for the program selected by PNS. You can:
 - Start up a program by input of automatic operation start signal PROD_START
 - Restart the program that has been suspended.
 Inputting the START signal restarts the program selected by PNS when \$SHELL_CFG.\$CONT_ONLY is set to ENABLED.
 - Input of CSTOPI cancels execution of the pro-grams selected by PNS when \$SHELL_CFG.\$USE_ABORT is set to ENABLED.

(10) Automatic operation start signal (input) PROD_START

Effective: When the command acceptance enabled signal (output) CMDENBL is turned on. See the description of CMDENBL in Section C.3.2 (1) for details.

Function: This input signal executes the selected program at the falling edge when the signal is turned off after being turned on.

C.3.2 Output Signals

This section describes the specifications of output signals for the peripheral device interface.

(1) Command acceptance enabled signal (output) CMDENBL

Turned on: When the remote conditions are satisfied and the system is not in the alarm status.

Turned off: When the remote conditions are not satisfied or the system is in the alarm status. The remote conditions are satisfied when all of the following are satisfied.

- The teach pendant is in the DISABLED status.
- The remote/local setting is set to REMOTE.
- Parameter \$RMT_MASTER is set to 0 (external interface).

- Signal *SFSPD is set to on, or in the normal status.
- (2) System ready signal (output) SYSRDY
Turned on: When power is applied to the motor of the robot.
Turned off: When power is not applied to the motor of the robot.
- (3) Program run signal (output) PROGRUN
Turned on: When the program is being executed.
Turned off: When the program is not being executed.
- (4) Held signal (output) HELD
This signal is used to check the status of the hold input.
Turned on: When the hold button on the teach pendant (or input) is being pressed down (or turned on).
Turned off: When the hold button on the teach pendant (or input) is not being pressed down (or is turned off).
- (5) Program paused signal (output) PAUSED
This signal is used together with output signal PROGRUN to determine whether a program can be restarted while it is being held.
Turned on: When a program is held and has not been restarted yet. While this signal is on, the program can be restarted and retains information such as that to return from a subprogram to the main program.
Turned off: When a program is being executed or is ready to start. If signal PROGRUN is on, the program is being executed. If signal PROGRUN is off, the program has not been executed and can be started from this status.
- (6) Alarm status signal (output) FAULT
Turned on: When the system is in the alarm status (or an alarm which can stop a program execution is detected.) The indicator lamp does not go on in warning.
Turned off: When the alarm status is released by an alarm release operation.
- (7) Reference point signal (output) ATPERCH
Turned on: When the robot is in the reference position specified in the parameter. (The reference point No.1 in reference point setup screen.)
Turned off: When the robot is not in the reference position specified in the parameter. (The reference point No.1 in reference point setup screen.) Up to three reference positions can be specified, but this signal is output when the robot is in the first reference position. For the other two reference positions, general-purpose signals can be assigned to output as such. (They can be set from the setup screen.)
- (8) Teach pendant enabled signal (output) TPENBL
Turned on: When the teach pendant is enabled.
Turned off: When the teach pendant is disabled.
- (9) Battery alarm signal (output) BATALM
Turned on: When the voltage of the battery for the CMOS memory backup drops below the reference.
Turned off: When the voltage of the battery for the CMOS memory backup is at the normal level.
- (10) Operating signal (output) BUSY
Turned on: When a program is being executed or is being processed from operation panels such as the teach pendant. (This has the same function as that of the BUSY lamp on the teach pendant.)

Turned off: When a program is not being executed nor is being processed from operation panels such as the teach pendant.

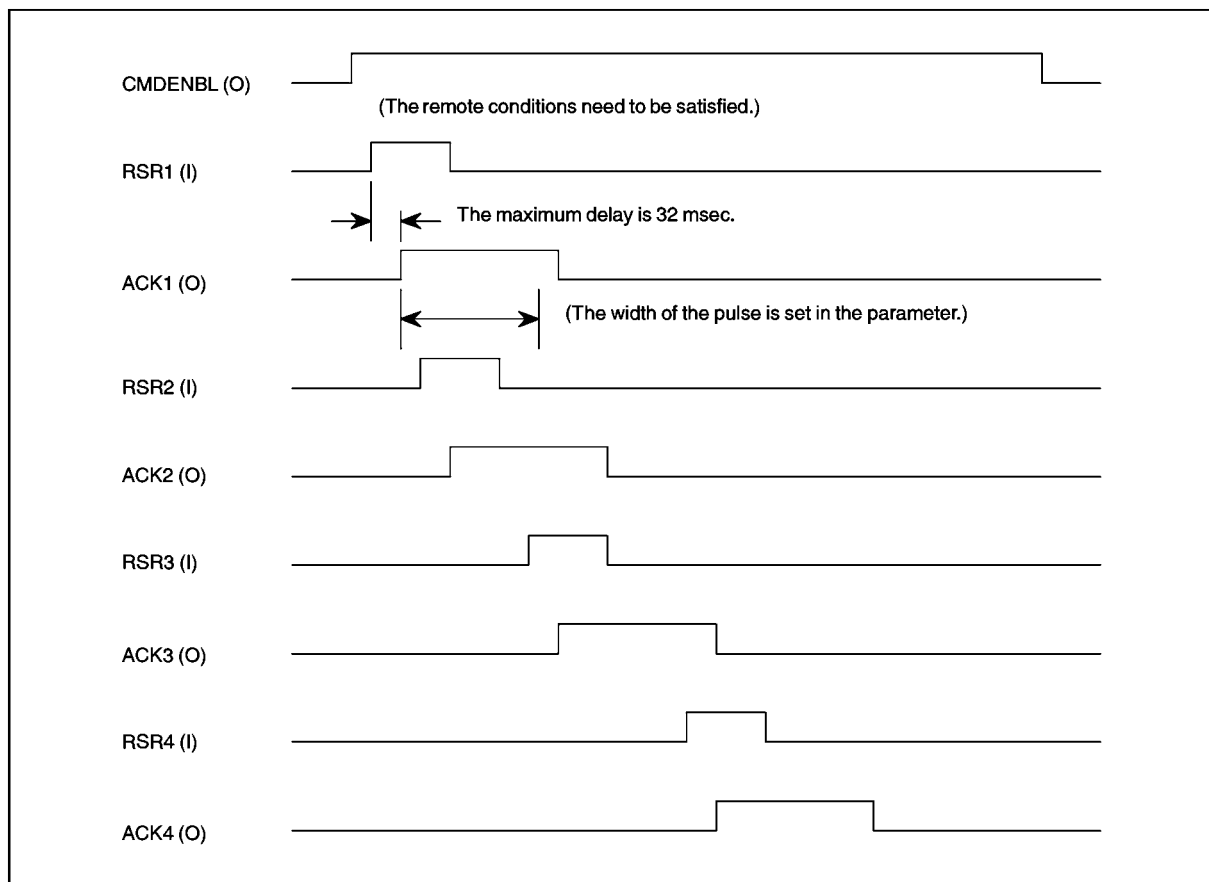
(11) RSR acknowledge signals (output) ACK1/ACK2/ACK3/ACK4

These signals are used together with the RSR function. They can be specified to be enabled or disabled from the RSR setup menu.

Turned on: When one of the signals from RSR1 to RSR4 is input and accepted. A pulse whose width is specified from the menu is output to acknowledge the signal.

Turned off: Normally. Since these signals are always output as pulses, they are normally in the off status.

The following chart shows the timing of the RSR input and ACK output.

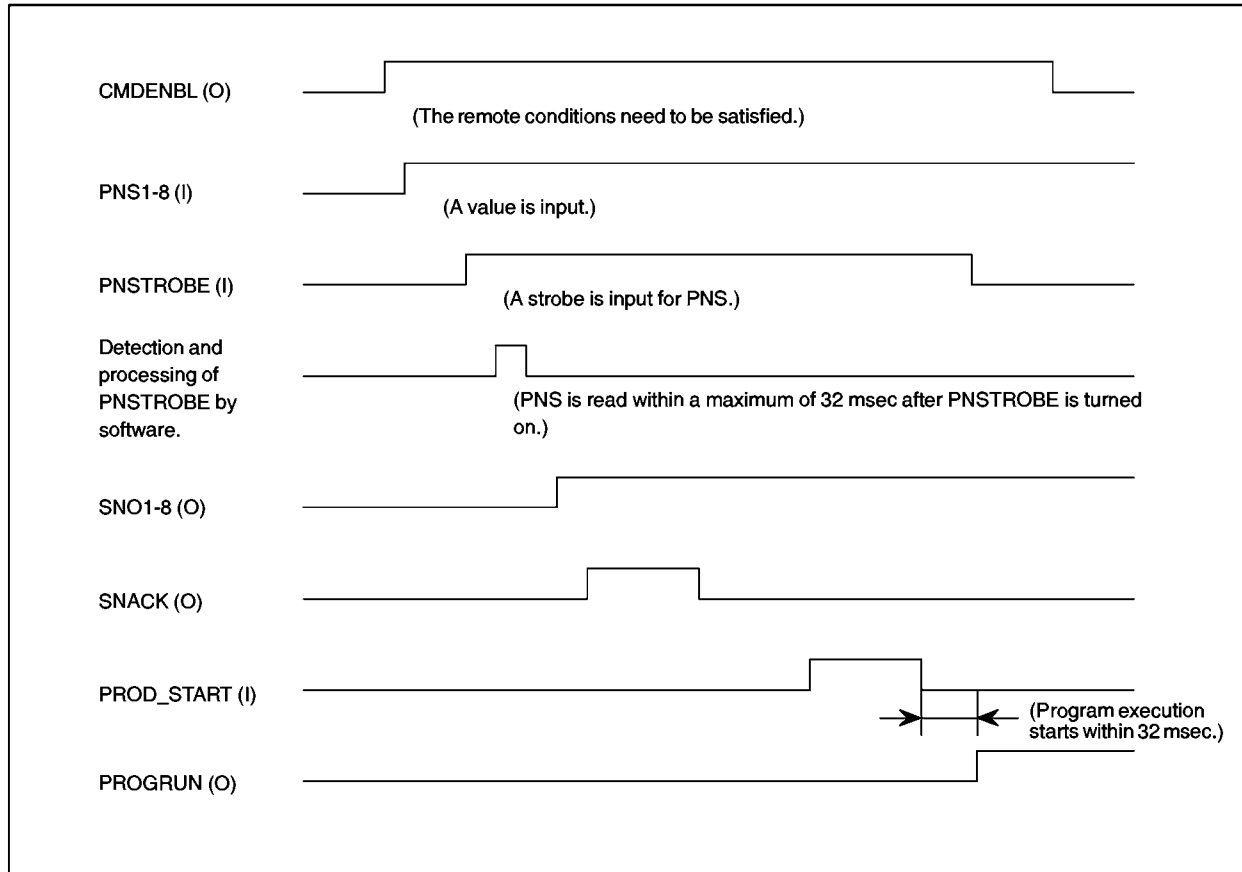


* Other RSR signals can be accepted even when the ACK signal is being output.

(12) PNS acknowledge signal (output) SNO/SNACK

These signals are used together with the PNS function.

Turned on: Whenever the PNS function is enabled. The selected program number is displayed in binary code (SN01 to SN08) on the teach pendant. If the number cannot be represented as an eight-bit number, it becomes zero. After selecting a program by PNS, a pulse is output from signal SNACK as a part of the PNS operation. The width of the pulse can be specified from the menu. See the timing chart below.



C.4 SPECIFICATIONS OF DIGITAL INPUT/OUTPUT

C.4.1 Overview

This section describes the external specifications of digital and analog input/output in the R-30iB controller.

C.4.2 Input/Output Hardware Usable in the R-30iB Mate Controller

The R-30iB Mate/R-30iB Mate Plus controller can use up to 512 digital input and output points or an equivalent number of analog input and output points. One analog input/output point uses the resources equivalent to those used by 16 digital I/O points.

The R-30iB can use a total of up to 512 I/O points.

The R-30iB controller can use the following I/O hardware.

- Process I/O printed board
- I/O unit model A

The process I/O board and the I/O unit model A can be used together.

C.4.3 Software Specifications

(1) RI/RO

These are signals sent to the connector at the wrist of the robot.

They cannot be assigned (redefined) and are fixed.

The standard format is eight inputs and eight outputs. The number of points that can be used for the connector at the wrist depends on the individual robot.

(2) DI/DO

The signal No. that is determined at hardware can be changed by software operation.

(3) Analog I/O

An analog I/O signal can access the analog I/O port (optional) on the process I/O board or the I/O port on the analog I/O module (used together with the I/O unit model A).

It reads and writes the digital value converted from the analog value of the I/O voltage. It means that the value does not always represent the real I/O voltage.

(4) Group I/O

Group I/O is a function, which can input or output multiple DI/DO signals as binary codes. Any number of continuous signals of up to 16 bits can be set for its use. It can be set in the menu DETAILS on the group I/O screen.

D OPTICAL FIBER CABLE

The R-30*i*B Mate/ R-30*i*B Mate Plus uses fiber optic cables for communication between the main board and servo amplifiers. Observe the following cautions when handling these fiber optic cables. Handle fiber optic cables with utmost care, especially when installing the unit.

(1) Protection during storage

When the electrical/optical conversion module (mounted on the printed) circuit board and the fiber optic cable are not in use, their mating surfaces must be protected with the lid and caps with which they are supplied. If left uncovered, the mating surfaces are likely to become dirty, possibly resulting in a poor cable connection.

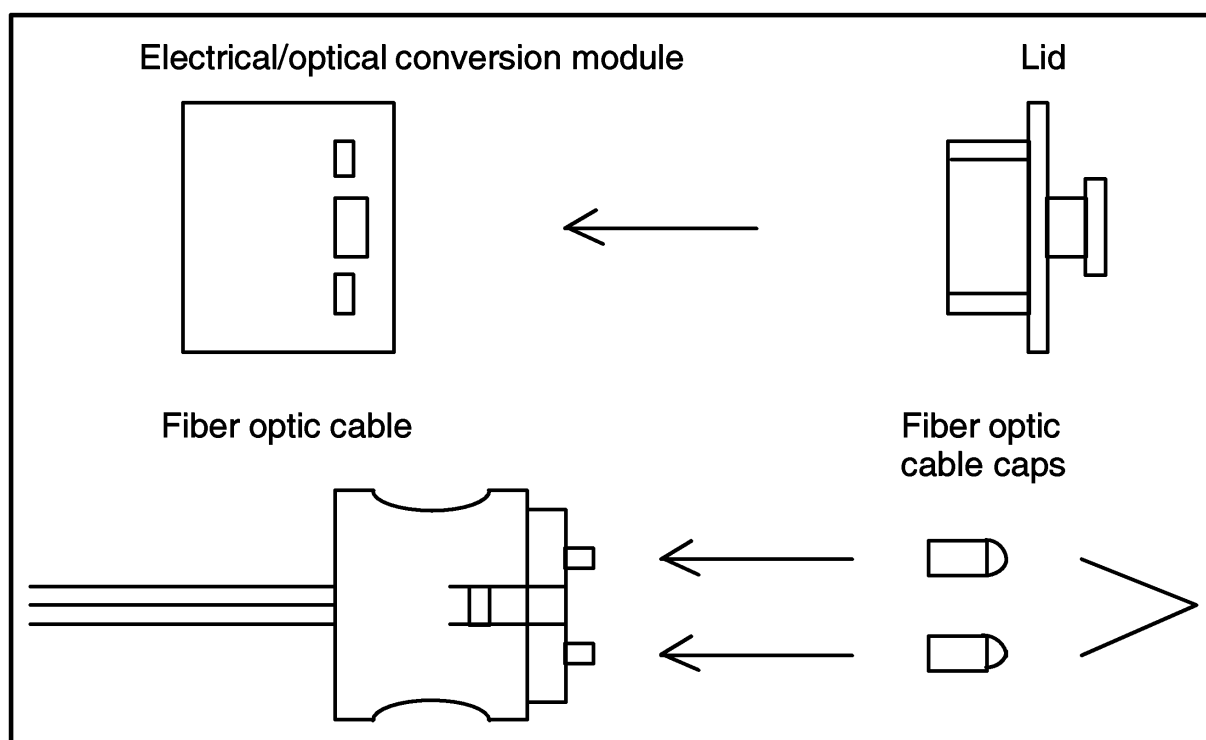


Fig.D (a) Protection of electrical/optical conversion module and fiber optic cable (when not in use)

(2) Fiber optic cable

External type

Fiber optic cord diameter:	ϕ 2.2 mm x 2 cords
Diameter of cable with reinforced cover:	ϕ 7.6 mm
Tensile strength:	
Cable with reinforced cover:	75 kg
Fiber optic cord:	7 kg per cord
Between fiber optic cord and connector:	2 kg
Minimum bending radius of fiber optic cord:	25 mm
Minimum bending radius of cable with reinforced cover:	50 mm
Bending resistance (cable with reinforced cover):	10 million bending cycles at room temperature (when the bending radius is 100 mm)
Flame resistance:	Equivalent to UL VW-1
Operating temperature:	-20 to 70°C

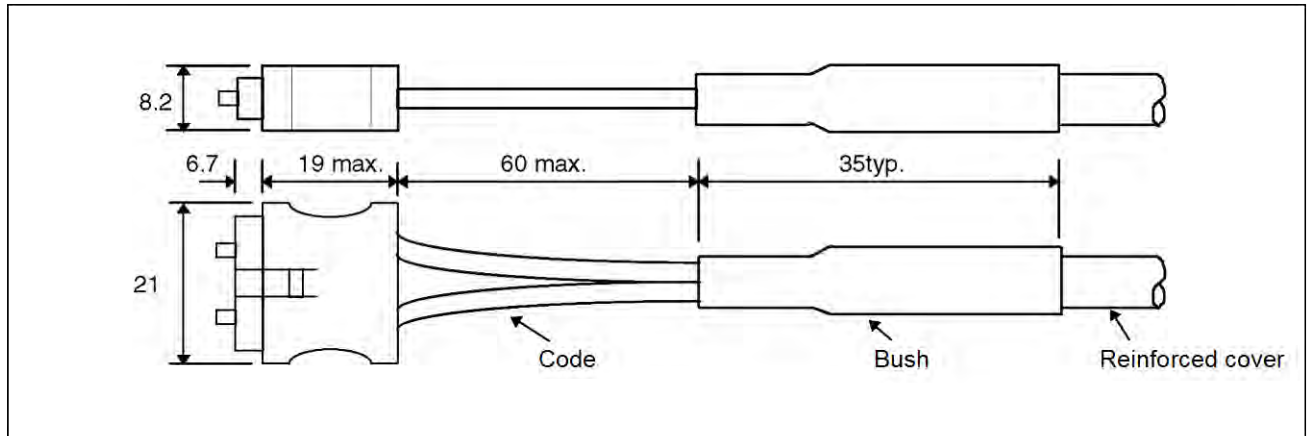


Fig.D (b) External dimensions of external optical cable Unit: mm

Internal type

Fiber optic cord diameter:	ϕ 2.2 mm x 2 cords
Tensile strength:	
Fiber optic cord:	7 kg per cord
Between fiber optic cord and connector:	2 kg
Minimum bending radius of fiber optic cord:	25 mm
Flame resistance:	Equivalent to UL VW-1
Operating temperature:	-20 to 70°C

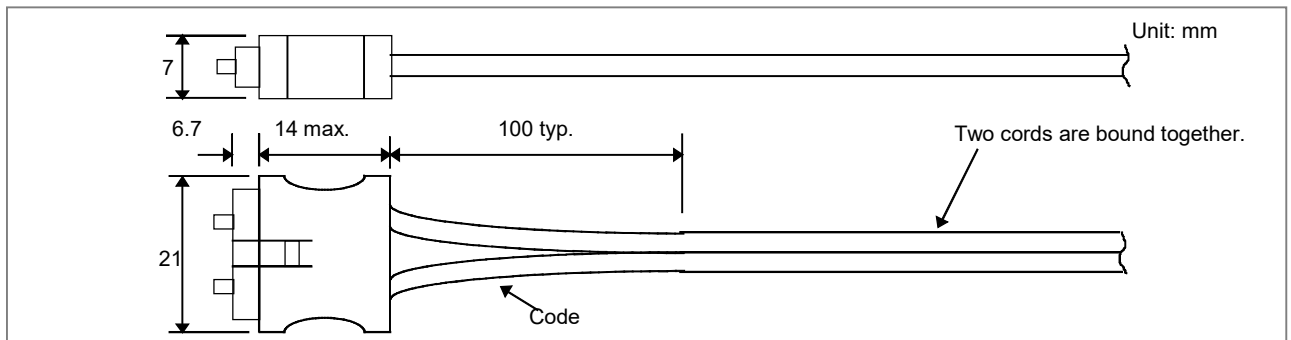


Fig.D (c) External dimensions of internal optical cable Unit: mm

- After it is connected, the optical connector is automatically locked by the lock levers on its top. To remove the connector, release the lock levers and pull the connector. (Do not pull on the fiber optic cord itself.)
- Although optical connectors cannot be connected in other than the correct orientation, always take note of the connector's orientation before making the connection.
- Take care to keep both parts of the optical connector (cable side and PCB side) clean. If they become dirty, wipe them with tissue paper or absorbent cotton to remove dirt. The tissue paper or absorbent cotton may be moistened with ethyl alcohol. Do not use any organic solvent other than ethyl alcohol.
- Fix the reinforcing cover by using a cable clamp, as shown in Fig. D(d), to prevent the weight of the fiber optic cable from being applied directly to the connecting part of the optical connector.
- Although the reinforcing cover of the external optical cable has sufficient mechanical strength, be careful not to drop heavy objects on the cable.

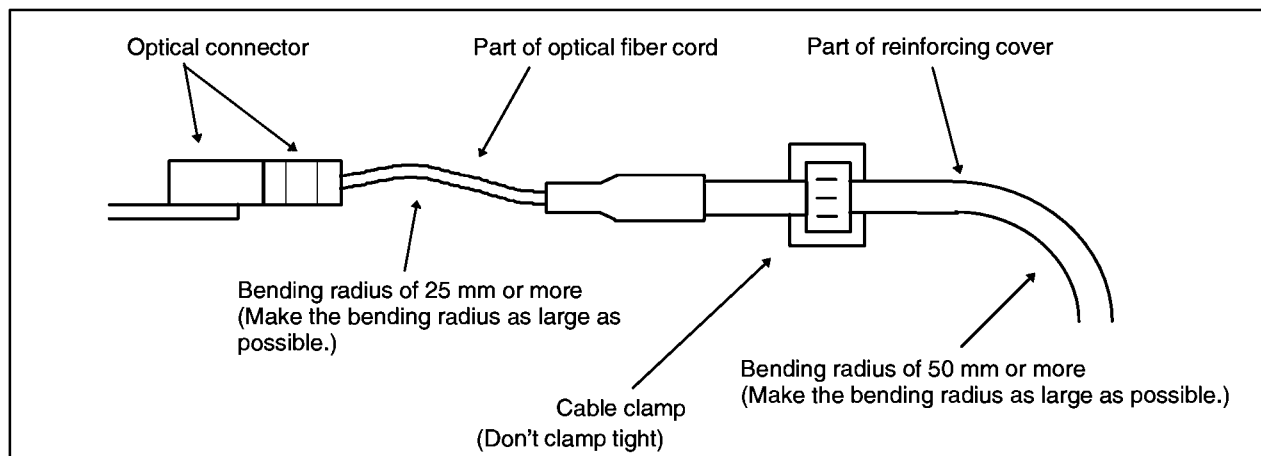


Fig.D (d) Fixing the cable with a clamp

- Any superfluous portion of the cable might be wound into loops. Should this prove necessary, make sure the diameter of each loop is at least 150 mm (for an external cable). Winding the cable into smaller loops can produce sharp curves that exceed the specified bend radius limit. Such bending can result in transmission loss, ultimately leading to a communication failure.
- When using a cable tie as a cable clamp, follow the instructions given below. Also, take care not to apply a bending force to one particular part of the cable when fixing it with a clamp. Failing to clamp the cable correctly might cut or damage it.

External cable:

Do not clamp the uncovered portion of the cable with a cable tie. When clamping the cable by the reinforcing cover, the clamping force is not an important factor to consider. However, ensure that the clamping force is as small as possible to ensure that the reinforcing cover is not deformed by the clamping.

If possible, the clamping force should be 5kg (111bs) or less.

Internal cable:

Lightly clamp the optical cable with a cable tie so that the cable shield is not deformed. Desirable clamping force is 1 to 2 kg (make sure that no force is applied to the cable).

Optical fiber cable for FSSB

Type	Specification	Max. length
Outside of cabinet	A66L-6001-0026#L to	50m (Slave to Slave:40m)
Inside of cabinet	A66L-6001-0023#L to	10m

E BRAKE RELEASE UNIT

E.1 SAFETY PRECAUTIONS



WARNING

- Support the robot arm by mechanical means to prevent it from falling down or rising up when brake is released. Before using the brake release unit, read the Operator's manual of the robot that tries to release the brake.
- Confirm that the robot is fixed tightly to the floor to prevent the falling down and unexpected movement of robot.
- Confirm that the outlet with earth is used for the power supply of brake release unit and earth of brake release unit is surely connected to earth of power supply. There is danger of getting an electric shock if earth is not connected.

E.2 CONFIRMATIONS BEFORE OPERATION

Confirm the followings before operation.

- (1) Confirm the exterior of the brake release unit and the power cable. Do not use it when there are damages in the unit and the cable.
- (2) Confirm that the power supply of the robot controller is disconnected.
- (3) There are Two types of brake release units according to the input voltage as shown in Table E.2 (a). Confirm the input voltage of the unit to refer to the input voltage label put to the unit (Fig.E.4).
- (4) Confirm that the voltage of power supply before connecting the power supply to the brake release unit. There is possibility to give the damaging to the brake or the brake release unit when the incorrect power supply is connected to the unit.

Table E.2 (a) Specification of Brake release unit

Brake release unit	Remarks
Brake release unit (AC 100V)	Input voltage AC100-115V, single phase
Brake release unit (AC 200V)	Input voltage AC200-240V, single phase

- (5) The brake release unit connection cable is different in each robot. Confirm the cable specification corresponding to the robot referring to Table E.2 (b).

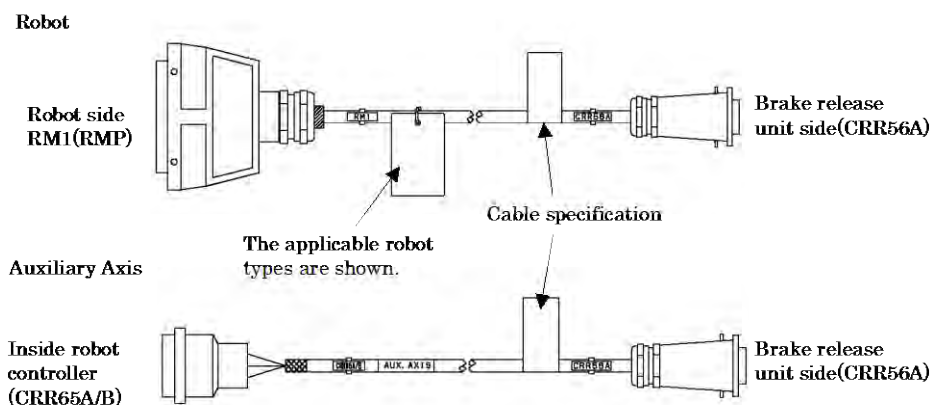


Fig.E.2 Brake release unit connection cable

Table E.2 (b) Specification of brake release unit connection cable

Controller	Applicable robot types	Specification of cable
R-30iB Mate	M-710iC, R-1000iA, R-2000iC, M-2iA, M-3iA,	A660-2005-T559
	M-10iA, M-10iD, M-20iA, M-20iB, ARC Mate 100iC, ARC Mate 100iD, ARC Mate 120iC, ARC Mate 0iB, R-0iB, M-1iA, LR Mate 200iD, ARC Mate 50iD	A660-2006-T881
	Aux. Axis	A660-2005-T711

E.3 OPERATION

E.3.1 In Case of Operating to the Robot

Operate the brake release unit according to the following procedures.

- (1) Support the robot arm by mechanical means to prevent it from falling down or rising up when brake is released. Refer to the Operator's manual for each robot.
- (2) Connect the Brake Release Unit connection cable to Brake Release Unit.
- (3) Disconnect the RM1 (RMP) connector from Robot, and connect the Brake Release Unit connection cable to the Robot. Keep the connection of Robot connection cable except RM1 (RMP) cable.
- (4) Connect the power cable of Brake release unit to power supply.
- (5) Press and hold the deadman switch in the middle position.
- (6) Press the brake switch '1'..'6' according to the axis that tries to release the brake, then brake will be released. (Refer to Table E.3.1) Two axes or more cannot be operated at the same time.

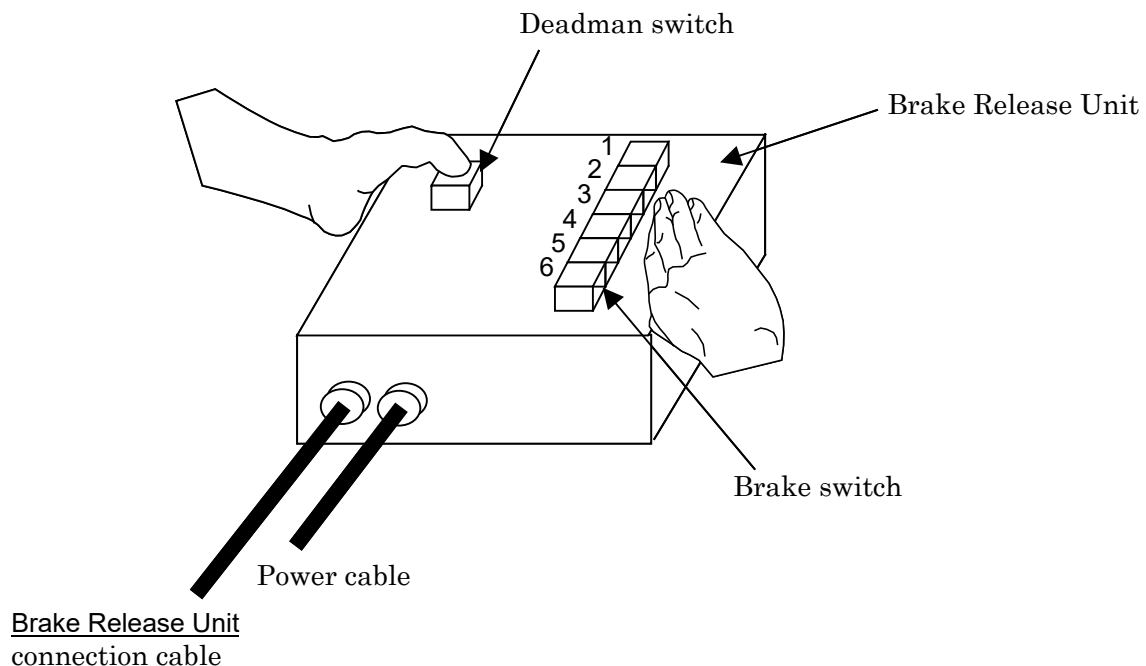


Fig.E.3.1 (a) Brake Release Unit

Table E.3.1 The relation between brake switch and robot axis

Robot Type	Brake Unit Button					
	1	2	3	4	5	6
Robots except below	J1	J2	J3	J4	J5	J6
LR Mate 200iD (5-Axis)	J1	J2	J3	-	J4	J5
In case of the auxiliary Axis	J1	-	-	-	-	-

- Refer to Fig.E.3.2 for the auxiliary axis.

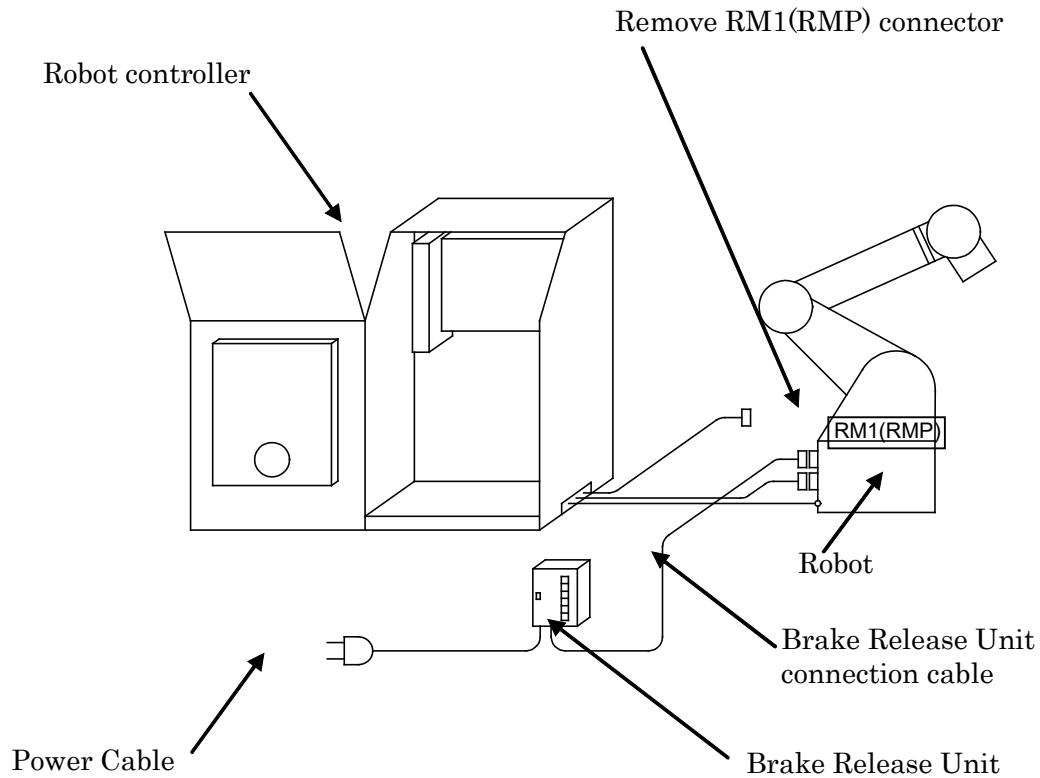


Fig.E.3.1 (b) How to connect Brake Release Unit(In case of operating to the Robot)

E.3.2 In Case of Operating to the Auxiliary Axis

Operate the brake release unit according to the following procedures.

- (1) Support the auxiliary Axis by mechanical means to prevent it from falling down or rising up when the brake is released.
- (2) Connect the Brake Release Unit connection cable to Brake Release Unit.
- (3) Disconnect the aux. axis brake connector (CRR65A/B), and connect the CRR65A/B connector to the Brake Release Unit connection cable. Keep the connection of all cables of aux. axis motor (power, Pulsecoder, brake).
- (4) Connect the power cable of Brake release unit to power supply.
- (5) Press and hold the deadman switch in the middle position.
- (6) Press the brake switch '1', then brake will be released.

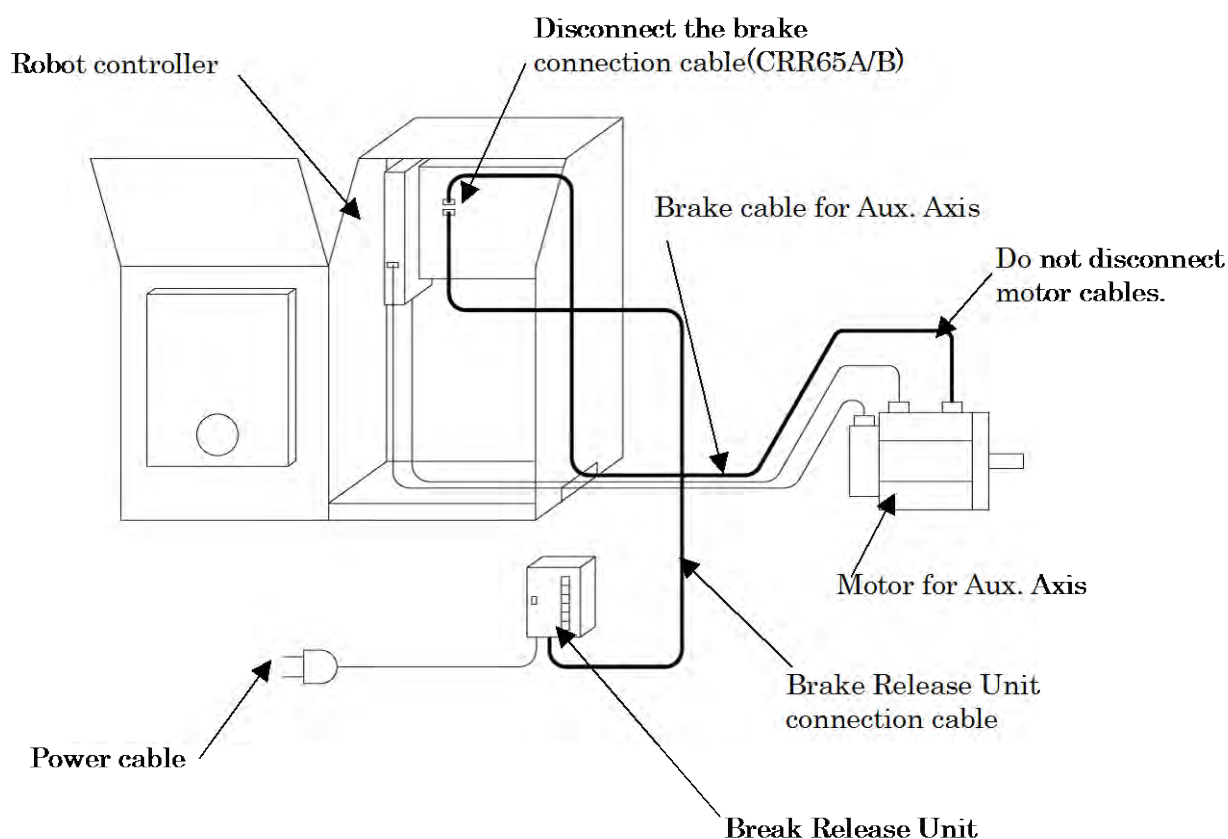


Fig.E.3.2 How to connect Brake Release Unit (In case of operating to the Aux. Axis)

E.4 HOW TO CONNECT THE PLUG TO THE POWER CABLE (IN CASE OF NO POWER PLUG)

Connect the plug to the power cable as follows. This plug is provided by customer.

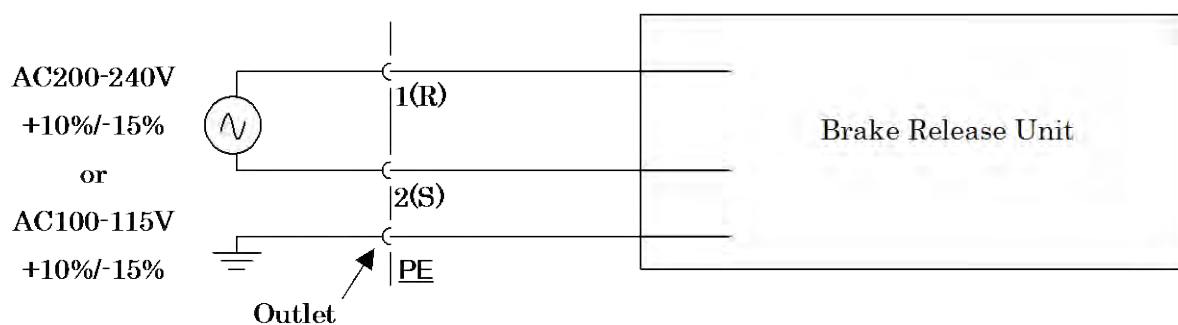
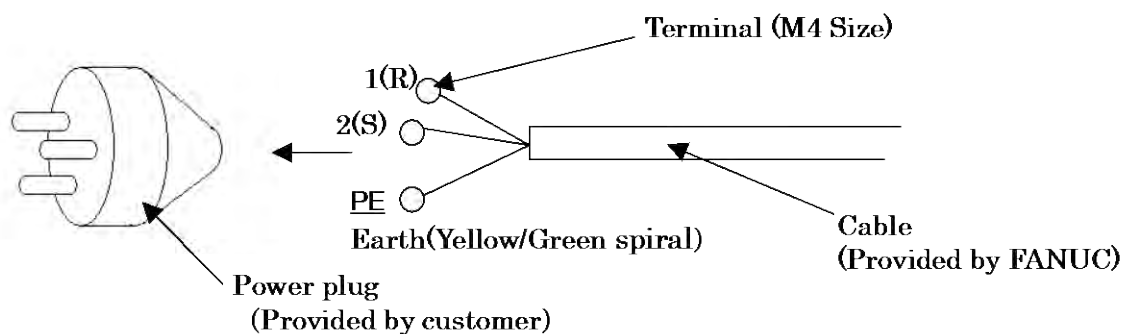


Fig.E.4 How to connect the plug to the power cable

 WARNING

- Only a specialist having the relevant expertise knowledge is permitted to connect the plug to the power cable.
- In the EU area, only plug complying with the relevant European product standard can be used.
- Do not install the plugs without protective earth pin.

E.5 DIMENSION

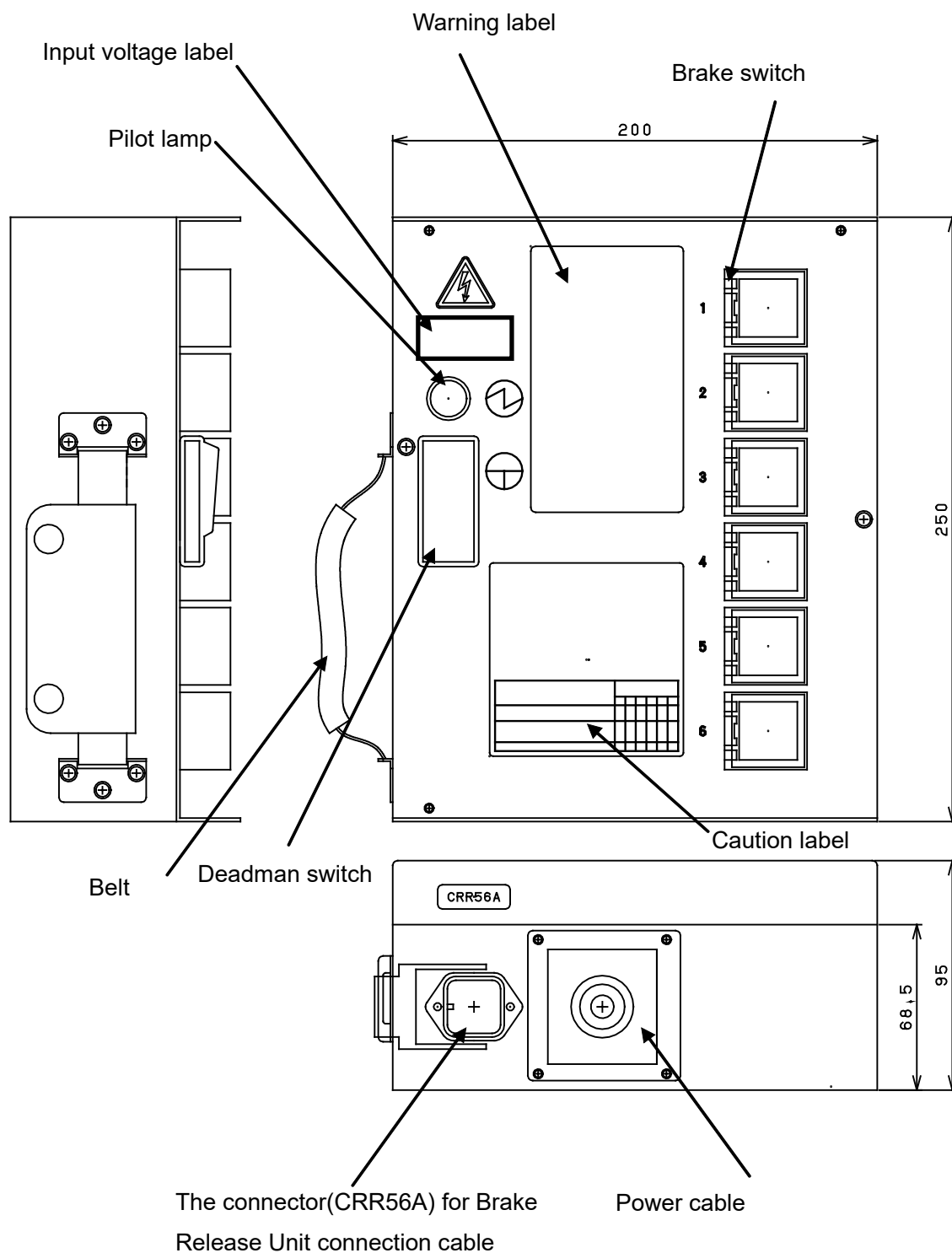


Fig.E.5 (a) Dimension of Brake Release Unit (Front view)

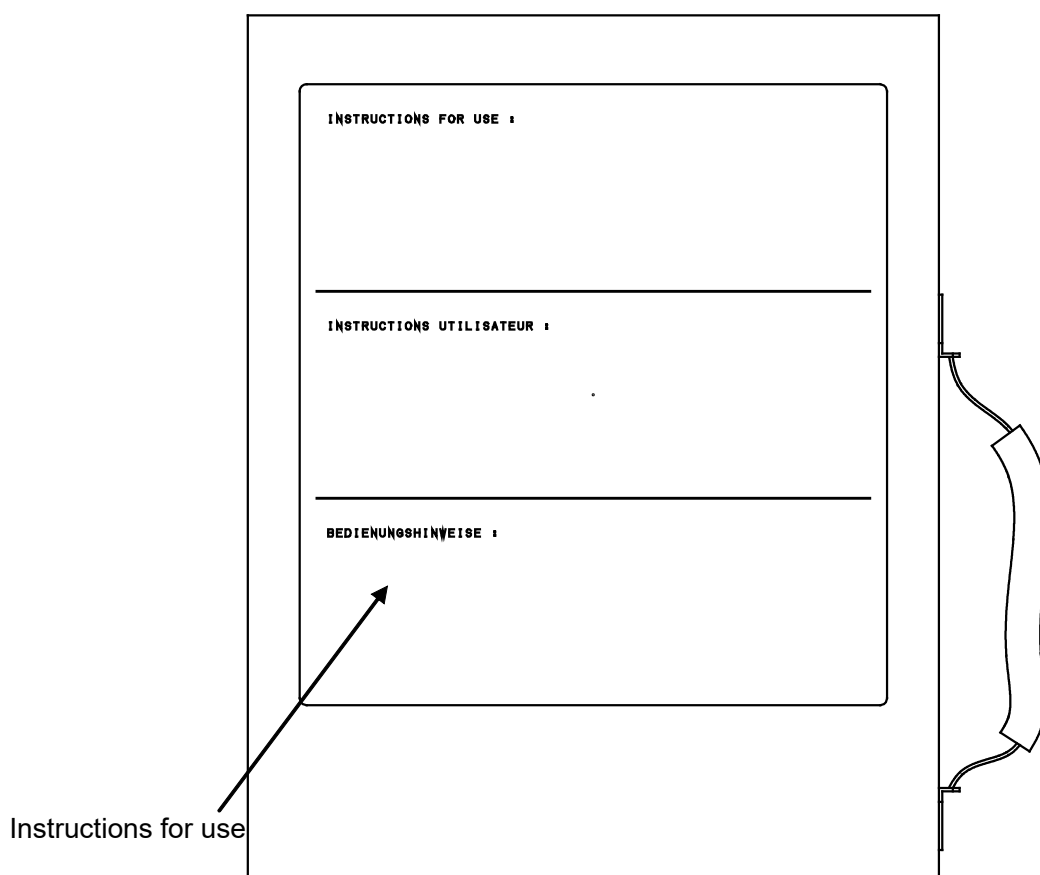


Fig.E.5 (b) Dimension of Brake Release Unit (Rear view)

E.6 FUSE

The fuses are mounted inside this unit. Please check the fuse when the pilot lamp doesn't light even if deadman switch is pressed. When the fuse is blown, exchange the fuse after finding the root cause of failure, and taking the appropriate countermeasures.

Manufacturer: Daito Communication Co.

Specification: P420H

Rating: 2A



WARNING

When the fuse is replaced, the power cable of brake release unit must be disconnected.

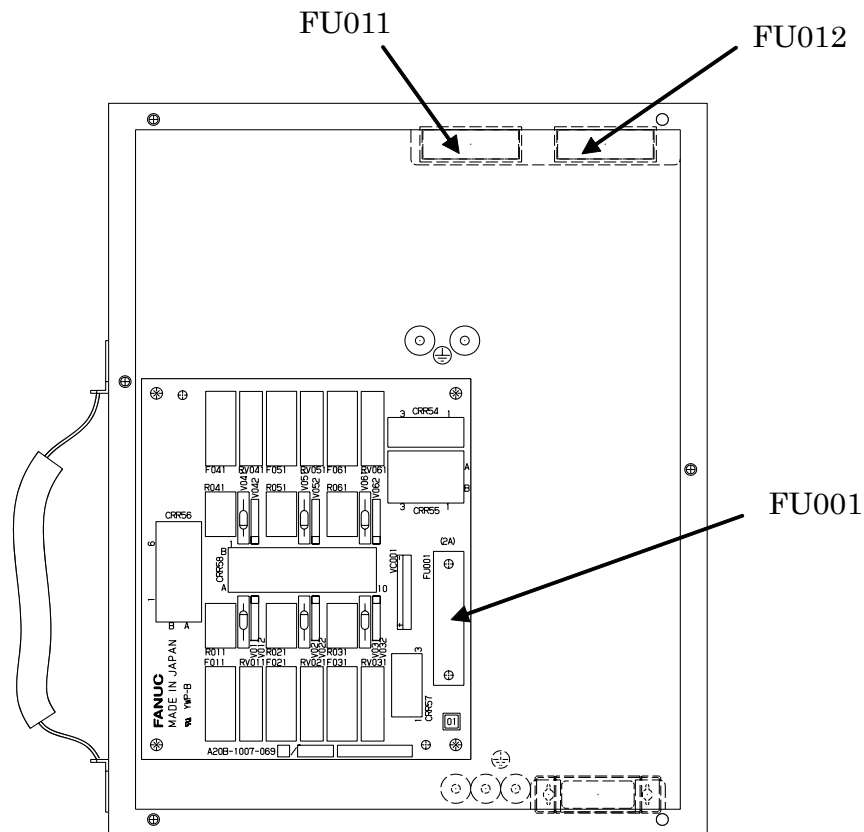


Fig.E.6 The location of fuses

E.7 SPECIFICATIONS

Input power supply

AC100-115V, 50/60Hz $\pm$ 1Hz, single phase, +10%/-15%, 1A

AC200-240V, 50/60Hz $\pm$ 1Hz, single phase, +10%/-15%, 1A

Weight

Brake Release Unit (AC 100V): 2.3 kg

Brake Release Unit (AC 200V): 3.5 kg

F TEACH PENDANT DISCONNECT FUNCTION (OPTION)

This appendix shows an instruction for Teach pendant disconnect function (Option).

F.1 CONFIGURATION

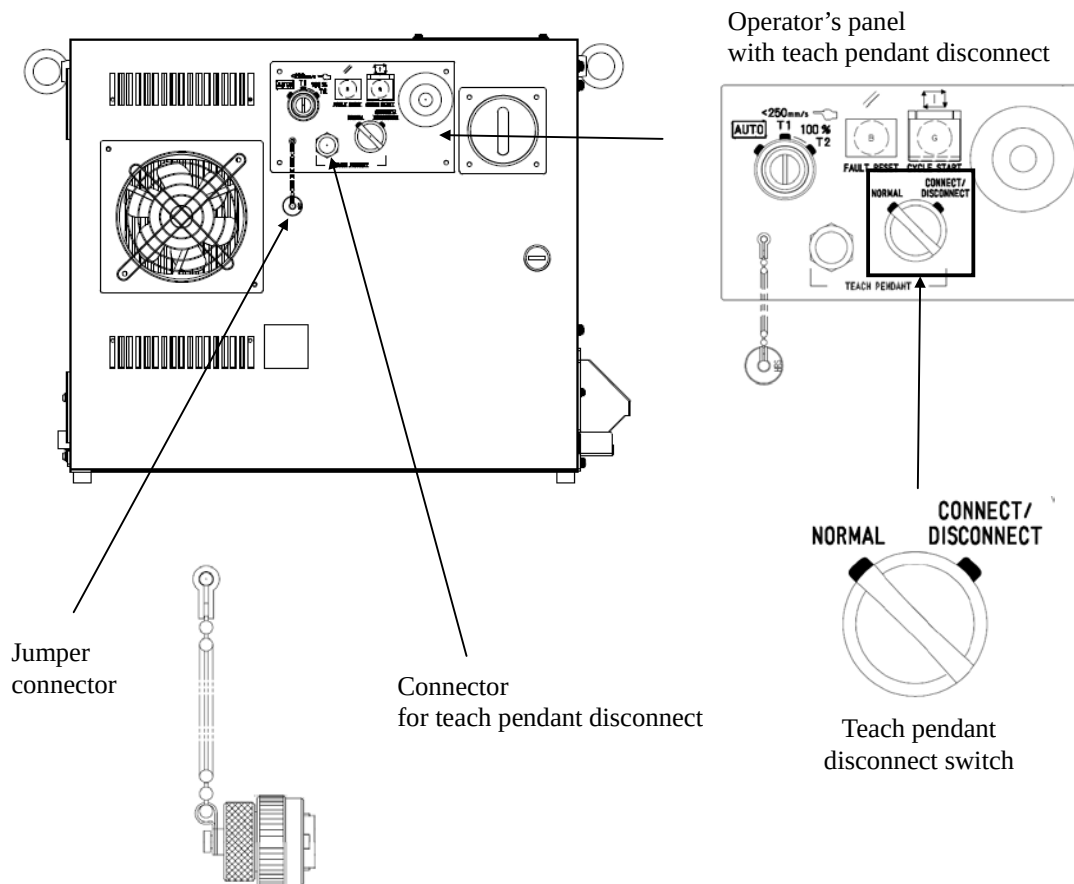


Fig.F.1 Teach pendant disconnect function

F.2 PROCEDURE OF TEACH PENDANT DISCONNECT

F.2.1 Teach Pendant Disconnect

- (1) Set AUTO mode.
- (2) Turn the disconnect switch to "Connect/Disconnect" position. (Robot stops because Operator's panel E-stop Alarm occurs and Power LED of the teach pendant is OFF.)
- (3) Disconnect the teach pendant cable.
- (4) Connect the jumper connector.
- (5) Turn the disconnect switch to "Normal" position.
- (6) Administrator should store the teach pendant and the teach pendant cable in the storage in order to avoid incorrect operation.

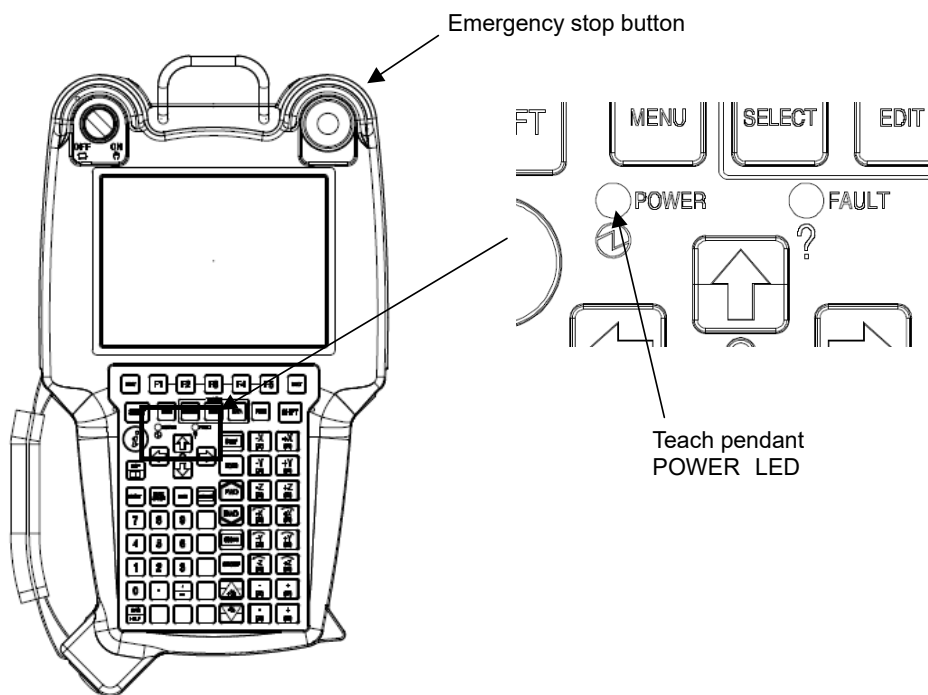
F.2.2 Teach Pendant Connect

- (1) Set AUTO mode.
- (2) Turn the disconnect switch to “Connect/Disconnect” position. (Robot stops because Operator’s panel E-stop Alarm occurs.)
- (3) Disconnect the jumper connector.
- (4) Connect the teach pendant cable with the teach pendant.
- (5) Turn the disconnect switch to “Normal” position.

WARNING

When the LED (POWER) on the teach pendant turned on, this teach pendant is connected to the robot controller and emergency stop button of the teach pendant is active.

When the LED (POWER) on the teach pendant turned off, This teach pendant is not connected to robot controller and emergency stop button of the teach pendant is not active.

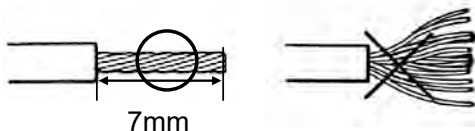


G INSTRUCTION FOR TERMINAL BLOCK

G.1 EXTERNAL EMERGENCY STOP SIGNAL INPUT/OUTPUT TERMINAL BLOCK

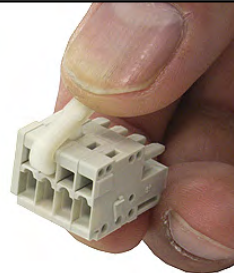
Stripping of Wire

Available wire size AWG 28 -14 (0.08 - 1.5mm²)



- Please check a strip length carefully.
- Please readjust a loose end.

Handling of the lever



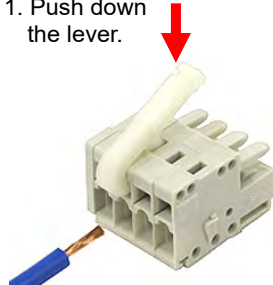
- Hold the connector, and push down the lever by finger.
- Don't handle the lever after fit the connector into PCB, otherwise PCB will be damaged by handling stress.



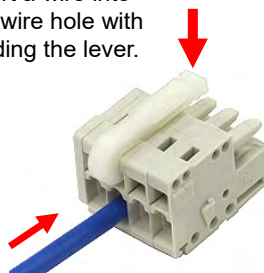
Operating Lever
Item No. 734-230

Wiring

1. Push down the lever.

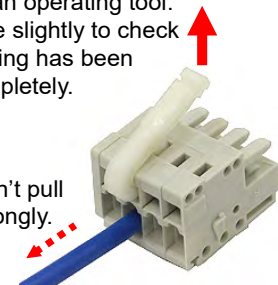


2. Insert a wire into the wire hole with holding the lever.

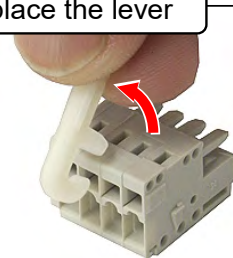


3. Release an operating tool. Pull a wire slightly to check if connecting has been done completely.

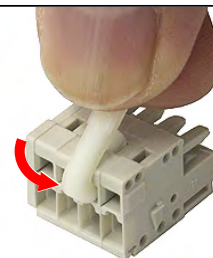
★ Don't pull strongly.



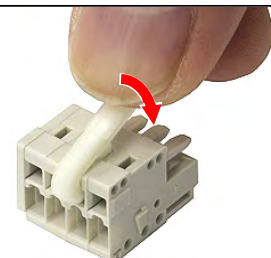
Replace the lever



1. Pull of the lever.



2. Hook the lever to the rectangle hole.



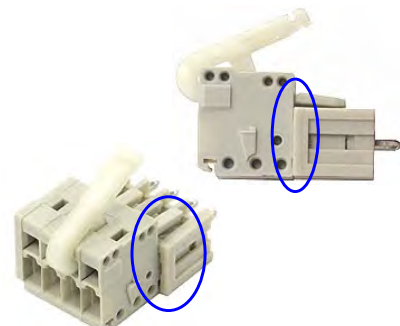
3. Push down the lever until click in.

Fit to header

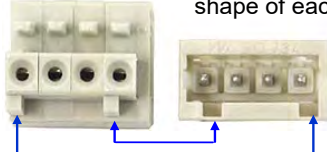
1. Push in the connector to the header.



2. Please check if the latch is hooked to header.



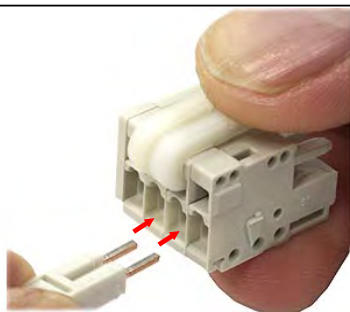
★ Be careful to fit the shape of each other.



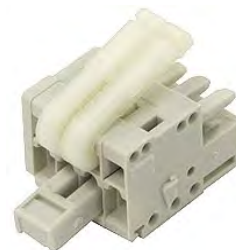
Installation of Jumper



1. Attach two levers to the connector.



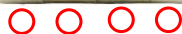
2. Hold down levers at the same time, then put the jumper into the connector.



★ Please confirm that the jumper is fully inserted.

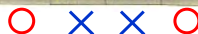
Availability of wires

Without jumpers



Max wire size 1.0mm² (AWG18)
(with Ferrule)

With jumpers

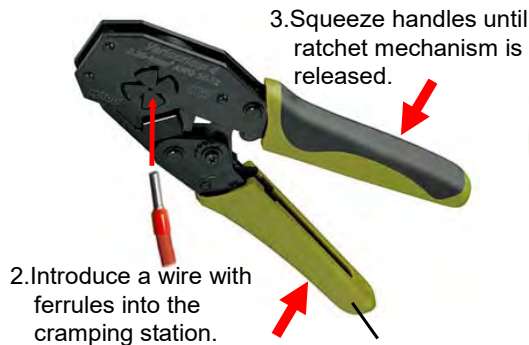


The wire cannot connect,
when attached the jumper.

Installation of Ferrules



1. Put a wire through the hole of ferrules.



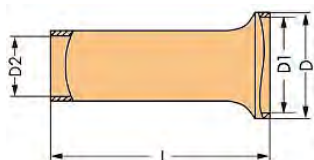
2. Introduce a wire with ferrules into the cramping station.

3. Squeeze handles until ratchet mechanism is released.



4. Please check if the wire crimped correctly.

Specifications of Ferrules



WAGO Item No.	Sleeve for mm ² (AWG)	Color	Stripped length(mm)	L (mm)	L1	D	D1	D2
216-301	0.25 (24)	Yellow	9.5	12.5	8.0	2.5	2.0	0.8
216-302	0.34 (24)	Turquoise	9.5	12.5	8.0	2.5	2.0	0.8
216-201	0.5 (22)	White	9.5	13.5	8.0	3.0	2.5	1.1
216-202	0.75 (20)	Gray	10.0	14.0	8.0	3.3	2.8	1.3
216-203	1.0 (18)	Red	10.0	14.5	8.0	3.6	3.0	1.5

★ **CAUTION!** Please make sure to use WAGO 206-204 to crimp the ferrules.

Pack-unit 100

G.2 TERMINAL CONVERTER BOARD TERMINAL BLOCK

Available wire

Solid conductor	0.2-1.5mm ²
Stranded conductor	0.2-1.5mm ²
Stripping length:	10(±0.5) mm

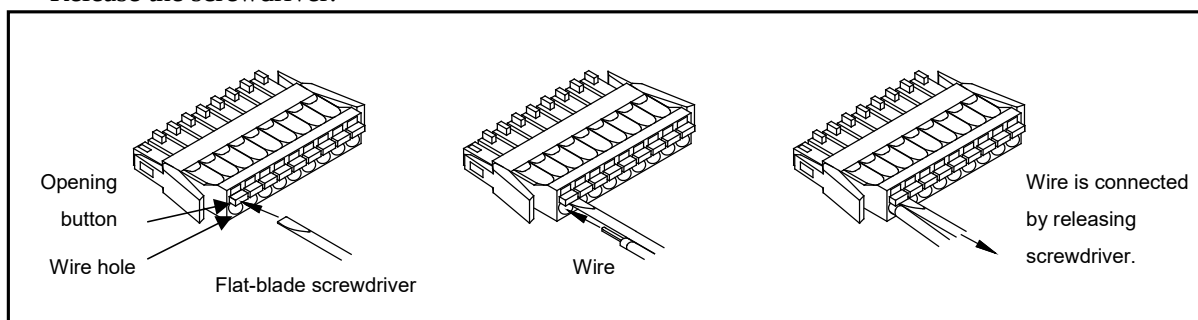
Ferrule is preferable to prevent breaking of wire and short of terminals.

Stranded conductor with ferrule: 0.25-0.75 mm²

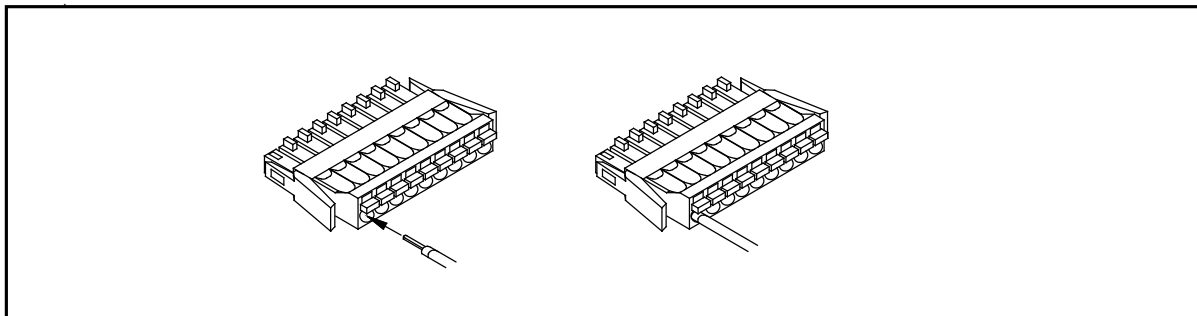
Wiring

There are two ways of wiring.

1. Insert a wire into wire hole with the opening button pushed by a flat-blade screwdriver.
Release the screwdriver.



2. Push a wire into wire hole.



Disconnection

Pull out a wire from wire hole with the opening button pushed by a flat-blade screwdriver.

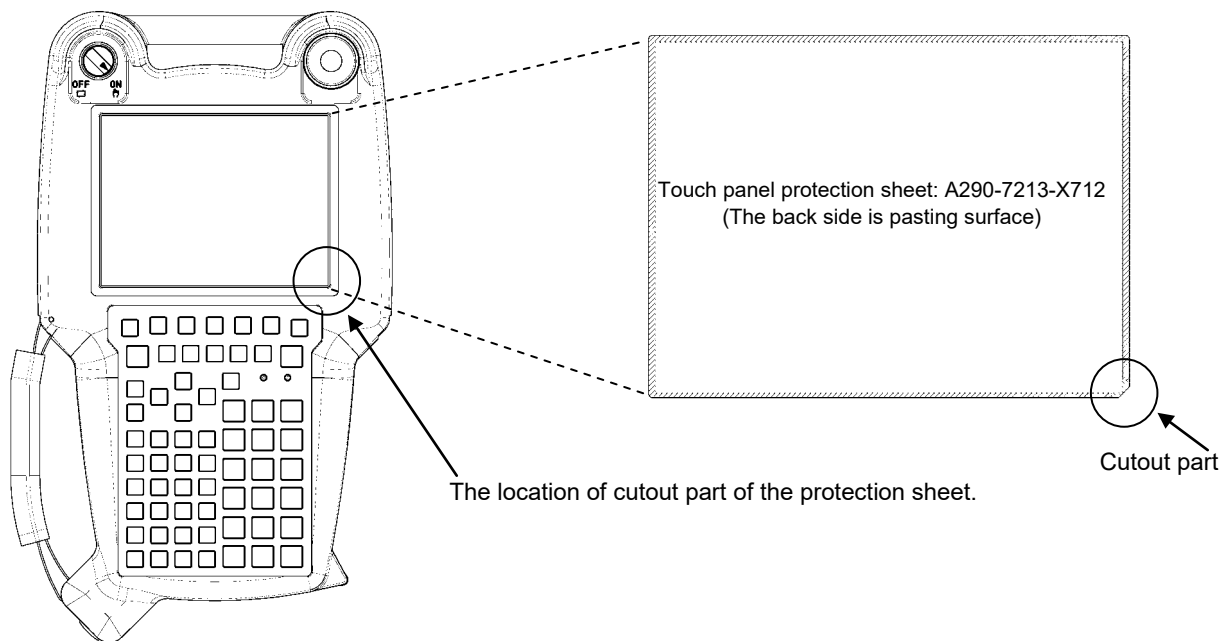
H

REPLACING THE PROTECTION SHEET

This appendix shows an instruction for replacing the protection sheet of the *iPendant* with touch panel.

Replacement procedure

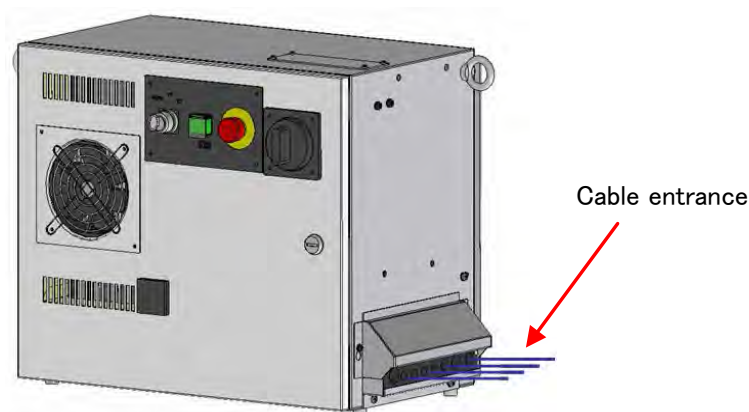
1. Remove old protect sheet.
2. Peel clear sheets pasted on both side of the new protect sheet.
3. Paste the protect sheet so that the cutout part is placed on the lower right portion.



SEALING OF THE CABLE ENTRANCE OF THE CABINET

I.1 CABLE ENTRANCE FOR Mate-CABINET

The external cables are connected through the cable entrance located on the rear surface of controller. The cable entrance consists of a hood, frame and Cable Seal Block with multiple circular shaped holes for cable sealing. There are different types of Cable Seal Block; the type used is determined by the controller and robot configuration. When all customer supplied cables are connected through the cable entrance, it may be necessary to increase the cable diameter of some cables to maintain an adequate seal at the cable entrance. It is also necessary to confirm Cable Seal Block has enough holes for all system and option cables. The number of available sealing holes for customer supplied cables varies depending on robot type and options. Reference the following illustrations.



CAUTION

If the cable diameter is not suitable for the hole size of Cable Seal Block, controller problems may occur because of insufficient environmental sealing of controller.

Without proper cable entrance sealing, airborne contaminants, both non-conductive and conductive may enter the interior of the controller. Foreign particulate entering the controller can have an impact on controller operation and reliability.

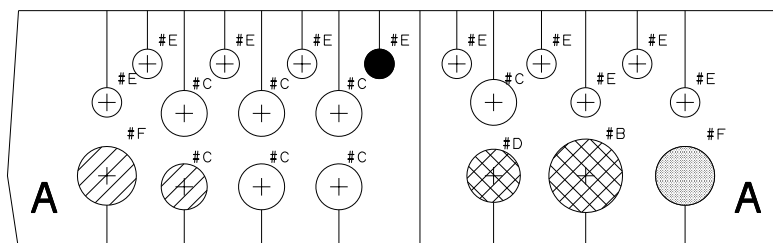
When customer supplied cables are connected through the cable seal entrance, the diameter of these cables should be adjusted to a suitable cable diameter to ensure proper controller cabinet sealing. Reference the following illustrations.

I.2 HOLES OF CABLE SEAL BLOCK FOR CABLE ENTRANCE

The cutout of Cable Seal block at the cable Entrance is shown as follows (Rear side view) .

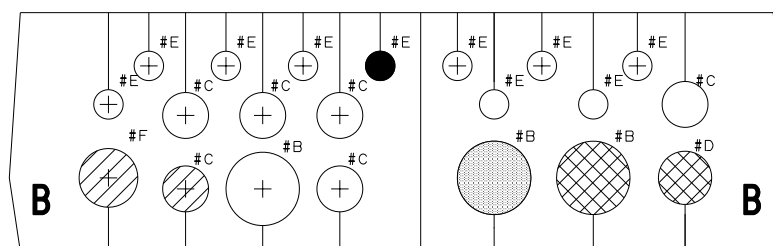
Cable Seal Block Type A (A230-0659-X013#A) :

LR Mate 200iC, LR Mate 200iD, M-1iA



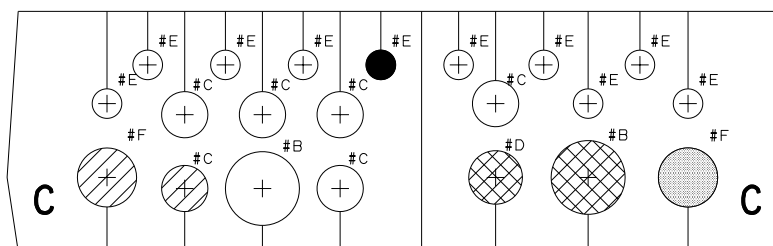
Cable Seal Block Type B (A230-0659-X013#B) :

ARC Mate 100iC, ARC Mate 100iD, ARC Mate 120iC, M-10iA, M-10iD, M-20iA, M-20iB, ARC Mate 0iB



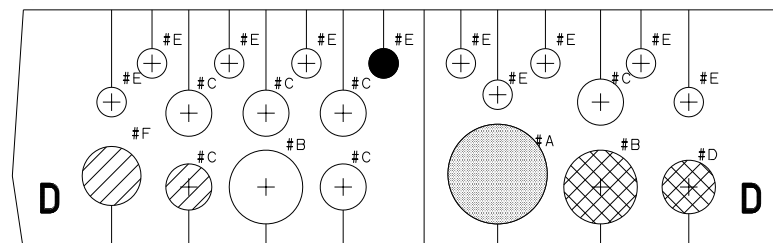
Cable Seal Block Type C (A230-0659-X013#C) :

ARC Mate50iD



Cable Seal Block Type D (A230-0659-X013#D) :

R-2000iC, R-1000iA, M-710iC



Robot power
(RM1)Robot plusecoder
(RP1)

Teach pendant cable



Power cable

Hole type	Diameter (mm)	Cable Seal Block type								Application (Include options)
		Type A		Type B		Type C		Type D		
		Qty	For Options	Qty	For Options	Qty	For Options	Qty	For Options	
#A	ϕ 27	0	0	0	0	0	0	1	0	Power Cable(RM1)
#B	ϕ 20	1	0	3	1	2	1	2	1	Pulse coder cable Thick, Flex(RP1) Power cable(RM1) Aux. cabinet connection cable (AC)
#C	ϕ 12.5	7	6	6	5	6	5	6	5	I/O, Power cable Link tracking, DeviceNet Thick cable Aux. cabinet connection cable (DC)
#D	ϕ 14.5	1	0	1	0	1	0	1	0	Pulse coder cable STD/Non-Flex(RP1)
#E	ϕ 8.5	10	9	10	9	10	9	10	9	Camera, EtherNet DeviceNet Thin cable Teach pendant cable Aux. cabinet connection cable (Optical)
#F	ϕ 16.0	2	0	1	0	2	0	1	0	Power cable, Power cable(RM1)

I.3 SUITABLE CABLE DIAMETER

The suitable cable diameter for option cables are shown on the following table.

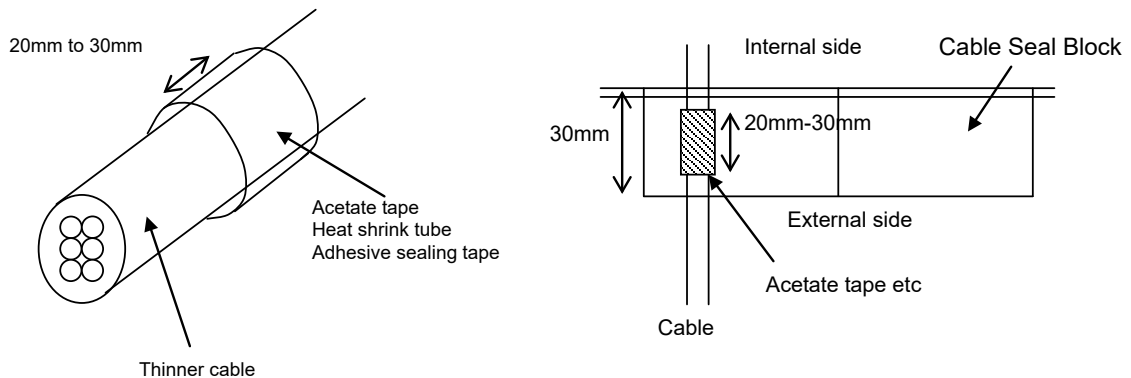
Hole type	Nominal (mm)	Tolerance (mm)	Suitable diameter (mm)	Qty			
				Cable Seal Block Type A	Cable Seal Block Type B	Cable Seal Block Type C	Cable Seal Block Type D
#B	φ 20	±1	φ 19–φ 21	0	1	1	1
#C	φ 12.5	±1	φ 11.5–φ 13.5	6	5	5	5
#E	φ 8.5	±1	φ 7.5–φ 9.5	9	9	9	9

NOTE

The holes for options(#B,#C,#E) are used for all options(Aux. axis, I/O, Network, Sensor). So confirm that the available holes are enough for all option cables.

I.4 ADJUST THE CABLE DIAMETER

To maintain proper sealing of controller enclosure, it may be necessary to adjust some cable diameters to work with an available cable port diameter. If the diameter of any cable is smaller than an available sealing port in the Cable Seal Block, increase the cable diameter to an appropriate diameter by applying acetate tape, adhesive sealing tape or heat shrink tubing over the cable jacket. If a foam type sealing tape is used, adjust the diameter to the compression state of foam for a particular cable diameter. The finished diameter of all cables should maintain an interference fit with Cable Seal Block. All unused cable ports must be plugged to ensure controller is sealed against contaminants. Sealing frame and hood must also be properly installed.



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REVISION RECORD

Edition	Date	Contents
07	Aug.,2017	• Addition of LR Mate 200iC/5WP, M-10iD/12, ARC Mate 100iD, CR-7iA/L, CR-7iA, CR-4iA • Addition of R-30iB Mate Plus • Correction of errors.
06	Mar.,2016	• Addition of M-20iB/25, M-710iC/20M • Addition of figure of process I/O board MA, MB. • Correction of errors.
05	Jul.,2015	• Addition of M-10iA/8L, ARC Mate 100iC/8L • Addition of specification of FROM/SRAM module
04	Apr.,2015	• Addition of M-10iA/7L/12S, M-20iA/12L, ARC Mate 100iC/7L/12S, ARC Mate 120iC/12L, R-2000iC/125L/165R/210R, M-710iC/45M/12L • Correction of errors.
03	Dec.,2013	• Addition of R-2000iC
02	Sep.,2013	• Addition of M-1iA, M-2iA, M-3iA, M-10iA, M-20iA, ARC Mate 50iD, ARC Mate 100iC, ARC Mate 120iC, ARC Mate 0iB, R-0iB, R-1000iA, M-710iC
01	Dec.,2012	

B-83525EN/07

